

Principle of Financial Computing

Kon Yi

September 23, 2026

Contents

1. Introduction	5
1.1. Class Information	5
1.2. Grading	5
1.3. Scope of the Course	6
1.4. Useful Journals	6
1.5. The Modelers' Hippocratic Oath	7
1.6. A First Look at the Markets and the Data	7
1.7. Analysis of Algorithms	11
2. Basic Financial Mathematics	14
2.1. The Time Line	14
2.2. Time Value of Money	14
2.3. Compounding	15
2.4. Present Value	18
2.5. Annuities	20
2.6. Amortization	20
2.7. Yields and the Internal Rate of Return	24
2.8. Numerical Methods for IRRs	25
2.9. Bonds	29
2.10. Day Count Conventions	33
2.11. Full Price, Accrued Interest, and the Clean Price	35
3. Bond Price Volatility	40
3.1. Price Volatility	40
3.2. Price Volatility of a Level-Coupon Bond	41
3.3. Macaulay Duration	42
3.4. What MD Does and Does Not Mean	45
3.5. Conversion to Annual Units	46
3.6. Modified Duration	47
3.7. Effective Duration	48
3.8. The Practices: Duration in Percentage Terms	50
3.9. Hedging	51
3.10. Convexity	52
3.11. Use of Convexity	54
3.12. The Practices: Convexity in Percentage Terms	55
3.13. Effective Convexity	55
3.14. A Numerical Caveat: Second Differences in Floating Point	56
3.15. Interest Rates and Bond Prices: Which Determines Which?	57
4. Term Structure of Interest Rates	59
4.1. What the Term Structure Is	59
4.2. Spot Rates	61
4.3. What Is Wrong with Yield-to-Maturity Discounting	62
4.4. Spot Rate Discount Methodology	63
4.5. Discount Factors	64
4.6. Extracting Spot Rates from the Yield Curve	65
4.7. Some Problems	66
4.8. Yield Spread and Static Spread	68

4.9. Of Spot Rate Curve and Yield Curve	69
4.10. Forward Rates	71
4.11. Spot Rates and Forward Rates	73
4.12. Forward Rates $\equiv$ Spot Rates $\equiv$ Yield Curve	75
4.13. Locking in the Forward Rate	76
4.14. Spot and Forward Rates under Continuous Compounding	78
4.15. Term Structure Theories	81
4.16. Duration, in Practice	85
4.17. Some Loose Ends on Dates	87
5. Fundamental Statistical Concepts	88
5.1. Moments	88
5.2. Correlation	92
5.3. Variance of a Sum	94
5.4. Conditional Expectation	96
5.5. The Normal Distribution	100
5.6. Moment Generating Functions	102
5.7. The Multivariate Normal Distribution	105
5.8. Generation of Univariate Normal Distributions	108
5.9. Generation of Bivariate Normal Distributions	112
5.10. The Lognormal Distribution	114
6. Option Basics	119
6.1. Calls and Puts	119
6.2. Exercise	120
6.3. American and European	121
6.4. Convenient Conventions	121
6.5. Payoff, Mathematically Speaking	122
6.6. Corporate Actions: Dividends and Splits	124
6.7. Short Selling	125
6.8. Covered Positions: Hedges	126
6.9. Covered Positions: Spreads	127
6.10. Covered Positions: Combinations	131
7. Arbitrage in Option Pricing	134
7.1. Arbitrage	134
7.2. The Portfolio Dominance Principle	135
7.3. The PV Formula Justified	135
7.4. Relative Option Prices	136
7.5. Put-Call Parity	137
7.6. Bounds from Parity	138
7.7. Early Exercise of American Calls	139
7.8. A General Result	140
8. Option Pricing Models	141
8.1. The Setting	141
8.2. The Binomial Option Pricing Model	142
8.3. A Call over a Single Period	143
8.4. Where Did the Risk Go?	144
8.5. Pseudo Probability	145
8.6. The Risk-Neutral Probability	145
8.7. The Multi-Period Model	146

8.8. Backward Induction	147
8.9. The Risk-Neutral Pricing Methodology	148
8.10. Self-Financing	149

1. Introduction

— Slides pp. 2–3 —

Lecture 1
September 10,
2026

You must go into finance, Amory.
— F. Scott Fitzgerald (1896–1940),
This Side of Paradise (1920)

*The two most dangerous words in Wall Street
vocabulary are “financial engineering.”*
— Wilbur Ross (2007)

The two epigraphs above already frame the tension this course lives in. Finance is where quantitative modelling pays off most visibly, and it is also where a misplaced faith in a model does the most damage. Our job is to learn the models *and* to keep track of what they assume.

§1.1. Class Information

— Slides pp. 4–6 —

The course follows the instructor’s own textbook, and everything administrative lives on the course web page rather than in these notes.

- **Textbook.** Yuh-Dauh Lyuu, *Financial Engineering & Computation: Principles, Mathematics, Algorithms*. Cambridge University Press, 2002.
- **Official web page.**

www.csie.ntu.edu.tw/~lyuu/finance1.html

- Lecture notes will be uploaded *before* class.
- Homeworks and teaching assistants will also be announced there.
- Do not mistake last year’s homeworks for this year’s!

- **Software.** Some of the software used in the course is collected at

www.csie.ntu.edu.tw/~lyuu/capitals.html

- **Questions.** Please ask many questions in class. In a large class this is the best way for the instructor to remember you.¹

§1.2. Grading

— Slides p. 7 —

Assessment is by **programming assignments** plus **one test**. Every homework is to be treated as an examination, so the academic-integrity rules are strict:

- You are expected to write your own codes and to turn in your source code.
- Do not copy or collaborate with fellow students.

¹“[A] science concentrator [...] said that in his eighth semester of [Harvard] college, there was not a single science professor who could identify him by name.” (*New York Times*, September 3, 2003.)

- Never ask your friends to write programs for you.
- Do not ask AI to code for you.
- Never give your codes to other students or publish your codes.

§1.3. Scope of the Course

— Slides pp. 8–9 —

§1.3.i. What This Course Is About

The course runs from financial theory down to running code, in roughly this order:

- Financial theories in pricing.
- Mathematical backgrounds.
- Derivative securities.
- Pricing models.
- Efficient algorithms in pricing financial instruments.
- Research problems.
- Help in finding your thesis directions.

The last two items are worth emphasising: pricing problems that are merely “computable” are usually not good enough, and the gap between a correct algorithm and a *fast* one is exactly where the open research questions sit.

§1.3.ii. What This Course Is *Not* About

Equally important is what will *not* be covered.

- **How to program.**² Programming ability is assumed, not taught — but it matters. A software bug cost Knight Capital Group, Inc. US\$457.6 million on August 1, 2012.³
- **Basic calculus, probability, combinatorics, and algebra.**
- **Details of the financial markets.**
- **How to be rich.**
- **How the markets will perform tomorrow.**
- **Professional behavior.**

Pricing theory tells you what a security is worth *given* a model of the market. It does not tell you what the market will do next.

§1.4. Useful Journals

— Slides pp. 10–13 —

The literature of the field is spread across finance, mathematics, operations research and computer science journals. The following list is the one given in class.

²See <http://www.csie.ntu.edu.tw/train/>.

³Kirilenko & Lo (2013).

- *Applied Mathematical Finance.*
- *Applied Mathematics and Computations.*
- *Communications on Pure and Applied Mathematics.*
- *Computers & Mathematics with Applications.*
- *European Journal of Finance.*
- *European Journal of Operational Research.*
- *Finance and Stochastics.*
- *Finance Research Letters.*
- *Financial Analysts Journal.*
- *Financial Management.*
- *Insurance: Mathematics and Economics.*
- *International Journal of Finance & Economics.*
- *International Journal of Theoretical and Applied Finance.*
- *Journal of Banking & Finance.*
- *Journal of Computational Finance.*
- *Journal of Derivatives.*
- *Journal of Economic Dynamics & Control.*
- *Journal of Empirical Finance.*
- *Journal of Finance.*
- *Journal of Financial Economics.*
- *Journal of Financial Markets.*
- *Journal of Fixed Income.*
- *Journal of Futures Markets.*
- *Journal of Financial and Quantitative Analysis.*
- *Journal of Portfolio Management.*
- *Journal of Real Estate Finance and Economics.*
- *Journal of Risk and Uncertainty.*
- *Management Science.*
- *Mathematical Finance.*
- *Quantitative Finance.*
- *Review of Financial Studies.*
- *Review of Derivatives Research.*
- *Review of Finance.*
- *Risk Magazine.*
- *SIAM Journal on Financial Mathematics.*
- *Stochastics and Stochastics Reports.*

§1.5. The Modelers' Hippocratic Oath

— Slides p. 14 —

Before writing down a single model, it is worth reading the oath proposed by Emanuel Derman and Paul Wilmott on January 7, 2009, in the wake of the financial crisis.

- I will remember that I didn't make the world, and it doesn't satisfy my equations.
- Though I will use models boldly to estimate value, I will not be overly impressed by mathematics.
- I will never sacrifice reality for elegance without explaining why I have done so.
- Nor will I give the people who use my model false comfort about its accuracy. Instead, I will make explicit its assumptions and oversights.
- I understand that my work may have enormous effects on society and the economy, many of them beyond my comprehension.

Intuition . A model is a deliberate simplification. The discipline is not in believing it, but in knowing precisely *where* it is wrong and *how much* that costs.

§1.6. A First Look at the Markets and the Data

— Slides p. 15 —

Two facts motivate everything that follows. First, the markets we want to price in are enormous, so even small pricing errors translate into large amounts of money. Second, the

price series they produce have a very definite statistical shape, and any pricing model has to be compatible with it.

§1.6.i. Outstanding U.S. Debts

The fixed-income market alone is measured in tens of trillions of dollars, and over the period below it roughly quadrupled. Note in particular how the **mortgage-related** and **asset-backed** columns grow — these are precisely the securities whose pricing requires the algorithmic machinery of this course.

Year 年份	Municipal 市政債	Treasury 美國公債	Mortgage- related 房貸相關	U.S. corporate 公司債	Fed agencies 聯邦機構債	Money market 貨幣市場	Asset- backed 資產擔保	Total 總計
85	859.5	1,437.7	372.1	776.5	293.9	847.0	0.9	4,587.6
86	920.4	1,619.0	534.4	959.6	307.4	877.0	7.2	5,225.0
87	1,010.4	1,724.7	672.1	1,074.9	341.4	979.8	12.9	5,816.2
88	1,082.3	1,821.3	772.4	1,195.7	381.5	1,108.5	29.3	6,391.0
89	1,135.2	1,945.4	971.5	1,292.5	411.8	1,192.3	51.3	7,000.0
90	1,184.4	2,195.8	1,333.4	1,350.4	434.7	1,156.8	89.9	7,745.4
91	1,272.2	2,471.6	1,636.9	1,454.7	442.8	1,054.3	129.9	8,462.4
92	1,302.8	2,754.1	1,937.0	1,557.0	484.0	994.2	163.7	9,192.8
93	1,377.5	2,989.5	2,144.7	1,674.7	570.7	971.8	199.9	9,928.8
94	1,341.7	3,126.0	2,251.6	1,755.6	738.9	1,034.7	257.3	10,505.8
95	1,293.5	3,307.2	2,352.1	1,937.5	844.6	1,177.3	316.3	11,228.5
96	1,296.0	3,459.7	2,486.1	2,122.2	925.8	1,393.9	404.4	12,088.1
97	1,367.5	3,456.8	2,680.2	2,346.3	1,022.6	1,692.8	535.8	13,102.0
98	1,464.3	3,355.5	2,955.2	2,666.2	1,296.5	1,978.0	731.5	14,447.2
99	1,532.5	3,281.0	3,334.2	3,022.9	1,616.5	2,338.2	900.8	16,026.4
00	1,567.8	2,966.9	3,564.7	3,372.0	1,851.9	2,661.0	1,071.8	17,056.1
01	1,688.4	2,967.5	4,125.5	3,817.4	2,143.0	2,542.4	1,281.1	18,565.3
02	1,783.8	3,204.9	4,704.9	3,997.2	2,358.5	2,577.5	1,543.3	20,170.1

Table 1.1.: Outstanding U.S. debts, in billions of U.S. dollars.

§1.6.ii. The Global OTC Derivatives Market

— Slides p. 16 —

Over-the-counter derivatives grew even faster, by close to an order of magnitude between 2000 and the 2008 crisis, after which the notional outstanding plateaued and then declined as trades were compressed and cleared centrally.

Remark 1.6.1. 衍生性商品 (derivative) 是價值取決於某個標的（利率、匯率、股價、信用等）的合約。**場外交易** (over-the-counter, OTC) 指買賣雙方私下議定、不經交易所撮合的交易，與合約標準化、由結算所集中結算的**交易所交易**相對。OTC 合約可客製化（金額、到期日、標的皆可議），代價是雙方直接承擔**對手方風險**；市場由少數大型銀行（dealer）造市，未平倉部位以利率交換（interest rate swap）為大宗。

Figure 1.1 的縱軸為**名目本金** (notional amount) ——僅是計算利息與收益的參考金額，並非真正易手的資金，故市場實際市值遠小於此。2008 年金融危機後，美國 Dodd-Frank 法與歐盟 EMIR 要求標準化的 OTC 商品（尤其信用違約交換 CDS 與利率交換）改採**中央結算** (central clearing)，並藉**壓縮** (compression) 沖銷互相抵銷的部位，名目餘額因而在 2013 年前後見頂並回落。

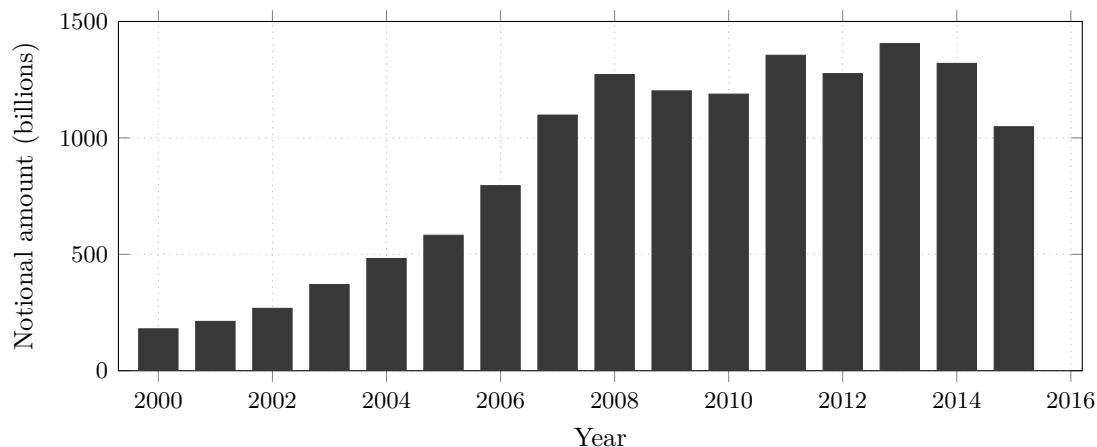


Figure 1.1.: The global OTC derivatives market. Bar heights are read off the chart shown in class.

§1.6.iii. The S&P 500 Index and Its Returns

— Slides pp. 17–19 —

Robert Shiller⁴ maintains a monthly series of the S&P 500 index reaching back to 1871. Plotted in levels (Figure 1.2), it is dominated by the post-1980 growth; the 1929 crash is barely visible on this scale.

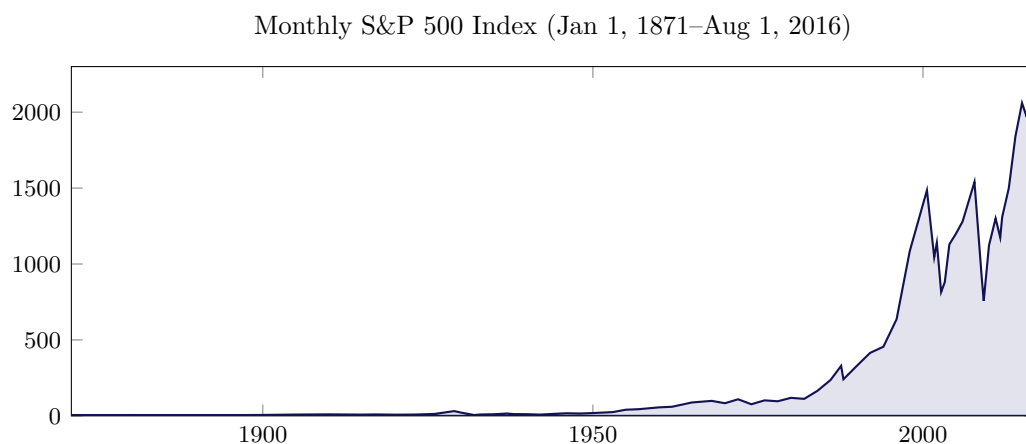


Figure 1.2.: The S&P 500 index in levels, reconstructed from the Shiller series.

Levels are not what we model, however. Prices are non-stationary and their scale is arbitrary, so we work with the **log return** over one period,

$$r_t = \ln \frac{S_t}{S_{t-1}},$$

where S_t is the index level at time t . Figure 1.3 plots the monthly log returns of the same series.

Remark 1.6.2. 為什麼不直接對價位建模？價位（levels）是**非定態**的——統計性質隨時間漂移，1900 年與 2000 年的「正常水準」天差地遠，無法用一組固定參數描述；而且指數從幾點起編是人為選擇，尺度本身沒有經濟意義。改用**對數報酬** $r_t = \ln(S_t/S_{t-1})$ 一次補齊三件事：

- S_t/S_{t-1} 是本期的成長倍數（漲 2% 即 1.02）。變化小時 $\ln(1+x) \approx x$ ，所以 r_t 近似百分比變化，且與絕對點數無關。

⁴Co-winner of the 2013 Nobel Prize in Economic Sciences.

- **可加性**：跨期報酬 $\ln(S_t/S_{t-k}) = r_t + r_{t-1} + \cdots + r_{t-k+1}$ ，相加即可，不必連乘。
- 報酬率序列大致定態——繞著 0 波動、沒有趨勢，正適合建模。

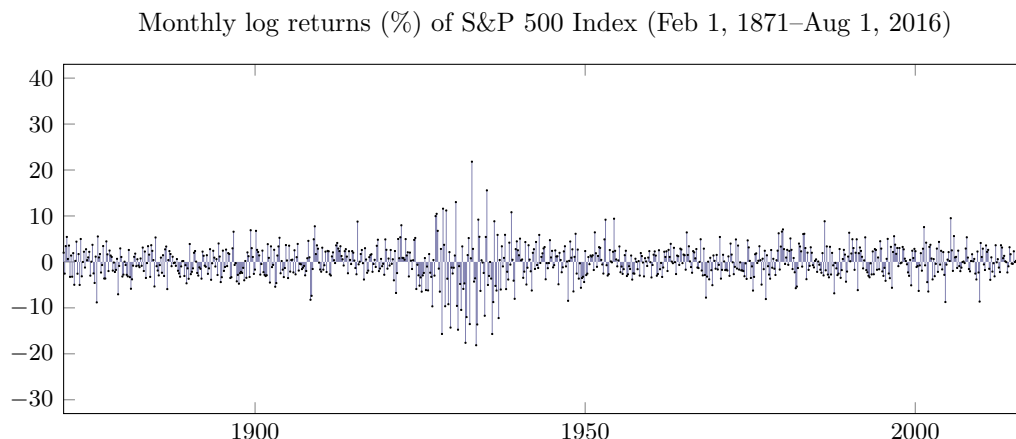


Figure 1.3.: Monthly log returns of the S&P 500 index. The series is a schematic reproduction of the plot shown in class: it reproduces the mean, the typical amplitude, the fat tails and the volatility cluster of the early 1930s, not the individual monthly observations.

Three features stand out, and they will constrain every model we build:

- The returns hover around zero and show no obvious trend, in sharp contrast with the levels.
- Their amplitude is *not* constant: quiet stretches alternate with violent ones (**volatility clustering**), the early 1930s being the extreme case.
- Large moves are far more frequent than a normal distribution would allow.

The last point is easiest to see from the empirical density (Figure 1.4): the body is a sharp peak near zero, but the tails stay noticeably above zero out to $\pm 30\%$.

Remark 1.6.3. 兩個關鍵術語：**波動聚集** (volatility clustering) 指抖動的幅度會成群出現——大波動旁邊通常還是大波動，平靜期也連成一片（1930 年代初最極端），也就是說波動率本身不是常數。**厚尾** (fat tails) 指極端漲跌遠比常態分佈預測的頻繁。看 Figure 1.4：曲線畫的是每月報酬率的出現頻率，中央為 0 附近又高又尖的峰（多數月份幾乎不動），但兩側尾巴延伸到 $\pm 30\%$ 仍明顯高於零，而常態分佈在那裡的機率早已可忽略。這種「尖峰厚尾」的形狀，正是接下來那句結論的依據。

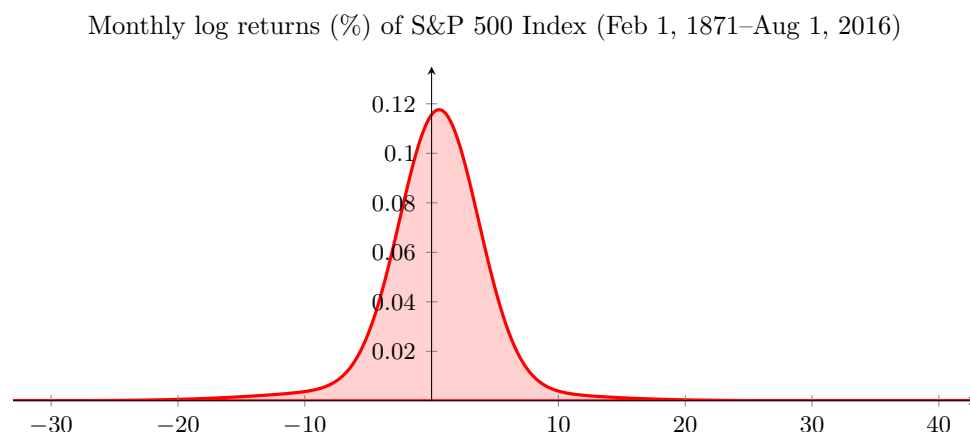


Figure 1.4.: Distribution of the monthly log returns of the S&P 500 index: a sharp central peak with tails much heavier than the normal.

Prices are not normally distributed, and their volatility is not constant. Any model that assumes otherwise is buying tractability with realism.
價格並非常態分佈，波動率也非定值；凡是假設相反的模型，都是以寫實性為代價，換取數學上的易處理性。

§1.7. Analysis of Algorithms

— Slides pp. 20–21 —

*I can calculate the motions
of the heavenly bodies,
but not the madness of people.*
— Isaac Newton (1642–1727)

*It is unworthy of excellent men
to lose hours like slaves
in the labor of computation.*
— Gottfried Wilhelm Leibniz (1646–1716)

Having seen how large the markets are and how awkward their data is, we now fix the vocabulary for the other half of the course: what it means for a pricing problem to be solvable, and at what cost.

§1.7.i. Computability and Algorithms

— Slides p. 22 —

Definition 1.7.1. An **algorithm** is a precise procedure that can be turned into a computer program.

Not every question admits one. **Uncomputable** problems are those for which no algorithm exists at all, no matter how much time we are willing to spend. Two standard examples — both, uncomfortably, about programs themselves:

- Does this program have infinite loops?
- Is this program bug free?

Everything else is **computable**, and computable problems split again according to how expensive the algorithm is: **intractable** problems admit algorithms in principle but not ones we can afford to run, while **tractable** problems admit algorithms that are practical at the sizes we care about. [Figure 1.5](#) summarises the picture.

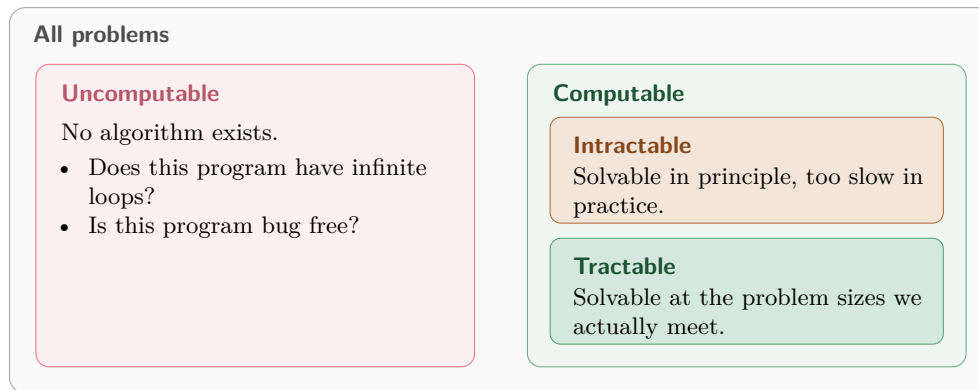


Figure 1.5.: The classification of problems. This course lives almost entirely in the bottom-right box — and much of the research effort consists of moving pricing problems into it.

Remark 1.7.2. The conventional formal criterion is how running time scales with the input size n . A problem is counted *tractable* when it admits an algorithm of *polynomial* time, $O(n^k)$ for some fixed k , and *intractable* when it admits none — every algorithm needs superpolynomial time, in most cases exponential. The rationale is practical: a polynomial bound keeps the cost manageable as n grows, whereas anything superpolynomial blows up.

§1.7.ii. Complexity

— Slides p. 23 —

To measure cost without committing to a machine or a language, we count operations rather than seconds.

Definition 1.7.3. A set of basic operations — $+$, $-$, $\times$, $/$, $\log$, x^y , e^x , $\dots$ — are assumed to take one unit of time each. The total number of such operations performed is the total work done by an algorithm, its **computational complexity**. The amount of memory space it uses is its **space complexity**.

Remark 1.7.4. Two conventions are worth making explicit.

- We concentrate on the *abstract* complexity of an algorithm instead of its detailed implementation. Constant factors from the language, compiler and hardware are deliberately ignored.
- Despite that, complexity is a good guide to an algorithm's *actual* running time. It is the coarse measure that survives a change of machine.

§1.7.iii. Common (Asymptotic) Complexities

— Slides p. 24 —

Let n stand for the “size” of the problem — the number of elements, the number of cash flows, the number of time periods, and so on. Complexity is then expressed as a function of n , and the standard names are as follows.

Name	Complexity	Remark
Linear	$O(n)$	
Quadratic	$O(n^2)$	
Cubic	$O(n^3)$	
Superpolynomial	e.g. $2^{O(\sqrt{n})}$	higher than any polynomial ⁵
Exponential time	$2^{O(n)}$	the usual worst case

Table 1.2.: Common asymptotic complexities, in increasing order of growth.

Note 1.7.5. The superpolynomial entry is not a curiosity. Several pricing problems that were only known to admit exponential-time algorithms have been pushed down to $2^{O(\sqrt{n})}$ — an improvement large enough to turn an unusable algorithm into a usable one, and precisely the kind of result this course aims at.

⁵E.g., T. Dai (B82506025, R86526008, D8852600) & Lyuu (2007); Lyuu & C. Wang (F95922018) (2011); H. Chiu (R98723059) (2012).

2. Basic Financial Mathematics

— Slides pp. 25–26 —

*In the fifteenth century
mathematics was mainly concerned with
questions of commercial arithmetic and
the problems of the architect.*

— Joseph Alois Schumpeter (1883–1950)

*I'm more concerned about
the return of my money than
the return on my money.*

— Will Rogers (1879–1935)

Everything in this chapter follows from one idea: a dollar today and a dollar next year are different goods, and the interest rate is the exchange rate between them. Once that exchange rate is pinned down, valuing a stream of payments is arithmetic — and, as we will see, arithmetic whose cost we can still argue about.

§2.1. The Time Line

— Slides p. 27 —

Before any formula, fix the convention for talking about time. Time is measured in **periods**; time 0 is now, and period k is the interval running from time $k - 1$ to time k . A cash flow “at time k ” arrives at the *end* of period k .

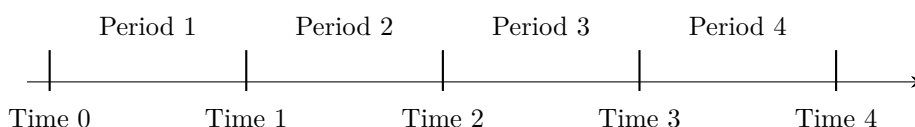


Figure 2.1.: The time line. Period k spans time $k - 1$ to time k .

§2.2. Time Value of Money

— Slides p. 28 —

Definition 2.2.1. Let r be the **interest rate** per period. The **future value** (FV) and **present value** (PV) of a single amount over n periods are related by

$$\begin{aligned} \text{FV} &= \text{PV}(1 + r)^n, \\ \text{PV} &= \text{FV} \times (1 + r)^{-n}. \end{aligned} \tag{2.1}$$

Intuition . $(1 + r)^n$ carries money forward in time and $(1 + r)^{-n}$ carries it backward. *Discounting* — the second operation — is the one we use constantly, because a security's price is quoted today while its payoffs arrive later.

$(1 + r)^n$ 把錢往未來搬， $(1 + r)^{-n}$ 把錢往現在搬。我們不斷用到的是後者——折現 (discounting) ——因為證券的價格報在今天，收益卻在日後才到。

The idea is old: it goes back to Fibonacci (1170–1240), and was put on a modern footing by Irving Fisher (1867–1947).

§2.3. Compounding

— Slides pp. 29–30 —

Equation (2.1) is stated per *period*, but rates are quoted per *year*. Reconciling the two is what compounding conventions are for — and getting the convention wrong is a classic source of pricing errors.

§2.3.i. Periodic Compounding

Suppose the *annual* interest rate r is compounded m times per annum. Each compounding period earns r/m , so one dollar grows along

$$1 \rightarrow \left(1 + \frac{r}{m}\right) \rightarrow \left(1 + \frac{r}{m}\right)^2 \rightarrow \left(1 + \frac{r}{m}\right)^3 \rightarrow \dots$$

Hence, after n years,

$$\text{FV} = \text{PV} \left(1 + \frac{r}{m}\right)^{nm}. \quad (2.2)$$

§2.3.ii. Common Compounding Methods

Method	m		
Annual	1	Weekly	52
Semiannual	2	Daily	365
Quarterly	4	Continuous	∞
Monthly	12		

§2.3.iii. Translating between Rate and Frequency

— Slides pp. 31–32 —

An annual interest rate of r compounded m times a year is “equivalent to” an interest rate of r/m per $1/m$ year. Read in reverse: if a loan asks for a return of 1% *per month*, the *annual* interest rate will be 12% with monthly compounding.

Example 2.3.1. The annual interest rate is 10% compounded twice per annum. Each dollar will grow to be

$$\left[1 + (0.1/2)\right]^2 = 1.1025$$

one year from now. The rate is therefore “equivalent to” an interest rate of 10.25% compounded once *per annum*, since

$$1 + 0.1025 = 1.1025.$$

The 10.25% in the example is the **effective annual rate**: the single annually-compounded rate that produces the same money after one year. The quoted 10% is only a **nominal rate**, and by itself it does not determine anything.

An interest rate is meaningless without the compounding frequency.

§2.3.iv. The Rule of 72

— Slides pp. 33–35 —

Let the annual interest rate be r with annual compounding. How many years T will it take for your money to double? The identity to solve is

$$(1+r)^T = 2, \quad \text{so} \quad T = \frac{\ln 2}{\ln(1+r)}.$$

Is there an easier way? The **rule of 72** is a heuristic that estimates T without a logarithm:

$$T \approx \frac{72}{r \text{ (\%)}}. \quad (2.3)$$

So it takes about $72/12 = 6$ years to double the GDP if the annual growth rate is 12%.

Reason. Expanding $\ln(1+r) = r - r^2/2 + O(r^3)$ gives, for small r ,

$$\frac{1}{\ln(1+r)} \approx \frac{1}{r(1-r/2)} \approx \frac{1}{r} + \frac{1}{2},$$

and therefore

$$\frac{\ln 2}{\ln(1+r)} \approx \frac{0.693}{r} + 0.3466. \quad \blacksquare$$

Remark 2.3.2. The leading constant is 0.693, i.e. 69.3 when r is measured in percent — so a “rule of 69.3” is the asymptotically correct one. The extra +0.3466 is a positive offset, so inflating the numerator to 72 compensates for it over the range of rates one actually meets. That 72 is also divisible by 2, 3, 4, 6, 8, 9, 12 is a large part of why it, rather than 70, became the folk rule.

Figure 2.2 plots the error of both variants: the true interest rate minus the approximation, in percent, against the number of years to double.

Remark 2.3.3. 這張圖把 Eq. (2.3) 反過來用：平常是「給利率 r ，估翻倍要幾年 T 」，這裡改成「給橫軸的 T （想要幾年翻倍），估需要多少利率」——把公式解成 $r \approx 72/T$ 。縱軸畫的是兩個利率估計值的差：真正在 T 年翻倍所需的 $r_{\text{真}} = 2^{1/T} - 1$ ，減去 Rule of 72 反解出的 $r_{\text{估}} = 72/T$ （兩者都乘 100 化成百分比）。所以縱軸單位是「利率的百分點」，不是「年」；橫軸的 T 在這裡是已知條件，不是被估計的量。

代入數字比較有感覺： $T = 1$ 年時，真正利率 100% 對 Rule of 72 的 72%，差 28 個百分點，極不準； $T = 8$ 年時，真正利率約 9.05% 對 9%，差只有 0.05，正是 caption 說的「超過約 8 年後極準」； $T = 20$ 年時差變成負的（約 -0.07 ），代表 Rule of 72 在長期會略為高估所需利率（“overshoot”）。

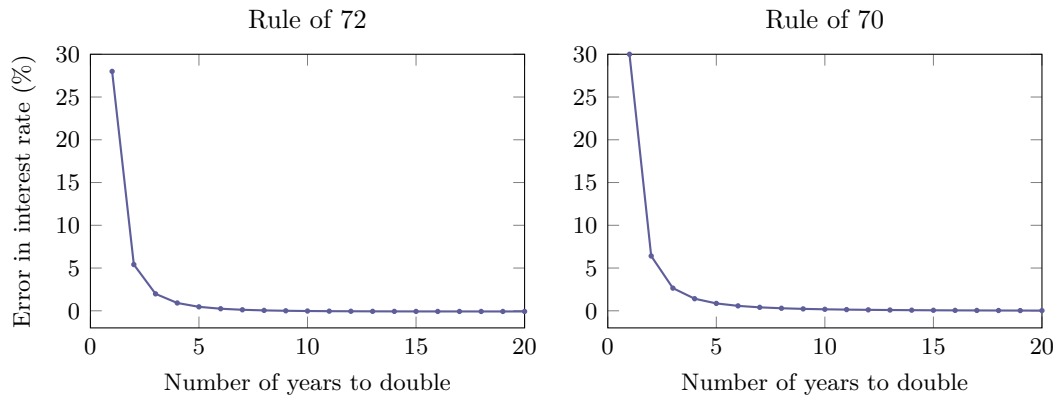


Figure 2.2.: How good is the rule of 72? Plotted is the true interest rate minus the approximation (in %). Both are badly off for very short doubling times and excellent beyond about eight years; the rule of 72 slightly *overshoots* there, the rule of 70 slightly *undershoots*.

§2.3.v. Continuous Compounding

— Slides pp. 36–37 —

Let $m \rightarrow \infty$ in Eq. (2.2), so that

$$\left(1 + \frac{r}{m}\right)^m \rightarrow e^r.$$

Then

$$FV = PV \times e^{rn}, \quad (2.4)$$

where $e = 2.71828\dots$ ¹

Continuous compounding is easier to work with, because the exponents simply add. Suppose the annual interest rate is r_1 for n_1 years and r_2 for the following n_2 years. Then the FV of one dollar will be

$$e^{r_1 n_1 + r_2 n_2}$$

after $n_1 + n_2$ years — no nested products, just a sum in the exponent.

§2.3.vi. Conversion between Compounding Methods

— Slides pp. 38–40 —

The word “equivalent” has already been used twice in quotation marks. Here is what it means.

Definition 2.3.4 (Equivalent rates). The **accumulation factor** $A(n)$ of a quoting convention is the amount one dollar grows to under it after n years:

$$A(n) = \begin{cases} e^{rn}, & r \text{ compounded continuously,} \\ \left(1 + \frac{r}{m}\right)^{mn}, & r \text{ compounded } m \text{ times per annum.} \end{cases}$$

Two quoted annual rates are **equivalent** if they turn one dollar into the same amount of money at every horizon — that is, if their accumulation factors agree as functions of n .

¹Jacob Bernoulli (1654–1705) in 1685.

Note 2.3.5. Checking one year is enough. Each accumulation factor above is a pure power of its own one-year factor, $A(n) = A(1)^n$, so $A_1(1) = A_2(1)$ already forces $A_1(n) = A_2(n)$ for every $n \geq 0$. Equivalently: two rates are equivalent exactly when they share the same **effective annual rate** $A(1) - 1$ of [Section 2.3.iii](#).

Let r_1 be the annual rate with continuous compounding and r_2 the *equivalent* rate compounded m times per annum. How are they related?

Proposition 2.3.6

The principle is that both interest rates must produce the same amount of money after one year, that is,

$$\left(1 + \frac{r_2}{m}\right)^m = e^{r_1}.$$

Therefore^a

$$\begin{aligned} r_1 &= m \ln\left(1 + \frac{r_2}{m}\right), \\ r_2 &= m(e^{r_1/m} - 1). \end{aligned}$$

^aAre they really equivalent? In what sense are they equivalent? Contributed by Mr. Chen, Tung-Li (D09922014) on February 24, 2023.

The same principle handles two periodic conventions. Suppose r_1 is the annual rate compounded m_1 times per annum and r_2 is the *equivalent* rate compounded m_2 times per annum. Then

$$\left(1 + \frac{r_1}{m_1}\right)^{m_1} = \left(1 + \frac{r_2}{m_2}\right)^{m_2}. \quad (2.5)$$

Observation . Every conversion in this subsection is the same one-line demand: *equal value after one year*. There is nothing to memorise beyond that.

§2.4. Present Value

— Slides p. 41 —

§2.4.i. The PV Formula

Definition 2.4.1. The PV of the cash flow $C_1, C_2, \dots, C_n$ at times $1, 2, \dots, n$ is

$$PV = \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \dots + \frac{C_n}{(1+y)^n}, \quad (2.6)$$

where y is the discount rate per period.

Remark 2.4.2. 我們想要衡量一筆包含未來的錢的現金流折算到現在價值多少，這條公式就是把單筆折現 $PV = FV(1+y)^{-n}$ 對每一筆現金流各做一次，再加起來：在 time k 收到的 C_k 要往回搬 k 期，所以除以 $(1+y)^k$ ——越晚到的錢折得越兇——加總便是整串現金流今天的總值。 y 是每期折現率，字母取自 *yield*，角色與前一章的 r 相同。

This formula and its variations are the engine behind most of financial calculations.² The difficulty is never the algebra; it is deciding what goes into it:

²Cochrane (2005), “Asset pricing theory all stems from one simple concept [...]: price equals expected discounted payoff.”

- What is y ?
- What are the C_i ?
- What is n ?

Note 2.4.3. Equation (2.6) is taken on faith for now. It will be justified later in the course from “sound” principles rather than assumed — see [Theorem 7.3.1](#), where the only principle used is that there is no free lunch.

§2.4.ii. Evaluating the PV Formula

— Slides pp. 42–44 —

Even a formula this simple raises the question the previous chapter set up: how much work does it actually take? The direct transcription is [Algorithm 2.1](#).

Algorithm 2.1: Evaluating the PV in Eq. (2.6) directly

```

1  $x := 0$ ;
2 for  $i = 1, 2, \dots, n$  do
3    $x := x + C_i / (1 + y)^i$ ;
4 return  $x$ ;
```

Computing $(1 + y)^i$ by repeated multiplication costs i operations, so the algorithm takes time proportional to $\sum_{i=1}^n i = O(n^2)$ if only $+$, $-$, $\times$ and $/$ are allowed. It can be improved to $O(n)$ by applying $a^b = e^{b \ln a}$ in step 3, since we agreed to count x^y as taking one unit of time.

But there is a better route, which needs no exponentiation at all ([Algorithm 2.2](#)).

Algorithm 2.2: Evaluating the PV by Horner’s rule

```

1  $x := 0$ ;
2  $d := 1 + y$ ;
3 for  $i = n, n - 1, \dots, 1$  do
4    $x := (x + C_i) / d$ ;
5 return  $x$ ;
```

Proposition 2.4.4 (Horner’s rule)

The underlying identity is

$$\left(\cdots \left(\left(\frac{C_n}{1+y} + C_{n-1} \right) \frac{1}{1+y} + C_{n-2} \right) \frac{1}{1+y} + \cdots \right) \frac{1}{1+y},$$

due to Horner (1786–1837) in 1819.

The algorithm takes $O(n)$ time, and it is the most efficient possible in terms of the absolute number of arithmetic operations.³ It can also be computed in parallel in $O(\log n)$ time using prefix sums.⁴

Two algorithms for the same formula, $O(n^2)$ against $O(n)$ — and the fast one is also the numerically simpler one. This is the pattern the whole course chases.

³A. Borodin & Munro (1975).

⁴Contributed by Mr. Chen, Yi-Feng (R08922077) on February 24, 2021.

§2.5. Annuities

— Slides pp. 45–47 —

§2.5.i. Annuities Certain

Definition 2.5.1. An **annuity** pays out the same C dollars at the end of each year for n years.

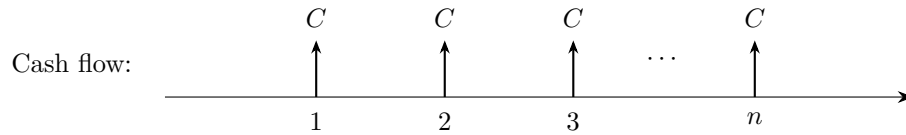


Figure 2.3.: The cash flow of an n -year annuity certain.

With a rate of r , specialising Eq. (2.6) to $C_i \equiv C$ and summing the geometric series gives

$$\text{PV} = \sum_{i=1}^n C(1+r)^{-i} = C \frac{1 - (1+r)^{-n}}{r},$$

while the FV at the end of the n th year is

$$\text{FV} = \sum_{i=0}^{n-1} C(1+r)^i = C \frac{(1+r)^n - 1}{r}.$$

The study of these sums goes back to Jan de Witt (1625–1672) in 1671 and Nicholas Bernoulli (1687–1759) in 1709.

§2.5.ii. General Annuities

Suppose instead that m payments of C dollars each are received per year (the **general annuity**), and let r be compounded m times per annum.⁵ Then the same geometric sum with period rate r/m and nm periods gives

$$\begin{aligned} \text{PV} &= \sum_{i=1}^{nm} C \left(1 + \frac{r}{m}\right)^{-i} = C \frac{1 - \left(1 + \frac{r}{m}\right)^{-nm}}{\frac{r}{m}}, \\ \text{FV} &= \sum_{i=0}^{nm-1} C \left(1 + \frac{r}{m}\right)^i = C \frac{\left(1 + \frac{r}{m}\right)^{nm} - 1}{\frac{r}{m}}. \end{aligned} \tag{2.7}$$

Remark 2.5.2. Annuity is paid at the end of every year, so note the indices of above formulas.

§2.6. Amortization

— Slides pp. 48–49 —

⁵If your r has a compounding frequency different from m , convert it with Eq. (2.5) first.

§2.6.i. The Amortization Method

Definition 2.6.1. **Amortization** is a method of repaying a loan through regular payments of interest *and* principal.

The mechanics are worth stating carefully, because every line of the schedule below follows from them:

- The size of the loan (the original balance) is reduced by the *principal* part of each payment.
- The *interest* part of each payment pays the interest incurred on the *remaining* principal balance.
- As the principal gets paid down over the term of the loan, the interest part of the payment diminishes.

An amortized loan is exactly a general annuity seen from the lender's side: the loan amount is the PV of the payment stream. Inverting Eq. (2.7) therefore gives the regular payment,

$$C = \text{loan amount} \times \frac{\frac{r}{m}}{1 - \left(1 + \frac{r}{m}\right)^{-nm}}. \quad (2.8)$$

For monthly payments, take $m = 12$.

Remark 2.6.2. 每期還款總金額一樣，只是把每期還款拆賬成上期剩餘本金造成的利息跟一部分本金。

Remark 2.6.3. 初看之下兩者的計息方式並不一樣：**攤還** (amortization) 貸款的利息是對剩餘本金計算的，而**年金** (annuity) 公式 (2.7) 裡根本沒有出現「餘額」，只是把每期的 C 折現後相加。兩者一致的理由是：折現本身就是「利息只算到該筆錢實際在外流通為止」的數學編碼。以下三點說的是同一件事。

(a) **這是恆等式，不是類比**。記 $i \triangleq r/m$ 、 $N \triangleq nm$ ，並令 B_k 為第 k 期付款後的剩餘本金。「利息按剩餘本金計、付款先付息再沖本」這句話的全部內容就是遞迴

$$B_k = B_{k-1}(1+i) - C, \quad B_0 = \text{loan amount}.$$

把它改寫成 $B_{k-1} = (C + B_k)/(1+i)$ 並反覆代入自己，得

$$B_0 = \sum_{k=1}^N \frac{C}{(1+i)^k} + \frac{B_N}{(1+i)^N}.$$

而 *fully amortized* 的意思正是 $B_N = 0$ ，末項消失，於是貸款額 = 付款流的**現值** (present value)，即 (2.7)。年金並不是用來近似攤還貸款的模型，它就是那條餘額遞迴式的封閉解。

(b) **折現如何「知道」餘額在變**。把 B_0 拆成 N 筆互不相干的零息小貸款，第 k 筆的金額恰為 $C(1+i)^{-k}$ 。它獨自滾 k 期複利後成為 $C(1+i)^{-k}(1+i)^k = C$ ，剛好被第 k 期的付款一次結清，此後即離開帳本、不再生息。每一筆都只在自己尚未被償還的期間生息，合起來就是「利息只對剩餘本金計算」。關鍵在於折現因子的指數是該筆付款自己的期數 k 而非全期 N ——已經還掉的錢不再出現在後面各項裡。

(c) **兩個假設要分開**。(a) 的恆等式對任意付款序列 $C_1, \dots, C_N$ 都成立，此時 $B_0 = \sum_k C_k(1+i)^{-k}$ ；「餘額計息」本身只保證這件事，與付款是否等額無關。真正額外用到的假設是等額付款 ($C_k \equiv C$)，它使上式成為等比級數、能求和成封閉解 (2.7)，我們也才能反解出 (2.8)。

§2.6.ii. Example: Home Mortgage

— Slides pp. 50–54 —

By paying down the principal consistently, the risk to the lender is lowered, and when the borrower sells the house, only the remaining principal is due the lender. Consider the equal-

payment case, i.e., fixed-rate, level-payment, fully amortized mortgages — these are called traditional mortgages in the U.S.⁶

Remark 2.6.4. 上段中譯：藉由持續償還本金，放款人承擔的風險逐步降低；當借款人賣掉房子時，只需清償當時的剩餘本金即可。以下只考慮等額付款的情形，也就是 **固定利率** (fixed-rate)、**等額付款** (level-payment)、**完全攤還** (fully amortized) 的房貸——這類房貸在美國被稱為 **傳統房貸** (traditional mortgage)。

註腳中譯：《經濟學人》(2016)：「多數國家的銀行以短期或浮動利率房貸來壓低自身風險；美國的借款人則享有較好的條件——便宜的 30 年期固定利率房貸，而且可以免費提前清償（即無提前清償違約金）。」

Example 2.6.5. Consider a 15-year, \$250,000 loan at 8.0% interest rate. Solve Eq. (2.7) with $PV = 250000$, $n = 15$, $m = 12$ and $r = 0.08$. That is,

$$250000 = C \frac{1 - \left(1 + \frac{0.08}{12}\right)^{-15 \times 12}}{\frac{0.08}{12}},$$

which gives a monthly payment of $C = 2389.13$.

Month	Payment	Interest	Principal	Remaining principal
				250,000.000
1	2,389.13	1,666.667	722.464	249,277.536
2	2,389.13	1,661.850	727.280	248,550.256
3	2,389.13	1,657.002	732.129	247,818.128
		...		
178	2,389.13	47.153	2,341.980	4,730.899
179	2,389.13	31.539	2,357.591	2,373.308
180	2,389.13	15.822	2,373.308	0.000
Total	430,043.438	180,043.438	250,000.000	

Table 2.1.: The amortization schedule of the 15-year, \$250,000 loan at 8%.

In every month:

- The principal and interest parts add up to \$2,389.13.
- The remaining principal is reduced by the amount indicated under the **Principal** heading.⁷ The **Principal** column forms a geometric sequence.⁸
- The interest is computed by multiplying the remaining balance of the previous month by $0.08/12$.

Note also that:

- The **Principal** column adds up to \$250,000 — it must, since the loan is fully amortized.
- The **Payment** column adds up to \$430,043.438! The difference, \$180,043.438, is the total interest paid over 15 years.
- If the borrower plans to pay off the mortgage right *after* the 178th month's monthly payment, he needs to pay another \$4,730.899.⁹

⁶The *Economist* (2016), "In most countries banks minimize their risk by offering short-term or floating-rate mortgages. American borrowers get a better deal: cheap 30-year fixed-rate mortgages that can be repaid early free."

⁷This column varies with r . Thanks to a lively class discussion on Feb 24, 2010. In fact, every column varies with r . Contributed by Ms. Wu, Japie (R01921056) on February 20, 2013.

⁸Contributed by Mr. Sun, Ao (R05922147) on February 22, 2017.

⁹Contributed by Ms. Wu, Japie (R01921056) on February 20, 2013.

Figure 2.4 shows the third bullet of Table 2.1 graphically: the payment is level, but its composition rotates from almost all interest to almost all principal.

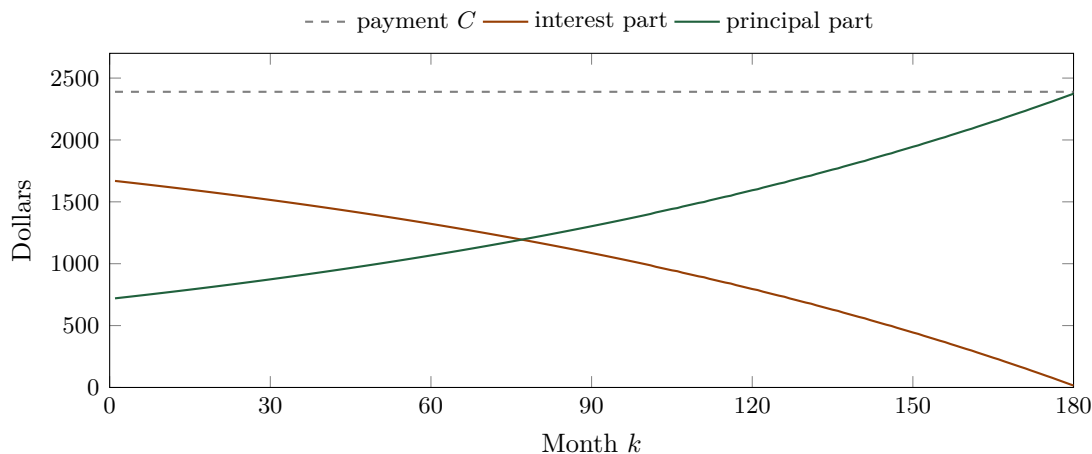


Figure 2.4.: Composition of the level monthly payment over the life of the loan, computed from the example above. The two parts always sum to C , and the principal part is a geometric sequence with ratio $1 + r/m$.

§2.6.iii. Calculating the Remaining Principal

— Slides pp. 55–57 —

Two ways to answer “how much is still owed after k payments?”

Method 1: run the schedule

- A month’s principal payment = monthly payment – (previous month’s remaining principal) $\times$ (monthly interest rate).
- A month’s remaining principal = previous month’s remaining principal – principal payment calculated above.
- Generate the amortization schedule until the particular month you are interested in.

This method is relatively slow — it costs $O(k)$ work — but it is universal in its applicability. It can, for example, accommodate prepayments and floating interest rates, neither of which has a closed form.

Method 2: discount the remaining cash flows

Right after the k th payment, the remaining principal is the PV of the future $nm - k$ cash flows because we have

$$B_{k-1}(1 + i) - C = B_k$$

and we can solve this to obtain

$$B_k = \sum_{i=1}^{nm-k} C \left(1 + \frac{r}{m}\right)^{-i} = C \frac{1 - \left(1 + \frac{r}{m}\right)^{-nm+k}}{\frac{r}{m}}. \quad (2.9)$$

This method is much faster — $O(1)$ instead of $O(k)$ — but it is limited in applications because it makes more assumptions: a level payment, a fixed rate, and no prepayment.

Exercise 2.6.6. Use Eq. (2.9) together with Eq. (2.8) to prove that the Principal column of Table 2.1 forms a geometric sequence.

The same trade-off as in Algorithm 2.2: the closed form is faster, the loop is more general. Which one you may use is decided by the contract, not by the mathematics.

§2.7. Yields and the Internal Rate of Return

— Slides pp. 58–59 —

So far the rate was given and we computed a value. The rest of the chapter runs the other way: the price is observed in the market, and we solve for the rate it implies.

§2.7.i. Bond Equivalent Yield and Mortgage Equivalent Yield

The term **yield** denotes the return of investment. Two widely used yields are the **bond equivalent yield** (BEY) and the **mortgage equivalent yield** (MEY). Recall Eq. (2.2), $FV = PV(1 + \frac{r}{m})^{nm}$:

- BEY corresponds to the r above that equates PV with FV when $m = 2$.
- MEY corresponds to the r above that equates PV with FV when $m = 12$.

Observation . BEY and MEY may differ numerically, but they refer to the same yield — they are the same economic fact quoted under two different compounding conventions, exactly as in Equation 2.5.

Remark 2.7.1. 殖利率 (yield)、債券等值殖利率 (bond equivalent yield, BEY)、房貸等值殖利率 (mortgage equivalent yield, MEY)。

上面兩點的意思是：市場給的事實只有「今天付 PV、 n 年後拿回 FV」，而年化利率並非事實而是報價——必須先選定一年複利幾次，才解得出

$$r = m \left[\left(\frac{FV}{PV} \right)^{1/(nm)} - 1 \right],$$

其值隨 m 而異。取 $m = 2$ 者稱 BEY，取 $m = 12$ 者稱 MEY。例如今今天付 100、一年後拿回 108.16，同一筆現金流可報價為 8.16% ($m = 1$ ，即**有效年利率** (effective annual rate, EAR))、8.00% (BEY) 或 7.87% (MEY)； m 愈大，達成同樣成長所需的**名目年利率** (nominal rate) 愈小。

兩個名字來自報價慣例——美國公債每半年付息 ($m = 2$)，房貸每月繳款 ($m = 12$)。因此跨市場比較前務必先換算，否則會誤判：房貸的 8% MEY 相當於債券的 8.13% BEY，兩個 8% 並不一樣。

§2.7.ii. The Internal Rate of Return

Definition 2.7.2. The **internal rate of return** (IRR) is the yield y which equates an investment's PV with its price P ,

$$P = \frac{C_1}{1+y} + \frac{C_2}{(1+y)^2} + \cdots + \frac{C_n}{(1+y)^n}.$$

Remark 2.7.3. **內部報酬率** (internal rate of return, IRR)；用在債券上時通常稱 **到期殖利率** (yield to maturity, YTM)。

上述定義口語一點的說法是：期初投入 P 元，此後在各個時點分別回收 $C_1, \dots, C_n$ 元； IRR 就是讓這些回收恰好抵掉當初那筆投入的利率。說得再具體些，令 B_k 表示第 k 期回收之後尚未收回的投資餘額：期初分文未收，故 $B_0 = P$ ；此後餘額每期先以利率 y 增值，再扣掉當期收到的 C_k 。IRR 就是使餘額在最後一期恰好歸零的那個 y ：

$$B_0 = P, \quad B_k = B_{k-1}(1 + y) - C_k, \quad B_n = 0.$$

把這條遞迴展開即回到上面的定義式。之所以稱為「內部」，是因為算它只用到交易本身的現金流，不必從外部指定折現率。

Table 2.1 正是這條遞迴的逐期紀錄：餘額每月以 $0.08/12$ 滾息、再減去 2,389.13 元，到第 180 個月剛好歸零。所以 8% 就是那筆房貸對放款人而言的 IRR——我們當初是先給定利率去求月付款，而 IRR 是同一條式子反過來問：給定價格與現金流，利率是多少？

IRR 的好處是自動把時間價值算進去。例如今天付 1,000、往後三年各收回 400，解 $1000 = \sum_{k=1}^3 400(1+y)^{-k}$ 得 $y = 9.70\%$ ；若同樣的 1,200 元要到第三年才一次收到，則只有 $1.2^{1/3} - 1 = 6.27\%$ 。總金額相同，但早收到的錢報酬率高得多。

惟下述再投資假設不可輕忽：承上例，若中途收回的 400 只能以 3% 再投資，實際年化報酬僅 7.33%，而非宣稱的 9.70%。

最後， y 在定義式中同時出現在 n 個不同的次方裡，求解 y 等同於解一個 n 次多項式的根，一般沒有公式解，只能用 [Section 2.8](#) 的數值方法逼近。

Remark 2.7.4. The IRR assumes all cash flows are reinvested at the *same* rate as the internal rate of return, because

$$FV = C_1(1+y)^{n-1} + C_2(1+y)^{n-2} + \dots + C_n.$$

So it must be used with caution. The issue disappears when there are no intermediate cash flows — which is one reason zero-coupon bonds ([Section 2.9.i](#)) are so convenient.

§2.8. Numerical Methods for IRRs

— Slides pp. 60–69 —

§2.8.i. The Root-Finding Problem

Define

$$f(y) \triangleq \sum_{t=1}^n \frac{C_t}{(1+y)^t} - P,$$

where P is the market price, and solve $f(y) = 0$ for a real $y \geq -1$.¹⁰

Proposition 2.8.1

If $C_t > 0$ for all t , then $f(y)$ is monotonically decreasing in y , so a unique real-number solution exists.

There is no closed form for y in general, so we need iterative methods. The three below trade off robustness against speed.

¹⁰Negative interest rates became a reality for German and Swiss bonds in 2015. In 2016, Sweden, Denmark, and Japan imposed negative interest rates on excess reserves. As many as 355 corporate bonds were issued with negative yields as of June of 2016. Even so, -100% should be a natural lower bound because why would anyone or financial institutions want to have every cent confiscated?

Remark 2.8.2. *Trade off A against B* 意為在 A 與 B 之間權衡取捨：想多要一點 B，就得犧牲一些 A。此處的兩個軸是**穩健性**（robustness，是否保證收斂、對起始值是否敏感）與**速度**（收斂階數），三者的對照見 Table 2.2。

二分法最穩健：只要初始區間滿足 $f(a)f(b) < 0$ 便保證收斂，但每步僅將區間減半，故最慢。牛頓法與割線法則都不保證收斂。例如 $f(x) = x^{1/3}$ 的牛頓疊代為 $x_{k+1} = x_k - 3x_k = -2x_k$ ，自任何非零起點出發都翻倍發散；而 $f(x) = x^3 - 2x + 2$ 自 $x_0 = 0$ 出發則在 $0 \leftrightarrow 1$ 之間永遠循環，一步都不前進。此外 $f'(x_k) \approx 0$ 時步長會爆炸，多根時亦可能收斂到非預期的那一個；割線法還多一個分母 $f(x_k) - f(x_{k-1})$ 趨零的風險。理論上能保證的只有局部收斂：起點充分接近單根時保證二次收斂，但定理並未告訴你「充分接近」究竟是多近。

不過就 IRR 這個問題而言，Proposition 2.8.1 其實已把風險排除了。當所有 $C_t > 0$ 時 f 不僅單調遞減，還滿足 $f''(y) = \sum_t t(t+1)C_t(1+y)^{-t-2} > 0$ 而為凸函數。凸函數的切線恆在曲線下方，故 $f(x_{k+1}) \geq 0$ ；又 f 遞減，這表示第一步之後每個迭代點都落在根的左側。此後 $f(x_k) \geq 0$ 而 $f'(x_k) < 0$ ，步長 $-f/f'$ 恆非負，迭代遂單調遞增且以根為上界，必定收斂。所以該命題不只是在說「根唯一」，更是在替牛頓法背書：對付息全為正的 IRR，牛頓法自任何 $y_0 > -1$ 出發都保證收斂。

實務上的求根程式（如 Brent's method）多半混用：先以二分法將根框住，再切換至割線或牛頓法加速，一旦迭代點跳出括號便退回二分法——保證來自二分法，速度來自另外兩者。

§2.8.ii. The Bisection Method

Start with a and b where $a < b$ and $f(a)f(b) < 0$. Then $f(\xi)$ must be zero for some $\xi \in [a, b]$. If we evaluate f at the midpoint $c \triangleq (a+b)/2$, either

- (1) $f(c) = 0$,
- (2) $f(a)f(c) < 0$, or
- (3) $f(c)f(b) < 0$.

In the first case we are done; in the second we continue the process with the new bracket $[a, c]$; and in the third we continue with $[c, b]$. The bracket is halved in the latter two cases, so after n steps ξ lies within a bracket of length $(b-a)/2^n$.

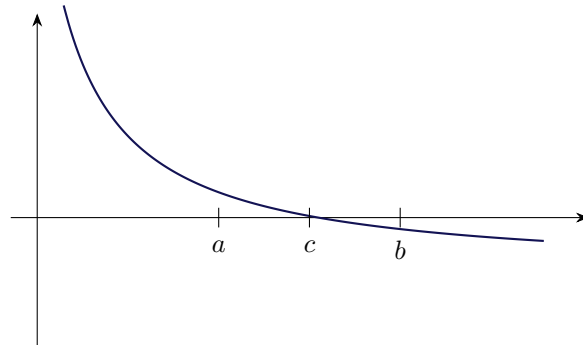


Figure 2.5.: The bisection method. f is decreasing with $f(a) > 0 > f(b)$, so the root is bracketed; evaluating at c halves the bracket.

§2.8.iii. The Newton–Raphson Method

It converges faster than the bisection method. Start with a first approximation x_0 to a root of $f(x) = 0$. Then

$$x_{k+1} \triangleq x_k - \frac{f(x_k)}{f'(x_k)}.$$

When computing yields,

$$f'(x) = - \sum_{t=1}^n \frac{t C_t}{(1+x)^{t+1}}. \quad (2.10)$$

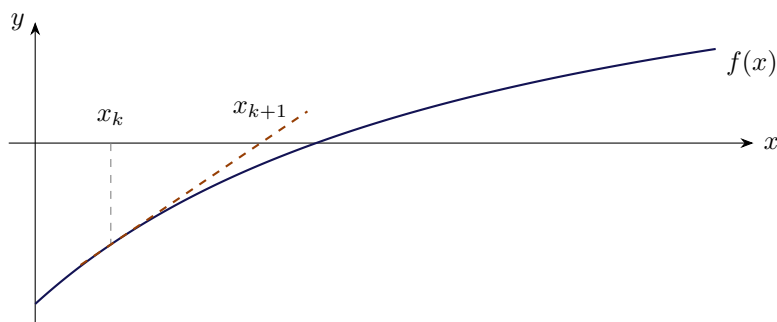


Figure 2.6.: The Newton–Raphson method. The tangent at x_k is followed to the axis to give x_{k+1} .

§2.8.iv. The Secant Method

A variant of the Newton–Raphson method: replace differentiation with difference. Start with two approximations x_0 and x_1 , then compute the $(k+1)$ st approximation with

$$x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})},$$

so $[f(x_k) - f(x_{k-1})]/(x_k - x_{k-1})$ replaces $f'(x_k)$.

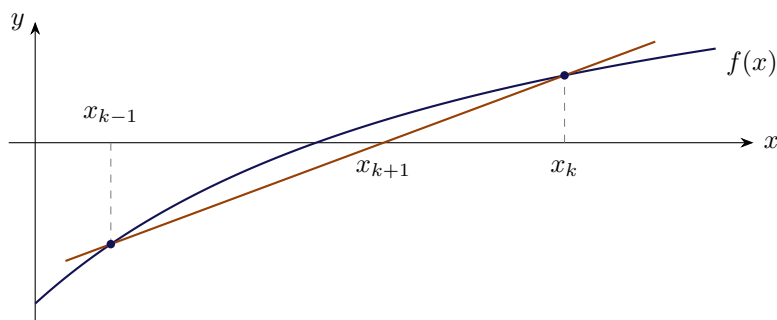


Figure 2.7.: The secant method. The chord through the two latest iterates is followed to the axis to give x_{k+1} ; compare Figure 2.6, which follows the tangent at a single iterate instead.

Its convergence rate is 1.618, slightly worse than the Newton–Raphson method’s 2. But the secant method does not evaluate $f'(x_k)$, which saves about 50% in computation efforts per iteration. For reference, the convergence rate of the bisection method is 1.

Method	Convergence rate	Needs f' ?	Convergence guaranteed?
Bisection	1	no	yes, once bracketed
Secant	1.618	no	no
Newton–Raphson	2	yes	no (see Proposition 2.8.1)

Table 2.2.: The three root-finding methods. Higher rate means fewer iterations, but each Newton–Raphson iteration costs roughly twice a secant one — and the last column is what that speed is bought with. Bisection cannot fail once $f(a)f(b) < 0$; the other two may diverge, cycle, or land on the wrong root.

Remark 2.8.3. 割線法 (secant method) 的幾何意義見 Figure 2.7：牛頓法在一個點上作切線、追到 x 軸得下一個近似值，割線法則改取最近兩個點連成的割線，追到 x 軸得下一個近似

值——名稱即由此而來。公式也是這樣來的：通過 $(x_{k-1}, f(x_{k-1}))$ 與 $(x_k, f(x_k))$ 的直線斜率為差商 $[f(x_k) - f(x_{k-1})]/(x_k - x_{k-1})$ ，令該直線的 y 座標為零解出交點，就得到上式。換言之，割線法就是把牛頓法裡的 $f'(x_k)$ 換成差商——用兩點的實際落差去估斜率，而不去要求函數可微或另外寫一份導數程式。

所謂省下約 50% 的計算量，關鍵在於 $f(x_{k-1})$ 在上一輪就已經算過、可直接沿用，故每輪只需一次新的函數求值；牛頓法每輪則要各算一次 f 與 f' ，而由 (2.10) 可見算 f' 的成本與算 f 相當。若改以每次函數求值（而非每輪疊代）來衡量效率，牛頓法為 $2^{1/2} \approx 1.414$ 、割線法為 1.618：割線法反而略勝一籌。

至於 1.618 這個怪數字，來自誤差遞迴 $e_{k+1} \approx C e_k e_{k-1}$ ：設 $e_{k+1} \sim A e_k^p$ 代入可得 $p^2 = p + 1$ ，故 $p = (1 + \sqrt{5})/2$ ，恰為黃金比例。

最後可與二分法對照：兩者都持有兩個點，但二分法只看 f 的正負號並取中點，割線法則利用 f 的數值大小外推。多用了資訊所以更快，代價是失去保證——割線法不保證收斂，且當 $f(x_k) \approx f(x_{k-1})$ 時分母趨近於零，數值上會不穩定。

§2.8.v. Systems of Nonlinear Equations

It is not easy to extend the bisection method to higher dimensions — there is no natural notion of “bracketing” a root in the plane. But the Newton–Raphson method extends directly.

Let (x_k, y_k) be the k th approximation to the solution of the two simultaneous equations

$$f(x, y) = 0, \quad g(x, y) = 0.$$

The $(k + 1)$ st approximation (x_{k+1}, y_{k+1}) satisfies the following *linear* equations,

$$\begin{bmatrix} \frac{\partial f(x_k, y_k)}{\partial x} & \frac{\partial f(x_k, y_k)}{\partial y} \\ \frac{\partial g(x_k, y_k)}{\partial x} & \frac{\partial g(x_k, y_k)}{\partial y} \end{bmatrix} \begin{bmatrix} \Delta x_{k+1} \\ \Delta y_{k+1} \end{bmatrix} = - \begin{bmatrix} f(x_k, y_k) \\ g(x_k, y_k) \end{bmatrix},$$

with unknowns

$$\Delta x_{k+1} \triangleq x_{k+1} - x_k, \quad \Delta y_{k+1} \triangleq y_{k+1} - y_k.$$

The above has a unique solution for $(\Delta x_{k+1}, \Delta y_{k+1})$ when the 2×2 matrix — the Jacobian — is invertible. Finally, set

$$x_{k+1} = x_k + \Delta x_{k+1}, \quad y_{k+1} = y_k + \Delta y_{k+1}.$$

Intuition . Each step replaces the nonlinear system by its linearisation at the current point and solves that exactly. In one dimension the linearisation is the tangent line of Figure 2.6; here it is the tangent plane.

Remark 2.8.4. 這節處理的是「 n 條方程式、 n 個未知數，且方程式非線性」的情形，前面求 IRR 不過是 $n = 1$ 的特例。金融上最常見的來源是**校準**（calibration）：以市場上觀測到的 n 個價格，反解出模型的 n 個未知參數——例如用兩檔債券的市價反推某個兩參數利率模型的兩個參數。

二分法之所以推不到高維，是因為它的保證來自中間值定理：一維上 f 一變號就把根夾住了。但在平面上 $f(x, y) = 0$ 的解集通常是一條曲線而非一個點，我們要的是它與 $g(x, y) = 0$ 那條曲線的交點；此時並沒有「把區域對半切、且能判斷根落在哪一半」的辦法，**框住根**（bracketing）這個概念本身就不存在。

至於那條矩陣方程式並非憑空而來，它就是一階泰勒展開後令其為零。在 (x_k, y_k) 附近，

$$f(x_{k+1}, y_{k+1}) \approx f(x_k, y_k) + \frac{\partial f}{\partial x} \Delta x_{k+1} + \frac{\partial f}{\partial y} \Delta y_{k+1},$$

g 亦然。我們希望新的一點恰好是解，於是令兩式右側皆為零、把常數項移到等號另一邊，寫成矩陣形式就是上式。

因此這整套做法與一維的牛頓法其實是同一件事。把該矩陣方程記為 $J\Delta = -F$ ，則 $\Delta = -J^{-1}F$ ，亦即 $\mathbf{x}_{k+1} = \mathbf{x}_k - J^{-1}F(\mathbf{x}_k)$ ：一維的「除以導數 $f'(x_k)$ 」在多維成了「左乘 Jacobian 的反矩陣」，而一維要求 $f'(x_k) \neq 0$ 也正對應到此處要求 J 可逆。繞這一圈的用意則在於，非線性方程組難解，線性方程組卻可直接以高斯消去法解出。

§2.9. Bonds

— Slides pp. 70–74 —

§2.9.i. Zero-Coupon Bonds

Definition 2.9.1. A **zero-coupon bond** (or **pure discount bond**) makes a single payment at maturity. By Eq. (2.1), the price of one that pays F dollars in n periods is

$$F/(1+r)^n, \quad (2.11)$$

where r is the interest rate per period.

Zero-coupon bonds can be used to meet future obligations as there is no reinvestment risk — the caveat attached to the IRR above simply does not arise.

Remark 2.9.2. **零息債券** (zero-coupon bond)，亦稱**純折價債券** (pure discount bond)； F 為**面值** (par value，即票面金額)， n 為到期期數。口語地說，它就是一張「中途完全不付息、到期一次付你一筆固定金額」的債券——正因為中途沒有任何**票息** (coupon)，才叫「零息」。它的報酬全部來自買價與面值之間的價差：由 (2.11) 可知，只要 $r > 0$ 價格便必然低於面值，即必定折價發行，故又稱純折價債券。

上段所謂沒有**再投資風險** (reinvestment risk)，指的正是先前 IRR 那條但書在此不成立：中途沒有現金流，就沒有錢需要再投資，也就無從假設再投資利率。因此零息債券的 IRR 就是你實際拿到的報酬，一分不差；這也是為何它適合用來支應未來某個時點的確定支出。

例中 20.83 這個數字採債券市場的報價慣例——以面額 100 為基準，故 20.83 意為面額 100 元的債券今天只賣 20.83 元。另可留意 $45.64\% \times 45.64\% \approx 20.83\%$ ：20 年期的價格恰為 10 年期價格的平方，因為折現是連乘而非相加。

Example 2.9.3. The interest rate is 8% compounded semiannually. A zero-coupon bond that pays the par value 20 years from now will be priced at $1/(1.04)^{40}$, that is, 20.83% of its par value.¹¹ It will be quoted as 20.83. If the bond matures in 10 years instead of 20, its price would be 45.64.

§2.9.ii. Level-Coupon Bonds

A **level-coupon bond** is described by its **coupon rate** and its **par value**, paid at maturity. Write F for the par value and C for the coupon.

Remark 2.9.4. **固定票息債券** (level-coupon bond，亦稱**附息債券**)； c 為**票面利率** (coupon rate)， F 為**面值** (par value)， C 為每期的**票息** (coupon)。口語地說，它就是「每期付你固定的 C 元，到期那一期除了 C 之外再一次還你 F 元」的債券，現金流見 Figure 2.8。留意原文的 *paid at maturity* 修飾的是 par value 而非 coupon rate：票息是每期都付，面值則是到期才一次付清。所謂 *level* (等額) 是指每期票息固定不變，以與浮動利率債券相區別。

¹¹One fifth!

票面利率與票息的換算為下文的 $C = Fc/m$ ： c 是年利率，乘上面值得一年的總票息，再除以一年付息次數 m 即為每期票息。例如面值 100、票面利率 8%、每半年付息一次，則 $m = 2$ 、每期票息 $C = 100 \times 0.08/2 = 4$ 元。

最要緊的是分清 c 與 r 這兩個利率： c 由合約寫死，決定你每期收到多少錢，發行後永不改變； r 則是市場折現率，隨市場浮動，決定這串現金流今天值多少錢。兩者的大小關係恰好決定債券售價高於、等於或低於面值，即**溢價** (premium)、**平價** (par) 與**折價** (discount) 三種情形：當 $c = r$ 時 $C = Fr/m$ ，代入 (2.12) 後前項化為 $F[1 - (1 + \frac{r}{m})^{-n}]$ ，與後項 $F(1 + \frac{r}{m})^{-n}$ 相加恰得 $P = F$ ；又 P 對 r 遞減，故

$$c = r \Rightarrow P = F \text{ (平價)}, \quad c < r \Rightarrow P < F \text{ (折價)}, \quad c > r \Rightarrow P > F \text{ (溢價)}.$$

而 $c = 0$ 即退化為前一小節的零息債券，價格恰為 (2.11)——零息債券只是本式的特例。

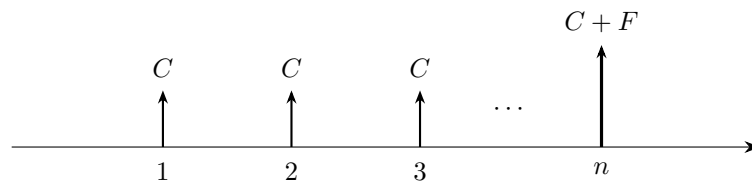


Figure 2.8.: The cash flow of a level-coupon bond.

Remark 2.9.5. Figure 2.8 隱含一項理想化的假設：今天恰好站在付息日上，故各筆現金流剛好落在 $1, 2, \dots, n$ 這些整數期上。真實的交割日多半落在兩個付息日之間，屆時每一筆現金流都要往前挪，指數由 i 變成 $\omega + i$ ——這正是 (2.15) 所處理的情形。

不過要留意，最後一期同時收到 C 與 F 並非理想化，真實債券的最後一次付息本就與本金同日支付。這一點在 (2.15) 中依然成立：末筆票息 ($i = n - 1$) 與 F 的折現指數同為 $\omega + n - 1$ ，可見放寬假設後兩者仍同時發生。

Coupon bonds can be thought of as a matching package of zero-coupon bonds, at least theoretically.^a

^a“You see, Daddy didn’t bake the cake, and Daddy isn’t the one who gets to eat it. But he gets to slice the cake and hand it out. And when he does, little golden crumbs fall off the cake. And Daddy gets to eat those,” wrote Tom Wolfe (1931–2018) in *Bonfire of the Vanities* (1987).

Remark 2.9.6. 這條 moral 的具體意思是：一張 n 期的付息債券，可視為 n 張面額 C 、到期日分別為 $1, 2, \dots, n$ 的零息債券，再加上一張面額 F 、到期日為 n 的零息債券，所組成的投資組合。下方的定價公式 (2.12) 正是把這個拆解逐項寫出來：第一項是 n 張票息零息債券的現值之和，第二項則是面值那一張的現值。

所謂 *at least theoretically*，是指這種拆解未必真能在市場上執行；但在美國公債市場它確實可行。自 1985 年起的 **STRIPS** 制度 (Separate Trading of Registered Interest and Principal of Securities) 允許交易商將一張公債的每一期票息與本金拆開、各自登記為獨立的零息債券單獨買賣，亦允許反向重組回原債券。雙向皆可執行，套利便會迫使「整張債券的價格」等於「各零息碎片價格之和」——這條 moral 因而不只是心算技巧，而是由無套利條件所保證。

這個觀點日後還有一個更要緊的用處：拆開之後每一張零息債券各有自己的到期日，理應各用該期限的即期利率折現。(2.12) 對所有期別一律套用單一的 r ，其實是一種簡化，正確的做法是改用一整條利率期限結構。

至於註腳的典故，出自 Tom Wolfe 的小說 *Bonfire of the Vanities* (1987)：書中債券業務員 Sherman McCoy 的妻子向女兒解釋爸爸的工作——蛋糕不是爸爸烤的，也不是爸爸吃的，但由爸爸負責切開、分送給大家；切的時候會掉下金色的碎屑，那些碎屑就歸爸爸。放在此處是一語雙關：交易商正是把債券「切片」(strip) 後分銷，自己既不創造現金流也不領取現金流，賺的就是買賣價差那些「碎屑」。

Splitting the cash flow of Figure 2.8 into an annuity of coupons plus a zero-coupon bond for the par value, and applying Eq. (2.7) and Eq. (2.11), gives the pricing formula

$$\begin{aligned} P &= \sum_{i=1}^n \frac{C}{\left(1 + \frac{r}{m}\right)^i} + \frac{F}{\left(1 + \frac{r}{m}\right)^n} \\ &= C \frac{1 - \left(1 + \frac{r}{m}\right)^{-n}}{\frac{r}{m}} + \frac{F}{\left(1 + \frac{r}{m}\right)^n}. \end{aligned} \quad (2.12)$$

Here n is the number of cash flows, m the number of payments per year, and r the annual rate compounded m times per annum. Note $C = Fc/m$ when c is the annual coupon rate.

Note 2.9.7. Because the sum collapses to a closed form, the price P can be computed in $O(1)$ time — no loop over the n coupons is needed.

§2.9.iii. Yields to Maturity

Definition 2.9.8. The **yield to maturity** (YTM) is the r that satisfies Eq. (2.12) with P being the bond price.

Example 2.9.9. For a 15% BEY, a 10-year bond with a coupon rate of 10% paid semiannually sells for

$$5 \times \frac{1 - [1 + (0.15/2)]^{-2 \times 10}}{0.15/2} + \frac{100}{[1 + (0.15/2)]^{2 \times 10}} = 74.5138$$

percent of par. So 15% is the yield to maturity if the bond sells for 74.5138.¹²

Remark 2.9.10. **到期殖利率** (yield to maturity, YTM) 的名稱可拆成兩半來理解：*yield* 指報酬率，*to maturity* 則指這個報酬率是在「一路持有到到期日」的前提下算出來的。換言之它回答的是：若今天以價格 P 買進、此後每期領取票息、到期領回面值且中途不賣出，這串現金流相當於年利率多少？

之所以要把持有期間寫進名字裡，是因為債券還有別的「殖利率」，各自假設不同的持有期間。**當期收益率** (current yield) = Cm/P 只比較票息收入與價格，完全不管到期時的資本利得或損失——以上例而言其值為 $10/74.5138 = 13.42\%$ ，低於 YTM 的 15%，差額正是價格將由 74.51 回升至面值 100 的那段資本利得。**贖回殖利率** (yield to call) 則假設持有至發行人提前贖回之日而非到期日；若中途賣出，實際得到的又是另一個持有期報酬率。既然數字取決於你假設持有多久，名字裡自然得講明是哪一種。

至於它與前面 IRR 的關係：YTM 就是把 (2.12) 反解 r ，亦即債券這筆投資的 IRR，只是債券市場習慣改稱 YTM。因此 IRR 的但書一併繼承：YTM 要成為你實際拿到的報酬，除了持有到期、發行人不違約之外，還須能把每一期收到的票息都以 r 再投資出去。零息債券沒有中途票息、無從再投資，其 YTM 恰為實際報酬——這正是前面說它「沒有再投資風險」的意思。

儘管有這些但書，YTM 仍是債券市場的標準報價，因為它把價格、票息與到期日壓縮成單一數字，使票面利率與年期各異的債券得以直接比較。

§2.9.iv. Price Behavior

— Slides pp. 75–79 —

(1) Direction

Bond prices fall when interest rates rise, and vice versa. This is immediate from Eq. (2.12) — every term is decreasing in r — and yet “only 24 percent answered the question correctly.”¹³

¹²Note that the yield 15% exceeds the coupon rate 10%.

¹³CNN, December 21, 2001.

Remark 2.9.11. 價格與市場利率反向的道理在於：票息 C 與面值 F 已由合約寫死，市場利率變動改變的不是你會收到多少錢，而是那些錢今天值多少。利率上升時，新發行的債券給得更多，你手上這張給得少的自然得降價才賣得掉；利率下跌則反之。註腳所引 2001 年 CNN 的調查中僅 24% 的人答對，正是因為多數人誤把「利率上升」聯想成「債券更值錢」——真正在變動的是折現率，而非票息。

(2) Premium, par, and discount

A level-coupon bond sells¹⁴

- at a **premium** (above its par value) when its coupon rate c is above the market interest rate r ;
- at **par** (at its par value) when its coupon rate is equal to the market interest rate;
- at a **discount** (below its par value) when its coupon rate is below the market interest rate.

Bonds selling at par, at a premium and at a discount are called **par bonds**, **premium bonds** and **discount bonds** respectively.

Yield (%)	Price (% of par)
7.5	113.37
8.0	108.65
8.5	104.19
9.0	100.00
9.5	96.04
10.0	92.31
10.5	88.79

Table 2.3.: A 9% coupon bond. The bond trades at par exactly when the yield equals the coupon rate (bold row), at a premium above, at a discount below.

Remark 2.9.12. 溢價 (premium)、平價 (par) 與折價 (discount)；依售價分類者分別稱為**平價債券** (par bonds)、**溢價債券** (premium bonds) 與**折價債券** (discount bonds)。三者的定義與判準已見 Remark 2.9.4： $c = r$ 時 $P = F$ ，且 P 對 r 遞減，故 $c > r$ 時售價高於面值（溢價）、 $c < r$ 時低於面值（折價）。

以下補上溢價與折價的金額。記 $a \triangleq [1 - (1 + \frac{r}{m})^{-n}] / \frac{r}{m}$ 為年金現值因子，則代入 $C = Fc/m$ 後，(2.12) 可寫成 $P = \frac{Fc}{m}a + F(1 + \frac{r}{m})^{-n}$ 。在同一式中把 c 換成 r ，右側恰化簡為 F ，於是有恆等式 $F = \frac{Fr}{m}a + F(1 + \frac{r}{m})^{-n}$ ——這正是平價的情形。兩式相減，末項相消，故

$$P - F = \frac{F(c - r)}{m}a = \sum_{i=1}^n \frac{F(c - r)}{m} \frac{1}{(1 + \frac{r}{m})^i},$$

末式是把 $a = \sum_{i=1}^n (1 + \frac{r}{m})^{-i}$ 展開的結果，意思一目了然：第 i 期你比市場行情多領（或少領） $F(c - r)/m$ 元，把這 n 筆差額各自折現回今天再加總，就是溢價（或折價）的金額。

以 Table 2.3 的 9% 債券驗證：殖利率 8% 時每期多領 0.5 元， $a = 17.292$ ，故 $P - F = 0.5 \times 17.292 = 8.65$ ，恰為表中 108.65 與面值 100 之差。另可留意，無論溢價或折價，到期時價格必然回到面值，故溢價債券的價格會隨時間逐步下滑至 F ——那是溢價的攤銷，並非虧損。

Remark 2.9.13. 在此說明如何讀 Table 2.3。表中描述的是一張 15 年期、票面利率 $c = 9\%$ 、每半年付息一次的債券，即 $n = 30$ 、 $m = 2$ 、每期票息 $C = Fc/m = 4.5$ 元；整張表

¹⁴Consult the text for proofs.

自始至終都是這同一張債券，票面利率固定為 9% 不變。變動的只有左欄的市場殖利率 r ，右欄則是把該 r 代入 (2.12) 算得的價格，並依債券市場慣例以面額 100 為基準表示（故 100.00 即等於面值）。例如第一列 $r = 7.5\%$ ： $a = [1 - (1.0375)^{-30}]/0.0375 = 17.829$ ， $P = 4.5 \times 17.829 + 100 \times (1.0375)^{-30} = 113.37$ ，與表中相符。粗體列即 $r = c = 9\%$ 的平價情形；其上方各列殖利率較低故溢價，下方各列殖利率較高故折價——留意表格說明裡的 *above* / *below* 指的是粗體列的上下方，而非殖利率的高低。這七列也就是 Figure 2.9 曲線上標出的七個點。

(3) Convexity

The price–yield relationship is not merely decreasing; it is *convex*. Concretely, take any yield change of size $\Delta > 0$. The price *gain* from a fall of Δ is larger than the price *loss* from a rise of that same Δ ,

$$P(r - \Delta) - P(r) > P(r) - P(r + \Delta),$$

and the gap widens as Δ grows.

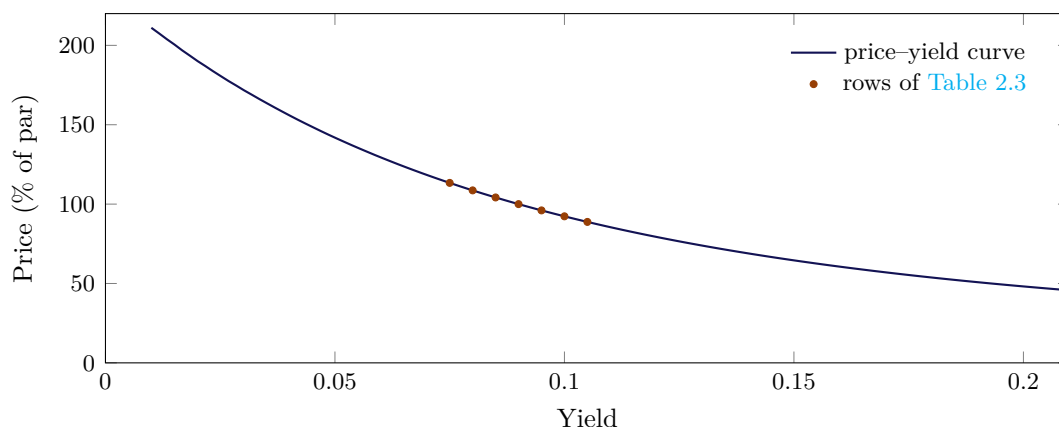


Figure 2.9.: Convexity of the price–yield curve, for the 15-year, 9% coupon bond paid semiannually that generates Table 2.3. The seven tabulated prices are marked; they lie on the curve.

Remark 2.9.14. 凸性即 $\partial^2 P / \partial r^2 > 0$ ：(2.12) 的每一項皆形如 $A(1 + \frac{r}{m})^{-i}$ ，其二階導數 $\frac{i(i+1)A}{m^2}(1 + \frac{r}{m})^{-i-2} > 0$ ，逐項為凸，總和自然為凸。其實際意涵是漲跌不對稱：由 Table 2.3，殖利率自 9% 降至 8% 價格漲 8.65，升至 10% 卻只跌 7.69，且幅度愈大這個不對稱愈明顯。換言之利率無論往哪個方向變動同樣幅度，持有人賺的總是多於賠的，故凸性本身即具價值，市場會為凸性較高的債券付出較高的價格——這是日後存續期間與凸性避險的基礎。

§2.10. Day Count Conventions

— Slides pp. 80–83 —

Every formula above assumed cash flows land on exact period boundaries. Real bonds settle on arbitrary calendar dates, so we need a rule converting two dates into a number of days. The rule is a *convention*, not a fact, and different markets use different ones.

§2.10.i. Actual/Actual

The first “actual” refers to the actual number of days in a month; the second refers to the actual number of days in a year.

Example 2.10.1. The number of days between June 17, 1992, and October 1, 1992, is 106: 13 days in June, 31 days in July, 31 days in August, 30 days in September, and 1 day in October.

Remark 2.10.2. 此處的「幾天」指的是兩個日期之間經過的天數，即「結束日減起始日」，而非兩端所涵蓋的日曆日數。以上例而言，6月只算13天（6/18至6/30，不含起始日6/17），10月則算1天（含結束日10/1），故為「不算頭、算尾」。

不過要留意，改採「算頭不算尾」會得到同一個答案（6月14天、10月0天，合計仍是106），因為兩者算的都是半開區間的長度。真正會算錯的是兩端都算（得107）與兩端都不算（得105）。

之所以採此慣例，是為了讓天數具有**可加性** (additivity)：任取中間日期 B ，恆有「 A 到 B 」的天數加「 B 到 C 」的天數等於「 A 到 C 」的天數。若兩端都算，銜接點 B 便會被重複計入，串接多段期間時就會逐段多算一天。下一小節的 (2.13) 也正是「結束日減起始日」的形式，可見兩種慣例在這一點上完全一致——差別只在每月、每年各算幾天，不在端點怎麼取。

§2.10.ii. 30/360

Each month has 30 days and each year 360 days. In general, the number of days from date (y_1, m_1, d_1) to date (y_2, m_2, d_2) is

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1) + (d_2 - d_1). \quad (2.13)$$

Example 2.10.3. The number of days between June 17, 1992, and October 1, 1992, is now 104: 13 days in June, 30 days in July, 30 days in August, 30 days in September, and 1 day in October.

Remark 2.10.4. (2.13) 看似繁瑣，其實只是「把每個日期換算成一個序號，再相減」。既然每月一律30天、每年一律360天，日期 (y, m, d) 在這套虛擬曆法上的序號就是 $360y + 30m + d$ ；兩個日期的序號相減即得該式，三項分別是「差幾年 $\times 360$ 」、「再差幾個月 $\times 30$ 」與「最後差幾號」。

關鍵在於後兩項允許為負，不必手動借位。以上例的 6/17 至 10/1 為例， $30 \times (10 - 6) + (1 - 17) = 120 - 16 = 104$ ；那個 -16 看似奇怪，但代數已自動處理了跨月的借位。

如此的好處是完全不必查曆法——不管某月是 28、29、30 還是 31 天，任兩日期的間隔都能純以算術求得，且每個「月」所孳生的利息一律相等。代價則是這套曆法裡沒有 31 號，故 d_1 或 d_2 為 31 時須先改成 30，即下文的調整。

If d_1 or d_2 is 31, we must change it to 30 *before* applying formula (2.13).¹⁵ Hence:

- There are 3 days between February 28 and March 1.
- There are 2 days between February 29 and March 1.
- There are 29 days between March 1 and March 31.

Remark 2.10.5. An equivalent formula to (2.13) without any adjustment is (check it)

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1 - 1) + \max(30 - d_1, 0) + \min(d_2, 30).$$

There are many variations on the “30/360” convention regarding 31, February 28, and February 29.¹⁶

Remark 2.10.6. 上式的四項各有明確意義，合起來正是「起始月的殘餘 + 中間的整月 + 結束月已過的部分」：

- $\max(30 - d_1, 0)$ ：起始月從 d_1 號到月底還剩幾天。若 $d_1 = 31$ 則為 0，等於把 31 號視同月底。

¹⁵This is the simplest of all the “30/360” variations (called the “30E/360” convention), used mainly in the Eurobond market (Kosowski & Neftci, 2015).

¹⁶Kosowski & Neftci (2015).

- $30 \times (m_2 - m_1 - 1)$ ：夾在起訖兩月之間的完整月份，每月 30 天。
- $\min(d_2, 30)$ ：結束月從月初到 d_2 號已過幾天。若 $d_2 = 31$ 則取 30。
- $360 \times (y_2 - y_1)$ ：整年的部分。

以 6/17 至 10/1 驗證： $\max(30 - 17, 0) = 13$ (6 月)、 $30 \times (10 - 6 - 1) = 90$ (7、8、9 三個整月)、 $\min(1, 30) = 1$ (10 月)，合計 104——恰好就是前例「13 天 + 30 + 30 + 30 + 1 天」那句話的代數寫法。

所謂 *without any adjustment*，是指 31 號的處理已被 $\max$ 與 $\min$ 吸收，不必如前文那樣先改數字再套公式，因此可直接寫成程式。至於 *check it*：令 $d' \triangleq \min(d, 30)$ 表示調整後的日，並注意恆等式 $\max(30 - d_1, 0) = 30 - d'_1$ ，則

$$\underbrace{30(m_2 - m_1 - 1)}_{\text{少算了一個月}} + \underbrace{(30 - d'_1)}_{\max(30-d_1,0)} + \underbrace{d'_2}_{\min(d_2,30)} = 30(m_2 - m_1) + (d'_2 - d'_1),$$

右側加上 $360(y_2 - y_1)$ 後正是調整後的 (2.13)，故兩式等價。

末句提醒的變體差異不可輕忽：本節採用的是歐式的 30E/360，對 d_1 與 d_2 一律截至 30，規則對稱；而美式的 30/360 (Bond Basis) 在 $d_2 = 31$ 但 $d_1 < 30$ 時並不把 d_2 改為 30，對二月底的處理也另有規定，算出的天數因而可能不同。故實務上合約必須載明採用哪一種慣例，否則利息金額會對不起來。

§2.11. Full Price, Accrued Interest, and the Clean Price

— Slides pp. 84–91 —

§2.11.i. The Full Price

In reality, the settlement date may fall on any day between two coupon payment dates. Let

$$\omega \triangleq \frac{\text{no. of days between the settlement and the next coupon payment date}}{\text{no. of days in the coupon period}}. \quad (2.14)$$

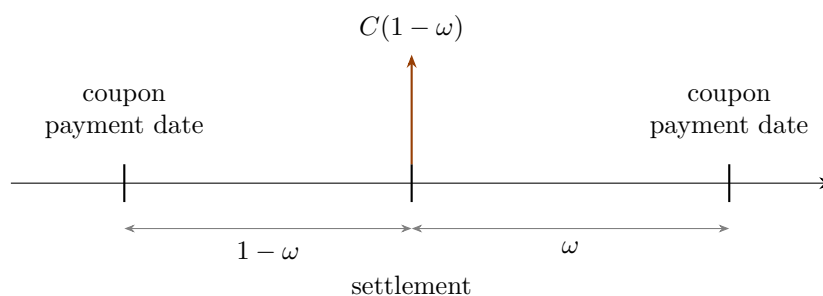


Figure 2.10.: A settlement date falling between two coupon payment dates. ω is the fraction of the coupon period still to run.

Discounting each remaining cash flow over its true (fractional) number of periods, the price is now calculated by

$$\begin{aligned} PV &= \frac{C}{\left(1 + \frac{r}{m}\right)^\omega} + \frac{C}{\left(1 + \frac{r}{m}\right)^{\omega+1}} + \cdots \\ &= \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{\omega+i}} + \frac{F}{\left(1 + \frac{r}{m}\right)^{\omega+n-1}}. \end{aligned} \quad (2.15)$$

This is the **full price**, also called the **dirty price** or **invoice price**.

Remark 2.11.1. **交割日** (settlement date) 是債券易手之日：賣方交出債券、買方付錢，此後所有尚未發生的現金流都歸買方。於是問題成了「這些剩餘現金流在交割日這一刻值多少」，而 (2.15) 算的正是這個數，稱為**全價** (full price)，亦稱**含息價** (dirty price) 或**發票價格** (invoice price)。

麻煩在於交割日多半不落在**付息日** (coupon payment date) 上，各筆現金流因而不位於整數期。(2.14) 的 ω 正是用來度量這個零頭：分子是交割日到下一個付息日的天數，分母是整個付息期間的天數，故 ω 表示「本期還剩多少比例沒走完」， $1 - \omega$ 則是已走完的部分（見 Figure 2.10）。

有了 ω ，(2.15) 做的事便與 (2.12) 相同——把每筆剩餘現金流各自折現回交割日再加總——只是每筆的期數都多帶一個 ω 。 $i = 0$ 是最近的那筆票息，只需再等 ω 期； $i = 1, \dots, n-1$ 是其後各筆，每筆再晚一整期；末筆票息與 F 同日支付，故指數同為 $\omega + n - 1$ 。取 $\omega = 1$ （交割日恰在付息日之後）即化回 (2.12)，可見後者只是本式的特例。

以 Example 2.11.4 的數字 ($C = 5$ 、 $F = 100$ 、 $n = 4$ 、 $\frac{r}{m} = 0.015$ 、 $\omega = 1/3$) 逐項展開：

i	距今期數 $\omega + i$	現金流	折現值
0	1/3	5	4.9752
1	4/3	5	4.9017
2	7/3	5	4.8293
3	10/3	5	4.7579
—	10/3	100	95.1583
合計			114.6225

而淨價加應計利息為 $111.2891 + 3.3333 = 114.6224$ ，兩者相符（尾差來自四捨五入）——這就是該例所謂「可用 (2.15) 驗證殖利率為 3%」的意思。

§2.11.ii. Accrued Interest and the Clean Price

The quoted price in the U.S./U.K. does *not* include the accrued interest; it is called the **clean price** or **flat price**. The buyer pays the invoice price: the quoted price *plus* the **accrued interest** (AI). AI equals

$$C \times \frac{\text{no. of days from the last coupon payment to the settlement date}}{\text{no. of days in the coupon period}} = C \times (1 - \omega).$$

Hence the yield to maturity is the r satisfying Eq. (2.15) when PV is the invoice price:

$$\text{clean price} + \text{AI} = \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{\omega+i}} + \frac{F}{\left(1 + \frac{r}{m}\right)^{\omega+n-1}}.$$

Intuition . The clean price is quoted precisely so that it does *not* saw-tooth: the full price jumps down by C on every coupon date, and stripping out the accrued interest removes that artefact, leaving a number that moves only with the yield.

Remark 2.11.2. invoice price 是整個 bond 剩餘的價值，包含賣方還沒領到的那一份 coupon，所以 coupon 應該照比例 $(1 - \omega)$ 分給賣方，也就是說 full price 扣掉 AI (分給賣方的利息) 才是 clean price (對買方來說 bond 對他真正有的價值)。

Remark 2.11.3. 先說清楚**全價** (full price) 是什麼：它就是 (2.15) 算出的那個數，也就是買方在交割日實際要掏出來的錢。它的毛病在於，縱使殖利率與債券本身都毫無變化，這個數字仍會隨付息週期上下起伏。以 $F = 100$ 、 $c = r = 10\%$ 、 $m = 2$ (故 $C = 5$) 為例，全價恰為 $100 \times 1.05^{1-\omega}$ ：剛付完息時 ($\omega = 1$) 為 100，其後隨下一筆票息逼近而一路爬升，至付息前夕

($\omega \rightarrow 0$) 達 105；票息一經發放，該筆現金流即自剩餘現金流中消失，全價隨之驟降回 100。如此週而復始，全價的時間序列遂呈鋸齒狀。於是看到「甲債券報 105、乙債券報 100」時，你分不出甲究竟真的比較貴，還是只是離付息日比較近而已。

市場的對策是把全價拆成兩塊：隨持有天數線性累積的那部分稱為**應計利息** (accrued interest, AI)，扣掉它之後的餘額稱為**淨價** (clean price)，亦稱 flat price——注意此處的 flat 與先前的平價 (par) 毫無關係，切勿混淆。承上例，淨價 = $100 \times 1.05^{1-\omega} - 5(1-\omega)$ ，整個付息週期內都停在 100 附近 ($\omega = 1/2$ 時為 99.97)，鋸齒已被濾掉。美、英市場報的就是淨價，它只隨殖利率與信用狀況變動。

AI 之所以等於 $C \times (1 - \omega)$ ，是因為票息按持有天數比例分攤：賣方持有了本期的 $1 - \omega$ ，就該分得這一期票息的 $1 - \omega$ 。但整筆 C 會在下一個付息日全額付給買方，故買方須在交割時先把屬於賣方的那一部份補上——這便是買方實付**發票價格** (invoice price) = 報價 + AI 的由來，而它正好等於全價。

於是很容易出錯的一點是：反解殖利率時，(2.15) 左邊的 PV 必須填發票價格而非報價，這正是本小節末那條 “clean price + AI = ...” 的用意。

§2.11.iii. Example (“30/360”)

Example 2.11.4. A bond with a 10% coupon rate and paying interest semiannually, with clean price 111.2891. The settlement date is July 1, 1993, and the maturity date is March 1, 1995.

There are 60 days between July 1, 1993, and the next coupon date, September 1, 1993, so $\omega = 60/180$. The accrued interest is

$$(10/2) \times \left(1 - \frac{60}{180}\right) = 3.3333$$

per \$100 of par value.

The yield to maturity is 3%. This can be verified by Eq. (2.15) with $\omega = 60/180$, $n = 4$, $m = 2$, $F = 100$, $C = 5$, $PV = 111.2891 + 3.3333$ and $r = 0.03$.

Remark 2.11.5. 疑問：例中驗證 YTM 時，不由分說直接取 $F = 100$ ；但若不知道這張債券的面額究竟是多少，YTM 不是就算不出來了嗎？

先說 $F = 100$ 是怎麼來的。它並非假設這張債券的面額恰好是 100 元，而是因為債券價格一律以面額的百分比報價：所謂 clean price 111.2891 意即「面額的 111.2891%」，而例中也明言 AI 是 *per \$100 of par value*。本章三處都採此慣例——零息債券例的 “quoted as 20.83”、Table 2.3 的欄位 “% of par”，以及本例。取 $F = 100$ 只是挑一個讓報價直接等於數值的單位罷了。

更要緊的是，YTM 根本不依賴面額。把 (2.15) 兩邊同除以 F 並代入 $C = Fc/m$ ，得

$$\frac{PV}{F} = \sum_{i=0}^{n-1} \frac{c/m}{\left(1 + \frac{r}{m}\right)^{\omega+i}} + \frac{1}{\left(1 + \frac{r}{m}\right)^{\omega+n-1}},$$

式中 F 已完全消失，只剩「價格佔面額的比例」、票面利率 c 、 ω 與 n 。面額加倍時票息與價格同步加倍，解出的 r 不變；故你不必知道面額究竟是 100、1,000 還是 1,000,000，只要知道價格相對於面額是多少——而那正是報價所提供的資訊。

真正需要用到實際面額的只有結算金額：買方應匯出 $\frac{\text{報價} + \text{AI}}{100} \times \text{實際面額}$ 。以本例而言，若買的是 1,000,000 面額，實付 $1.146224 \times 1,000,000 = 1,146,224$ 元，但殖利率仍是 3%。

§2.11.iv. Price Behavior (2) Revisited

Previously we said a bond sells at par if the yield to maturity equals the coupon rate. But that assumed the settlement date is *on* a coupon payment date.

Suppose the settlement date for a bond selling at par¹⁷ falls between two coupon payment dates. Then its yield to maturity is *less* than the coupon rate.¹⁸ The reason: exponential growth to C is replaced by linear growth, hence overpaying the accrued interest.

Remark 2.11.6. 疑問：前面才說「殖利率等於票面利率 $\Leftrightarrow$ 賣在平價」，但這只在交割日是在付息日才會是對的，為何交割日一落在兩個付息日之間，平價的債券殖利率反而低於票面利率？原文只丟下一句「指數成長被線性成長取代」與「見課本習題」。

情境：關鍵在「賣在平價」講的是哪個價——註腳已註明是報價（即淨價）等於面值，但買方實付的是淨價加 AI。於是真正該問的是：這筆實付金額，與「殖利率真的等於 c 時本應支付的金額」相比，究竟是多還是少？

脈絡：先講清楚前提——「賣在平價」是假設而非推導結果，註腳已言明它指報價（淨價）等於面值。於是買方實付

$$\underbrace{F}_{\text{淨價 (依假設等於面額)}} + \underbrace{C(1-\omega)}_{\text{AI}} = F + F \frac{c}{m}(1-\omega) = F \left[1 + \frac{c}{m}(1-\omega) \right],$$

此處用到票息 $C = Fc/m$ 由合約決定、與殖利率無關。（留意 F 在此身兼二職：它既是 (2.15) 裡的面額，又因假設而等於報價。）

接著問：若殖利率真的等於票面利率 c ，本應付多少？把 (2.15) 提出 $(1 + \frac{c}{m})^{1-\omega}$ ，得

$$PV = \left(1 + \frac{c}{m}\right)^{1-\omega} \left[\sum_{k=1}^n \frac{C}{\left(1 + \frac{c}{m}\right)^k} + \frac{F}{\left(1 + \frac{c}{m}\right)^n} \right],$$

中括號恰為 (2.12) 在殖利率等於 c 時的值，即 F ；故本應支付 $F(1 + \frac{c}{m})^{1-\omega}$ 。

令 $x \triangleq 1 - \omega$ 表示本期已走完的比例，兩者遂可並排寫成

$$\underbrace{F \left(1 + \frac{c}{m}\right)^x}_{\text{本應支付}} \quad \text{與} \quad \underbrace{F \left[1 + \frac{c}{m}x\right]}_{\text{慣例實付}}.$$

後者正是前者在 $x = 0$ 與 $x = 1$ 兩端之間的直線內插——兩式在 $x = 0$ 時同為 F （剛付完息，AI = 0），在 $x = 1$ 時同為 $F(1 + \frac{c}{m})$ （付息前夕，AI = C ），兩端完全吻合，差別只在中間。而 $(1 + \frac{c}{m})^x$ 為凸函數，凸函數的圖形在兩端點之間必位於該連線之下（見 Figure 2.11），故對 $0 < x < 1$ 恆有

$$\left(1 + \frac{c}{m}\right)^x < 1 + \frac{c}{m}x.$$

亦即實付金額高於「殖利率等於 c 時應付的金額」；又價格對殖利率遞減，買方付得愈多代表殖利率愈低，故實際殖利率必然低於票面利率。

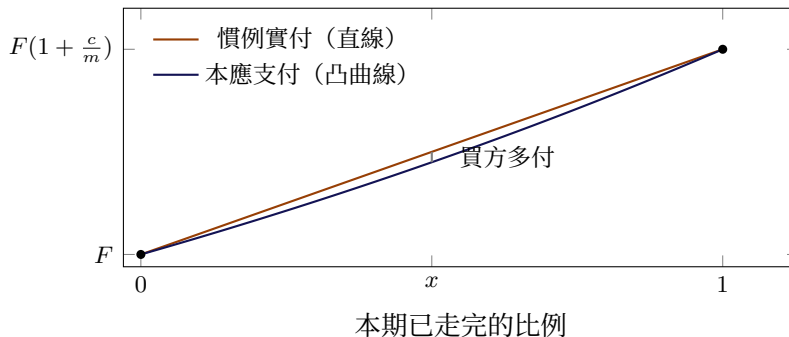


Figure 2.11.: 慣例的線性累積與正確的指數累積。兩者在期初與期末完全一致，中間則因直線位於凸曲線之上而使買方略為多付。（為看得清楚，圖中的 c/m 遠大於實際。）

¹⁷The quoted price equals the par value.

¹⁸See Exercise 3.5.6 of the textbook for proof.

代個數字看規模—— $F = 100$ 、 $c = 10\%$ 、 $m = 2$ 、 $\omega = 1/2$ 時，本應付 $100 \times 1.05^{0.5} = 102.4695$ ，慣例卻要付 $100 + 2.5 = 102.5$ ，多付約 0.03；偏差在期中最大，兩端則歸零。

歸根結柢，這是 AI 採「按天數線性分攤」所付出的代價。該約定之所以線性，是為了讓買賣雙方不必對任何利率有共識就能算出同一個數字（公式裡根本沒有 r ）；換來的偏差就記在這裡。這也正是下方 moral 的意思：平價、溢價、折價那些漂亮的敘述，都是「交割日恰在付息日上」這個乾淨模型底下的結論。

The tidy statements about premium, par and discount are statements about a model with clean period boundaries. Day count conventions are where that model meets the calendar — and the tidy statements need amending.

3. Bond Price Volatility

— Slides pp. 92–93

Lecture 2
September 17,
2026

*Well, Beethoven, what is this?*¹

— Attributed to Prince Anton Esterházy

*You mean to tell me that
the success of my program and my re-election
hinges on the Federal Reserve
and a bunch of f—king bond traders?*

— Bill Clinton to Robert Rubin, January 7, 1993

The previous chapter answered the question “what is a bond worth?”. Given a yield y we can compute a price P , and given a price we can solve back for the yield. This chapter answers the next question, which is the one a risk manager actually loses sleep over: *if the yield moves, how much does the price move?*

The whole chapter is an exercise in Taylor expansion. Write the price as a function $P(y)$ of the yield; then

$$\Delta P = \frac{\partial P}{\partial y} \Delta y + \frac{1}{2} \frac{\partial^2 P}{\partial y^2} (\Delta y)^2 + \dots$$

The first-order term, suitably normalised, is called **duration**; the second-order term, suitably normalised, is called **convexity**. Everything else — Macaulay duration, modified duration, effective duration, dollar duration, $D\%$, effective convexity — is one of these two numbers in a different set of units, or computed by a different method. Keeping that map in mind is the best defence against the terminology.

Duration is the first derivative of price with respect to yield; convexity is the second. Everything in this chapter is bookkeeping around those two facts.
存續期間 (duration) 就是價格對殖利率的一階導數，凸性 (convexity) 就是二階導數；本章其餘的一切，都只是圍繞這兩件事的記帳工作。

§3.1. Price Volatility

— Slides pp. 94–95

Volatility here measures how bond prices respond to interest rate changes. It is the key input to the risk management of interest rate-sensitive securities: a desk that holds \$1 billion of bonds needs to know, in dollars, what a one-basis-point move costs it.

The question to make precise is this: *what is the sensitivity of the percentage price change to changes in interest rates?* Note the two deliberate choices hidden in that sentence. We measure the response in *percentage* terms, so that bonds of different sizes can be compared, and we measure it against a change in the *yield*, which is the single number we have agreed to summarise the whole discount curve by.

¹Mass in C major, Op. 86 (1808).

Definition 3.1.1 (Price volatility). The **price volatility** of a security with price P and yield y is

$$-\frac{\frac{\partial P}{\partial y}}{P}. \quad (3.1)$$

Intuitively, for a small yield move Δy ,

$$\frac{\Delta P}{P} \approx -\text{price volatility} \times \Delta y. \quad (3.2)$$

Remark 3.1.2. (3.1) 的兩個設計都有理由。

為何要除以 P ? $\partial P/\partial y$ 的單位是「元 per 單位殖利率」，面額 100 的債券和面額 1,000 的同一檔債券會得到差十倍的數字。除以 P 之後得到的是百分比變化對殖利率變化的敏感度，與部位大小無關，才能拿不同債券互相比較。

為何要加負號？價格對殖利率遞減 ($\partial P/\partial y < 0$)，加上負號後，**價格波動性** (price volatility) 對一般債券是正數，講「這檔債券的波動性是 8」比講「是 -8」自然。代價是 (3.2) 裡必須把負號再寫回去。

Note 3.1.3. (3.1) is a *local* statement: it is exact only in the limit $\Delta y \rightarrow 0$. How badly it fails for a finite Δy is precisely what convexity, in Section 3.10, will measure.

§3.2. Price Volatility of a Level-Coupon Bond

— Slides p. 96 —

For a level-coupon bond the derivative can be computed in closed form. Recall from the previous chapter that with n periods to maturity, a coupon C per period, par value F , and a *periodic* yield y ,

$$P = \frac{C}{y} \left[1 - (1+y)^{-n} \right] + F(1+y)^{-n}. \quad (3.3)$$

Proposition 3.2.1

The price volatility of a level-coupon bond is

$$-\frac{(C/y)n - (C/y^2)[(1+y)^{n+1} - (1+y)] - nF}{(C/y)[(1+y)^{n+1} - (1+y)] + F(1+y)}. \quad (3.4)$$

Here F is the par value and C is the coupon payment per period.

Proof. Differentiate (3.3) term by term:

$$\frac{\partial P}{\partial y} = -\frac{C}{y^2} + \frac{C}{y^2}(1+y)^{-n} + \frac{C}{y}n(1+y)^{-n-1} - nF(1+y)^{-n-1}.$$

Multiplying through by $(1+y)^{n+1}$ clears every negative power:

$$(1+y)^{n+1} \frac{\partial P}{\partial y} = \frac{C}{y}n - \frac{C}{y^2}[(1+y)^{n+1} - (1+y)] - nF,$$

which is the numerator of (3.4). Multiplying (3.3) by the same factor,

$$(1+y)^{n+1}P = \frac{C}{y}[(1+y)^{n+1} - (1+y)] + F(1+y),$$

which is the denominator. The quotient is therefore $(\partial P/\partial y)/P$, and (3.4) is its negative, as (3.1) requires. ■

Two observations about the formula.

- **It simplifies with $C = Fc/m$.** If the bond pays an annual coupon rate c with m payments per year, then $C = Fc/m$, and F divides out of numerator and denominator alike. What is left depends only on c/m , y and n — as it must, since price volatility is a percentage quantity and cannot depend on how big the bond is.
- **It is positive.** For the bond above,

$$-\frac{\partial P / \partial y}{P} > 0,$$

because P is a sum of positive cash flows discounted by $(1+y)^{-i}$, every one of which falls as y rises. Higher yield, lower price: the sign is not an accident but a restatement of the discounting formula.

§3.3. Macaulay Duration

— Slides p. 97 —

Definition 3.3.1 (Macaulay duration). The **Macaulay duration**² (MD) is a weighted average of the times to an asset's cash flows. The weights are the cash flows' PVs divided by the asset's price. Formally, for cash flows $C_1, \dots, C_n$ at times $1, \dots, n$,

$$\text{MD} \triangleq \frac{1}{P} \sum_{i=1}^n \frac{C_i}{(1+y)^i} i. \quad (3.5)$$

The definition really is an average. Put

$$w_i \triangleq \frac{1}{P} \frac{C_i}{(1+y)^i},$$

the fraction of today's price that is accounted for by the i th cash flow. Since $P = \sum_i C_i(1+y)^{-i}$ we have $\sum_i w_i = 1$, and (3.5) reads $\text{MD} = \sum_i w_i i$. So MD is the *centre of mass* of the cash flows when each is given the weight of its present value.

Remark 3.3.2. 麥考利存續期間 (Macaulay duration, MD) 在問一句話：這檔債券的錢，平均要等幾期才拿得到？

i 是第 i 筆現金流距今幾期（不是到期日，到期日是 n ）。直接把 $1, \dots, n$ 平均，等於不管每期收到的錢多寡，通通當同一回事；但債券到期那期通常連本金帶票息一起付，金額遠大於中途單純的票息，不該跟其他期一樣輕。所以要用每筆現金流的現值佔總價格的比重 w_i 來加權——金額愈大、時間愈近，權重愈大，MD 就愈被拉向那一期。這就是 (3.5) 是**加權平均** (weighted average)、 w_i 叫**現值權重** (PV weight) 的原因。

下面例子裡 $w_5 = 0.744$ ，於是 $\text{MD} = 4.34$ ：比單純平均的 3 大得多（因為多數價值壓在最後一期），又比到期日 5 略小（因為中途先收到票息）。把每期現值想成掛在時間軸上的砝碼，MD 就是這根槓桿的平衡點，即**重心** (centre of mass)，見 Figure 3.1。

這個數字有用，是因為錢等得愈久、折現折得愈兇，價格對殖利率也就愈敏感——所以「平均要等幾期」同時也是「殖利率一動，價格會動多少」的度量，這正是下一小節 (3.6) 要證明的事情。

單位是「期」，換算成年要除以每年付息次數 k (Section 3.5)。

²Frederick Macaulay (1882–1970) in 1938.

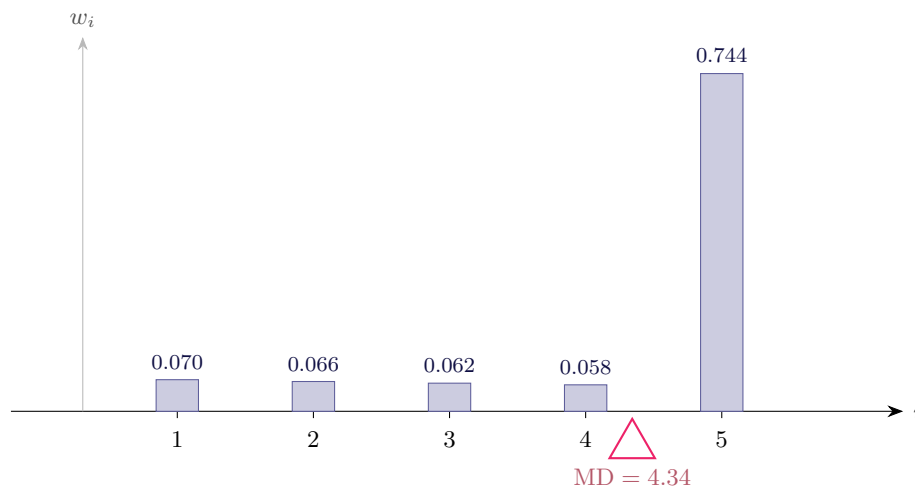


Figure 3.1.: Macaulay duration as a centre of mass. The bars are the PV weights w_i of a five-period bond with $F = 100$, period coupon rate 8% and yield $y = 6\%$; the balance point sits at $MD = 4.34$ periods, short of the maturity $n = 5$ because the four coupons pull it left.

Example 3.3.3 (Not on the slides). A five-period bond with $F = 100$, coupon $C = 8$ per period and periodic yield $y = 6\%$ has

$$P = \sum_{i=1}^4 \frac{8}{1.06^i} + \frac{108}{1.06^5} = 108.4247,$$

with PV weights $w_1, \dots, w_5 = 0.0696, 0.0657, 0.0620, 0.0584, 0.7443$. Hence

$$MD = 1(0.0696) + 2(0.0657) + 3(0.0620) + 4(0.0584) + 5(0.7443) = 4.3422 \text{ periods.}$$

Almost three quarters of the price is the final principal repayment, which is why MD sits so close to $n = 5$.

Question 3.3.4. What if $C_i = (1 + c)^i$ for some constant c , letting $n \rightarrow \infty$ and assuming $c > y$?^a

^aContributed by Mr. Chen, Yu-Hsing (B06901048, R11922045) on March 3, 2023.

Answer. Write $q \triangleq (1 + c)/(1 + y) > 1$, so that $C_i(1 + y)^{-i} = q^i$. Then

$$P = \sum_{i=1}^n q^i = \frac{q(q^n - 1)}{q - 1} \xrightarrow{n \rightarrow \infty} \infty,$$

and the unnormalised numerator $\sum_{i=1}^n i q^i$ diverges faster still. Their ratio behaves like n for large n , so $MD \rightarrow \infty$ as well.

The example is not a curiosity but a warning. The weighted-average reading of (3.5) presumes that the weights w_i sum to a finite number; when the coupons grow faster than the discount factor shrinks, the price itself is infinite and there is nothing left to average. Any security whose cash flows grow at a rate exceeding its own discount rate has no finite price and no finite duration. ■

§3.3.i. MD as a Measure of Price Volatility

— Slides p. 98 —

Here is the fact that makes MD worth defining: it is the bridge between the “average waiting time” picture of (3.5) and the “price sensitivity” picture of (3.1).

Theorem 3.3.5

The Macaulay duration, in periods, equals

$$\text{MD} = -(1+y) \frac{\partial P}{\partial y} \frac{1}{P}. \quad (3.6)$$

Proof. From $P = \sum_{i=1}^n C_i(1+y)^{-i}$,

$$\frac{\partial P}{\partial y} = - \sum_{i=1}^n i C_i(1+y)^{-i-1} = - \frac{1}{1+y} \sum_{i=1}^n i C_i(1+y)^{-i}.$$

Multiply both sides by $-(1+y)/P$:

$$-(1+y) \frac{\partial P}{\partial y} \frac{1}{P} = \frac{1}{P} \sum_{i=1}^n \frac{C_i}{(1+y)^i} i = \text{MD}. \quad \blacksquare$$

Intuition . (3.6) says $\text{MD} = (1+y) \times \text{price volatility}$. Two quantities defined for entirely different reasons — one a statement about *time*, the other a statement about *sensitivity* — turn out to differ only by the factor $1+y$.

The reason is visible in the proof: differentiating $(1+y)^{-i}$ brings down exactly the factor i , which is the time index. Long cash flows are discounted more steeply, so a security’s sensitivity to y is governed by how far out its money sits.

§3.3.ii. MD of a Level-Coupon Bond

— Slides p. 99 —

Proposition 3.3.6

The MD of a level-coupon bond is

$$\text{MD} = \frac{1}{P} \left[\sum_{i=1}^n \frac{iC}{(1+y)^i} + \frac{nF}{(1+y)^n} \right]. \quad (3.7)$$

It can be simplified to

$$\text{MD} = \frac{c(1+y)[(1+y)^n - 1] + ny(y-c)}{cy[(1+y)^n - 1] + y^2}, \quad (3.8)$$

where c is the *period* coupon rate.

(3.7) is just (3.5) with the cash flows of a level-coupon bond written out: a coupon C in each of periods $1, \dots, n$, plus the par value F at period n .

For (3.8), write $C = Fc$ and let $A(n) \triangleq \sum_{i=1}^n (1+y)^{-i} = [1 - (1+y)^{-n}]/y$ be the ordinary annuity factor, so (3.3) reads $P = F[cA(n) + (1+y)^{-n}]$. The bracket in (3.7) factors the same way:

$$\sum_{i=1}^n \frac{iC}{(1+y)^i} + \frac{nF}{(1+y)^n} = F \left[c \sum_{i=1}^n \frac{i}{(1+y)^i} + \frac{n}{(1+y)^n} \right].$$

So the numerator and the denominator P of (3.7) carry the *same* factor F , and it cancels — this is why F disappears. C disappears too, but only because it was never independent: it enters solely through $C = Fc$, and once F is gone all that is left of it is the rate c . Summing the remaining arithmetic–geometric series and simplifying gives (3.8), which — exactly as with (3.4) — depends only on c , y and n , never on the bond’s size.

Two consequences are worth stating separately.

- **A coupon bond has MD < n.** If $c > 0$ then some of the weight w_i sits at times $i < n$, so the centre of mass in Figure 3.1 is pulled to the left of n .
- **A zero-coupon bond has MD = n.** With $c = 0$ there is a single cash flow, $w_n = 1$, and the average of one number is that number. Its duration *is* its term to maturity.

§3.4. What MD Does and Does Not Mean

§3.4.i. When the Formulas Apply

— Slides p. 100 —

Note 3.4.1. Formulas (3.6) and (3.7) hold only if the coupon C , the par value F , and the maturity n are all independent of the yield y — that is, only if the cash flow is independent of yields.

This is a far stronger assumption than it looks, and the slides’ argument for why deserves unpacking.

Suppose the market yield declines. Mechanically, (3.8) then reports a *longer* MD: lower y discounts the distant cash flows less severely, so the weights w_i shift towards large i and the centre of mass moves right. For a plain bond that is the end of the story, and a longer MD does mean greater price volatility.

But now suppose the security’s maturity *itself* shortens when yields fall. Then the actual price volatility, as originally defined in (3.1), may *decrease* at the very moment the formula says the duration lengthened. Formula and reality point in opposite directions.

Remark 3.4.2. 白話翻譯一下上面這段在講什麼。

MD 的公式 ((3.6)、(3.7)) 都偷偷假設了一件事： C 、 F 、 n 這些現金流的內容不會因為 y 變動而跟著變——也就是「不管殖利率變成怎樣，這張債券該付多少錢、哪一期付，永遠不變」。一般公司債、公債確實如此，但可贖回債券 (callable bond) 跟房貸抵押證券不是：殖利率一降，發行人可能提前贖回、屋主可能提前還款 (prepayment) 再融資，現金流本身就跟著 y 一起變了。

現在把「公式算出來的」跟「實際上發生的」分開看：

- **公式算出來的：**套用 (3.8)，殖利率 y 下降 → 折現折得比較輕 → 較晚那幾期的權重 w_i 變大 → MD (加權平均的期數) 變長。這一步純粹是代公式，跟債券實際上會不會被贖回無關。
- **實際上發生的：**如果這張債券在殖利率下降時被提前贖回或提前還款，它的壽命反而縮短了。壽命一短，價格對利率的敏感度 ((3.1) 那個真正的定義) 通常也跟著下降。

為什麼壽命短，敏感度就低？最乾淨的例子是零息債： $P = F(1 + y)^{-n}$ ，算一下 (3.1) 就得到價格波動性 $= n/(1 + y)$ ——敏感度就是到期期數 n 除以 $1 + y$ ， n 愈小，敏感度愈低，一目了然。直覺上也對： n 愈小，這筆錢愈快回到你手上， y 的變動只需要「折現少數幾期」，累積效果自然小； n 愈大，同一個 y 的變動要連乘（或連加，在指數上）更多次，效果就被放大。一般附息債也是同樣道理：Section 3.3.i 已經證明 MD 就是 $(1 + y) \times$ 價格波動性，而 MD 本身是現金流到期期數的加權平均，到期日一旦提前，這個加權平均自然跟著縮小，價格波動性也跟著等比例縮小。

§3.4.ii. How Not To Think about MD

— Slides p. 101 —

The MD has its origin in measuring the length of time a bond investment is outstanding, and the name still carries that connotation. But it should be seen mainly as measuring *price volatility*, for the simple reason that the temporal reading breaks down in cases the volatility reading handles without complaint.

- **Duration can be longer than maturity.** A leveraged inverse floater — a note whose coupon is, say, $20\% - 2 \times$ the reference rate — behaves like a long position in a fixed-rate bond financed by a short position in a floater. Its price sensitivity can exceed that of *any* fixed-rate bond of the same maturity.
- **Duration can be negative.** An interest-only (IO) strip of a mortgage pool loses value when rates fall, because prepayments cut short the very interest payments it owns. Its price *rises* with yields, so $\partial P / \partial y > 0$ and its duration is negative. Figure 3.2 on p. 49 shows exactly such a price curve.

Neither of these makes sense under the maturity interpretation: no security is outstanding for longer than its maturity, and none is outstanding for a negative length of time. Many, if not most, duration-related terms can only be comprehended as measuring volatility.

Remark 3.4.3. 簡單說：MD 本質上是「利率敏感度」(sensitivity)，「加權平均到期時間」只是現金流固定不變時剛好等於敏感度的副產品；現金流一旦隨 y 變化，MD 就可能跑出時間該有的範圍。

- **槓桿反向浮動利率債券 (leveraged inverse floater)：**票息 $= 20\% - 2 \times$ 參考利率。等同於「多一張固定利率債 (fixed-rate bond)、用借來的錢放空一張浮動利率債 (floater)」——浮動債每期重設票息，久期 (duration) 近乎 0，固定債久期則正常。用槓桿多買固定債、空浮動債對沖，等於把固定債的利率敏感度用槓桿倍數放大，MD 自然能超過同期限的一般定息債。
- **IO strip (interest-only strip, 純利息證券)：**只分到房貸池的利息，不分本金。 y (連帶房貸利率) 一下降，屋主**提前還款** (prepayment, 也叫 refinance 轉貸) 的比例上升；本金一旦提前還完，後面本該收的利息就直接消失，不只是折現折得少。這個「現金流消失」的損失蓋過折現效果，導致價格跟 y 反向變動， $\partial P / \partial y > 0$ ，MD 變負。

Read “duration” as “price sensitivity measured in units of time”, never as “time”.

把 duration 讀成「以時間為單位表示的價格敏感度」，而不是「時間」。

§3.5. Conversion to Annual Units

— Slides p. 102 —

So far everything has been measured in *periods*, with y the periodic yield. Market conventions quote years and annual yields, so both formulas need converting. Let k be the compounding

frequency per annum and let y now denote the *annual* yield, so that the periodic yield is y/k and cash flow i arrives at time i/k years.

For the MD to be year-based, modify (3.7) to

$$\frac{1}{P} \left[\sum_{i=1}^n \frac{i}{k} \frac{C}{(1 + \frac{y}{k})^i} + \frac{n}{k} \frac{F}{(1 + \frac{y}{k})^n} \right]. \quad (3.9)$$

Nothing has changed except that each time index i has been re-expressed as i/k years. (n itself is still a count of *periods*, not years; only the time label in the numerator has been converted to i/k and n/k years, while the discounting exponent stays in periods.) Formula (3.6) likewise becomes

$$\text{MD} = -\left(1 + \frac{y}{k}\right) \frac{\partial P}{\partial y} \frac{1}{P}. \quad (3.10)$$

By definition, then,

$$\text{MD (in years)} = \frac{\text{MD (in periods)}}{k}.$$

Remark 3.5.1. 注意 MD(in periods) 是指每期利率 $\frac{y}{k}$ 算出來的 md，然後 MD(in year) 只是把時間的單位從期換到年，利率還是一樣。三個式子講的是同一件事，只是單位不同，但實務上最常出錯的就是單位。

(3.9) 這一步很直覺： i/k 、 n/k 只是把「第 i 期」重新標成「第 i/k 年」， k 是常數可以直接從加總裡拿出來，所以整個式子自然就是 MD(periods)/ k ，沒有懸念。

(3.10) 看起來像是另外冒出來的一個公式，但它不是重新定義出來湊答案的，而是可以從 (3.6) 用連鎖律 (chain rule) 硬算出來，結果剛好也等於 MD(periods)/ k 。推導如下。

把每期的殖利率記成 $y_p = y/k$ (也就是 (3.6) 裡原本那個 y ，這裡改名是為了不跟年殖利率 y 搞混)。因為 P 其實是 y_p 的函數， y 只是透過 $y_p = y/k$ 間接影響 P ，連鎖律給出

$$\frac{\partial P}{\partial y} = \frac{\partial P}{\partial y_p} \cdot \frac{\partial y_p}{\partial y} = \frac{1}{k} \frac{\partial P}{\partial y_p}.$$

直覺上： y 動 1 個單位，真正拿去折現的 $y_p = y/k$ 只被拖著動 $1/k$ 個單位，所以價格對「年殖利率 y 」的敏感度，天生就比對「期殖利率 y_p 」的敏感度小了 k 倍——這就是那個 $1/k$ 的來源，跟前面的係數完全無關。

把這個關係代進 (3.10)：

$$\begin{aligned} -\left(1 + \frac{y}{k}\right) \frac{\partial P}{\partial y} \frac{1}{P} &= -(1 + y_p) \cdot \frac{1}{k} \frac{\partial P}{\partial y_p} \cdot \frac{1}{P} = \frac{1}{k} \underbrace{\left[-(1 + y_p) \frac{\partial P}{\partial y_p} \frac{1}{P} \right]}_{= \text{MD(periods)}, \text{ 即 (3.6)}} = \frac{\text{MD(periods)}}{k}. \end{aligned}$$

注意係數 $(1 + y/k)$ 只是原本的 $(1 + y_p)$ 換了個寫法而已，並沒有跟著 k 做任何調整；真正帶出 $1/k$ 的，是求導變數從 y_p 換成 y 時連鎖律多乘出來的那個因子。兩者是獨立的兩件事，只是恰好都跟 k 有關，才會讓最後結果乾淨地等於 MD(periods)/ k ——跟直接把加總式除以 k ((3.9) 的做法) 殊途同歸，這正是三個式子彼此一致、講的是同一件事的原因。

§3.6. Modified Duration

— Slides p. 103 —

The factor $1 + y$ in (3.6) is a nuisance if what one actually wants is the sensitivity itself. Dividing it out gives the quantity traders quote.

Definition 3.6.1 (Modified duration).

$$\text{modified duration} \triangleq -\frac{\partial P}{\partial y} \frac{1}{P} = \frac{\text{MD}}{1 + y}. \quad (3.11)$$

So modified duration is exactly the price volatility (3.1) of the first section, under a new name. By the Taylor expansion,

$$\text{percent price change} \approx -\text{modified duration} \times \text{yield change}. \quad (3.12)$$

Note 3.6.2. Taylor-expanding P to first order, $P(y+\Delta y) \approx P(y) + \frac{\partial P}{\partial y} \Delta y$; dividing by $P(y)$ and substituting (3.11) gives exactly (3.12) — the same computation that produced (3.2). “price volatility” and “modified duration” are two names for that one term.

Note 3.6.3. Modified duration equals MD under continuous compounding. With continuous compounding $P = \sum_i C_i e^{-yt_i}$, so $\partial P / \partial y = -\sum_i t_i C_i e^{-yt_i}$ and the ratio $-(\partial P / \partial y) / P$ is the PV-weighted average of the t_i with no correction factor at all. The $1+y$ in (3.11) is an artifact of discrete compounding; it is the discrete analogue of the fact that $\frac{d}{dy} e^{-yt} = -t e^{-yt}$ exactly, whereas $\frac{d}{dy} (1+y)^{-i} = -i(1+y)^{-i-1}$ picks up one extra factor of $(1+y)^{-1}$.

§3.6.i. Example

— Slides p. 104 —

Example 3.6.4. Consider a bond whose modified duration is 11.54 with a yield of 10%. If the yield increases instantaneously from 10% to 10.1%, the approximate percentage price change is

$$-11.54 \times 0.001 = -0.01154 = -1.154\%.$$

Remark 3.6.5. 注意單位：殖利率從 10% 升到 10.1%，變化量是 0.1 個百分點，也就是 $\Delta y = 0.001$ ，即十個基點 (basis point, bp)，不是 0.1。

§3.6.ii. Modified Duration of a Portfolio

— Slides p. 105 —

Proposition 3.6.6

By calculus, the modified duration of a portfolio equals

$$\sum_i \omega_i D_i, \quad (3.13)$$

where D_i is the modified duration of the i th asset and ω_i is the market value of that asset expressed as a percentage of the market value of the portfolio.

Proof. Let the portfolio value be $V = \sum_i P_i$ and $\omega_i = P_i / V$. Then

$$-\frac{1}{V} \frac{\partial V}{\partial y} = -\frac{1}{V} \sum_i \frac{\partial P_i}{\partial y} = \sum_i \frac{P_i}{V} \left(-\frac{1}{P_i} \frac{\partial P_i}{\partial y} \right) = \sum_i \omega_i D_i. \quad \blacksquare$$

§3.7. Effective Duration

— Slides p. 106 —

Yield changes may alter the cash flow, or the cash flow may be too complex for simple formulas — exactly the callable bonds and mortgage securities that broke (3.7) in Section 3.4. We need a general numerical formula for volatility, one that asks the pricing model rather than a closed form.

Definition 3.7.1 (Effective duration).

$$\text{effective duration} \triangleq \frac{P_- - P_+}{P_0(y_+ - y_-)}, \quad (3.14)$$

where P_- is the price if the yield is decreased by Δy , P_+ is the price if the yield is increased by Δy , P_0 is the initial price, y is the initial yield, and Δy is small.

So $y_{\pm} = y \pm \Delta y$ and $y_+ - y_- = 2\Delta y$: (3.14) is the **central difference** approximation to $-(\partial P / \partial y) / P$. The prices P_- , P_0 , P_+ are obtained by re-running the pricing model three times — with the cash flows allowed to change in response to the yield, which is the whole point.

— Slides p. 107 —

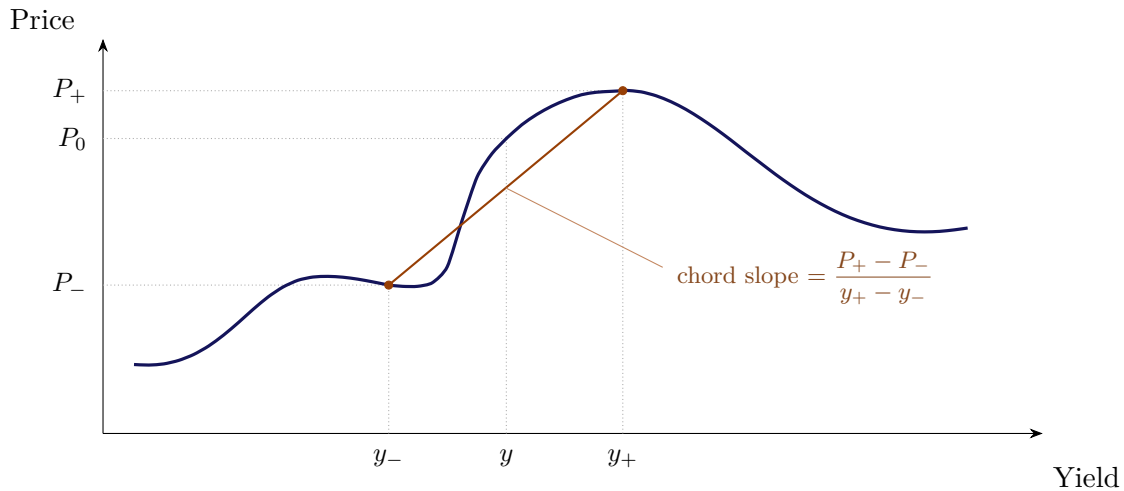


Figure 3.2.: Effective duration reads the slope of the chord through (y_-, P_-) and (y_+, P_+) , not of the tangent at y . The curve is traced from the plot shown in class. It is deliberately badly behaved—a plateau on the left, a steep rise through the bracket, and a peak just past y_+ —and it is *increasing* in the yield over $[y_-, y_+]$, so this security's duration there is negative. No closed-form duration formula can handle such a price curve.

Remark 3.7.2. 為什麼要定義 effective duration? (3.7) 的算法是先把價格寫成 y 的封閉公式，再對 y 解析微分。一旦現金流本身會隨 y 改變——可贖回債被贖回、房貸提前還款——這樣的封閉公式根本寫不出來，(3.7) 也就沒有意義（正是 Section 3.4 點出的情形）。effective duration 的辦法是繞開微分：把定價模型當黑箱，讓 y 分別偏移 $\pm\Delta y$ 、現金流也跟著一起變動，重新定價三次得到 P_- , P_0 , P_+ ，再用連接 (y_-, P_-) 與 (y_+, P_+) 的割線斜率去近似 (y, P_0) 處切線斜率 $-\partial P / \partial y$ 。也就是說，(3.14) 不管定價公式長什麼樣、甚至有沒有公式，都能算——這正是「一個泛用的數值公式」的意思。（順帶一提：一般中央差分寫成 $(P_+ - P_-) / (y_+ - y_-)$ 去近似 $\partial P / \partial y$ ；(3.14) 的分子把順序反過來寫成 $P_- - P_+$ ，分母卻維持 $y_+ - y_-$ 不變，這不是筆誤，而是把 duration 定義裡本來就有的那個負號直接吃進分子，等價於在標準中央差分前面乘上 -1 。）

這張圖在講什麼？橫軸是殖利率 y 、縱軸是價格 P ；虛線標出 (y_-, P_-) 、 (y, P_0) 、 (y_+, P_+) 三個點。曲線刻意畫得不規則——左邊一段平台、中間陡然爬升、過了 y_+ 後反而往下掉——用來代表 MBS、**純息分割債券** (interest-only strip, IO) 這類現金流會隨 y 改變的證券，這種形狀一般封閉公式完全處理不了。橙色直線是連接 (y_-, P_-) 與 (y_+, P_+) 的割線，它的斜率（取負號、除以 P_0 ）正是 (3.14) 在算的量。在圖上區間 $[y_-, y_+]$ 裡曲線是遞增的，即 $P_- < P_+$ ，於是 $P_- - P_+ < 0$ ，代入 (3.14) 得到 effective duration < 0 ——這就是 Section 3.4 「duration 可以是負的」那個 IO strip 例子的具體樣子。

— Slides p. 108 —

One can compute the effective duration of just about any financial instrument: all it takes is a pricing routine that can be called three times. An alternative is to use the one-sided difference

$$\frac{P_0 - P_+}{P_0 \Delta y}, \quad (3.15)$$

which is more economical — two model runs instead of three, and P_0 is usually already known — but theoretically less accurate.

Note 3.7.3. The accuracy claim is the standard one from numerical differentiation. Expanding $P(y \pm \Delta y)$ about y ,

$$\frac{P_- - P_+}{2\Delta y} = -\frac{\partial P}{\partial y} - \frac{1}{6} \frac{\partial^3 P}{\partial y^3} (\Delta y)^2 + \dots, \quad \frac{P_0 - P_+}{\Delta y} = -\frac{\partial P}{\partial y} - \frac{1}{2} \frac{\partial^2 P}{\partial y^2} \Delta y + \dots.$$

The central difference loses the first-order error term to symmetry: its leading error is $O((\Delta y)^2)$ against $O(\Delta y)$ for the one-sided version. Halving Δy buys a factor of four in one case and only a factor of two in the other.

Remark 3.7.4. P_0 是現在的價格，通常本來就知道；中央差分 (3.14) 分子不含 P_0 ，真正要多算的是 P_- 與 P_+ 兩次；單邊差分 (3.15) 直接拿現成的 P_0 當基準，只需多算一次 P_+ ，所以少一次定價。

不準在哪裡：把 y 對稱地往兩邊推，二階（曲率）誤差在 P_- 與 P_+ 裡符號相反、相減即抵消，只剩三階項，誤差 $O((\Delta y)^2)$ ；只往一個方向推就沒有這種抵消，誤差停在 $O(\Delta y)$ 。把 Δy 減半，中央差分誤差縮到 1/4、單邊只縮到 1/2——這正是「省一次定價」換來的代價。

§3.8. The Practices: Duration in Percentage Terms

— Slides p. 109 —

Duration is usually expressed in percentage terms — call it $D_\%$ — for quick mental calculation.³ The percentage price change expressed in percentage terms is then approximated by

$$-D_\% \times \Delta r \quad (3.16)$$

when the yield increases instantaneously by $\Delta r\%$.

Remark 3.8.1. 先說清楚一個容易誤會的地方： $D_\%$ 不是把 D 換算成百分比形式後得到的另一個數字。它只是 (3.16) 這條式子裡的係數；式子裡真正換了單位的是 Δr （百分點）跟價格變化（百分比），不是係數本身。下面會證明 $D_\%$ 其實跟原來的 duration 數值完全相同——證完之後的說明會解釋為什麼這件事一點都不奇怪。

Example 3.8.2. Suppose $D_\% = 10$ and $\Delta r = 2$. Price will drop by 20%, as $10 \times 2 = 20$.

Exercise 3.8.3. $D_\%$ in fact equals the original duration. Prove it!

Proof. Let D be the modified duration, so that $\Delta P/P \approx -D \Delta y$ with Δy in decimals. Writing the yield change in percentage points, $\Delta r = 100 \Delta y$, and the price change in percent, $100 \Delta P/P$, gives

$$100 \frac{\Delta P}{P} \approx -D (100 \Delta y) = -D \Delta r.$$

Comparing with (3.16) yields $D_\% = D$. The two factors of 100 — one on each side — cancel exactly, which is why the convention costs nothing. ■

³Neftci (2008): “Market professionals do not like to use decimal points.”

Remark 3.8.4. $D\%$ 是哪一種 duration？不是新的第五種量。前面算出來的那個數字 D ——這裡是 modified duration (3.11)，若現金流會隨 y 變就是 effective duration (3.14)—— $D\%$ 就是同一個 D ，數值完全不變，只是換一套單位讀它：把 Δy （小數）換成以百分點記的 $\Delta r = 100 \Delta y$ ，把 $\Delta P/P$ （小數）換成以百分比記的 $100 \Delta P/P$ 。

§3.9. Hedging

— Slides p. 110 —

Hedging offsets the price fluctuations of the position to be hedged by the hedging instrument in the opposite direction, leaving the total wealth unchanged.

Remark 3.9.1. **部位** (position) 指你手上持有的一筆資產或負債；要避險的部位 (the position to be hedged) 就是你原本就有、想保護價值的那筆持倉，例如一個債券投資組合。避險工具 (hedging instrument) 是額外配上去的另一筆部位，專門用來抵銷前者的風險，通常是另一種對 y 敏感度相反的證券。

「往相反方向抵銷」的意思是： y 一變動，原部位賺（賠）多少，避險工具那邊就賠（賺）差不多多少；兩筆加在一起，總財富幾乎不受 y 變動影響——這就是 leaving the total wealth unchanged。「差不多多少」要怎麼配、配多少，就是本節接下來要算的東西。

Percentages are the wrong unit for that job: two positions of different size with equal and opposite *modified* durations do not cancel. What must cancel is the change in *dollars*.

Definition 3.9.2 (Dollar duration).

$$\text{dollar duration} \triangleq \text{modified duration} \times \text{price} = -\frac{\partial P}{\partial y}. \quad (3.17)$$

The approximate dollar price change is then

$$\text{price change} \approx -\text{dollar duration} \times \text{yield change}. \quad (3.18)$$

One can hedge a bond portfolio with a dollar duration D by bonds with a dollar duration $-D$: the two first-order terms cancel and the combined position is, to first order, insensitive to y .

Intuition . Modified duration is a *rate*; dollar duration is a *quantity*. Hedging is an accounting identity about quantities, so the hedge ratio must be computed in dollar durations.

Note 3.9.3. The hedge is only first-order and only instantaneous. It fails for large moves — that is the second-order term, i.e. convexity — and it fails when the two legs' yields do not move together, which no single- y framework can express.

Remark 3.9.4. 「first-order」「second-order」呼應本章最開頭的泰勒展開 $\Delta P = (\partial P / \partial y) \Delta y + \frac{1}{2} (\partial^2 P / \partial y^2) (\Delta y)^2 + \dots$ ：一階項只跟 Δy 成正比，也就是 duration 那一項；二階項跟 $(\Delta y)^2$ 成正比，也就是 convexity 那一項。

怎麼看出避險只做到一階：(3.18) 這條式子本身就只有一階項，完全沒寫 $(\Delta y)^2$ 那一塊。配 D 與 $-D$ 兩筆 dollar duration，只保證兩邊代入 (3.18) 算出來的一階項互相抵銷；至於兩邊真正的二階項 (convexity)，公式裡從沒要求它們相等，通常也真的不相等，所以加總後還留著一段沒被抵銷的 $(\Delta y)^2$ 尾巴。 Δy 很小時這尾巴可以忽略，避險看起來完美； Δy 一大，這段二階尾巴就顯現出來，避險失靈——這就是「fails for large moves」。

至於「兩腳殖利率沒有一起動」：腳 (leg) 就是避險這筆交易的其中一邊——被避險的部位是一腳，拿來避險的工具是另一腳，兩腳各自有自己的價格、自己的殖利率。這是另一個完全不同的限制，跟 order 高低無關：整套 D 、 $-D$ 的算法都假設這兩腳共用同一個 y （利率平行移動）， y 一變兩腳一起變、變一樣多。如果兩腳的殖利率各自變動、不同步（例如一邊是公司債、一邊是公債，信用利差自己在變），就沒有單一個 y 可以同時代入兩腳的一階分析了。

§3.10. Convexity

— Slides p. 111 —

Duration is the first derivative. The second derivative gets its own name.

Definition 3.10.1 (Convexity).

$$\text{convexity (in periods)} \triangleq \frac{\partial^2 P}{\partial y^2} \frac{1}{P}. \quad (3.19)$$

Exercise 3.10.2. The convexity of a level-coupon bond is positive. Prove it!

Proof. Write the coupons and the par value as separate terms,

$$P = \sum_{i=1}^n C_i (1+y)^{-i} + F(1+y)^{-n},$$

where $C_i = C$ is the coupon paid at period i and F is the par value returned at maturity. Differentiating term by term,

$$\frac{\partial P}{\partial y} = - \sum_{i=1}^n i C_i (1+y)^{-i-1} - n F (1+y)^{-n-1},$$

and once more,

$$\frac{\partial^2 P}{\partial y^2} = \sum_{i=1}^n i(i+1) C_i (1+y)^{-i-2} + n(n+1) F (1+y)^{-n-2} > 0,$$

since $C_i \geq 0$, $F \geq 0$, not all of them are zero, and $1+y > 0$. Dividing by $P > 0$ leaves the sign unchanged. The argument uses nothing about level coupons: *any* security with non-negative, yield-independent cash flows has positive convexity. ■

For a bond with positive convexity, the price rises more for a rate decline than it falls for a rate increase of equal magnitude. (也就是說，凸性為正的債券，殖利率下降時價格漲的幅度，會大於殖利率上升同樣幅度時價格跌的幅度。) That asymmetry is the content of [Figure 3.3](#): the true price curve bends away from its own tangent, and it bends *upwards*, so the tangent understates the price on both sides.

— Slides p. 112 —

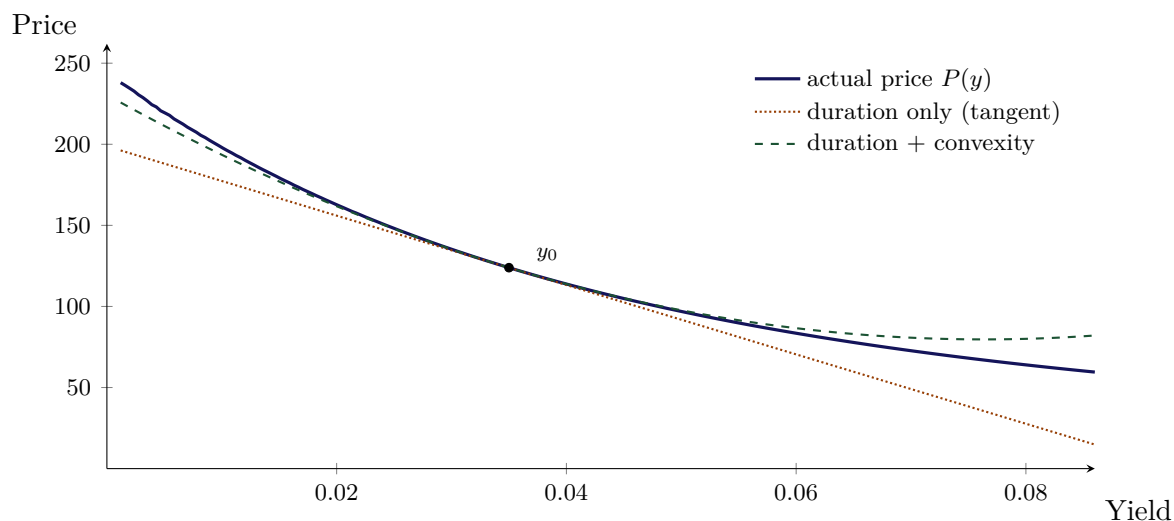


Figure 3.3.: Price versus yield for a level-coupon bond ($n = 30$, $C = 4.8$, $F = 100$), expanded about $y_0 = 3.5\%$. The tangent — the duration-only approximation — lies below the true curve everywhere, understating the gain from a rate fall and overstating the loss from a rate rise. Adding the convexity term recovers almost all of the discrepancy over the plotted range.

Remark 3.10.3. 把這張圖讀成三句話：

- 實線是真正的價格曲線，對殖利率遞減且向上凸。
- 點線是只用 duration 的線性近似，也就是曲線在 y_0 的切線。由於曲線是凸的，切線恆在曲線下方：殖利率跌時低估漲幅，殖利率漲時高估跌幅——兩邊都對持有人有利，這就是「正凸性是好事」的圖解。
- 虛線加進了凸性的二次項，在相當寬的範圍內幾乎與實線重合。

注意切線與真實曲線的落差隨 $|\Delta y|$ 呈平方成長： $\Delta y = 1\%$ 時幾乎看不出來， $\Delta y = 4\%$ 時已經差了幾十塊。這就是為什麼小幅度時可以只用 duration，大幅度時非補上凸性不可。

§3.10.i. Consequences and Units

— Slides p. 113 —

So between two bonds with the same price and duration, the one with a higher convexity is more valuable.⁴

Suppose there are k periods per annum. Convexity measured in periods and convexity measured in years are related by

$$\text{convexity (in years)} = \frac{\text{convexity (in periods)}}{k^2}. \quad (3.20)$$

Note 3.10.4. This is the chain rule, applied twice. Let $u = y/k$ be the periodic yield, so the price is $P(u)$, and let $\tilde{P}(y) \triangleq P(y/k)$ be the same price viewed as a function of the annual yield y . Since $u = y/k$ is linear in y , $du/dy = 1/k$ is constant, and

$$\frac{\partial \tilde{P}}{\partial y} = \frac{\partial P}{\partial u} \cdot \frac{du}{dy} = \frac{1}{k} \frac{\partial P}{\partial u},$$

⁴Do you spot a problem here (Christensen & Sørensen, 1994)?

which is $\text{MD}(\text{years}) = \text{MD}(\text{periods})/k$ up to the $(1+u)$ -vs.-1 normalising factor in (3.6)–(3.10). Differentiating once more, and noting that $\partial P/\partial u$ is again a function of $u = y/k$, the chain rule applies a *second* time:

$$\frac{\partial^2 \tilde{P}}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{1}{k} \frac{\partial P}{\partial u} \right) = \frac{1}{k} \cdot \frac{\partial^2 P}{\partial u^2} \cdot \frac{du}{dy} = \frac{1}{k^2} \frac{\partial^2 P}{\partial u^2}.$$

Dividing both sides by the same P gives exactly (3.20): the k^2 is two factors of k^{-1} , one from each application of the chain rule, not an independent convention.

More generally, this is an induction on the order r of the derivative. The base case $r = 1$ is the MD computation above. For the inductive step, suppose $\partial^r \tilde{P}/\partial y^r = k^{-r} \partial^r P/\partial u^r$. The right-hand side is a function of $u = y/k$ composed with $y \mapsto y/k$, so differentiating both sides once more in y picks up one further factor $du/dy = 1/k$:

$$\frac{\partial^{r+1} \tilde{P}}{\partial y^{r+1}} = k^{-r} \cdot \frac{\partial^{r+1} P}{\partial u^{r+1}} \cdot \frac{du}{dy} = k^{-(r+1)} \frac{\partial^{r+1} P}{\partial u^{r+1}}.$$

Hence any quantity that is an r th derivative with respect to the yield converts between periods and years by a factor of k^{-r} : k^{-1} for duration ($r = 1$), k^{-2} for convexity ($r = 2$).

Remark 3.10.5. 那個腳註的「問題」在哪裡？

表面上的推論是：兩檔債券今天價格相同、duration 相同，但一檔凸性較高。殖利率不論漲跌，凸性高的那檔表現都不會比較差（漲得多、跌得少）。既然如此，它應該比較值錢——但它們今天價格明明一樣。這聽起來像是白吃的午餐，也就是**套利**（arbitrage）機會。

Christensen 與 Sørensen（1994）點出的正是這個矛盾：在**無套利**（no-arbitrage）的**期限結構**（term structure）模型裡，這種情形不可能持續存在。真正的機制是凸性要付錢買——市場會把凸性高的債券的殖利率壓低（或說價格推高），高凸性帶來的好處恰好被較低的收益率抵銷。

更根本的問題出在推論的隱含前提：整條殖利率曲線平行移動。這個假設本身就與無套利相容不了（若曲線只能平行移動，就能建構出套利部位）。一旦改用真正的期限結構模型，「同價同 duration 但凸性較高」的兩檔債券根本不會同時存在於市場上。

§3.11. Use of Convexity

— Slides p. 114 —

The approximation $\Delta P/P \approx -\text{duration} \times \text{yield change}$ works for small yield changes. For larger yield changes, keep the next term of the Taylor expansion:

$$\begin{aligned} \frac{\Delta P}{P} &\approx \frac{\partial P}{\partial y} \frac{1}{P} \Delta y + \frac{1}{2} \frac{\partial^2 P}{\partial y^2} \frac{1}{P} (\Delta y)^2 \\ &= -\text{duration} \times \Delta y + \frac{1}{2} \times \text{convexity} \times (\Delta y)^2. \end{aligned} \tag{3.21}$$

Recall Figure 3.3: the dashed curve is (3.21), and the gap between it and the dotted tangent is the convexity term doing its work.

Note 3.11.1. Because convexity is positive for an ordinary bond, the second term is positive whichever way y moves. The correction always *raises* the predicted price relative to the duration-only estimate — which is exactly the asymmetry stated on p. 111 of the slides.

§3.12. The Practices: Convexity in Percentage Terms

— Slides p. 115 —

Convexity is likewise usually expressed in percentage terms — call it $C_{\%}$ — for quick mental calculation. The percentage price change expressed in percentage terms is approximated by

$$-D_{\%} \times \Delta r + C_{\%} \times (\Delta r)^2/2 \quad (3.22)$$

when the yield increases instantaneously by $\Delta r\%$.

Example 3.12.1. Price will drop by 17% if $D_{\%} = 10$, $C_{\%} = 1.5$ and $\Delta r = 2$, because

$$-10 \times 2 + \frac{1}{2} \times 1.5 \times 2^2 = -20 + 3 = -17.$$

Exercise 3.12.2. $C_{\%}$ equals convexity divided by 100. Prove it!

Proof. Start from (3.21) with Δy in decimals and substitute $\Delta y = \Delta r/100$:

$$100 \frac{\Delta P}{P} \approx -D(100 \Delta y) + \frac{1}{2} \text{conv} \frac{(100 \Delta y)^2}{100} = -D \Delta r + \frac{1}{2} \frac{\text{conv}}{100} (\Delta r)^2.$$

Comparing with (3.22) gives $D_{\%} = D$ as before and $C_{\%} = \text{conv}/100$. ■

Remark 3.12.3. $D_{\%} = D$ 這件事哪裡容易搞錯？直覺上很容易以為「掛了 %」代表這個數字本身被換成了百分比形式，所以 $D_{\%}$ 應該跟 D 長得不一樣（比方乘或除以 100）。但 $D_{\%}$ 、 $C_{\%}$ 從來不是把 D 、conv 本身換算單位，它們只是 (3.16)、(3.22) 這兩條式子裡的係數；式子裡真正換單位的是變量：輸入端 $\Delta y \rightarrow \Delta r = 100\Delta y$ ，輸出端 $\Delta P/P \rightarrow 100 \Delta P/P$ 。係數本身要變成多少，要看它前面到底黏著幾個 Δy 。

§3.13. Effective Convexity

— Slides p. 116 —

Just as effective duration replaced the closed-form first derivative by a difference quotient, effective convexity replaces the second.

Definition 3.13.1 (Effective convexity).

$$\text{effective convexity} \triangleq \frac{P_+ + P_- - 2P_0}{P_0(0.5 \times (y_+ - y_-))^2}, \quad (3.23)$$

where P_- is the price if the yield is decreased by Δy , P_+ the price if it is increased by Δy , P_0 the initial price, y the initial yield, and Δy small.

Since $0.5 \times (y_+ - y_-) = \Delta y$, the denominator is just $P_0(\Delta y)^2$, and (3.23) is the standard central second difference $[P_+ + P_- - 2P_0]/(\Delta y)^2$ normalised by P_0 . Effective convexity is most relevant when a bond's cash flow is interest rate sensitive — the callable bonds and mortgage securities for which (3.19) cannot even be written down, and for which the convexity often turns out *negative*.

Remark 3.13.2. 這條式子在算什麼？沿用本章最開頭的泰勒展開 $\Delta P = P'\Delta y + \frac{1}{2}P''(\Delta y)^2 + \frac{1}{6}P'''(\Delta y)^3 + \dots$ （這裡 ' 表示對 y 微分），在 y 兩側各展開一次：

$$P_+ = P_0 + P'\Delta y + \frac{1}{2}P''(\Delta y)^2 + \frac{1}{6}P'''(\Delta y)^3 + \dots, \quad P_- = P_0 - P'\Delta y + \frac{1}{2}P''(\Delta y)^2 - \frac{1}{6}P'''(\Delta y)^3 + \dots$$

effective duration 是把這兩式相減，讓一階項 $P'\Delta y$ 留下來、其他項互相抵銷或變高階小量。effective convexity 反過來，把兩式相加：一階、三階這些奇數階的項符號相反、正好消掉，只留下偶數階：

$$P_+ + P_- = 2P_0 + P''(\Delta y)^2 + O((\Delta y)^4) \implies \frac{P_+ + P_- - 2P_0}{(\Delta y)^2} = P'' + O((\Delta y)^2).$$

再除以 P_0 ，換成跟 (3.19) 一樣「除以 P 」的正規化，就得到 (3.23)。所以這條公式跟 effective duration 是同一招的延伸：effective duration 用兩點的差逼近一階導數，effective convexity 用三點的加權和逼近二階導數。

幾何意義。 y 是 y_- 、 y_+ 的中點，所以連接 (y_-, P_-) 與 (y_+, P_+) 的割線，在 y 這一點的高度正好是兩端點的平均 $(P_- + P_+)/2$ 。因此

$$P_+ + P_- - 2P_0 = 2 \left[\frac{P_- + P_+}{2} - P_0 \right]$$

就是「割線在 y 處的高度」與「曲線實際在 y 處的高度 P_0 」之間差距的兩倍，見 Figure 3.4。曲線向上凸（跟 Figure 3.3 一樣）時，曲線落在割線下方，這個差距是正的，effective convexity 隨之為正；曲線若在這段反而向下凹，差距變負，算出來的 effective convexity 就是負的——這正是 MBS、可贖回債券的凸性常常是負的原因。

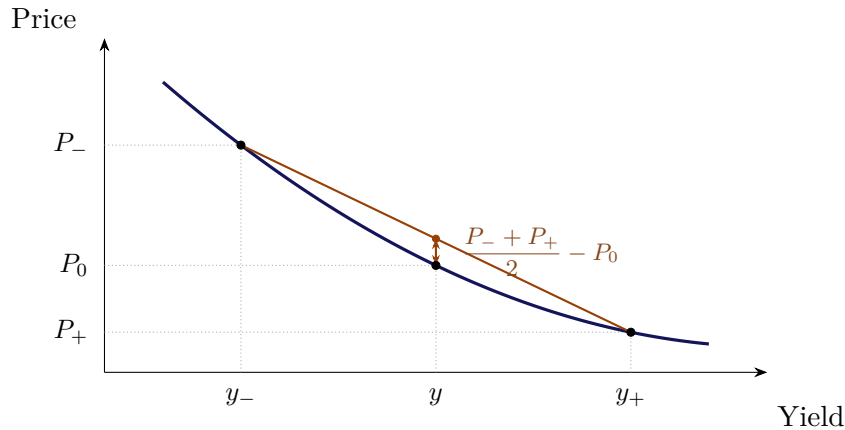


Figure 3.4.: The chord through (y_-, P_-) and (y_+, P_+) sits above the curve at the midpoint y whenever the curve is convex there. Effective convexity is, up to the factor $2/[(\Delta y)^2 P_0]$, exactly that vertical gap $(P_- + P_+)/2 - P_0$.

How to choose the right Δy is a delicate matter.

§3.14. A Numerical Caveat: Second Differences in Floating Point

— Slides p. 117 —

The last sentence is not a throwaway remark, and the slides make the point with a deliberately trivial example. Take $f(x) = x^2$, whose second derivative at $x = 1$ is exactly 2, and form the central second difference⁵

$$\frac{(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2}{(\Delta x)^2}.$$

⁵The slide writes the target as $d^2 f(x)^2 / dx^2$; the quantity actually approximated is $d^2 f / dx^2 = 2$.

In exact arithmetic the numerator is $2(\Delta x)^2$ and the quotient is 2 for *every* Δx — there is no truncation error at all, since f is a quadratic. In floating point the answer is a disaster.

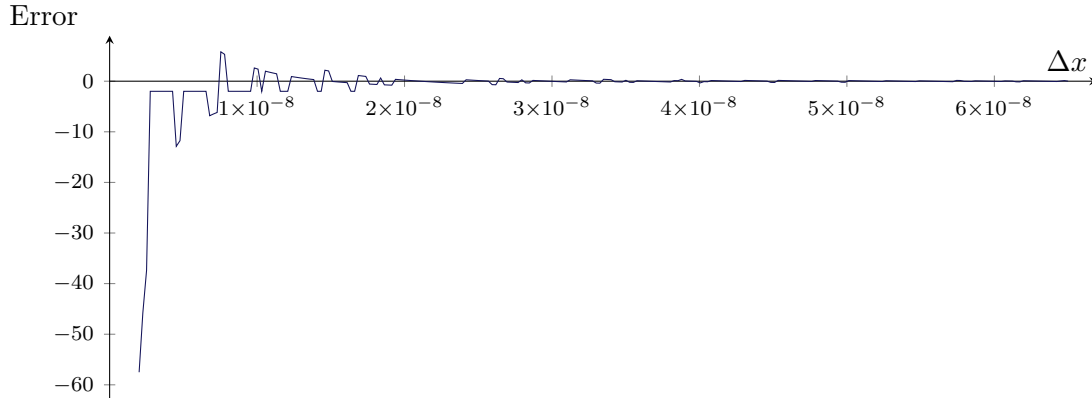


Figure 3.5.: The difference between $[(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2]/(\Delta x)^2$ and its exact value 2, computed in IEEE double precision. The flat stretches at -2 are the cases where the numerator underflows to exactly zero; the spikes are the cases where it survives as one or two units in the last place.

Remark 3.14.1. 這個現象叫做**災難性抵銷** (catastrophic cancellation)，值得一步一步看清楚。

取 $\Delta x \approx 10^{-8}$ 。在雙精度浮點數中，1 附近的**機器精度** (machine epsilon) 是 $\varepsilon = 2^{-52} \approx 2.22 \times 10^{-16}$ ，也就是說 1 附近能分辨的最小間隔就是這個數量級。

而 $(1 + \Delta x)^2 + (1 - \Delta x)^2 - 2$ 的真值是 $2(\Delta x)^2 \approx 2 \times 10^{-16}$ ——恰好落在 ε 的同一個數量級。換句話說，我們要算的整個答案都藏在最後一個位元裡。多數情況下捨入直接把它抹成 0，商數變成 0，誤差就是 $0 - 2 = -2$ (圖中那些平坦段)；偶爾殘留一兩個 ulp，除以 $(\Delta x)^2 \approx 10^{-16}$ 之後就被放大成 ± 10 甚至 -57 的巨大誤差 (圖中的尖峰)。

推論： Δy 不是愈小愈好。把**截斷誤差** (truncation error) 與**捨入誤差** (round-off error) 寫在一起，設 u 為相對捨入誤差的量級：

$$\underbrace{\frac{1}{12}|P^{(4)}|(\Delta y)^2}_{\text{截斷誤差, 隨}\Delta y\text{ 遞減}} + \underbrace{\frac{4u|P|}{(\Delta y)^2}}_{\text{捨入誤差, 隨}\Delta y\text{ 遞增}}.$$

兩者相權，二階差分的最佳步長約為 $\Delta y \sim \varepsilon^{1/4} \approx 1.2 \times 10^{-4}$ ；一階 (中央) 差分則是 $\Delta y \sim \varepsilon^{1/3} \approx 6 \times 10^{-6}$ 。所以 (3.23) 的 Δy 取 1 個 basis point (10^{-4}) 左右通常是合理的，取 10^{-8} 則會得到一堆雜訊。

This numerical issue is common in financial engineering but does not have general solutions yet.⁶

§3.15. Interest Rates and Bond Prices: Which Determines Which?

— Slides p. 118 —

Question 3.15.1. Do interest rates determine bond prices, or do bond prices determine interest

⁶See pp. 872ff of the slides.

rates?^a

^aContributed by Mr. Wang, Cheng (R01741064) on March 5, 2014.

If you have one, you have the other. Given a price, the yield is the root of (3.3); given a yield, the price is its value. So they are just two names given to the same thing: the **cost of fund (資金成本)**.

The difference is one of habit rather than of substance:

- Traders most likely work with prices.
- Banks most likely work with interest rates.

Price and yield are one quantity in two coordinate systems. Duration and convexity are the derivatives that translate between them.

價格與殖利率是同一個量的兩套座標；duration 與 convexity 就是在兩者之間換算的導數。

4. Term Structure of Interest Rates

— Slides pp. 119–120 —

*Why is it that the interest of money is lower,
when money is plentiful?*

— Samuel Johnson (1709–1784)

If you have money, don't lend it at interest.

*Rather, give [it] to someone
from whom you won't get it back.*

— Thomas Gospel 95

Every formula so far has carried a single interest rate. One y discounted all of a bond's cash flows in (2.6); one y was perturbed to get duration and convexity in the last chapter. That was a convenient fiction, and this chapter retires it.

Money borrowed for three months and money borrowed for thirty years do not cost the same, and the pattern of rates across maturities — the **term structure** — is itself the object to be measured. The plan is:

- (1) say what the term structure is and what its standard shapes are;
- (2) replace the single y by a whole schedule of **spot rates** $S(1), S(2), \dots$, one per maturity, and show why yield-to-maturity discounting was inconsistent all along;
- (3) extract those spot rates from observed bond prices (**bootstrapping**), and see where that procedure runs into the data;
- (4) use the curve to measure credit risk (**spreads**) and relate it back to the yield curve we started from.

Remark 4.0.1. 信用風險 (credit risk)：發行人到期還不出錢（違約）的風險。公司債可能違約，殖利率因此比公債高，這多出來的部分就是**利差** (spread)，細節見 [Section 4.8](#)。

§4.1. What the Term Structure Is

— Slides p. 121 —

The term structure is concerned with how interest rates change with maturity. Concretely, the set of yields to maturity for bonds form the term structure — subject to two conditions that do all the work:

- The bonds must be of equal quality.
- They differ solely in their terms to maturity.

The term structure is fundamental to the valuation of fixed-income securities.

Remark 4.1.1. 期限結構 (term structure) 就是「殖利率隨到期日怎麼變化」畫出來的一條曲線——橫軸是到期日，縱軸是殖利率。

具體做法是拿一群債券各自的到期殖利率當縱軸上的點，但只有兩個條件都成立時，畫出來的才算是期限結構：品質相同、只差到期日。這樣才能保證點跟點之間的殖利率差異純粹來自到期日不同；若拿信用品質不同的債券比，測到的其實是信用風險 ([Section 4.8](#))，畫出來的就不是期限結構。

最後一句「fundamental to valuation」是說：債券價格是把每期現金流分別折現再加總，不同到期日理當用不同的折現率，而期限結構正是提供這些折現率的來源——這也是本章稍後要用一整組 spot rate $S(1), S(2), \dots$ 取代單一 y 的原因。

— Slides p. 122 —

Three names get used loosely and are worth separating once and for all:

- The term “**term structure**” often refers exclusively to the yields of *zero-coupon* bonds. This is the convention adopted here, and the reason will become clear in [Section 4.4](#): only a zero has a single, unambiguous date attached to it.
- A **yield curve** plots the yields to maturity of *coupon* bonds against maturity.
- A **par yield curve** is constructed from bonds trading near par.

Note 4.1.2. 這三條曲線是用同一個市場的資料，算出來的三種不同東西。

一檔 coupon bond 的 yield 為什麼會「混雜」不同到期日的利率：它的 YTM 是用同一個折現率 y 去折現所有期數的現金流、再反解出來的單一數字；coupon 越高，越早期的現金流在價格裡佔比越重，YTM 就越偏向那些早期的折現率。換句話說，同樣 n 年到期、但 coupon 不同的兩檔債券，YTM 一般並不相同——同一個到期日底下，YTM 其實不只一個值，會隨挑到哪個 coupon 而變。這就是「yield curve 是 term structure 被混雜後的版本」的意思。

par yield curve 就是從這些不同的 coupon 裡挑出一個標準答案：要求債券恰好 trade at par，即 $P = F$ 。由價格公式 (3.3)，

$$P = \frac{C}{y} \left[1 - (1 + y)^{-n} \right] + F(1 + y)^{-n},$$

令 $P = F$ ，兩邊減去 $F(1 + y)^{-n}$ 、再除以共同的 $[1 - (1 + y)^{-n}]$ ，直接解出 $F = C/y$ ，也就是 coupon rate C/F 剛好等於 yield y 。所以「trade at par」不是巧合，而是代數上唯一決定了 coupon 該取多少：對每個到期日，par yield curve 只採用「coupon 恰好等於自己 yield」的那一檔債券，不再讓 coupon 隨便挑——這就是把「混雜」固定下來的意思。實務上很少有債券剛好以 par 交易，所以會取 trading near par 的債券近似，此時 $C/F \approx y$ 只是近似相等，不是恰好相等。

市面上大多數所謂的「yield curve」，包括 [Figure 4.1](#) 裡的美國公債利率曲線，其實都是 par yield curve。

§4.1.i. The Yield Curve of U.S. Treasuries

— Slides p. 123 —

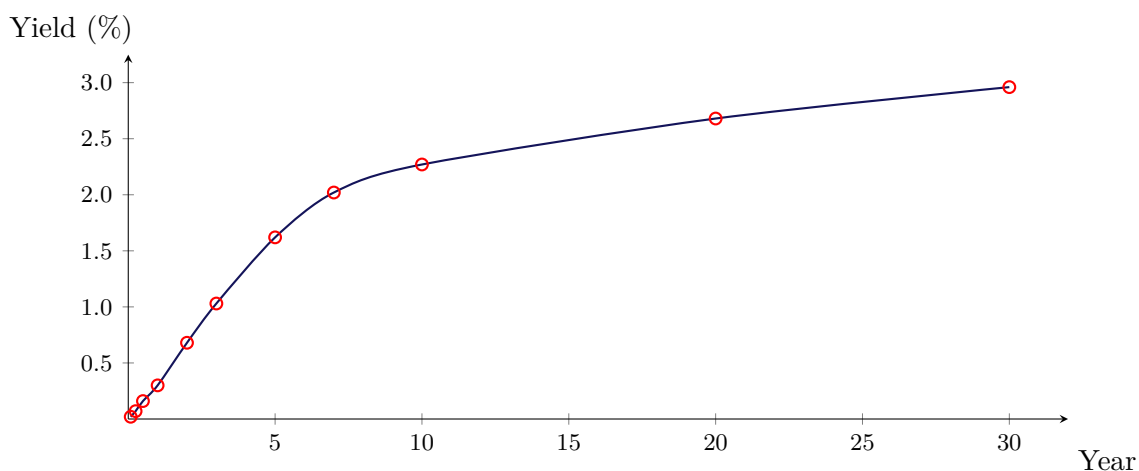


Figure 4.1.: Yield curve of U.S. Treasuries as of July 24, 2015, as shown in class. The circles are the eleven quoted maturities — 1 and 3 and 6 months, then 1, 2, 3, 5, 7, 10, 20 and 30 years — and the line is an interpolation through them. Note how unevenly the quotes are spaced: eight of the eleven sit inside the first ten years.

Remark 4.1.3. 這張圖有兩個訊息，一個明顯、一個要留意。

明顯的是形狀：短端幾乎貼地（1 個月只有 0.02%），長端接近 3%，是標準的**正常**（normal）曲線。這是 2015 年的美國——金融海嘯後聯準會把短率壓在零附近已近七年。

要留意的是橫軸上的點有多稀疏。市場只在十一個到期日上報價，其中八個擠在前十年；10 年到 20 年之間整整十年沒有任何資料點。

§4.1.ii. Four Typical Shapes

— Slides p. 124 —

- A **normal** yield curve is upward sloping.
- An **inverted** yield curve is downward sloping.
- A **flat** yield curve is flat.
- A **humped** yield curve is upward sloping at first but then turns downward sloping.

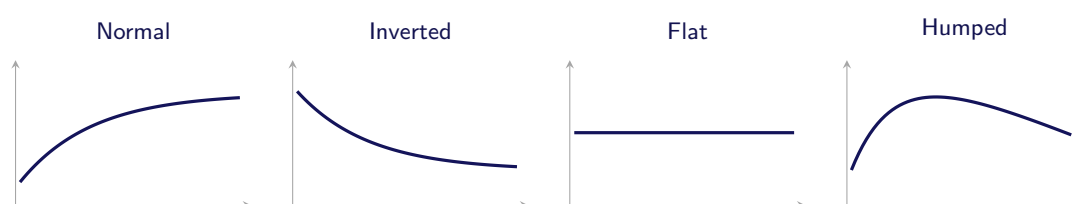


Figure 4.2.: The four shapes named in class. The horizontal axis is maturity and the vertical axis is yield in every panel.

Remark 4.1.4. 這四種形狀不只是分類，市場普遍把它們讀成對未來的訊息：

- **正常**（normal）：最常見。長天期資金較貴，通常解讀為經濟擴張、或投資人對長期持有要求額外補償。
- **倒掛**（inverted）：**短率**（short rate）高於長率。歷史上多次出現在景氣衰退之前，因此被當作預警訊號——市場預期央行日後必須降息。
- **平坦**（flat）：常是正常與倒掛之間的過渡狀態。
- **駝峰**（humped）：中期利率最高，兩端較低。

倒掛那一條的推理值得補一句。長天期利率大致是「今天的短率」與「市場預期未來各期短率」的平均：把錢鎖十年，跟每期滾動短天期投資十年，預期報酬應該相當，否則資金會往報酬高的那邊跑到兩者拉平為止（這正是之後預期理論的核心）。既然長率是預期短率的平均，長率低於今天的短率就只有一種解釋——市場預期未來的短率會比現在低，也就是預期央行日後會降息。而央行通常是在景氣轉弱時才降息，倒掛因此被當成衰退的預警訊號。

要強調的是：這些都是經驗上的關聯，不是本課程要證明的定理。曲線形狀為何如此，是 Section 4.7 之後幾節（預期理論等）才會處理的問題；此處只需認得這四個名字。

§4.2. Spot Rates

— Slides p. 125 —

Definition 4.2.1 (Spot rate). The i -period **spot rate** $S(i)$ is the yield to maturity of an i -period zero-coupon bond. The PV of one dollar i periods from now is by definition

$$[1 + S(i)]^{-i}. \quad (4.1)$$

Three immediate remarks, all of them from the slides:

- (4.1) is the price of an i -period zero-coupon bond, by (2.11). There is nothing to prove: a zero has one cash flow, so its yield and its discount rate are the same number by construction.

- The one-period spot rate $S(1)$ is called the **short rate**.
- The **spot rate curve** — that is, the term structure, per our convention — is the plot of spot rates against maturity:

$$S(1), S(2), \dots, S(n).$$

Intuition . A spot rate is a *price*, not a forecast. $S(5) = 4\%$ says that one dollar delivered five periods from now trades today at 1.04^{-5} dollars; it says nothing about what the short rate will be in five periods' time.

即期利率 (spot rate) 是價格，不是預測。 $S(5) = 4\%$ 的意思是「五期後的一塊錢今天值 1.04^{-5} 元」，而不是「五期後的短率會是 4% 」。

§4.3. What Is Wrong with Yield-to-Maturity Discounting

— Slides p. 126 —

In the bond price formula (2.6),

$$\sum_{i=1}^n \frac{C}{(1+y)^i} + \frac{F}{(1+y)^n},$$

every cash flow is discounted at the same yield y . Now consider two **riskless bonds**¹ with *different* yields to maturity, the difference arising purely because their cash flows differ:

$$PV_1 = \sum_{i=1}^{n_1} \frac{C}{(1+y_1)^i} + \frac{F}{(1+y_1)^{n_1}}, \quad (4.2)$$

$$PV_2 = \sum_{i=1}^{n_2} \frac{C}{(1+y_2)^i} + \frac{F}{(1+y_2)^{n_2}}. \quad (4.3)$$

Note 4.3.1. 無風險債券 (riskless bond, 亦稱 default-free bond) 指的是承諾的現金流必定如期足額收到，發行人不會違約；實務上以公債（如美國國庫券）為代表。

這個假設在這裡不是裝飾，而是整個論證的關鍵：

- 兩檔債券都無風險，才能說它們在第 i 期付的那一塊錢是同一件商品——同一天、同樣金額、同樣確定。既然是同一件商品，就不該有兩個價格。
- 若兩者信用不同， $y_1 \neq y_2$ 反而理所當然：差額就是**信用價差** (credit spread)，是對違約風險的補償，談不上矛盾。把風險排除後，殖利率的差異就只剩下現金流的時間分佈可以解釋，這正是下一頁要指出的問題所在。

要注意「無風險」只針對違約而言，不表示沒有**利率風險**：市場利率變動時，債券價格照樣會漲跌（見 Table 2.3），只是持有到期必定拿得到 C 與 F 。

— Slides p. 127 —

The yield-to-maturity methodology discounts their *contemporaneous* cash flows with different rates. But shouldn't they be discounted at the same rate?

¹**Riskless** (or **default-free**) means the promised cash flows are certain to arrive in full and on time — the issuer cannot default. Government bonds such as U.S. Treasuries are the standard proxy.

Remark 4.3.2. 這一頁的投影片只丟出一個問句就結束了，但這是本章的樞紐，值得補完。

把 (4.2) 與 (4.3) 的第 i 期 (i 同時小於 n_1 與 n_2) 拿出來對照：兩檔債券都在同一天付出同樣的 C 元，兩者都無風險，可是前者用 $(1 + y_1)^{-i}$ 折現、後者用 $(1 + y_2)^{-i}$ 折現。既然 $y_1 \neq y_2$ ，同一天、同樣金額、同樣無風險的一塊錢，被賦予了兩個不同的現值。

這違反**一價法則** (law of one price)：同一件商品不能有兩個價格。而且這不只是美感問題——真有兩個價格就有**套利** (arbitrage) 空間：從便宜的那邊買進未來的一塊錢、往貴的那邊賣出，鎖住無風險利潤。

問題出在哪裡？ y 是「把一整串現金流壓成一個數字」的內部報酬率，它同時被當成第 1 期到第 n 期的折現率使用。但這串現金流的時間分佈（票息高低、到期長短）會影響這個平均值，於是不同的債券得到不同的 y 。換句話說， y 是債券的屬性，而折現率應該是日期的屬性。

解方就在下一節：每個日期給它自己的折現率 $S(i)$ ，與是哪一檔債券付這筆錢無關。**期限結構** (term structure) 平坦時 ($S(1) = S(2) = \dots$) 兩種作法才會一致；這也解釋了為什麼 y 一直「差不多堪用」——它是對真實折現曲線的一個加權平均近似。

§4.4. Spot Rate Discount Methodology

— Slides p. 128 —

The repair is to stop thinking of a bond as one instrument with one yield. A cash flow $C_1, C_2, \dots, C_n$ is equivalent to a *package of zero-coupon bonds*, with the i th bond paying C_i dollars at time i .

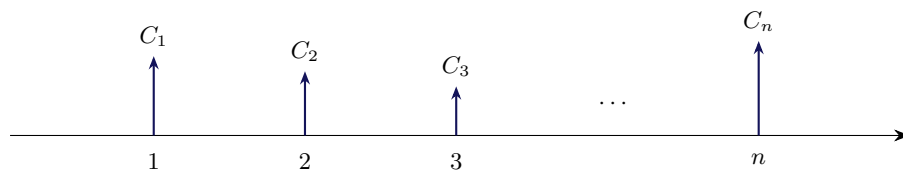


Figure 4.3.: A cash flow seen as a package of zero-coupon bonds. Each arrow is a separate security, to be priced by its own spot rate.

— Slides p. 129 —

Pricing each arrow at the spot rate matching *its own* date and adding up, a level-coupon bond has the price

$$P = \sum_{i=1}^n \frac{C}{[1 + S(i)]^i} + \frac{F}{[1 + S(n)]^n}. \quad (4.4)$$

- This pricing method incorporates information from the term structure.
- It discounts each cash flow at the matching spot rate.

The discount rate belongs to the *date*, not to the *bond*.
折現率屬於日期，不屬於債券。

Note 4.4.1. Compare (4.4) with (2.6). They have the same shape; the only change is that the single y has been replaced by a schedule $S(1), \dots, S(n)$. Every complaint of the previous section evaporates: two riskless bonds now discount their common time- i dollar by the same factor, because that factor no longer knows which bond it came from.

§4.5. Discount Factors

— Slides p. 130 —

It is worth giving those common factors a name, because they, rather than the rates, are what the pricing formula actually uses.

Definition 4.5.1 (Discount factors). In general, any riskless security having a cash flow $C_1, C_2, \dots, C_n$ should have a market price of

$$P = \sum_{i=1}^n C_i d(i), \quad (4.5)$$

where

$$d(i) \triangleq [1 + S(i)]^{-i}, \quad i = 1, 2, \dots, n,$$

are called the **discount factors**.

- $d(i)$ is the PV of one dollar i periods from now.
- The above formula will be justified later in the course, rather than assumed; the argument is [Theorem 7.3.1](#).
- The discount factors are often interpolated to form a continuous function called the **discount function**.

Note 4.5.2. 最後一點的意思是： $d(1), d(2), \dots, d(n)$ 只在整數期上有定義，但現實中要折現的日期不見得落在這些格點上（下一期票息可能是 2.5 期後才付）。把這串離散的點連成一條曲線 $d(t)$ (t 為任意實數)，就得到**折現函數** (discount function)。

舉個例子。設市場給出

$$S(1) = 3\%, \quad S(2) = 3.5\%, \quad S(3) = 4\%,$$

亦即

$$d(1) = 1.03^{-1} = 0.97087, \quad d(2) = 1.035^{-2} = 0.93351, \quad d(3) = 1.04^{-3} = 0.88900.$$

現在要問：2.5 期後的 1,000 元今天值多少？表上沒有 $d(2.5)$ ，只能內插。最簡單的線性內插是

$$d(2.5) \approx \frac{d(2) + d(3)}{2} = 0.91125,$$

於是 $PV \approx 1,000 \times 0.91125 = 911.25$ 元。

順帶一提，內插的對象會影響答案：若改成先內插利率 $S(2.5) \approx 3.75\%$ 再換算，得

$$d(2.5) = 1.0375^{-2.5} = 0.91207,$$

PV 變成 912.07 元，兩者差了約 0.82 元。差距雖小，但在名目金額動輒上億的部位上並非可以忽略；實務上因此不用直線，而是對 d (或 $\ln d$) 配適三次樣條等較平滑的曲線。

Remark 4.5.3. 為什麼**折現因子** (discount factor) 要多取一個名字？因為 (4.5) 對 $d(i)$ 是線性的，對 $S(i)$ 卻不是。

$d(i)$ 是價格（今天買一張「 i 期後付一元」的憑證要花多少錢）， $S(i)$ 只是把同一件事換成年化報酬率的說法。凡是關於「一籃子現金流值多少」的敘述，用 d 寫出來都是加權求和，線性代數就能處理；換成 S 就會冒出 $[1 + S(i)]^{-i}$ 這種非線性的東西。後面的 bootstrapping、避險、以及由債券價格反推曲線的迴歸，全都是在 d 的座標下進行的。

另外，用 d 寫，**無套利** (no-arbitrage) 的條件也特別乾淨。所謂套利，是指一個不必掏錢、卻穩賺不賠的部位（正式定義見 [Section 7.1](#)）；此處可交易的就是那 n 張零息債券，一組買賣就是一組 $d(i)$ 的加權和。於是無套利等價於一句話： $d(i) > 0$ ——不會有人把未來的一塊錢白送給你。

至於常見的 $1 > d(1) > d(2) > \dots$ （愈晚的錢愈不值錢），那是額外假設「現金可以零成本保存」之後才有的結論，不是無套利本身的內容。保管現金其實要成本，這也正是負利率（2015 年起的德國、瑞士公債）能夠存在而不被套利掉的原因。

§4.6. Extracting Spot Rates from the Yield Curve

— Slides p. 131 —

The market does not quote $S(1), S(2), \dots$ directly; beyond the very short end it quotes coupon bonds. Spot rates must therefore be *extracted*, and the extraction is recursive.

Remark 4.6.1. 「beyond the very short end」指的是殖利率曲線最左端以外的部分。最短的那一段（美國約一年以內的 Treasury bills）本身就是零息債券，價格與 $d(1)$ 是同一件事，所以 $S(1)$ 可以直接讀出來；但兩年、五年、十年、三十年這些期限，市場上掛牌的都是每期付息的**附息債券**（coupon bond），沒有純粹「 n 期後付一元」的憑證在交易。

這就是必須反推的原因：一張 n 期附息債券的價格是 $d(1), \dots, d(n)$ 的加權和，單一個價格裡混著 n 個未知數。但若 $S(1), \dots, S(n-1)$ 已經由更短的債券定出來，那麼這個加權和裡只剩 $d(n)$ 未知，於是一式一未知、可以解出來。

- Start with the short rate $S(1)$. Note that short-term Treasuries are zero-coupon bonds, so $S(1)$ can simply be read off.
- Compute $S(2)$ from the two-period coupon bond price P by solving

$$P = \frac{C}{1 + S(1)} + \frac{C + 100}{[1 + S(2)]^2}. \quad (4.6)$$

The point is that (4.6) contains exactly one unknown: $S(1)$ is already known, so the first term is a number, and $S(2)$ can be solved for in closed form.

— Slides p. 132 —

Inductively, we are given the market price P of the n -period coupon bond and

$$S(1), S(2), \dots, S(n-1).$$

Then $S(n)$ can be computed from (4.4), repeated here:

$$P = \sum_{i=1}^n \frac{C}{[1 + S(i)]^i} + \frac{F}{[1 + S(n)]^n}.$$

The running time can be made to be $O(n)$. The procedure is called **bootstrapping**.

§4.6.i. Why $O(n)$

The naive reading of the induction costs $O(n^2)$: at step n one re-sums $n-1$ discounted coupons. The saving comes from noticing that the sum only ever grows by one term — provided we accumulate *discount factors* rather than discounted coupons, since each bond has its own coupon.

Write $A(k) \triangleq \sum_{i=1}^k d(i)$ for the running annuity factor. The n -period bond with coupon C_n and par F satisfies

$$P_n = C_n A(n-1) + (C_n + F) d(n),$$

so that

$$d(n) = \frac{P_n - C_n A(n-1)}{C_n + F}, \quad S(n) = d(n)^{-1/n} - 1, \quad A(n) = A(n-1) + d(n). \quad (4.7)$$

Each step is $O(1)$ work, so the whole curve costs $O(n)$.

Algorithm 4.1: Bootstrapping the spot rate curve

Input: prices $P_1, \dots, P_n$ and coupons $C_1, \dots, C_n$ of bonds maturing in $1, \dots, n$ periods;
par value F

Output: spot rates $S(1), \dots, S(n)$

```

1  $A := 0$ ;
2 for  $k = 1, 2, \dots, n$  do
3    $d := (P_k - C_k A) / (C_k + F)$ ;
4    $S(k) := d^{-1/k} - 1$ ;
5    $A := A + d$ ;
6 return  $S(1), \dots, S(n)$ ;
```

Note 4.6.2. The k th root costs one unit of time under the convention of [Section 1.7.ii](#), where x^y was declared a basic operation. Without that convention the count would carry an extra factor for the exponentiation — a good illustration of how much a complexity claim depends on the cost model agreed in advance.

§4.6.ii. A Worked Example

Example 4.6.3 (Not on the slides). Suppose annual periods, $F = 100$, and that the market quotes *par* bonds — bonds priced at 100, so that each one's coupon rate equals its own yield — at

$$y_1 = 3.5\%, \quad y_2 = 4.0\%, \quad y_3 = 4.3\%, \quad y_4 = 4.5\%.$$

For a par bond, $P_k = F$ and $C_k = Fy_k$, so (4.7) simplifies to $d(k) = [1 - y_k A(k-1)] / (1 + y_k)$. Running [Algorithm 4.1](#):

k	y_k	$A(k-1)$	$d(k)$	$S(k)$
1	3.5%	0	0.966184	3.5000%
2	4.0%	0.966184	0.924378	4.0100%
3	4.3%	1.890561	0.880830	4.3204%
4	4.5%	2.771391	0.837596	4.5301%

Checking the second row against (4.6) directly: $100 - 4/1.035 = 96.135266$, and $(104/96.135266)^{1/2} - 1 = 4.0100\%$ as claimed.

Remark 4.6.4. 這張表有兩件事值得盯著看。

第一， $S(k)$ 一律高於同期的 y_k ($k = 1$ 時相等)，而且差距隨到期日拉長而擴大：1、10、20、30 個基點。這不是巧合，而是 [Section 4.9](#) 要證明的定理——曲線正常時必然如此。

第二， $d(k)$ 嚴格遞減 ($0.966 > 0.924 > 0.881 > 0.838$)，符合前一節說的無套利要求。若某組報價算出來的 d 不遞減，那不是曲線奇特，而是資料有問題。

§4.7. Some Problems

— Slides p. 133 —

[Algorithm 4.1](#) is clean; the data are not.

- Treasuries of the same maturity might be selling at different yields (the **multiple cash flow problem**).
- Some maturities might be missing from the data points (the **incompleteness problem**).
- Treasuries might not be of the same quality.

- Interpolation and fitting techniques are needed in practice to create a smooth spot rate curve.²

Remark 4.7.1. 逐條說明：

(一) **多重現金流問題** (multiple cash flow problem)。同一到期日可能有好幾檔公債，票息不同。依上一節的分析，票息不同的債券本來就會有不同的 y ——這正是我們一開始要拋棄 y 的理由。麻煩在於它們解出來的 $S(n)$ 也會互相矛盾，於是必須決定採信哪一檔，或做某種平均。

(二) **不完整問題** (incompleteness problem)。Algorithm 4.1 要求每一期 $1, 2, \dots, n$ 都有一檔債券，現實中不可能。Figure 4.1 裡 10 年到 20 年之間沒有任何報價，半年期的格點更是大半空著。缺的格子只能猜。

(三) **品質不齊**。新發行券 (on-the-run) 流動性遠優於舊券 (off-the-run)，殖利率因而偏低；稅負待遇、能否作為附買回擔保品也都造成差異。這些價差與期限無關，卻全被算進曲線裡。

(四) **插補** (interpolation)。前三項的結果是：實務上得到的不是一條曲線，而是一團帶雜訊且有缺口的點。要變成可用的 $d(\cdot)$ ，必須挑一個插補或擬合方法——而那個腳註的力道很重：通常沒有經濟學上的理由。選哪條曲線，往往只是選哪一種數學上的平滑性。

Note 4.7.2. 第 (三) 條的三種價差各有來歷。

流動性。價格與殖利率反向，「殖利率低」即「大家願意出更高的價錢」。隨時能脫手、買賣價差極窄，本身就是有價值的方便，投資人願意為它多付錢；流動性差的券則必須便宜一點賣，多給的那一截就是**流動性溢酬** (liquidity premium)。交易集中在最新標售的那一檔，它一旦被下一次標售取代就沒人碰了。

稅負。殖利率是稅前的，投資人在意的是稅後。同一到期日、票息不同的兩檔債券，現金流的稅務性質不同——高票息每年課一次利息所得，折價券的報酬大半以到期價差實現、稅率與時點都不一樣——於是稅前價格被推到不同水準。

附買回。**附買回** (repo) 是「賣出並約定買回」，實質上是拿債券當擔保品借現金，價差即利息。若某檔券特別多人想借（例如放空者必須交割它），它的 repo 利率會低於一般擔保品，持有人得以用低於市場的成本取得資金。持有它既然多賺這一點，大家就願意接受它債券本身的殖利率低一些；不被接受當擔保品的券則反過來要賣便宜。

§4.7.i. Which One?

— Slides p. 134 —

The incompleteness problem is worth seeing rather than reading about. Here are the same eleven observations of Figure 4.1, joined three different ways.

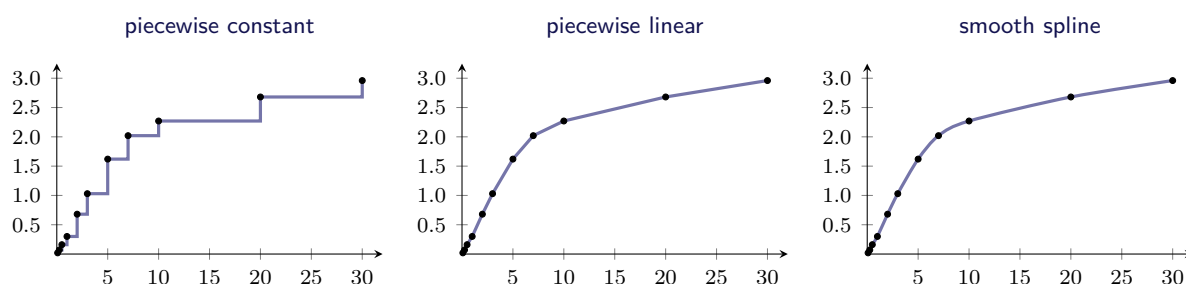


Figure 4.4.: Which one? Three curves through the identical eleven quotes of Figure 4.1; the horizontal axis is the maturity in years and the vertical axis the yield in percent, as before. Every one of them reprices all observed bonds exactly; they disagree about everything in between.

²Often without economic justifications.

The data fix the curve only where the data are. Everywhere else the curve is an assumption — and any security priced off that stretch inherits it.
 資料只能決定有資料的地方；其餘部分是假設。凡是用那段曲線定價的商品，都繼承了這個假設。

§4.8. Yield Spread and Static Spread

Everything so far assumed riskless cash flows. A corporate bond promises the same shape of cash flow but may not deliver, and the market charges for that. The two measures below quantify the charge; they differ in whether they use the term structure or ignore it.

§4.8.i. Yield Spread

— Slides p. 135 —

Consider a risky bond with the cash flow $C_1, C_2, \dots, C_n$ and selling for P .

- Calculate the IRR of the risky bond.
- Calculate the IRR of a riskless bond with comparable maturity.
- The **yield spread** (殖利率價差/利差) is their difference.

§4.8.ii. Static Spread

— Slides p. 136 —

Were the risky bond riskless, it would fetch

$$P^* = \sum_{t=1}^n \frac{C_t}{[1 + S(t)]^t}. \quad (4.8)$$

But as risk must be compensated, in reality $P < P^*$.

Definition 4.8.1 (Static spread). The **static spread** is the amount s by which the spot rate curve has to shift in parallel to price the risky bond:

$$P = \sum_{t=1}^n \frac{C_t}{[1 + s + S(t)]^t}. \quad (4.9)$$

Unlike the yield spread, the static spread explicitly incorporates information from the term structure.

Remark 4.8.2. 兩者差在哪裡，用一句話講：**殖利率價差** (yield spread) 拿風險債券的 IRR 去減另一檔債券的 IRR；**靜態價差** (static spread) 則是把整條無風險曲線平移到剛好能解釋這個價格。

殖利率價差的毛病承襲自 Section 4.3：兩個 IRR 各自是各自現金流的加權平均，票息與期限只要不同，比較就不乾淨。靜態價差把每一期的現金流都對到它自己的 $S(t)$ ，再問「所有期別同時加上多少，價格才對得上」，因此拿掉了現金流形狀造成的干擾。

先確認 (4.9) 這個 s 真的存在、而且只有一個。把右邊看成 s 的函數 $\varphi(s)$ ：票息 C_t 皆為正，每一項 $C_t[1 + s + S(t)]^{-t}$ 都隨 s 遞減，故 φ 連續且嚴格遞減。又 $\varphi(0) = P^* > P$ （不加價差會定出太高的價格），而 $s \rightarrow \infty$ 時 $\varphi(s) \rightarrow 0 < P$ 。由中間值定理，恰有一個 $s > 0$ 滿足 $\varphi(s) = P$ ：價格與靜態價差是一一對應的，報其中一個等於報另一個。期限結構平坦時 ($S(t) \equiv S$)，(4.9) 退化成 $P = \sum C_t / (1 + S + s)^t$ ，此時靜態價差恰好等於殖利率價差——兩者的差異完全來自曲線的斜率。

「靜態」二字則是留給後面的伏筆：它假設現金流固定不變。一旦債券附有贖回權或提前還款權 (Section 3.13 那些商品)，現金流會隨利率改變，就需要**選擇權調整價差** (option-adjusted spread, OAS)。

Example 4.8.3 (Not on the slides). Take the spot curve bootstrapped above, $S(1) = 3.5000\%$, $S(2) = 4.0100\%$, $S(3) = 4.3204\%$, and a three-period risky bond paying 6, 6, 106. Then (4.8) gives $P^* = 104.7114$. Suppose it actually trades at $P = 95$.

- Its IRR solves $95 = \sum_t C_t(1+y)^{-t}$, giving $y = 7.9380\%$. Against the comparable riskless three-period par yield of 4.3%, the *yield spread* is 363.8 basis points.
- Solving (4.9) gives a *static spread* of $s = 364.7$ basis points.

The two differ by about a basis point here because the curve is only mildly sloped over three periods — by Remark 4.8.2 a flat curve would make them agree exactly; on a steep curve, or a long bond, the gap is far larger.

§4.9. Of Spot Rate Curve and Yield Curve

— Slides p. 137 —

We began with the yield curve and replaced it with the spot rate curve. How do the two relate? Write y_i for the yield to maturity of the i -period coupon bond.

- $S(k) \geq y_k$ if $y_1 < y_2 < \dots$ (yield curve is normal).
- $S(k) \leq y_k$ if $y_1 > y_2 > \dots$ (yield curve is inverted).
- $S(k) \geq y_k$ if $S(1) < S(2) < \dots$ (spot rate curve is normal).
- $S(k) \leq y_k$ if $S(1) > S(2) > \dots$ (spot rate curve is inverted).
- If the yield curve is flat, the spot rate curve coincides with the yield curve.

Throughout, y_k is the **par yield**: the coupon rate at which the k -period bond trades at par. In particular $y_1 = S(1)$, the one-period bond being zero-coupon.

The last three bullets follow from one small lemma, which is worth isolating because it is the precise sense in which “a yield is an average of spot rates”; the first two need one more step.

Lemma 4.9.1

Let a security with non-negative cash flows $C_1, \dots, C_k$, not all zero, be priced by (4.5), and let y_k be its yield to maturity. Then

$$\min_{1 \leq i \leq k} S(i) \leq y_k \leq \max_{1 \leq i \leq k} S(i).$$

Proof. Put $f(y) \triangleq \sum_{i=1}^k C_i(1+y)^{-i}$, which is strictly decreasing for $y > -1$, and let $\bar{S} \triangleq \max_i S(i)$. Since $(1+\bar{S})^{-i} \leq [1+S(i)]^{-i}$ for every i ,

$$f(\bar{S}) \leq \sum_{i=1}^k C_i [1+S(i)]^{-i} = P = f(y_k),$$

and f decreasing gives $y_k \leq \bar{S}$. The other inequality is symmetric. ■

The lemma sandwiches y_k between the smallest and the largest spot rate the bond touches. Whenever the hypothesis names the *spot* curve, that sandwich already identifies which end $S(k)$ sits at, and the third, fourth and fifth bullets drop out:

- If the spot rate curve is normal then $\max_{i \leq k} S(i) = S(k)$, so $y_k \leq S(k)$.
- If it is inverted then $\min_{i \leq k} S(i) = S(k)$, so $y_k \geq S(k)$.
- If the curve is flat, $\min = \max$ and y_k is squeezed to the common value; spot and yield curves coincide.

The first two bullets assume instead that the *yield* curve is monotone, and the lemma says nothing about where $S(k)$ then lies: a normal yield curve does not force a normal spot curve.³ So they have to be proved on their own, by induction along the curve.

Proposition 4.9.2

若 $y_1 < y_2 < \dots$ (yield curve 為 normal)，則每個 k 皆有 $S(k) \geq y_k$ ；若 $y_1 > y_2 > \dots$ (inverted)，則每個 k 皆有 $S(k) \leq y_k$ 。

Proof. 以下只證 normal 的情形；inverted 只需把每一個不等號反向。以下 rate 皆設為正。

記號。除了 (4.7) 的 annuity factor $A(k) = \sum_{i \leq k} d(i)$ ——以 spot rate 逐期折現的那一個——再記

$$a_k(y) \triangleq \sum_{i=1}^k (1+y)^{-i} = \frac{1 - (1+y)^{-k}}{y}$$

為同一筆 annuity 改用單一 flat rate y 折現的因子，亦即 Remark 2.9.12 的 annuity present value factor。折現率愈高、現值愈低，故 a_k 對 y 嚴格遞減；這是後面唯一用到的性質。

策略。直接比較 $S(k)$ 與 y_k 無從下手，因為 y_k 是前 k 期整段曲線的綜合結果，不是第 k 期一個人的事。辦法是把待證的「 $S(k) \geq y_k$ 」改寫成一個只涉及前 $k-1$ 期的等價條件（第二步），如此它便能由前一個到期日的結論推出，於是沿到期日歸納（第三步）。第零步與第一步只是把要用的兩條等式擺好。

第零步： y_k 是怎麼來的。曲線 $d(1), \dots, d(n)$ 已由 bootstrapping 定出，於是任何現金流的價格都算得出來。考慮 k 期、par value 為 1、每期付 coupon c 的 bond，它的價格是 c 的一次函數

$$P(c) = cA(k) + d(k).$$

$A(k) > 0$ 故 P 嚴格遞增，又 $P(0) = d(k) < 1$ 而 $c \rightarrow \infty$ 時 $P \rightarrow \infty$ ，因此方程 $P(c) = 1$ 有唯一解

$$y_k = \frac{1 - d(k)}{A(k)} > 0.$$

這個唯一的 coupon rate 就是 par yield y_k ：它完全由曲線決定，沒有挑選的餘地，而那張 bond 是據此算出來的合成物，不必真的在市場上交易。反過來說，「它的價格是 1」並非觀察或假設，而是我們解 c 時所要求的條件。

（把 coupon 固定成使 bond 以 par 成交的那一個，是為了讓 y_k 有定義。同一到期日、coupon 不同的 bond，YTM 本來就不同——即 Section 4.7 的 multiple cash flow problem——若不指定 coupon，「 $y_1 < y_2 < \dots$ 」根本不是一個意義明確的條件。）

第一步：同一張 bond 的兩條定價式。把 y_k 代回，第一條等式就只是 $P(y_k) = 1$ 的展開：

$$1 = y_k A(k) + d(k) = y_k A(k-1) + (1 + y_k) d(k). \quad (4.10)$$

第二個等號把第 k 期的最後一張 coupon 從 $A(k)$ 拆出來、與同時到達的 principal 併在一起，好讓「前 $k-1$ 期」與「第 k 期」各據一項。

第二條等式來自 YTM 的定義：YTM 是能還原同一個價格的那個單一 flat rate。這張 bond 以 par 成交，而依 Remark 2.9.12，par 等價於 coupon rate 等於 YTM，故它的 YTM 恰為 y_k ，於是

$$1 = y_k a_{k-1}(y_k) + (1 + y_k)(1 + y_k)^{-k}. \quad (4.11)$$

兩式並未把 spot rate 與 YTM 混用：(4.10) 全程以曲線折現， y_k 在其中只是 coupon 的金額；(4.11) 全程以 flat rate 折現， y_k 在其中才兼任 discount rate。兩式共用的僅是左邊那個價格 1——也正因如此才能相減。

³ $S(1), \dots, S(4) = 0.5\%, 1\%, 16\%, 15.5\%$ gives par yields 0.5000%, 0.9975%, 13.7364%, 13.7853%, which increase although $S(4) < S(3)$.

第二步：相減，換成只涉及前 $k - 1$ 期的條件。兩式左邊同為 1，相減得

$$0 = y_k [A(k-1) - a_{k-1}(y_k)] + (1 + y_k) [d(k) - (1 + y_k)^{-k}]. \quad (4.12)$$

(手法與 Remark 2.9.12 導出 premium 金額時相同：兩條同值的定價式相減，差異自然浮現。) 兩個中括號因此不同號。而待證的 $S(k) \geq y_k$ 即 $d(k) \leq (1 + y_k)^{-k}$ ，說的正是第二個中括號 ≤ 0 ，故

$$S(k) \geq y_k \iff A(k-1) \geq a_{k-1}(y_k). \quad (4.13)$$

右邊只牽涉前 $k - 1$ 期，這正是要的形式。它的意思也很直白：總價已被釘在 1，曲線若對前 $k - 1$ 期的 coupon 折現得不比 flat rate y_k 吝嗇，那麼前面讓掉的就必須在第 k 期補回來， $S(k)$ 只能高於 y_k 。

第三步：歸納。 $k = 1$ 時一期 bond 沒有中間 coupon， $y_1 = S(1)$ ，結論以等號成立。

設 $S(k-1) \geq y_{k-1}$ ，即 $d(k-1) \leq [1 + y_{k-1}]^{-(k-1)}$ 。第零步用在到期日 $k - 1$ 上給出 $1 = y_{k-1}A(k-1) + d(k-1)$ ，解出 annuity factor 後

$$A(k-1) = \frac{1 - d(k-1)}{y_{k-1}} \geq \frac{1 - [1 + y_{k-1}]^{-(k-1)}}{y_{k-1}} = a_{k-1}(y_{k-1}) \geq a_{k-1}(y_k),$$

其中第一個不等號用歸納假設，第二個用 $y_{k-1} < y_k$ 與 a_{k-1} 遞減。這恰是 (4.13) 的右邊，故 $S(k) \geq y_k$ 。■

整段論證只在一個地方用到 normal 的假設：第三步把 $A(k-1)$ 的下界從 y_{k-1} 換成 y_k 。這也說明為什麼假設非得是關於整條曲線不可——單獨一個 k 的資訊推不出任何結論。

§4.9.i. Shapes

— Slides p. 138 —

- The spot rate curve often has the same shape as the yield curve.
 - If the spot rate curve is inverted (normal, resp.), then the yield curve is inverted (normal, resp.).
- But this is only a trend, not a mathematical truth.⁴

Note 4.9.3. The asymmetry is worth stating plainly, because the slides put theorem and trend in the same list.

The lemma above is a *theorem*: given the shape of the spot curve, the position of y_k relative to $S(k)$ is settled at every k individually. The claim on this page is different — it is about the *shape* of the whole yield curve, i.e. about how y_k compares with y_{k+1} , and that comparison involves two different bonds with two different sets of weights. Nothing forces the weighting to behave monotonically, and the textbook exhibits a spot curve that is monotone while the induced yield curve is not.

The practical reading: use the spot curve to price, and treat the yield curve's shape as a summary for conversation, not as an input.

§4.10. Forward Rates

— Slides p. 139 —

The yield curve contains information regarding future interest rates currently “expected” by the market. To extract it, compare two ways of holding money from time 0 to time j .

- **The maturity strategy.** Invest \$1 for j periods to end up with $[1 + S(j)]^j$ dollars at time j .

⁴See a counterexample in the text.

- **The rollover strategy.** Invest \$1 in bonds for i periods and at time i invest the proceeds in bonds for another $j - i$ periods, where $j > i$. This will have

$$[1 + S(i)]^i [1 + S(i, j)]^{j-i}$$

dollars at time j , where $S(i, j)$ denotes the $(j - i)$ -period spot rate i periods from now.

The first number is known today. The second is not: $S(i, j)$ is a rate that will only be quoted at time i . The question the market implicitly answers is what value of $S(i, j)$ would make the two strategies tie.

Remark 4.10.1. rollover strategy 是這樣一件事：兩種策略都是今天投入一元、都要在時點 j 把錢拿回來，差別只在中途要不要換手。

maturity strategy 一次買一張 j 期的 zero-coupon bond，錢鎖到 j 期，結果 $[1 + S(j)]^j$ 今天就已確定。rollover strategy 則先買一張較短的 i 期 bond ($i < j$)，到第 i 期拿回本利和 $[1 + S(i)]^i$ ，再把這筆錢全數投入一張剩餘 $j - i$ 期的 bond，持到 j 期。所謂 rollover (展期) 就是指到期後把錢原封不動轉進下一張。

記號 $S(i, j)$ 要注意：兩個參數是起點與終點，不是起點與期限。它指「站在第 i 期時，期限 $j - i$ 期的 spot rate」，因此今天的 $S(j)$ 在這個記號下就是 $S(0, j)$ 。

關鍵的不對稱在於： $S(i)$ 與 $S(j)$ 都是今天報價表上的數字， $S(i, j)$ 卻要等到第 i 期才會被報出來。所以 maturity strategy 的終值今天是已知數，rollover strategy 的終值今天還是未知數——兩者唯一的差別就在這裡。於是可以反問： $S(i, j)$ 要是多少，兩條路才會打成平手？那個數就是下面的 forward rate。

— Slides p. 140 —

Definition 4.10.2 (Forward rate). When $S(i, j)$ equals

$$f(i, j) \triangleq \left[\frac{(1 + S(j))^j}{(1 + S(i))^i} \right]^{1/(j-i)} - 1, \quad (4.14)$$

we will end up with the same $[1 + S(j)]^j$ dollars. The $f(i, j)$ are the **(implied) forward (interest) rates** — more precisely, the $(j - i)$ -period forward rate i periods from now.

(4.14) is just the tie condition solved for the unknown:

$$[1 + S(i)]^i [1 + f(i, j)]^{j-i} = [1 + S(j)]^j \iff [1 + f(i, j)]^{j-i} = \frac{(1 + S(j))^j}{(1 + S(i))^i}.$$

As expected, $f(0, j) = S(j)$: with $i = 0$ the rollover strategy is the maturity strategy.

§4.10.i. The Time Line

— Slides p. 141 —

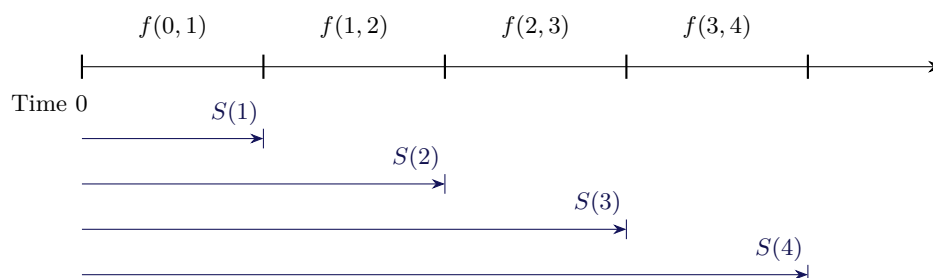


Figure 4.5.: The time line. The spot rates below run from today out to each maturity; the one-period forward rates above cover the individual legs between consecutive dates.

Remark 4.10.3. 這張圖是整節的骨架，值得盯著看清楚兩者的分工。

下半部的 $S(k)$ 都從 time 0 出發——每一條都是「從今天走到第 k 期」的平均利率，所以彼此重疊。上半部的 $f(k, k+1)$ 則彼此不重疊，各自只負責一小段。

用速度打比方： $S(k)$ 是走完前 k 期的平均速度， $f(k, k+1)$ 是第 $k+1$ 期的瞬時速度。平均速度由瞬時速度累積而成——這正是 (4.15) 要說的事；而平均速度上升，必然是因為當下的瞬時速度高於既有的平均，這正是 Section 4.11 的不等式。

§4.10.ii. Forward Rates and Future Spot Rates

— Slides p. 142 —

Note 4.10.4. We did not assume any *a priori* relation between $f(i, j)$ and the future spot rate $S(i, j)$ — that is the subject of the term structure theories in Section 4.15. We merely looked for the future spot rate that, *if realized*, will equate the two investment strategies.

The $f(i, i+1)$ are the **instantaneous forward rates** or **one-period forward rates**.

Remark 4.10.5. $f(i, j)$ 與 $S(i, j)$ 是兩個不同的東西，這是本章最常被誤解的地方。

遠期利率 (forward rate) $f(i, j)$ 是今天的數字：由 (4.14) 從今天查得到的 $S(i)$ 與 $S(j)$ 算出來，今天就已完全確定。 $S(i, j)$ 是未來的數字：第 i 期市場真正報出來的那個 $(j-i)$ 期 spot rate，今天還只是隨機變數。

兩者容易混為一談，是因為 forward rate 的定義寫成「當 $S(i, j)$ 等於 $f(i, j)$ 時兩策略打成平手」。那是一個條件句： $f(i, j)$ 回答的是「若要打平， $S(i, j)$ 該是多少」，並非斷言 $S(i, j)$ 真的會等於它。到了第 i 期，實現的 $S(i, j)$ 通常不等於 $f(i, j)$ ，而兩種策略孰優孰劣正是由這個落差決定。

這也是為什麼前面「market expects」那句話加了引號： f 是市場定價隱含的數字，至於 $f(i, j)$ 是否等於市場對 $S(i, j)$ 的期望值，是一個需要額外假設的理論命題 (Section 4.15 的無偏預期理論就是這樣一個假設)，而不是推導出來的結果。

Section 4.13 會更進一步： $f(i, j)$ 不只是算出來的數字，而是今天就能用現有 bond 鎖定的真實利率——這與「預測未來」仍是兩回事。

§4.11. Spot Rates and Forward Rates

— Slides p. 143 —

Proposition 4.11.1

When the spot rate curve is normal, the forward rate dominates the spot rates,

$$f(i, j) > S(j) > \cdots > S(i).$$

When the spot rate curve is inverted, the forward rate is dominated by the spot rates,

$$f(i, j) < S(j) < \cdots < S(i).$$

Proof. Suppose the curve is normal, so $S(i) < S(j)$ for $i < j$. From (4.14),

$$[1 + f(i, j)]^{j-i} = \frac{(1 + S(j))^j}{(1 + S(i))^i} > \frac{(1 + S(j))^j}{(1 + S(j))^i} = [1 + S(j)]^{j-i},$$

where the inequality replaces $S(i)$ in the denominator by the larger $S(j)$. Taking $(j-i)$ th roots gives $f(i, j) > S(j)$. The inverted case reverses every inequality. ■

Example 4.11.2 (Not on the slides). Take the curve bootstrapped in [Section 4.6](#), namely $S(1) = 3.5000\%$, $S(2) = 4.0100\%$, $S(3) = 4.3204\%$, $S(4) = 4.5301\%$. The one-period forward rates are

$$f(0, 1) = 3.5000\%, \quad f(1, 2) = 4.5225\%, \quad f(2, 3) = 4.9440\%, \quad f(3, 4) = 5.1617\%.$$

Each sits above the spot rate for the same terminal date, and the forward curve rises faster than the spot curve — the marginal rate outruns the average it is pulling up.

— Slides p. 144 —

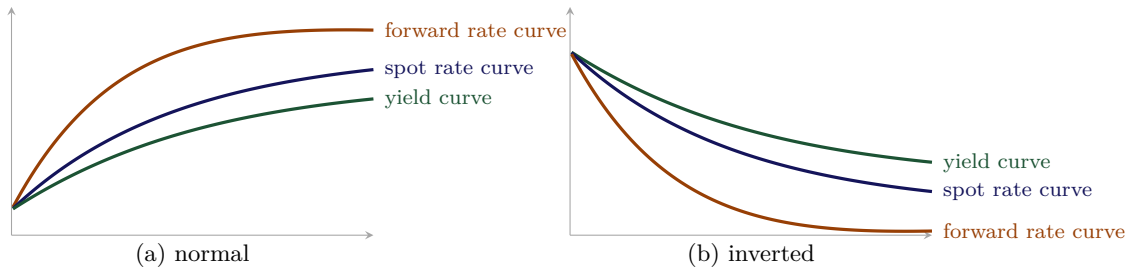


Figure 4.6.: The three curves never cross: forward above spot above yield when the *spot rate* curve is normal, and exactly the reverse when it is inverted. Horizontal axis maturity, vertical axis rate.

Remark 4.11.3. 先說三條曲線為何能畫在同一張圖上。 $f(i, j)$ 有兩個變數，圖上畫的是它的對角線切片 $k \mapsto f(k-1, k)$ ，即固定 $j-i=1$ 的 one-period forward rate。這沒有損失資訊，因為 (4.14) 串接起來會 telescoping：

$$[1 + f(i, j)]^{j-i} = \frac{(1 + S(j))^j}{(1 + S(i))^i} = \prod_{k=i}^{j-1} [1 + f(k, k+1)],$$

任何 $f(i, j)$ 都是那條曲線上各點的幾何平均。另兩條同樣是切片： $S(k)$ 就是 $S(0, k)$ （固定今天出發）， y_k 則是把 coupon 固定成 par 後唯一的那一條。三者各被固定掉一個自由度，剩下的參數都讀成到期期限，因此共用橫軸。

再說順序，關鍵是**每一條都是它上面那一條的平均**，而平均的窗口一路回溯到今天。

其一，(4.15) 在 $i=0$ 時說

$$[1 + S(k)]^k = \prod_{m=1}^k [1 + f(m-1, m)],$$

即 $1 + S(k)$ 是前 k 個 one-period forward 之 gross factor 的幾何平均（連續複利下 (4.19) 即字面上的算術平均）。滑動平均要上升，新進的那一項就必須高於平均本身，故 spot curve 為 normal 時 $f(k-1, k) > S(k)$ ——這正是 [Proposition 4.11.1](#)。

其二， y_k 是單一折現率卻要處理 bond 的所有現金流，依 [Lemma 4.9.1](#) 它夾在 $\min_{i \leq k} S(i)$ 與 $\max_{i \leq k} S(i)$ 之間，實為 $S(1), \dots, S(k)$ 以各期現金流現值為權重的加權平均；coupon 落在較早、較低的日期，故 normal 時 $y_k \leq S(k)$ ，zero-coupon bond 取等號。

兩者疊起來即

$$\underbrace{f(k-1, k)}_{\text{只第 } k \text{ 期}} > \underbrace{S(k)}_{\text{平均 } 1 \sim k \text{ 期}} \geq \underbrace{y_k}_{\text{再按 coupon 加權}}.$$

窗口愈寬就算進愈多較早、較低的 rate，值被拉得愈低；inverted 時較早的 rate 較高，同一機制把每個平均往上拉，順序於是整串反轉。兩條不等式對每個 k 逐點成立，故三條曲線從不交叉。

最後，前提必須掛在 *spot curve* 的單調性上：上述兩條不等式是定理，而「yield curve 的形狀跟著 *spot curve* 一樣單調」只是傾向（見 Section 4.9 那則 note）。若只知道 yield curve 為 normal，*spot curve* 仍可能非單調，圖上的順序便未必成立。

§4.12. Forward Rates $\equiv$ Spot Rates $\equiv$ Yield Curve

— Slides p. 145 —

The three curves are not merely ordered; they carry exactly the same information. The FV of \$1 at time n can be derived in two ways.

- Buy n -period zero-coupon bonds and receive

$$[1 + S(n)]^n.$$

- Buy one-period zero-coupon bonds today and a series of such bonds at the forward rates as they mature. The FV is

$$[1 + S(1)][1 + f(1, 2)] \cdots [1 + f(n - 1, n)].$$

Remark 4.12.1. 為什麼兩種算法的 FV 一定相同？可以從代數與經濟兩面看。

代數面：telescoping。先注意 $1 + S(1) = 1 + f(0, 1)$ ，再把 (4.14) 取 $j = i + 1$ 代進每一個因子：

$$\prod_{k=1}^n [1 + f(k - 1, k)] = \frac{(1 + S(1))^1}{(1 + S(0))^0} \cdot \frac{(1 + S(2))^2}{(1 + S(1))^1} \cdots \frac{(1 + S(n))^n}{(1 + S(n - 1))^{n-1}} = [1 + S(n)]^n.$$

相鄰分子分母兩兩相消，只剩頭尾。換句話說，forward rate 本來就是定義成讓 rollover 與 maturity 打平的那個數（見 (4.14) 下方的 tie condition）；這裡只是把 n 段「每段各自打平」串起來，整條路自然也打平。

經濟面：無套利。第二種做法用的是今天就鎖定的 $f(k - 1, k)$ ，而不是屆時才報價的未來 spot rate $S(k - 1, k)$ 。Section 4.13 會看到，每一段 forward loan 都能用今天交易的零息債券複製出來，所以這條 rollover 路徑的 FV 在今天就已完全確定，和第一種一樣是無風險的。兩個無風險策略、同樣今天投入 \$1、同樣在時點 n 收錢，終值若不同，就買進終值高的、放空終值低的，今天淨投入為零、時點 n 穩賺差額——這是**套利**（arbitrage）。因此兩者必須相等。

要小心的是：若第二種做法改成「到期後按當時的市場利率再投資」，FV 就變成隨機的 $[1 + S(1)][1 + S(1, 2)] \cdots [1 + S(n - 1, n)]$ ，一般不會等於 $[1 + S(n)]^n$ 。相等是 forward rate 的性質，不是未來 spot rate 的性質。

— Slides p. 146 —

Theorem 4.12.2

Since the two are identical,

$$S(n) = \left\{ [1 + S(1)][1 + f(1, 2)] \cdots [1 + f(n - 1, n)] \right\}^{1/n} - 1. \quad (4.15)$$

Hence the forward rates — specifically the one-period forward rates — determine the spot rate curve.

Other equivalencies can be derived similarly, such as

$$f(T, T + 1) = \frac{d(T)}{d(T + 1)} - 1. \quad (4.16)$$

Proof of (4.16). By definition $d(k) = [1 + S(k)]^{-k}$, so

$$\frac{d(T)}{d(T+1)} = \frac{(1 + S(T+1))^{T+1}}{(1 + S(T))^T} = 1 + f(T, T+1)$$

by (4.14) with $i = T, j = T+1$. ■

Spot rates, forward rates, discount factors and the yield curve are four coordinate systems for one object. Any one of them can be computed from any other; none contains extra information.

即期利率、遠期利率、折現因子、殖利率曲線，是同一個東西的四套座標。任一套都能算出其餘三套，沒有誰多帶了資訊。

§4.13. Locking in the Forward Rate

— Slides p. 147 —

So far $f(n, m)$ is an arithmetic construct. The following trade turns it into a rate one can actually contract at today, using nothing but zero-coupon bonds that trade now.

- Buy one n -period zero-coupon bond for $1/(1 + S(n))^n$ dollars.
- Sell $(1 + S(m))^m/(1 + S(n))^n$ m -period zero-coupon bonds.
- No net initial investment, because the cash inflow equals the cash outflow, both being $1/(1 + S(n))^n$.
- At time n there will be a cash inflow of \$1.
- At time m there will be a cash outflow of $(1 + S(m))^m/(1 + S(n))^n$ dollars.

Why the initial cost is zero. Each m -period zero sells for $(1 + S(m))^{-m}$, so selling $(1 + S(m))^m/(1 + S(n))^n$ of them raises

$$\frac{(1 + S(m))^m}{(1 + S(n))^n} \times \frac{1}{(1 + S(m))^m} = \frac{1}{(1 + S(n))^n},$$

which is exactly what the n -period zero costs. ■

Remark 4.13.1. 這一頁要說的是： $f(n, m)$ 不只是從 spot curve 算出來的數字，今天就能用市場上現成的零息債券做出一筆利率為 $f(n, m)$ 的借款。這組交易（買 n 期 zero、同時放空適當數量的 m 期 zero）之後會被稱為 **locking trade**：lock 指「鎖定」，因為它讓人在今天就把未來 n 到 m 這段期間的利率鎖死在 $f(n, m)$ ，不必等到時點 n 再看市場報價 $S(n, m)$ 。

兩條腿各做什麼。買一張 n 期零息債券，是今天把錢借出去、第 n 期拿回 \$1；賣出（放空） m 期零息債券，則是今天借錢進來，第 m 期每張還 \$1。單看任何一條腿，今天都有現金進出。

數量怎麼挑。賣出的張數 $(1 + S(m))^m/(1 + S(n))^n$ 是刻意選的：它讓放空拿到的錢剛好付得起那張 n 期債券（上面的 proof）。於是今天兩邊互相抵銷，一毛錢都不出。

剩下的現金流。把時點 0 抵銷掉之後，只剩第 n 期收到 \$1、第 m 期付出 $(1 + S(m))^m/(1 + S(n))^n$ 。這就是「第 n 期借進 \$1、第 m 期還本帶息」的借款，只是今天就已經簽好。還款金額除以 \$1 就是 $m - n$ 期的 gross factor，由 (4.14) 它正好等於 $(1 + f(n, m))^{m-n}$ ，所以這筆借款的利率就是 $f(n, m)$ （下一頁 Figure 4.7）。

這為什麼重要。未來的 spot rate $S(n, m)$ 今天還不知道，但 $f(n, m)$ 今天就能鎖定，而且所有條件都已定死，沒有任何不確定性。這也是上一節「兩種算法 FV 相同」的無套利論證所依靠的事實：rollover 路徑每一段用的都是這樣鎖定的利率，所以它和直接買 n 期債券一樣是無風險的。反過來，如果市場上另有一筆 forward loan 的利率不等於 $f(n, m)$ ，就可以一邊用它、一邊用這組零息債券做反向操作，賺取無風險利潤。

— Slides p. 148 —

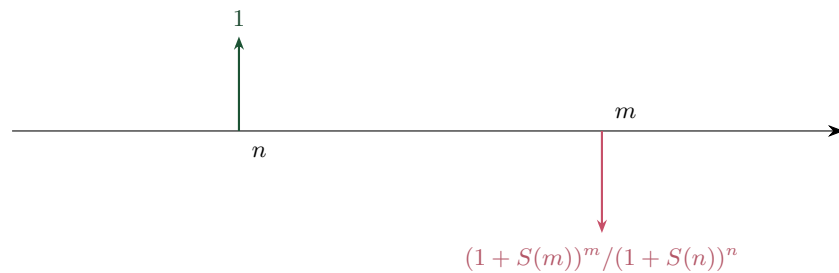


Figure 4.7.: The cash flow locked in today. Nothing changes hands at time 0; one dollar arrives at time n and is repaid at time m .

This implies the interest rate between times n and m equals $f(n, m)$, because by (4.14)

$$\frac{(1 + S(m))^m}{(1 + S(n))^n} = (1 + f(n, m))^{m-n}.$$

§4.13.i. Forward Loans

— Slides p. 149 —

We had generated the cash flow of a type of forward contract called the **forward loan**. Agreed upon today, it enables one to

- borrow money at time n in the future, and
- repay the loan at time $m > n$ with an interest rate equal to the known forward rate $f(n, m)$.

Question 4.13.2. Can the spot rate curve be arbitrarily drawn?^a

^aContributed by Mr. Dai, Tian-Shyr (B82506025, R86526008, D88526006) in 1998.

Answer. 不行。每一筆 forward loan 都能用市場上交易的 zero-coupon bond 複製出來，所以曲線隱含的 forward rate 是真的能成交的利率——它們本身也必須不能被套利。

假設某個 n 使得 $f(n, n+1) < 0$ ，由 (4.16) 這等價於 $d(n) < d(n+1)$ ，也就是較晚到期的 zero 反而比較貴。這時賣出一張 $(n+1)$ 期 zero、買進一張 n 期 zero：今天淨收入 $d(n+1) - d(n) > 0$ ，時點 n 收到 \$1，時點 $n+1$ 要付出 \$1。把時點 n 收到的那 \$1 以現金形式保管到時點 $n+1$ ，剛好用來支付後面的義務。結果是今天白拿一筆正的錢，將來沒有任何負債——這就是 arbitrage。

因此曲線必須滿足

$$1 > d(1) \geq d(2) \geq \dots, \quad \text{等價地} \quad (1 + S(n))^n \text{ 對 } n \text{ 非遞減。}$$

(取 $n = 0$ 、 $d(0) = 1$ 用同一個論證就得到 $d(1) < 1$) 特別地，inverted curve 不能下降得任意快；Section 4.14 會給出同一個界的連續時間版本。 ■

Remark 4.13.3. 先說為什麼會冒出這個問題。到目前為止，spot rate curve 一直被當成輸入：給任何一組 $S(1), S(2), \dots$ ，都能用 (4.14) 算出一組 $f(i, j)$ ，代數上從來不會卡住。如果 f 只是算出來的數字，那 spot curve 畫成什麼形狀似乎都可以。

但上一節剛證明了 $f(n, m)$ 不只是數字：用 zero-coupon bond 就能在今天把它鎖定成一筆真的 forward loan。一旦 forward rate 是能成交的價格，它就得接受市場的檢驗——例如「 n 到 $n+1$ 之間借錢還能拿到利息補貼」這種 forward rate 若出現，就會被套利。所以問題其實是在問：forward loan 的可交易性反過來對 spot curve 加了什麼限制？答案就是上面的單調性條件：

$(1 + S(n))^n$ 不能隨 n 變小。這也呼應前面「曲線是 normal 還是 inverted」的討論：inverted 是允許的，但只能緩慢下降，不能降到讓某段 forward rate 變負。

再說上面那段論證偷偷用了一個假設：現金可以無成本地保管。正因為存著鈔票的報酬率是 0，任何低於 0 的遠期利率才會被套利掉。

現實中這個假設並不完全成立——大額現金的保管、保險與運送都要錢，而且央行可以對準備金課負利率。2014 年之後的歐元區與日本公債確實出現過負的即期與遠期利率，而且持續多年。所以正確的說法是：曲線不能任意畫，但下界不是 0，而是 $-(\text{持有現金的成本})$ 。

§4.13.ii. Synthetic Bonds

— Slides p. 150 —

Read the locking trade as an identity between portfolios. We had seen that

$$\text{forward loan} = n\text{-period zero} - [1 + f(n, m)]^{m-n} \times m\text{-period zero}.$$

Thus

$$n\text{-period zero} = \text{forward loan} + [1 + f(n, m)]^{m-n} \times m\text{-period zero}.$$

We have created a **synthetic** zero-coupon bond with forward loans and other zero-coupon bonds. This is useful if the n -period zero is unavailable or illiquid.

Remark 4.13.4. 這些式子在說什麼。把每個金融商品看成「各時點收付多少錢」的一張清單，式子就是在加減這些清單：+ 是持有 (long)，- 是放空 (short)，係數是張數，= 表示兩邊每個時點的現金流都一樣。令 $K = [1 + f(n, m)]^{m-n}$ ，逐期列出來：

	時點 n	時點 m
n -period zero	+1	0
$K \times m$ -period zero	0	+ K
forward loan	+1	- K

第一式：第一列減第二列，得到第三列——這就是上一節的做法（買 n 期 zero、放空 K 張 m 期 zero，今天淨成本 0）。第二式：把同一件事移項，第三列加第二列，得回第一列。

搞這個幹嘛。第二式告訴你：就算市場上買不到（或很少成交） n 期 zero，只要能交易 m 期 zero 和 forward loan，就能自己**合成** (synthesize) 一張現金流一模一樣的 n 期 zero。現金流一樣，價格就必須一樣，否則可以套利；驗算也對得上：forward loan 今天不用錢，組合成本 $K d(m) = d(n)$ 。

Note 4.13.5. This is the first appearance of a move that will run through the rest of the course: an identity between cash flows is an identity between *prices*, so any instrument that can be replicated must cost what its replicating portfolio costs. Here it fills a hole in the maturity spectrum — exactly the incompleteness problem of [Section 4.7](#), attacked by construction rather than by interpolation.

§4.14. Spot and Forward Rates under Continuous Compounding

— Slides p. 151 —

Every formula above simplifies under continuous compounding, because products of gross rates become sums of rates. Recall (2.4): one dollar grows to e^{rn} rather than $(1 + r)^n$.

- The pricing formula:

$$P = \sum_{i=1}^n C e^{-iS(i)} + F e^{-nS(n)}. \quad (4.17)$$

- The market discount function:

$$d(n) = e^{-nS(n)}. \quad (4.18)$$

- The spot rate is an *arithmetic average* of forward rates:

$$S(n) = \frac{f(0,1) + f(1,2) + \cdots + f(n-1,n)}{n}. \quad (4.19)$$

推導(4.19). 與 Section 4.12 同一個論證：\$1 放到時點 n ，有兩種方式。直接買 n 期 zero，連續複利下得到 $e^{nS(n)}$ ；逐期 rollover、每期用今天鎖定的 one-period forward rate，得到

$$e^{f(0,1)} e^{f(1,2)} \cdots e^{f(n-1,n)} = e^{f(0,1)+f(1,2)+\cdots+f(n-1,n)}.$$

兩者都是今天就確定的無風險終值，無套利下必須相等：

$$e^{nS(n)} = e^{f(0,1)+f(1,2)+\cdots+f(n-1,n)}.$$

兩邊取 $\ln$ 得 $nS(n) = f(0,1) + \cdots + f(n-1,n)$ ，再除以 n 即得。

也可以逐項看：one-period 的 tie condition $e^{(k-1)S(k-1)} e^{f(k-1,k)} = e^{kS(k)}$ 取 $\ln$ 得 $f(k-1,k) = kS(k) - (k-1)S(k-1)$ ，對 $k = 1, \dots, n$ 相加時中間項兩兩相消 (telescoping)，只剩 $nS(n) - 0 \cdot S(0) = nS(n)$ 。 ■

Remark 4.14.1. 把 (4.19) 與離散版的 (4.15) 並排看，就知道連續複利的好處在哪裡。

離散版是**幾何平均** (geometric mean)： $1 + S(n)$ 等於 n 個 $1 + f$ 相乘再開 n 次方。連續版是**算術平均** (arithmetic mean)： $S(n)$ 就是 n 個 f 直接相加除以 n 。

原因只有一個： $e^a e^b = e^{a+b}$ 。指數把乘法換成加法，於是「利率的累積」變成單純的求和，微分與積分也才用得上——這就是為什麼從這裡之後的理論工作幾乎一律採用連續複利。

— Slides p. 152 —

- The formula for the forward rate:

$$f(i,j) = \frac{jS(j) - iS(i)}{j - i}. \quad (4.20)$$

- The one-period forward rate:

$$f(j, j+1) = -\ln \frac{d(j+1)}{d(j)}. \quad (4.21)$$

推導(4.20). 與離散版 (4.14) 同一個 tie condition：\$1 放到時點 j ，maturity strategy 得 $e^{jS(j)}$ ；rollover strategy 先放 i 期得 $e^{iS(i)}$ ，再以鎖定的 $f(i,j)$ 放 $j-i$ 期，得 $e^{iS(i)} e^{(j-i)f(i,j)}$ 。兩者相等：

$$e^{iS(i)+(j-i)f(i,j)} = e^{jS(j)}.$$

取 $\ln$ 得 $iS(i) + (j-i)f(i,j) = jS(j)$ ，移項後除以 $j-i$ 即得。 ■

推導(4.21). 由 (4.18)， $d(k) = e^{-kS(k)}$ ，所以

$$\frac{d(j+1)}{d(j)} = \frac{e^{-(j+1)S(j+1)}}{e^{-jS(j)}} = e^{-[(j+1)S(j+1)-jS(j)]} = e^{-f(j,j+1)},$$

最後一步是 (4.20) 取 $i = j$ 、 $j \rightarrow j+1$ (分母 $j-i = 1$)。兩邊取 $\ln$ 再加負號即得。 ■

— Slides p. 153 —

Definition 4.14.2 (Instantaneous forward rate curve).

$$f(T) \triangleq \lim_{\Delta T \rightarrow 0} f(T, T + \Delta T) = S(T) + T \frac{\partial S}{\partial T}. \quad (4.22)$$

Proof. By (4.20) with $i = T$ and $j = T + \Delta T$,

$$f(T, T + \Delta T) = \frac{(T + \Delta T)S(T + \Delta T) - T S(T)}{\Delta T},$$

which is the difference quotient of the function $T \mapsto T S(T)$. Its limit is $\frac{d}{dT}[T S(T)] = S(T) + T S'(T)$. ■

Two consequences are recorded on the slide:

- $f(T) > S(T)$ if and only if $\partial S / \partial T > 0$, i.e. the spot rate curve is normal. This is the continuous-time restatement of Section 4.11, and (4.22) makes it immediate: $f - S = T \partial S / \partial T$.
- If $S(T) < -T (\partial S / \partial T)$, then $f(T) < 0$.

Remark 4.14.3. 第二條正是前面「曲線不能任意畫」的連續版本。

離散版是 Section 4.13 裡那題 question 的答案：forward rate 不能是負的，否則借長賣短、中間持有現金就能套利。連續版把「一期 forward rate」換成「瞬間 forward rate $f(T)$ 」，其餘一字不改。兩邊逐行對照：

	離散	連續
條件	$f(n, n+1) \geq 0$	$f(T) \geq 0$
用 d 寫	$d(n) \geq d(n+1)$	$d(T) = e^{-TS(T)}$ 遞減
用 S 寫	$(1 + S(n))^n$ 遞增	$T S(T)$ 遞增

每一欄的三行互相等價：離散欄由 (4.16)， $1 + f(n, n+1) = d(n)/d(n+1)$ ，而 $d(n) = (1 + S(n))^{-n}$ ；連續欄由 (4.22)， $f(T) = \frac{d}{dT}[T S(T)]$ ，而 $d(T) = e^{-TS(T)}$ 。最後一行的對應也很直接： $(1 + S(n))^n$ 是 \$1 放 n 期的終值， $e^{TS(T)}$ 是連續版的終值， $T S(T)$ 就是它的指數。所以兩邊說的是同一件事——\$1 放得愈久，終值不能變少。

§4.14.i. An Example

— Slides p. 154 —

Example 4.14.4. Let the interest rates be continuously compounded, and suppose the spot rate curve is⁵

$$S(T) \triangleq 0.08 - 0.05 e^{-0.18T}.$$

Then by (4.22) the forward rate curve is

$$\begin{aligned} f(T) &= S(T) + T S'(T) \\ &= 0.08 - 0.05 e^{-0.18T} + 0.009 T e^{-0.18T}, \end{aligned}$$

since $S'(T) = 0.009 e^{-0.18T}$.

⁵J. Hull & White (1994).

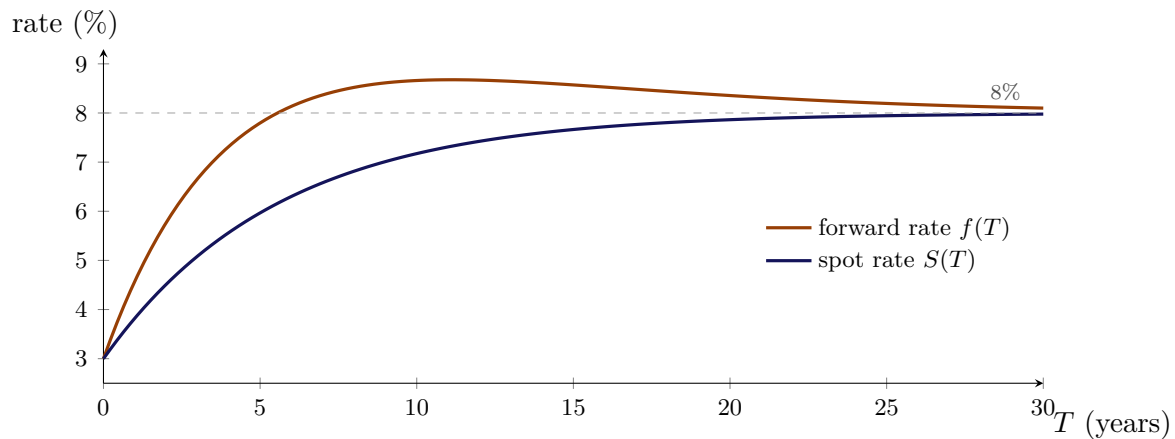


Figure 4.8.: The Hull–White spot curve and the forward curve it implies. The spot curve rises monotonically to its asymptote of 8%; the forward curve overshoots it, peaking at 8.68% near $T = 11.1$ before falling back. Both start at 3%, since $f(0) = S(0)$.

Remark 4.14.5. 遠期曲線衝過長期水準再折返，這件事值得解釋——它不是畫錯了。

$S(T)$ 是 $[0, T]$ 上所有瞬時利率的平均。既然平均要從 3% 一路爬到 8%，而起點被壓在 3%，那麼中段的瞬時利率就必須高於 8%，才拉得動整條平均線。等到平均值接近 8%、不再需要被拉動時， f 才降回 8%。

這正是「邊際領先平均」的標準圖像，也是實務上常說的：遠期曲線是即期曲線的放大版——即期曲線的任何斜率，在遠期曲線上都被乘上 T 放大了（見 (4.22)）。

§4.15. Term Structure Theories

Everything up to here was arithmetic: given prices, the forward rates follow by identity. The remaining question is economic — what, if anything, does $f(i, j)$ say about the *future* spot rate $S(i, j)$? Answering it requires an assumption, and the candidate assumptions are the term structure theories.

§4.15.i. Unbiased Expectations Theory

— Slides p. 155 —

Theorem 4.15.1 (Unbiased expectations theory)

The forward rate equals the average future spot rate,

$$f(a, b) = E[S(a, b)]. \quad (4.23)$$

- It does *not* imply that the forward rate is an accurate predictor for the future spot rate.
- It implies the maturity strategy and the rollover strategy produce the same result at the horizon “on average”.

Remark 4.15.2. 先釐清定理本身： $f(a, b)$ 是今天就算得出來的數字， $S(a, b)$ 則要到時點 a 才會在市場上報出來，今天看它是隨機變數。這個理論假設前者等於後者的期望值。它是一個關於市場的假設，不是前面那種由無套利推出來的定理。

第一點：「無偏」不等於「準」。 $f(a, b) = E[S(a, b)]$ 只是說 forward rate 沒有系統性地偏高或偏低：把很多次的預測誤差 $S(a, b) - f(a, b)$ 平均起來是 0。但任何一次實際報出的 $S(a, b)$ 都可能離 $f(a, b)$ 很遠，例如今天算出 $f(1, 2) = 5\%$ ，一年後的實際利率是 3% 或 7% 都跟理論不衝突。

第二點：兩種策略「平均」打平。回想 Section 4.10 的兩種策略。maturity strategy 的終值今天已經確定；rollover strategy 的終值要用到未來的 $S(a, b)$ ，所以是隨機的。forward rate 的定義是：若 rollover 後半段的利率剛好等於 $f(a, b)$ ，兩者就打平。這個理論再假設 $f(a, b)$ 是 $S(a, b)$ 的期望值，於是 rollover 的終值平均而言等於 maturity 的終值；實際走出來的某一條路徑則可能高也可能低。「on average」加了引號，是因為這句話只在粗略意義下成立：終值裡的 $S(a, b)$ 是出現在 $(1 + S(a, b))^{b-a}$ 裡，而一般來說 $E[(1 + S)^k] \neq (1 + E[S])^k$ 。下一小節的「bad」expectations theory 就是在這種細節上出問題。

— Slides p. 156 —

The theory ties the shape of today's curve to the market's expectations.

Proposition 4.15.3

Under (4.23), a normal spot rate curve is due to the fact that the market expects the future spot rate to rise. Conversely, the spot rate is expected to fall if and only if the spot rate curve is inverted.

Proof. First, $f(j, j+1) > S(j+1)$ if and only if $S(j+1) > S(j)$. Indeed, by (4.14),

$$1 + f(j, j+1) = \frac{(1 + S(j+1))^{j+1}}{(1 + S(j))^j},$$

so $f(j, j+1) > S(j+1)$ is equivalent to $(1 + S(j+1))^j > (1 + S(j))^j$, i.e. to $S(j+1) > S(j)$. Substituting (4.23) gives

$$E[S(j, j+1)] > S(j+1) > \cdots > S(1) \quad \text{if and only if} \quad S(j+1) > \cdots > S(1). \quad \blacksquare$$

Remark 4.15.4. 第一個小結論不需要任何理論——它純粹是 (4.14) 的代數重寫，也就是 Section 4.11 的特例。理論的作用只在最後一步：把算出來的 $f(j, j+1)$ 換成期望值 $E[S(j, j+1)]$ 。

那串不等式為什麼等於「預期利率上升」？關鍵是把每個符號讀成「哪個時點的一期利率」：

- $S(1) = S(0, 1)$ 是今天的一期利率；
- $S(j, j+1)$ 是時點 j 的一期利率，今天未知， $E[S(j, j+1)]$ 就是市場對它的預期。

「預期未來利率上升」的意思就是：預期時點 j 的一期利率高於今天的一期利率，即 $E[S(j, j+1)] > S(1)$ 。那串不等式的頭尾正是這件事：

$$\underbrace{E[S(j, j+1)]}_{\text{預期的未來一期利率}} > S(j+1) > \cdots > \underbrace{S(1)}_{\text{今天的一期利率}},$$

中間的 $S(j+1) > \cdots > S(1)$ 就是「spot curve 是 normal」。所以整句讀成：曲線 normal $\iff$ 預期的未來一期利率高過今天的一期利率。

直覺上：由 (4.15)， $S(j+1)$ 是 $S(1), f(1, 2), \dots, f(j, j+1)$ 的（幾何）平均；在此理論下這些 f 就是各期的預期一期利率。所以長天期 spot rate 就是「今天的利率加上未來各期預期利率」的平均。平均值隨期限上升（normal），唯一的原因就是後面加進來的預期利率比較高——市場預期利率會漲。

所以這條命題的內容其實是：在**無偏預期理論**（unbiased expectations theory）之下，「殖利率曲線正常」與「市場預期短率將上升」是同一件事。這也是新聞上「倒掛的殖利率曲線預示降息」那句話的理論根據——但別忘了它建立在假設之上，而下一小節正是要說明這一類假設有多脆弱。

§4.15.ii. A “Bad” Expectations Theory

— Slides p. 157 —

Consider the natural-sounding strengthening: the expected returns⁶ on all possible riskless bond strategies are equal for all holding periods. It turns out to be internally inconsistent.

Over a *two*-period horizon, buying a two-period bond and rolling over one-period bonds must tie:

$$(1 + S(2))^2 = (1 + S(1)) E[1 + S(1, 2)]. \quad (4.24)$$

After rearrangement,

$$\frac{1}{E[1 + S(1, 2)]} = \frac{1 + S(1)}{(1 + S(2))^2}. \quad (4.25)$$

Remark 4.15.5. 這個理論在說什麼。先固定一個**持有期間** (holding period) H ：今天投入 \$1，看時點 H 手上有多少。在這 H 期內，用無違約風險的債券可以有很多玩法：買 H 期 zero 持有到期、買短的一路 rollover、買更長的 zero 然後在時點 H 提前賣掉……這個理論主張：只要持有期間相同，所有這些策略的預期 one-plus return 都相等，而且這對每一個 H 都成立。其中 **one-plus return** = 期末價值 / 期初投入，算的是整個持有期間：例如兩期 zero 持有兩期是 $(1 + S(2))^2$ ，不是 $1 + S(2)$ 。

背後的想法是：若投資人只看期望報酬、不在乎風險，那麼哪個策略的期望報酬比較高，大家就會去搶，價格被推到直到打平為止。注意「無風險」只是指不會違約；rollover 或提前賣出時，未來的利率與價格今天仍未知，所以報酬還是隨機的，才需要取期望值。

所以什麼時候「要相等」？只比同一個持有期間的策略：

- $H = 2$ (這一頁)：兩期 zero 持有到期 vs. 一期 zero rollover 兩次；
- $H = 1$ (p. 158–159)：一期 zero 持有到期 (報酬確定是 $1 + S(1)$) vs. 兩期 zero 持有一期後賣掉 (賣價未知)。

p. 159 說「the theory says the returns are equal」，就是因為那兩個策略的持有期間都是 1，理論要求它們預期報酬相同。

跟 unbiased expectations theory 的關係。不是先相信 unbiased expectations theory 再去推廣它。單看 $H = 2$ 那條式子，它其實就是 unbiased expectations theory 用在 $f(1, 2)$ 上 ($E[S(1, 2)] = f(1, 2)$)；單看 $H = 1$ 那條式子，它就是下一小節的 local expectations theory。兩條各自都可以成立，這個「bad」theory 的問題是把「投資人不在乎風險」這個直覺同時套到所有持有期間上。接下來三頁要證明：兩條式子不可能同時成立，除非未來利率毫無不確定性。所以結論不是「unbiased expectations theory 錯了」，而是「expectations theory 只能挑一個持有期間來要求，不能全都要」。

證明的策略。先用 $H = 2$ 得到一條等式 (這一頁)，再用 $H = 1$ 得到另一條 (下一頁)，兩條合起來會跟 Jensen's inequality 衝突。

兩期的持有期間。今天投入 \$1，看時點 2 的結果：

- 買兩期 zero 持有到期：終值 $(1 + S(2))^2$ ，今天就確定；
- 買一期 zero，時點 1 再以當時的利率 $S(1, 2)$ 滾一期：終值 $(1 + S(1))(1 + S(1, 2))$ ，因為 $S(1, 2)$ 今天未知，只能取期望值。

理論說兩者預期相等，就是 (4.24)。到這裡還只是 unbiased expectations theory 的兩期版本 ($E[1 + S(1, 2)] = 1 + E[S(1, 2)]$)，沒有任何問題。

— Slides p. 158 —

Now consider two *one*-period strategies.

- Strategy one buys a two-period bond for $(1 + S(2))^{-2}$ dollars and sells it after one period. Its price then will be $(1 + S(1, 2))^{-1}$, so the expected return is

$$E[(1 + S(1, 2))^{-1}] / (1 + S(2))^{-2}.$$

⁶More precisely, the one-plus returns.

- Strategy two buys a one-period bond with a return of $1 + S(1)$.

— Slides p. 159 —

The theory says the returns are equal:

$$\frac{1 + S(1)}{(1 + S(2))^2} = E \left[\frac{1}{1 + S(1, 2)} \right].$$

Combining this with (4.25) yields

$$E \left[\frac{1}{1 + S(1, 2)} \right] = \frac{1}{E[1 + S(1, 2)]}. \quad (4.26)$$

— Slides p. 160 —

Proposition 4.15.6

(4.26) is impossible save for a certain economy — that is, unless $S(1, 2)$ is known today. (除非是一個確定性的經濟——也就是未來利率 $S(1, 2)$ 今天就已知、沒有任何隨機性——否則 (4.26) 不可能成立。這裡的 save for 是「除了」)。

Proof. Jensen's inequality states that $E[g(X)] > g(E[X])$ for any nondegenerate random variable X and strictly convex function g (i.e. $g''(x) > 0$). Use

$$g(x) \triangleq (1 + x)^{-1},$$

for which $g''(x) = 2(1 + x)^{-3} > 0$ whenever $x > -1$. With $X = S(1, 2)$,

$$E \left[\frac{1}{1 + S(1, 2)} \right] > \frac{1}{1 + E[S(1, 2)]} = \frac{1}{E[1 + S(1, 2)]},$$

which contradicts (4.26) unless $S(1, 2)$ is degenerate, i.e. deterministic. ■

Remark 4.15.7. 這個結果的教訓比它的算式重要，值得講清楚。

錯在哪裡？不在於某一個式子算錯，而在於這個理論要求太多：它同時宣稱所有期間長度、所有策略的期望報酬都相等。兩期視角給出 (4.24)，一期視角給出另一條式子，兩者合起來就逼出 (4.26)——一個只有在未來利率毫無不確定性時才成立的等式。

根源是「期望值」與「取倒數」不可交換。債券價格是利率的凸函數 ($1/(1+x)$ 凸)，所以「先取期望再折現」與「先折現再取期望」必然不同，差額就是 **Jensen 不等式** (Jensen's inequality) 留下的缺口。這與 Section 3.10 的凸性其實是同一件事的兩個面貌：那裡凸性讓持有人「漲賺得比跌賠得多」，這裡凸性讓 $E[1/(1+X)] > 1/E[1+X]$ 。

出路是只挑一種視角來要求，別全都要——這就是下一小節的**局部預期理論** (local expectations theory)。

§4.15.iii. Local Expectations Theory

— Slides p. 161 —

Theorem 4.15.8 (Local expectations theory)

The expected rate of return of any bond over a single period equals the prevailing one-

period spot rate:

$$\frac{E[(1 + S(1, n))^{-(n-1)}]}{(1 + S(n))^{-n}} = 1 + S(1) \quad \text{for all } n > 1. \quad (4.27)$$

Remark 4.15.9. (4.27) 就是「今天買一張 n 期 zero、持有一期後賣掉」的預期 one-plus return (賣價 / 買價)：

- 分母 $(1 + S(n))^{-n}$ ：這張 n 期 zero 今天的價格 (買價)；
- 分子：到了時點 1，它變成剩 $n - 1$ 期的 zero，當時的價格是 $(1 + S(1, n))^{-(n-1)}$ ，其中 $S(1, n)$ 是時點 1 才報出的利率，今天未知，所以取期望值 (預期賣價)；
- 右邊 $1 + S(1)$ ：直接買一期 zero 持有到期的報酬，今天就確定。

兩邊比的是比值，與買多少無關：左邊用面額 \$1 的 zero 算，右邊若也用面額 \$1 的一期 zero，買價 $(1 + S(1))^{-1}$ 、到期拿 \$1，比值同樣是 $1 + S(1)$ ，跟「投入 \$1 拿回 $1 + S(1)$ 」一樣。所以整條式子說：持有一期的話，買哪一張 zero 預期報酬都一樣，都等於一期利率。 $n = 2$ 就是 p. 159 那條式子；這裡只要求持有期間 $H = 1$ ，不再同時要求 $H = 2$ ，因此不會碰到上一小節的 Jensen 矛盾。

等價的寫法是把分母移過去：

$$\underbrace{(1 + S(n))^{-n}}_{\text{今天價格}} = \frac{E[\text{時點1 的價格}]}{1 + S(1)},$$

即今天的價格 = 下一期的預期價格用一期利率折現回來。(today's price of every bond is the expectation of its price next period, discounted at the known one-period rate. That is precisely the recursive structure a tree or lattice model needs, which is why the theory reappears later in the course as the *risk-neutral* pricing convention.)

§4.16. Duration, in Practice

— Slides p. 162 —

We had assumed parallel shifts in the spot rate curve — the caveat flagged back in [Section 3.6.ii](#). Now that the curve is a genuine object with n separate coordinates, that assumption can be relaxed.

To handle more general shifts, define a vector $[c_1, c_2, \dots, c_n]$ that characterizes the shift:

- Parallel shift: $[1, 1, \dots, 1]$.
- Twist: $[1, 1, \dots, 1, -1, \dots, -1]$, $[1.8, 1.6, 1.4, 1, 0, -1, -1.4, \dots]$, etc.
- ...

At least one c_i should be 1 as the reference point.

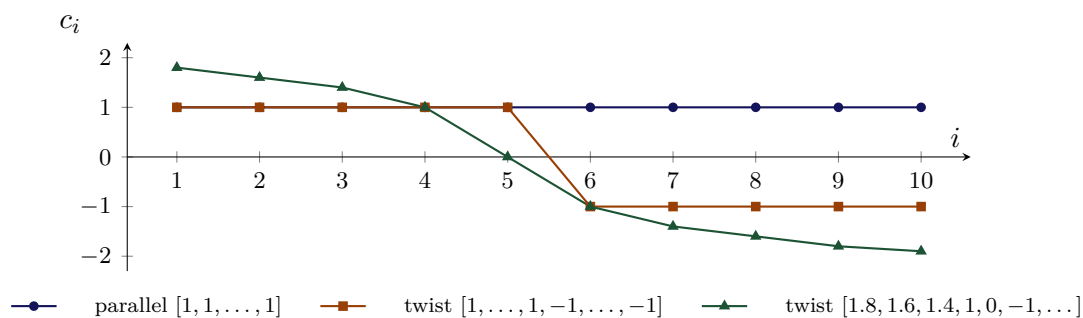


Figure 4.9.: Shift vectors. Each describes how much the spot rate at maturity i moves per unit of the common factor y ; a parallel shift is the special case in which every maturity moves alike.

Remark 4.16.1. 問題在哪。之前算 duration 時說「利率上升 Δy ，價格掉多少」，暗中假設曲線上每一期的 spot rate 都一起上升 Δy ，也就是整條曲線平移 (parallel shift)。但實際上曲線常常不是這樣動的：可能短天期漲、長天期跌（曲線變平或倒掛），也可能中間動得多、兩端動得少。

這一頁的做法。把「曲線怎麼動」拆成兩部分：

$$S(i) \longrightarrow S(i) + y c_i,$$

- y ：一個數字，代表動多大；
- $[c_1, \dots, c_n]$ ：一個向量，代表動的形狀——第 i 期跟著 y 動幾倍。

以 $y = 0.1\%$ 為例：

- $[1, 1, \dots, 1]$ ：每一期都 $+0.1\%$ ，就是原本的平移；
- $[1, \dots, 1, -1, \dots, -1]$ ：前幾期 $+0.1\%$ 、後幾期 -0.1% ，短端漲長端跌，曲線被「扭」了 (twist)；
- $[1.8, 1.6, 1.4, 1, 0, -1, \dots]$ ：第 1 期 $+0.18\%$ 、第 4 期 $+0.1\%$ 、第 5 期不動、再往後轉為下跌，是比較平滑的扭轉。

上圖就是把這三個向量畫出來：橫軸是期別 i ，縱軸是 c_i 。

為什麼要有一個 $c_i = 1$ 。 y 和 c_i 只以乘積出現， $[c_i]$ 全部乘 2、 y 除以 2 是同一種變動，所以要固定尺度。規定某一期的 $c_i = 1$ ， y 就有了明確意義：那一期的 spot rate 動了多少。

下一頁再用這個 y 定義 duration： y 動一點點時，價格變動的百分比。它是原本 duration 的推廣：若形狀選平移 $[1, \dots, 1]$ 、且曲線本來就是平的 ($S(i)$ 全等於某個 r)，價格變成 $P(y) = \sum_i C_i / (1 + r + y)^i$ ，這正是把殖利率從 r 移到 $r + y$ 的價格函數，於是 $-\frac{1}{P} \frac{\partial P}{\partial y} \Big|_{y=0} = \frac{1}{P} \sum_i \frac{i C_i}{(1 + r)^{i+1}}$ ，恰好就是 (3.11) 的 modified duration。

— Slides p. 163 —

Definition 4.16.2 (Duration under a general shift). Let

$$P(y) \triangleq \sum_i \frac{C_i}{(1 + S(i) + y c_i)^i} \quad (4.28)$$

be the price associated with the cash flow $C_1, C_2, \dots$, and define duration as

$$-\frac{\partial P(y)/P(0)}{\partial y} \Big|_{y=0} \quad \text{or} \quad -\frac{P(\Delta y) - P(-\Delta y)}{2P(0) \Delta y}.$$

Modified duration equals the above when

$$[c_1, c_2, \dots, c_n] = [1, 1, \dots, 1], \quad S(1) = S(2) = \dots = S(n).$$

Remark 4.16.3. 這一節把第三章的 duration 放回正確的位置，值得把三件事對齊。

(一) y 換了角色。(4.28) 裡的 y 不再是殖利率，而是一個抽象的衝擊大小；真正決定各期利率怎麼動的是向量 $[c_i]$ 。 y 動一單位，第 i 期的即期利率就動 c_i 單位。

(二) 為什麼要有「至少一個 $c_i = 1$ 」。 y 與 $[c_i]$ 只以乘積 $y c_i$ 出現，把 $[c_i]$ 全部乘以 2、 y 全部除以 2，模型完全一樣。固定某個 $c_i = 1$ 就是挑一個參考期別來定尺度，讓算出來的 duration 有明確的單位：「當那一期的利率動 1 時，價格動幾個百分比」。

(三) 第二個公式就是中央差分。它與 (3.14) 的 effective duration 是同一個數值技巧，只是這裡擾動的是整條曲線而非單一殖利率——連同 Section 3.13 之後那段關於 Δy 取多大的討論，也照樣適用。

實務上把不同的 $[c_i]$ 各算一個 duration，就得到**關鍵利率存續期間** (key rate duration)：取 $[c_i]$ 為只在某一期為 1、其餘為 0 的向量，便可量出部位對曲線上各個點的個別暴險，而不是一個總數。一個 duration 為零的部位，仍可能對**扭轉** (twist) 有巨大暴險。twist 指曲線

繞著某個期別「轉動」：一端上升、另一端下降，改變的是曲線的斜率而不是整體高低（例如短端 -0.1% 、長端 $+0.1\%$ ，曲線變陡）。例如持有長天期債券、同時放空短天期債券，把數量配到平移下的 duration 為零；平移時兩邊損益互相抵銷，但遇到上述變陡的 twist，長債因長端利率上升而跌價、放空的短債因短端利率下降而漲價，兩邊同時虧錢。

§4.17. Some Loose Ends on Dates

— Slides p. 164 —

Every formula in this chapter has indexed time by integers $1, 2, \dots, n$. Converting those to actual dates raises a list of unglamorous questions that nonetheless have to be answered before any of it can be coded:

- Holidays.
- Weekends.
- Business days ($T + 2$, etc.).
- Shall we treat a year as 1 year whether it has 365 or 366 days?

Note 4.17.1. These are the same concerns as the day count conventions of [Section 2.10](#), now compounded by the settlement calendar. None of them is deep, and all of them are mandatory: a discount factor $d(i)$ is only as well defined as the date i names, and two desks using different holiday calendars will bootstrap two different curves from the same quotes.

（翻譯：這些問題和 [Section 2.10](#) 的 day count convention 是同一類，只是現在還多疊了一層交割日曆（settlement calendar）的問題。它們都不深奧，卻每一個都非處理不可：折現因子 $d(i)$ 的定義只能跟「第 i 期」指的是哪一天一樣明確；兩個交易櫃檯若用不同的假日日曆，拿同一組報價 bootstrap 出來的曲線就會不同。其中 $T + 2$ 指成交後第 2 個營業日才交割（錢券互換），所以現金流的實際日期還要跳過週末與假日來算。）

5. Fundamental Statistical Concepts

— Slides pp. 165–166 —

*There are three kinds of lies:
lies, damn lies, and statistics.*

— Misattributed to Benjamin Disraeli (1804–1881)

*If 50 million people believe a foolish thing,
it's still a foolish thing.*

— George Bernard Shaw (1856–1950)

*One death is a tragedy,
but a million deaths are a statistic.*

— Josef Stalin (1879–1953)

Everything so far has been deterministic. A bond's cash flows were known, the curve was observed, and the only uncertainty admitted — in the term structure theories of [Section 4.15](#) — was smuggled in through an expectation operator that was never defined.

This chapter supplies the vocabulary that the rest of the course runs on. Nothing in it is deep; what matters is that every later model, from the binomial tree to Black–Scholes to Monte Carlo pricing, is assembled from these pieces. Three of them will do most of the work:

- **Conditional expectation**, which is how a model says “given what we know today”.
- **The normal distribution**, which is what we will assume *log* returns follow.
- **The lognormal distribution**, which is what prices then follow — and the reason ([5.29](#)) below turns into the Black–Scholes formula later on.

The chapter is built in layers, and each section uses only what came before it. [Section 5.1–Section 5.3](#) develop means, variances and covariances from the linearity of expectation, including the Cauchy–Schwarz inequality, Chebyshev's inequality and the law of large numbers. [Section 5.4](#) defines conditional expectation and proves the tower law. [Section 5.5–Section 5.7](#) introduce densities, the normal distribution and its moment generating function, and the multivariate normal distribution. [Section 5.8](#) and [Section 5.9](#) turn this theory into algorithms for simulation, and prove the central limit theorem along the way. Finally, [Section 5.10](#) combines everything into the lognormal model for prices, ending with the expected payoff of a call.

Note 5.0.1. The epigraphs are a warning worth keeping. [Figure 1.4](#) already showed that real returns are *not* normal: the tails are far too heavy. We use the normal distribution anyway, because it is the only one that stays tractable under the operations we need. The discipline recommended by the Modelers' Hippocratic Oath in [Section 1.5](#) applies exactly here — know precisely where the assumption is wrong.

§5.1. Moments

— Slides p. 167 —

The expectation $E[X]$ tells us where a random variable is centred, but nothing about how widely it is spread or how it moves together with other random variables. Moments fill in that picture. Throughout, we take for granted the two basic properties of expectation: it is *linear*, $E[aX + bY] = aE[X] + bE[Y]$, and *monotone*, $X \leq Y$ implies $E[X] \leq E[Y]$. Apart from one limiting argument in [Lemma 5.1.4](#), everything in this section is derived from these two facts.

§5.1.i. Variance and Covariance

For $n \geq 1$, the **n th moment** of X is $E[X^n]$, and its **n th central moment** is $E[(X - \mu)^n]$, where $\mu = E[X]$. The mean is the first moment. The second central moment measures spread and gets its own name.

Definition 5.1.1. The **variance** of a random variable X is defined as

$$\text{Var}[X] \triangleq E[(X - E[X])^2]. \quad (5.1)$$

The **covariance** between random variables X and Y is

$$\text{Cov}[X, Y] \triangleq E[(X - \mu_X)(Y - \mu_Y)], \quad (5.2)$$

where μ_X and μ_Y are the means of X and Y , respectively. Random variables X and Y are **uncorrelated** if $\text{Cov}[X, Y] = 0$.

The covariance is positive when X and Y tend to be above their means at the same time, and negative when one tends to be above its mean while the other is below. Taking $Y = X$ in (5.2) recovers $\text{Cov}[X, X] = \text{Var}[X]$: variance is covariance with oneself. So most facts about variance are special cases of facts about covariance.

Proposition 5.1.2

For random variables X, Y, Z with finite variances and constants a, b, c :

1. (Computational form) $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$; in particular $\text{Var}[X] = E[X^2] - (E[X])^2$.
2. (Symmetry) $\text{Cov}[X, Y] = \text{Cov}[Y, X]$.
3. (Bilinearity) $\text{Cov}[aX + bY, Z] = a \text{Cov}[X, Z] + b \text{Cov}[Y, Z]$, and likewise in the second argument.
4. (Constants do not matter) $\text{Cov}[X + c, Y] = \text{Cov}[X, Y]$ and $\text{Cov}[c, Y] = 0$.
5. (Scaling of variance) $\text{Var}[a + bX] = b^2 \text{Var}[X]$.

Proof. 1. Expand the product and use linearity, remembering that μ_X and μ_Y are constants:

$$E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y = E[XY] - \mu_X \mu_Y.$$

2. The definition is symmetric in X and Y .

3. The mean of $aX + bY$ is $a\mu_X + b\mu_Y$, so $(aX + bY) - E[aX + bY] = a(X - \mu_X) + b(Y - \mu_Y)$. Multiply by $Z - \mu_Z$ and take expectations. The second argument follows by symmetry.

4. $X + c$ has mean $\mu_X + c$, so $(X + c) - E[X + c] = X - \mu_X$ and nothing changes. A constant equals its own mean, so $c - E[c] = 0$.

5. By items 3 and 4, $\text{Var}[a + bX] = \text{Cov}[a + bX, a + bX] = \text{Cov}[bX, bX] = b^2 \text{Cov}[X, X]$. ■

Items 2 and 3 say that covariance behaves like a dot product of vectors, with the variance playing the role of squared length. This analogy is worth keeping in mind: the Cauchy–Schwarz inequality in Section 5.2 and the formula for the variance of a sum in Section 5.3 are both the random-variable versions of familiar facts about dot products. Item 5 says that shifting a random variable does not change its spread, while scaling it by b scales the standard deviation by $|b|$.

§5.1.ii. What Variance Controls

Variance is defined as an average squared distance from the mean. The next results make precise the idea that a small variance forces X to stay close to its mean.

Lemma 5.1.3 (Markov's inequality)

If $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Proof. Let $\mathbf{1}_{\{X \geq a\}}$ be the random variable that equals 1 when $X \geq a$ and 0 otherwise. Then $a \mathbf{1}_{\{X \geq a\}} \leq X$: when $X \geq a$ the left side is $a \leq X$, and otherwise it is $0 \leq X$. Taking expectations and using monotonicity gives $a P(X \geq a) \leq E[X]$. ■

Lemma 5.1.4

If $X \geq 0$ and $E[X] = 0$, then $X = 0$ with probability one.

Proof. By Lemma 5.1.3, $P(X \geq 1/n) \leq n E[X] = 0$ for every $n \geq 1$. The event $\{X > 0\}$ is the union of the increasing events $\{X \geq 1/n\}$, so $P(X > 0) = \lim_{n \rightarrow \infty} P(X \geq 1/n) = 0$. ■

Proposition 5.1.5 1. $\text{Var}[X] \geq 0$, and $\text{Var}[X] = 0$ if and only if X is a constant with probability one.

2. (Chebyshev's inequality) If $\sigma = \sqrt{\text{Var}[X]} > 0$ and $k > 0$, then

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Proof. 1. $\text{Var}[X]$ is the expectation of the non-negative random variable $(X - \mu)^2$, so it is non-negative. If it is zero, Lemma 5.1.4 gives $(X - \mu)^2 = 0$, that is, $X = \mu$, with probability one. Conversely, a constant has zero variance by item 5 of Proposition 5.1.2 with $b = 0$.

2. Apply Lemma 5.1.3 to $(X - \mu)^2 \geq 0$ with $a = k^2\sigma^2$: $P((X - \mu)^2 \geq k^2\sigma^2) \leq \sigma^2/(k^2\sigma^2)$. ■

Chebyshev's inequality holds for *every* distribution with finite variance, which is why it is weak: it only says that at most 1/4 of the probability lies more than two standard deviations from the mean. For a normal distribution the true figure is about 4.6%. The gap between the two is the room that heavy tails can occupy.

The third and fourth central moments, scaled by σ , describe the shape of the distribution beyond its spread:

$$\text{skewness} = \frac{E[(X - \mu)^3]}{\sigma^3}, \quad \text{kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4}.$$

Skewness measures asymmetry, and kurtosis measures how much of the variance comes from rare, large deviations. Both are unchanged by the scaling $X \mapsto a + bX$ with $b > 0$, for the same reason as item 5 of Proposition 5.1.2. Section 5.6 shows that the normal distribution has kurtosis 3.

§5.1.iii. Independence and Uncorrelatedness

Uncorrelatedness is often confused with a much stronger notion.

Definition 5.1.6. Random variables X and Y are **independent** if, for all sets A and B ,

$$P(X \in A, Y \in B) = P(X \in A) P(Y \in B).$$

In plain words: knowing the value of X tells you nothing at all about Y . The following form of the definition is the one used in calculations.

Proposition 5.1.7

X and Y are independent if and only if

$$E[f(X)g(Y)] = E[f(X)]E[g(Y)]$$

for all functions f and g for which the expectations exist.

Proof. Suppose X and Y are independent and discrete. Independence with $A = \{x\}$ and $B = \{y\}$ gives $P(X = x, Y = y) = P(X = x)P(Y = y)$, so

$$E[f(X)g(Y)] = \sum_{x,y} f(x)g(y)P(X=x)P(Y=y) = \left(\sum_x f(x)P(X=x)\right)\left(\sum_y g(y)P(Y=y)\right),$$

which is $E[f(X)]E[g(Y)]$. With densities, independence means the joint density factors as $f_X(x)f_Y(y)$, and the same computation works with integrals in place of sums.

Conversely, take $f = \mathbf{1}_A$ and $g = \mathbf{1}_B$, the functions that equal 1 on A (resp. B) and 0 elsewhere. Then $E[\mathbf{1}_A(X)\mathbf{1}_B(Y)] = P(X \in A, Y \in B)$ and $E[\mathbf{1}_A(X)] = P(X \in A)$, so the identity is exactly the definition of independence. ■

Corollary 5.1.8

Independent random variables with finite means are uncorrelated.

Proof. First, $E[XY]$ is finite: applying Proposition 5.1.7 to the non-negative functions $|x|$ and $|y|$ gives $E[|XY|] = E[|X|]E[|Y|] < \infty$.¹ Now take $f(x) = x$ and $g(y) = y$: $E[XY] = E[X]E[Y]$, so $\text{Cov}[X, Y] = E[XY] - E[X]E[Y] = 0$ by the computation in item 1 of Proposition 5.1.2, which uses only that $E[X]$, $E[Y]$ and $E[XY]$ are finite. ■

No second moments are needed here: for dependent random variables, finite variances are what guarantee that $\text{Cov}[X, Y]$ is finite (by the Cauchy–Schwarz inequality of Section 5.2), but independence already makes $E[XY]$ finite.

The proof shows exactly how much weaker uncorrelatedness is. $\text{Cov}[X, Y] = 0$ is the single identity $E[XY] = E[X]E[Y]$, which is Proposition 5.1.7 for *one* choice of f and g . Independence demands it for all f and g , including nonlinear ones such as x^2 or e^x . So the converse of the corollary fails.

Example 5.1.9. Let $X \sim N(0, 1)$ and $Y = X^2$. With $f(x) = x$ and $g(y) = y$, we get $E[XY] = E[X^3] = 0 = E[X]E[X^2]$, so X and Y are uncorrelated. (The odd moments of X vanish because its density is symmetric about 0.) But with $f(x) = x^2$ and $g(y) = y$, we get $E[X^2Y] = E[X^4] = 3$ while $E[X^2]E[Y] = 1$, so X and Y are not independent. (The value $E[X^4] = 3$ is computed in Section 5.6.) Of course they are not: Y is computed directly from X .

Section 5.2 explains what uncorrelatedness *does* say: that the best straight line through the data is flat.

¹For non-negative f and g the identity in Proposition 5.1.7 holds even when the expectations are infinite; the proof is the same.

§5.2. Correlation

— Slides p. 168 —

Covariance mixes two things: how strongly two variables move together, and how large they are. Measuring a price in cents instead of dollars multiplies every covariance with it by 100, although nothing about the co-movement has changed. Correlation removes the size.

Definition 5.2.1. The **standard deviation** of X is the square root of the variance,

$$\sigma_X \triangleq \sqrt{\text{Var}[X]}.$$

The **correlation** (or **correlation coefficient**) between X and Y is

$$\rho_{X,Y} \triangleq \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}, \quad (5.3)$$

provided both have nonzero standard deviations.²

§5.2.i. Invariance under Rescaling

Proposition 5.2.2

For constants a, c and nonzero b, d ,

$$\rho_{a+bX, c+dY} = \text{sign}(bd) \rho_{X,Y}.$$

In particular, correlation is unchanged by changing the units or the origin of either variable.

Proof. By Proposition 5.1.2, $\text{Cov}[a + bX, c + dY] = bd \text{Cov}[X, Y]$, while $\sigma_{a+bX} = |b| \sigma_X$ and $\sigma_{c+dY} = |d| \sigma_Y$. Dividing, the ratio is $bd/(|b||d|) = \text{sign}(bd)$ times $\rho_{X,Y}$. ■

§5.2.ii. The Range of Correlation

The fact that ρ lies between -1 and 1 comes from the following inequality, the random-variable version of $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$.

Lemma 5.2.3 (Cauchy–Schwarz inequality)

For random variables U and V with $E[U^2]$ and $E[V^2]$ finite,

$$(E[UV])^2 \leq E[U^2] E[V^2].$$

If $E[U^2] > 0$, equality holds if and only if $V = t_0 U$ with probability one for some constant t_0 .

Proof. First, $E[UV]$ is finite, because $|UV| \leq \frac{1}{2}(U^2 + V^2)$. If $E[U^2] = 0$, then $U = 0$ with probability one by Lemma 5.1.4, and both sides are zero. Otherwise, for every real number t , the random variable $(V - tU)^2$ is non-negative, so

$$0 \leq q(t) \triangleq E[(V - tU)^2] = E[V^2] - 2t E[UV] + t^2 E[U^2].$$

This is a quadratic in t that opens upward and never goes below zero, so it has at most one real root and its discriminant is not positive: $4(E[UV])^2 - 4E[U^2] E[V^2] \leq 0$.

²Wilmott (2009): “the correlations between financial quantities are notoriously unstable.” It may even break down “at high-frequency time intervals” (Budish, Cramton, & Shim, 2015).

If equality holds, the discriminant is zero and q has a double root $t_0 = E[UV]/E[U^2]$. Then $E[(V - t_0U)^2] = 0$, so $V = t_0U$ with probability one by [Lemma 5.1.4](#). Conversely, if $V = t_0U$, both sides equal $t_0^2(E[U^2])^2$. ■

Theorem 5.2.4

$-1 \leq \rho_{X,Y} \leq 1$. Moreover $|\rho_{X,Y}| = 1$ if and only if $Y = a + bX$ with probability one for some constants a and $b \neq 0$, that is, Y is an **affine** function of X ;^a in that case $\rho_{X,Y}$ has the sign of b .

^aAn *affine* function (仿射函数) is one of the form $f(x) = a + bx$: a linear function plus a constant. Its graph is any straight line, not necessarily one through the origin. Strictly speaking, a *linear* function also requires $a = 0$, but $\rho = \pm 1$ does not; for example, $Y = 3 + 2X$ has $\rho = 1$.

Proof. Apply [Lemma 5.2.3](#) to $U = X - \mu_X$ and $V = Y - \mu_Y$. Then $E[UV] = \text{Cov}[X, Y]$, $E[U^2] = \sigma_X^2 > 0$ and $E[V^2] = \sigma_Y^2$, so $\text{Cov}[X, Y]^2 \leq \sigma_X^2 \sigma_Y^2$. Dividing by $\sigma_X^2 \sigma_Y^2$ gives $\rho_{X,Y}^2 \leq 1$.

Equality holds if and only if $Y - \mu_Y = t_0(X - \mu_X)$, that is, $Y = a + bX$ with $b = t_0$ and $a = \mu_Y - t_0\mu_X$. Here $b \neq 0$, since otherwise Y would be constant and $\sigma_Y = 0$. Finally, [Proposition 5.2.2](#) with X in place of Y gives $\rho_{X, a+bX} = \text{sign}(b) \rho_{X,X} = \text{sign}(b)$. ■

So $\rho = \pm 1$ describes a perfect straight-line relationship, and every other value lies strictly in between. The next subsection says what the values in between mean.

§5.2.iii. Correlation and the Best Straight Line

Suppose we want to predict Y by a straight-line function $a + bX$ of X , and we measure the quality of the prediction by the mean squared error $E[(Y - a - bX)^2]$.

Proposition 5.2.5

Assume $\sigma_X > 0$. The mean squared error $E[(Y - a - bX)^2]$ is minimised by

$$b^* = \frac{\text{Cov}[X, Y]}{\text{Var}[X]} = \rho_{X,Y} \frac{\sigma_Y}{\sigma_X}, \quad a^* = \mu_Y - b^* \mu_X,$$

and the minimum value is

$$E[(Y - a^* - b^*X)^2] = \sigma_Y^2 (1 - \rho_{X,Y}^2). \quad (5.4)$$

Proof. For fixed b , write $W = Y - bX$. Then $E[(W - a)^2] = \text{Var}[W] + (E[W] - a)^2$, by item 1 of [Proposition 5.1.2](#) applied to $W - a$, together with $\text{Var}[W - a] = \text{Var}[W]$ from item 5. This is smallest when $a = E[W] = \mu_Y - b\mu_X$, and the minimum is $\text{Var}[W]$. By bilinearity,

$$\text{Var}[Y - bX] = \text{Var}[Y] - 2b \text{Cov}[X, Y] + b^2 \text{Var}[X],$$

a quadratic in b that opens upward, with its minimum at $b^* = \text{Cov}[X, Y]/\text{Var}[X]$. Substituting b^* gives

$$\text{Var}[Y] - \frac{\text{Cov}[X, Y]^2}{\text{Var}[X]} = \sigma_Y^2 \left(1 - \frac{\text{Cov}[X, Y]^2}{\sigma_X^2 \sigma_Y^2} \right) = \sigma_Y^2 (1 - \rho_{X,Y}^2). \quad \blacksquare$$

Equation (5.4) gives the cleanest reading of ρ . Before seeing X , the best constant prediction of Y is μ_Y , with mean squared error σ_Y^2 . The best straight line in X reduces this error by the factor $1 - \rho^2$. So ρ^2 is the *fraction of the variance of Y explained by a straight line in X* . When

$|\rho| = 1$ the line explains everything, in agreement with [Theorem 5.2.4](#). When $\rho = 0$ the best slope is $b^* = 0$: the line is flat and explains nothing.

This is the precise sense in which uncorrelated means “no *linear* relationship”. It does not rule out other relationships. In [Example 5.1.9](#) the points (X, Y) lie on the parabola $y = x^2$, which is symmetric about $x = 0$, so the best straight line through it is flat, even though Y is completely determined by X . [Section 5.4](#) shows that dropping the restriction to straight lines leads to the conditional expectation, which predicts $Y = X^2$ perfectly.

Remark 5.2.6. 開頭那個腳註不是客套話，值得展開。

(一) **相關係數不穩定**。 ρ 是用歷史樣本估出來的，而它本身會隨時間漂移——尤其在市場承壓時，資產之間的相關係數往往一起往 1 靠攏。用平時估出來的 ρ 去算危機時的投資組合風險，會系統性地低估損失；2008 年的結構型商品定價就是栽在這裡。

(二) **高頻下會崩解**。把觀測間隔縮短到毫秒等級，兩個本來高度相關的資產（例如 S&P 500 期貨與其 ETF）算出來的相關係數會急遽下降，這就是所謂的 **Epps 效應** (Epps effect)。原因是兩邊的成交並非同時發生，取樣間隔一短，非同步造成的雜訊就蓋過真正的共同波動。Budish、Cramton 與 Shim (2015) 指出，這個在極短時間尺度上反覆出現的價差，正是高頻交易「軍備競賽」的獵物。

(三) **相關係數只看得到直線關係**。做風險管理時，常用「兩個資產相關係數很低」來說明 diversification（分散投資）有效。但由 (5.4)， ρ 量的只是直線能解釋多少；真正出事的時候，資產之間的關係往往是非線性的——平常看起來各走各的，一遇到危機卻一起大跌。這種關係，就像 [Example 5.1.9](#) 的拋物線一樣，相關係數看不出來。

結論： ρ 是一個取樣間隔與樣本期間的函數，報告時必須說清楚是用什麼頻率、什麼期間算的；而且即使 ρ 很小，也不代表兩個資產沒有關係。

§5.3. Variance of a Sum

— Slides p. 169 —

A portfolio is a weighted sum of asset returns, so the variance of a weighted sum is the first formula of risk management. Because covariance is bilinear ([Proposition 5.1.2](#)), it is also a direct computation.

Proposition 5.3.1

The variance of a weighted sum of random variables equals

$$\text{Var} \left[\sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}[X_i, X_j]. \quad (5.5)$$

It becomes

$$\sum_{i=1}^n a_i^2 \text{Var}[X_i] \quad (5.6)$$

when the X_i are uncorrelated.^a

^aBienaymé (1853).

Proof. Since variance is covariance with oneself, apply bilinearity in each argument:

$$\text{Var} \left[\sum_i a_i X_i \right] = \text{Cov} \left[\sum_i a_i X_i, \sum_j a_j X_j \right] = \sum_i \sum_j a_i a_j \text{Cov}[X_i, X_j].$$

If the X_i are uncorrelated, every off-diagonal term vanishes and only the $i = j$ terms, $a_i^2 \text{Var}[X_i]$, survive. ■

The same argument with two different weight vectors gives the covariance of two weighted sums,

$$\text{Cov}\left[\sum_i a_i X_i, \sum_j b_j X_j\right] = \sum_i \sum_j a_i b_j \text{Cov}[X_i, X_j]. \quad (5.7)$$

§5.3.i. The Covariance Matrix

The double sums are easier to handle in matrix notation. Collect the random variables into a column vector $\mathbf{X} = (X_1, \dots, X_n)^\top$ and define its **covariance matrix**

$$C \triangleq [\text{Cov}[X_i, X_j]]_{1 \leq i, j \leq n},$$

whose diagonal holds the variances. With weights $\mathbf{a} = (a_1, \dots, a_n)^\top$, (5.5) and (5.7) read

$$\text{Var}[\mathbf{a}^\top \mathbf{X}] = \mathbf{a}^\top C \mathbf{a}, \quad \text{Cov}[\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{X}] = \mathbf{a}^\top C \mathbf{b}.$$

- Proposition 5.3.2** 1. C is symmetric and **positive semidefinite**: $\mathbf{a}^\top C \mathbf{a} \geq 0$ for every $\mathbf{a} \in \mathbb{R}^n$.
 2. C is **positive definite** ($\mathbf{a}^\top C \mathbf{a} > 0$ for every $\mathbf{a} \neq \mathbf{0}$) if and only if no nontrivial combination $\mathbf{a}^\top \mathbf{X}$ is constant with probability one.
 3. If A is an $m \times n$ matrix and $\mathbf{U} = A\mathbf{X}$, then the covariance matrix of $\mathbf{U}$ is $ACA^\top$.

Proof. 1. Symmetry is item 2 of Proposition 5.1.2, and $\mathbf{a}^\top C \mathbf{a} = \text{Var}[\mathbf{a}^\top \mathbf{X}] \geq 0$.

2. By Proposition 5.1.5, $\mathbf{a}^\top C \mathbf{a} = \text{Var}[\mathbf{a}^\top \mathbf{X}] = 0$ exactly when $\mathbf{a}^\top \mathbf{X}$ is constant with probability one.

3. $U_k = \sum_i A_{ki} X_i$ is a weighted sum with weights given by the k th row of A . By (5.7), $\text{Cov}[U_k, U_l] = \sum_{i,j} A_{ki} C_{ij} A_{lj}$, which is the (k, l) entry of $ACA^\top$. ■

Item 2 has a concrete meaning: C fails to be positive definite exactly when one of the variables is redundant, in the sense that some combination of them carries no risk at all. Item 3 is what Section 5.9 uses to manufacture correlated random variables from independent ones.

§5.3.ii. Averages and the Law of Large Numbers

The uncorrelated case (5.6) has a consequence that the whole of Monte Carlo simulation rests on.

Theorem 5.3.3 (Weak law of large numbers)

Let $X_1, X_2, \dots$ be uncorrelated with common mean μ and common variance σ^2 , and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then $E[\bar{X}_n] = \mu$, $\text{Var}[\bar{X}_n] = \sigma^2/n$, and for every $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0 \quad (n \rightarrow \infty).$$

Proof. The mean follows from linearity. By (5.6) with $a_i = 1/n$, $\text{Var}[\bar{X}_n] = n \cdot \sigma^2/n^2 = \sigma^2/n$. Chebyshev's inequality (Proposition 5.1.5) with $k\sigma/\sqrt{n} = \varepsilon$ gives the bound. ■

So the average of many independent draws settles down to the mean, and the standard error shrinks like $\sigma/\sqrt{n}$: to gain one more decimal digit of accuracy, one needs 100 times as many draws. This is exactly how Monte Carlo pricing estimates an expectation, and why it is slow.

The same formula also shows the limits of diversification. If an equally weighted portfolio holds n assets, each with variance σ^2 and every pair with correlation ρ , then (5.5) gives

$$\text{Var}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n^2}\left(n\sigma^2 + n(n-1)\rho\sigma^2\right) = \frac{\sigma^2}{n} + \left(1 - \frac{1}{n}\right)\rho\sigma^2 \rightarrow \rho\sigma^2.$$

The uncorrelated part of the risk can be diversified away; the common part $\rho\sigma^2$ cannot. (The formula also shows that such a ρ cannot be negative when n is large, since the variance must stay non-negative: a common correlation ρ among n variables requires $\rho \geq -1/(n-1)$.)

§5.4. Conditional Expectation

— Slides p. 170 —

Definition 5.4.1. “ $X | I$ ” denotes X conditional on the information set I ; the information set can be another random variable’s value, or the past values of X , say. The **conditional expectation**

$$E[X | I]$$

is the expected value of X conditional on I . It is itself a *random variable*.

The definition above is informal: it does not say how to compute $E[X | I]$, nor in what sense it is “the” expected value. Before proving anything we make it precise, first in the simplest case and then in general. Throughout, X is assumed to have a finite mean, $E[X] < \infty$.

§5.4.i. The Elementary Case

By the **elementary case** we mean the following situation: the information set I consists of the value of a single random variable Y , and Y is *discrete*, taking finitely or countably many values y , each with $P(Y = y) > 0$. In this case $E[X | I]$, which we then write $E[X | Y]$, can be computed by an explicit formula. Once we learn that $Y = y$, the natural forecast of X is its average under the conditional distribution,

$$g(y) \triangleq E[X | Y = y] = \sum_x x P(X = x | Y = y) = \sum_x x \frac{P(X = x, Y = y)}{P(Y = y)},$$

defined for every y with $P(Y = y) > 0$. This is an ordinary number for each fixed y . The conditional expectation is what we get by plugging in the random value of Y :

$$E[X | Y] \triangleq g(Y).$$

Before Y is observed we do not know which y will occur, so $g(Y)$ is random; this is the sense in which $E[X | I]$ is a random variable. When X and Y have a joint density $f(x, y)$, the analogous formula $g(y) = \int x f(x, y) dx / \int f(x, y) dx$ is the natural guess. This is *not* the elementary case, since $P(Y = y) = 0$ for every y and the division by $P(Y = y)$ above no longer makes sense; justifying this formula is one of the reasons for the general definition below.

Example 5.4.2. Roll a fair die and let X be the result. Suppose we are only told whether X is even, and let $Y = 1$ if it is and $Y = 0$ otherwise. Then

$$E[X | Y = 1] = \frac{2 + 4 + 6}{3} = 4, \quad E[X | Y = 0] = \frac{1 + 3 + 5}{3} = 3.$$

So $E[X | Y]$ is the random variable that equals 4 with probability $\frac{1}{2}$ and 3 with probability $\frac{1}{2}$. It is coarser than X : it only resolves as much as Y does.

§5.4.ii. The General Definition

The formula for g breaks down when $P(Y = y) = 0$ for every y , and it says nothing when the information is something like “all prices up to time t ”. What survives in every case is the following property, which we first check in the elementary case.

Lemma 5.4.3

In the elementary case, with $E[X | Y] = g(Y)$ as defined above, for every bounded function h ,

$$E[E[X | Y] h(Y)] = E[X h(Y)]. \quad (5.8)$$

Proof. Group the outcomes by the value of Y :

$$E[g(Y) h(Y)] = \sum_y g(y) h(y) P(Y = y) = \sum_y h(y) \sum_x x P(X = x, Y = y) = \sum_{x,y} x h(y) P(X = x, Y = y),$$

and the last sum is $E[X h(Y)]$. ■

Read (5.8) as a statement about bets. Any bet whose payoff $W = h(Y)$ can be settled using only the information in Y pays the same on average whether it is scaled by the true outcome X or by the forecast $E[X | Y]$. In other words, as far as anyone who only knows Y can tell, $E[X | Y]$ and X are the same. This becomes the definition.

Say that a random variable W is **determined by I** if its value is known once the information in I is revealed.³

Definition 5.4.4. $E[X | I]$ is a random variable M such that

1. M is determined by I , and
2. $E[MW] = E[XW]$ for every bounded W determined by I .

Such an M always exists; this is a theorem of measure theory (the Radon–Nikodym theorem), which we take for granted. It is also unique, which is what entitles us to say “the conditional expectation”.

Proposition 5.4.5 (Uniqueness)

If M_1 and M_2 both satisfy Definition 5.4.4, then $M_1 = M_2$ with probability one.

Proof. Let $D = M_1 - M_2$. It is determined by I , and subtracting the two versions of condition 2 gives $E[DW] = 0$ for every bounded W determined by I . Take $W = \mathbf{1}_{\{D > 0\}}$, which equals 1 when $D > 0$ and 0 otherwise; it is determined by I because D is. Then $E[D \mathbf{1}_{\{D > 0\}}] = 0$. But $D \mathbf{1}_{\{D > 0\}} \geq 0$, so by Lemma 5.1.4 it is zero with probability one, that is, $P(D > 0) = 0$. The same argument with $W = \mathbf{1}_{\{D < 0\}}$ gives $P(D < 0) = 0$. ■

The payoff of the abstract definition is that every property below is proved the same way: write down a candidate, and check the two conditions of Definition 5.4.4. Uniqueness then says the candidate *is* the conditional expectation.

§5.4.iii. Basic Properties

³In measure-theoretic probability, I is a σ -field $\mathcal{G}$, and “determined by I ” means “ $\mathcal{G}$ -measurable”. When I is the value of Y , the random variables determined by I are exactly the functions $h(Y)$.

Proposition 5.4.6

Let X and Z have finite means, and let a, b be constants.

1. (Linearity) $E[aX + bZ | I] = a E[X | I] + b E[Z | I]$.
2. (Known quantities come out) If W is bounded and determined by I , then $E[WX | I] = W E[X | I]$. In particular $E[W | I] = W$.
3. (Irrelevant information is ignored) If X is independent of I , then $E[X | I] = E[X]$.
4. (Monotonicity) If $X \geq 0$, then $E[X | I] \geq 0$. Consequently $X \leq Z$ implies $E[X | I] \leq E[Z | I]$.

Proof. Write $M_X = E[X | I]$ and $M_Z = E[Z | I]$, and let V denote an arbitrary bounded random variable determined by I .

1. The candidate $aM_X + bM_Z$ is determined by I , and $E[(aM_X + bM_Z)V] = a E[M_X V] + b E[M_Z V] = a E[XV] + b E[ZV] = E[(aX + bZ)V]$.

2. The candidate WM_X is determined by I , since both factors are. The product WV is bounded and determined by I , so it is a legitimate test variable for M_X : $E[(WM_X)V] = E[M_X(WV)] = E[X(WV)] = E[(WX)V]$. For the special case, take $X = 1$, whose conditional expectation is clearly 1.

3. The candidate is the constant $E[X]$, which is determined by any I . Since V is determined by I and X is independent of I , X and V are independent, so $E[XV] = E[X] E[V] = E[E[X] V]$.

4. Take $V = \mathbf{1}_{\{M_X < 0\}}$. Then $M_X V \leq 0$, while $E[M_X V] = E[XV] \geq 0$ because $X \geq 0$. So $E[M_X V] = 0$ with $M_X V \leq 0$, forcing $M_X V = 0$, that is, $P(M_X < 0) = 0$. For the consequence, apply this to $Z - X \geq 0$ and use linearity. ■

Items 2 and 3 are the two extremes. If you already know W , conditioning on I cannot tell you anything more about it, so it behaves like a constant. If I carries no information about X , the best you can do is the unconditional mean. Everything else lies between the two.

§5.4.iv. The Tower Law**Theorem 5.4.7** (Law of iterated conditional expectations, or the tower law)

$$E[X] = E[E[X | I]]. \quad (5.9)$$

Proof. The constant $W = 1$ is bounded and determined by any I , so condition 2 of [Definition 5.4.4](#) gives $E[E[X | I] \cdot 1] = E[X \cdot 1]$. ■

In the elementary case, (5.9) reads

$$E[X] = \sum_y E[X | Y = y] P(Y = y),$$

which is the **law of total expectation**: the overall average is the weighted average of the group averages, weighted by the size of each group. In the die example, $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 3 = 3.5 = E[X]$.

— Slides p. 171 —

Theorem 5.4.8

If I_2 contains at least as much information as I_1 , then

$$E[X | I_1] = E[E[X | I_2] | I_1]. \quad (5.10)$$

Proof. Let $M_2 = E[X | I_2]$. We check that the candidate $M_1 = E[X | I_1]$ satisfies the two conditions for being $E[M_2 | I_1]$. It is determined by I_1 by definition. Now let V be bounded and determined by I_1 . Because I_2 contains everything in I_1 , V is also determined by I_2 , so it is a legitimate test variable for both M_1 and M_2 :

$$E[M_1 V] = E[X V] = E[M_2 V].$$

The outer equalities are condition 2 for M_1 and for M_2 , respectively. By uniqueness, $M_1 = E[M_2 | I_1]$. ■

(5.9) is the special case where I_1 is the trivial information set (knowing nothing), since conditioning on nothing is the ordinary expectation. The other order of conditioning is even simpler: $E[E[X | I_1] | I_2] = E[X | I_1]$, by item 2 of Proposition 5.4.6, because $E[X | I_1]$ is determined by I_1 and hence by I_2 . Either way, when two conditionings are stacked, *the smaller information set wins*.

Here I_1 contains price information up to time t_1 , and I_2 contains price information up to a later time $t_2 > t_1$. In general,

$$I_1 \subseteq I_2 \subseteq \dots$$

means the players never forget past data, so the information sets are increasing over time.⁴

§5.4.v. Conditional Expectation as the Best Forecast

Section 5.2.iii showed that the best *straight line* $a+bX$ for predicting Y has slope $\text{Cov}[X, Y] / \text{Var}[X]$. Conditional expectation answers the same question without the restriction to straight lines.

Theorem 5.4.9

Suppose $E[X^2] < \infty$. Among all random variables Z determined by I with $E[Z^2] < \infty$, the mean squared error $E[(X - Z)^2]$ is smallest for $Z = E[X | I]$. More precisely,

$$E[(X - Z)^2] = E[(X - M)^2] + E[(M - Z)^2], \quad M = E[X | I]. \quad (5.11)$$

Proof. Expand $X - Z = (X - M) + (M - Z)$:

$$E[(X - Z)^2] = E[(X - M)^2] + 2E[(X - M)(M - Z)] + E[(M - Z)^2].$$

The random variable $V = M - Z$ is determined by I , so condition 2 of Definition 5.4.4 gives $E[XV] = E[MV]$, that is, $E[(X - M)V] = 0$, and the cross term vanishes.⁵ The last term is non-negative and equals zero when $Z = M$. ■

Equation (5.11) is Pythagoras' theorem. The forecast error $X - M$ is orthogonal to everything determined by I , so M is the “foot of the perpendicular” from X onto the space of forecasts that use only I . The regression line of Section 5.2.iii is the foot of the perpendicular onto the smaller space of straight lines $a + bX$, which is why it can do worse. For $X \sim N(0, 1)$ and $Y = X^2$, the best line is flat and explains nothing, whereas $E[Y | X] = X^2$ predicts Y perfectly.

⁴Hirsa & Neftci (2014). This idea is formalised by sigma fields and filtrations in probability theory.

⁵Condition 2 was stated for bounded V . It extends to V with $E[V^2] < \infty$ by truncating V at $\pm n$ and letting $n \rightarrow \infty$; we skip the details.

§5.4.vi. Conditional Variance

Define the **conditional variance** by analogy with (5.1),

$$\text{Var}[X | I] \triangleq E[(X - E[X | I])^2 | I] = E[X^2 | I] - (E[X | I])^2.$$

The second form follows by expanding the square and using item 2 of Proposition 5.4.6 to pull out $E[X | I]$, just as in the unconditional case.

Proposition 5.4.10 (Law of total variance)

If $E[X^2] < \infty$, then

$$\text{Var}[X] = E[\text{Var}[X | I]] + \text{Var}[E[X | I]]. \quad (5.12)$$

Proof. Let $M = E[X | I]$; by the tower law $E[M] = E[X]$. Taking expectations of the second form of $\text{Var}[X | I]$ and using the tower law on $E[X^2 | I]$,

$$E[\text{Var}[X | I]] = E[X^2] - E[M^2], \quad \text{Var}[M] = E[M^2] - (E[X])^2.$$

Adding the two gives $E[X^2] - (E[X])^2 = \text{Var}[X]$. ■

The total uncertainty in X splits into two parts: the uncertainty that remains *after* learning I , on average, and the uncertainty in the forecast $E[X | I]$ itself, which comes from not yet knowing what I will say. Since both parts are non-negative, $\text{Var}[E[X | I]] \leq \text{Var}[X]$: forecasts are always less volatile than the thing they forecast. In the die example, $\text{Var}[X] = \frac{35}{12}$, the forecast takes the values 3 and 4 with equal probability so $\text{Var}[E[X | Y]] = \frac{1}{4}$, and within each group the variance is $\frac{8}{3}$; indeed $\frac{8}{3} + \frac{1}{4} = \frac{35}{12}$.

Remark 5.4.11. 這一節補了很多數學，這裡把它們串起來，看它們在後面會怎麼用到。

(一) $E[X | I]$ 為什麼是隨機變數。從骰子的例子就看得出來：在還不知道是奇數還是偶數之前，「條件期望值」可能是 3 也可能是 4，所以它本身是個隨機變數。一般的定義 (Definition 5.4.4) 則是反過來刻畫它：只要是能用 I 結算的賭注，用 $E[X | I]$ 算出來的平均報酬都跟用 X 算的一樣。後面所有性質都是從這個定義，加上唯一性推出來的。

(二) (5.10) 的意思。用白話講：今天對「明天的預測」做預測，得到的就是今天的預測。如果你今天就能確定明天會把預測往上調，那今天就應該直接調上去。之後會定義鞅 (martingale)：一個過程 M_t 若對所有 $t_1 \leq t_2$ 都滿足 $E[M_{t_2} | I_{t_1}] = M_{t_1}$ ，就叫做鞅。由 (5.10) 馬上可以得到：對任何 X ， $M_t = E[X | I_t]$ 都是一個鞅。無套利定價說的正是「折現後的價格是某個機率下的條件期望值」，所以折現後的價格是鞅。

(三) 條件期望值是最好的預測。Theorem 5.4.9 說 $E[X | I]$ 是只用 I 能做到的最小均方誤差預測。Section 5.2.iii 的回歸直線只在直線裡面找最好的，條件期望值則是在所有函數裡面找。這兩者的差距，正是前面說「相關係數看不到非線性關係」的原因。

(四) 為什麼要寫出 $I_1 \subseteq I_2 \subseteq \dots$ 。它把「資訊只會增加、不會忘記」這個假設明確寫出來。這也是 (5.10) 能用的前提：證明裡用到「由 I_1 決定的東西，也由 I_2 決定」。數學上這一串資訊集叫做過濾 (filtration)，是隨機過程裡描述「隨時間知道什麼」的方式。

§5.5. The Normal Distribution

The random variables so far could be discrete or continuous; nothing depended on the difference. From here on we work mostly with continuous distributions, so we first collect the tools for handling densities.

§5.5.i. Distribution Functions and Densities

The **distribution function** of a random variable X is $F_X(x) \triangleq P(X \leq x)$. It increases from 0 to 1 as x runs from $-\infty$ to ∞ . X has a **density** f_X if

$$P(a < X \leq b) = \int_a^b f_X(x) dx \quad \text{for all } a < b,$$

in which case $f_X = F'_X$ wherever f_X is continuous. Expectations are then computed by integrals instead of sums, $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$, and every result proved so far holds with the same proof.

We will often need the density of a transformed variable $Y = g(X)$. The rule is to go through the distribution function.

Proposition 5.5.1 (Change of variables)

Let X have density f_X , and let g be differentiable and strictly monotone with inverse $h = g^{-1}$. Then $Y = g(X)$ has density

$$f_Y(y) = f_X(h(y)) |h'(y)|.$$

Proof. If g is increasing, then $Y \leq y$ exactly when $X \leq h(y)$, so $F_Y(y) = F_X(h(y))$. Differentiating by the chain rule gives $f_Y(y) = f_X(h(y)) h'(y)$, and $h' > 0$. If g is decreasing, then $Y \leq y$ exactly when $X \geq h(y)$, so $F_Y(y) = 1 - F_X(h(y))$ and $f_Y(y) = -f_X(h(y)) h'(y)$, with $h' < 0$. ■

The factor $|h'(y)|$ accounts for stretching: if g spreads a small interval of x -values over a longer interval of y -values, the same probability is spread more thinly, and the density drops accordingly.

§5.5.ii. Definition and Standardisation

— Slides p. 172 —

Definition 5.5.2. A random variable X has the **normal distribution** with mean μ and variance σ^2 if its probability density function is

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}. \quad (5.13)$$

This is expressed by $X \sim N(\mu, \sigma^2)$. The **standard normal distribution** has zero mean, unit variance, and the following distribution function:

$$\text{Prob}[X \leq z] \triangleq N(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx. \quad (5.14)$$

We write $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$ for the standard normal density, so that $N' = \phi$. The definition calls μ and σ^2 the mean and the variance, and it calls (5.13) a density; all three claims need checking. The first step reduces everything to the standard case.

Proposition 5.5.3 (Standardisation)

$X \sim N(\mu, \sigma^2)$ if and only if $Z = (X - \mu)/\sigma \sim N(0, 1)$. Consequently $X = \mu + \sigma Z$ and

$$P(X \leq x) = N\left(\frac{x - \mu}{\sigma}\right).$$

Proof. Apply [Proposition 5.5.1](#) to $g(x) = (x - \mu)/\sigma$, whose inverse $h(z) = \mu + \sigma z$ has $h' = \sigma$: the density of Z is $\sigma \cdot e^{-(\sigma z)^2/(2\sigma^2)}/(\sigma\sqrt{2\pi}) = \phi(z)$. The converse is the same computation with g and h exchanged. ■

So a single function $N(\cdot)$ serves every normal distribution; this is why it appears in every option pricing formula.

- Proposition 5.5.4**
1. $\int_{-\infty}^{\infty} \phi(z) dz = 1$, so (5.13) is a density.
 2. If $Z \sim N(0, 1)$, then $E[Z] = 0$ and $\text{Var}[Z] = 1$. Hence $X \sim N(\mu, \sigma^2)$ has mean μ and variance σ^2 .
 3. (Symmetry) $N(-z) = 1 - N(z)$ for every z .

Proof. **1.** Let $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx$. Then I^2 is a double integral over the plane, which we evaluate in polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$:

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi \left[-e^{-r^2/2} \right]_0^{\infty} = 2\pi.$$

So $I = \sqrt{2\pi}$ and $\int \phi = 1$.

2. $z \phi(z)$ is an odd function, so $E[Z] = \int z \phi(z) dz = 0$. For the variance, note that $\phi'(z) = -z \phi(z)$ and integrate by parts:

$$E[Z^2] = \int_{-\infty}^{\infty} z \cdot z \phi(z) dz = \left[-z \phi(z) \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z) dz = 0 + 1.$$

For general $X = \mu + \sigma Z$, use linearity and item 5 of [Proposition 5.1.2](#).

3. ϕ is even, so $P(Z \leq -z) = P(Z \geq z) = 1 - N(z)$. ■

Together, standardisation and symmetry answer every question about normal probabilities with a table of $N(z)$ for $z \geq 0$. For instance, $P(|X - \mu| \leq k\sigma) = N(k) - N(-k) = 2N(k) - 1$, which is about 68.3%, 95.4% and 99.7% for $k = 1, 2, 3$. Compare this with Chebyshev's guarantee in [Proposition 5.1.5](#), which for $k = 2$ promises only 75%.

Note 5.5.5. The integral in (5.14) has no closed form in elementary functions, so $N(\cdot)$ is always evaluated numerically. Every option pricing formula in this course will end up calling it, which makes a fast and accurate $N(\cdot)$ a genuine implementation concern rather than a detail.

§5.6. Moment Generating Functions

— Slides p. 173 —

Definition 5.6.1. The **moment generating function** of a random variable X is defined as

$$\theta_X(t) \triangleq E[e^{tX}].$$

Since $e^{tX} > 0$, the expectation always makes sense, but it may be $+\infty$. We always have $\theta_X(0) = 1$. We say that the moment generating function *exists* if $\theta_X(t) < \infty$ for every t in some interval $(-h, h)$ with $h > 0$. All the results below assume this.

§5.6.i. Moments from Derivatives

Proposition 5.6.2

If θ_X exists, then every moment $E[X^n]$ is finite,

$$\theta_X(t) = \sum_{n=0}^{\infty} \frac{E[X^n]}{n!} t^n \quad \text{for } |t| < h, \quad (5.15)$$

and consequently $\theta_X^{(n)}(0) = E[X^n]$ for every $n \geq 0$.

Proof. Expand the exponential: $e^{tX} = \sum_{n \geq 0} t^n X^n / n!$. Taking the expectation term by term gives (5.15).⁶ The right-hand side of (5.15) is a power series in t , and the coefficient of t^n in the Taylor series of θ_X at 0 is $\theta_X^{(n)}(0)/n!$. Comparing coefficients gives $\theta_X^{(n)}(0) = E[X^n]$. ■

This is where the name comes from: θ_X packs all the moments of X into one function, and differentiating n times at 0 pulls out the n th moment. In particular,

$$E[X] = \theta_X'(0), \quad \text{Var}[X] = \theta_X''(0) - (\theta_X'(0))^2.$$

It is often easier to differentiate $\ln \theta_X$ instead. Using $\theta_X(0) = 1$,

$$(\ln \theta_X)'(0) = \theta_X'(0) = E[X], \quad (\ln \theta_X)''(0) = \theta_X''(0) - (\theta_X'(0))^2 = \text{Var}[X]. \quad (5.16)$$

§5.6.ii. Transformations and Sums

- Proposition 5.6.3**
1. (Affine maps) For constants a and b , $\theta_{a+bX}(t) = e^{at} \theta_X(bt)$.
 2. (Independent sums) If X and Y are independent, then $\theta_{X+Y}(t) = \theta_X(t) \theta_Y(t)$.

Proof. 1. $E[e^{t(a+bX)}] = e^{at} E[e^{(bt)X}]$.

2. Apply Proposition 5.1.7 with $f(x) = e^{tx}$ and $g(y) = e^{ty}$: $E[e^{t(X+Y)}] = E[e^{tX} e^{tY}] = E[e^{tX}] E[e^{tY}]$. ■

Item 2 is the main reason moment generating functions are useful. The distribution of $X + Y$ is given by a convolution integral, which is usually hard to compute. The moment generating function turns that convolution into a plain multiplication.

To go back from θ_{X+Y} to the distribution of $X + Y$ we need one more fact, which we quote without proof.

Theorem 5.6.4 (Uniqueness)

If $\theta_X(t) = \theta_Y(t) < \infty$ for all t in some interval $(-h, h)$ with $h > 0$, then X and Y have the same distribution.

So a moment generating function that exists identifies the distribution. To show that some random variable has a certain distribution, it is enough to compute its moment generating function and recognise it.

§5.6.iii. The Normal Distribution

⁶Swapping E and $\sum$ needs justification. For $|t| < h$, every partial sum is bounded in absolute value by $e^{|tX|} \leq e^{tX} + e^{-tX}$, whose expectation $\theta_X(t) + \theta_X(-t)$ is finite. The dominated convergence theorem then allows the swap. The same bound shows $E[|X|^n] < \infty$, because $|tX|^n/n! \leq e^{|tX|}$.

Theorem 5.6.5

The moment generating function of $X \sim N(\mu, \sigma^2)$ is

$$\theta_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right). \quad (5.17)$$

Proof. First let $Z \sim N(0, 1)$. Complete the square in the exponent, $tz - \frac{1}{2}z^2 = \frac{1}{2}t^2 - \frac{1}{2}(z - t)^2$:

$$\theta_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz - z^2/2} dz = e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2} dz = e^{t^2/2}.$$

The last integral equals 1 because $e^{-(z-t)^2/2}/\sqrt{2\pi}$ is the $N(t, 1)$ density. For general $X \sim N(\mu, \sigma^2)$, [Proposition 5.5.3](#) gives $X = \mu + \sigma Z$. By item 1 of [Proposition 5.6.3](#),

$$\theta_X(t) = e^{\mu t} \theta_Z(\sigma t) = e^{\mu t} e^{\sigma^2 t^2/2}. \quad \blacksquare$$

As a check, $\ln \theta_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2$, so (5.16) gives mean μ and variance σ^2 , as it should. The normal distribution is exactly the distribution whose $\ln \theta$ is a polynomial of degree two; all its higher-order structure is fixed by the mean and the variance.

Corollary 5.6.6 1. If $X \sim N(\mu, \sigma^2)$ and $b \neq 0$, then $a + bX \sim N(a + b\mu, b^2\sigma^2)$.

2. If $X_1, \dots, X_n$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$, then $\sum_i X_i \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$.

Proof. 1. By [Proposition 5.6.3](#), $\theta_{a+bX}(t) = e^{at} \exp(\mu b t + \frac{1}{2}\sigma^2 b^2 t^2) = \exp((a + b\mu)t + \frac{1}{2}b^2\sigma^2 t^2)$, which is the moment generating function of $N(a + b\mu, b^2\sigma^2)$. Apply [Theorem 5.6.4](#).

2. By independence, $\theta_{\sum X_i}(t) = \prod_i \exp(\mu_i t + \frac{1}{2}\sigma_i^2 t^2) = \exp((\sum_i \mu_i)t + \frac{1}{2}(\sum_i \sigma_i^2)t^2)$. Apply [Theorem 5.6.4](#) again. $\blacksquare$

Corollary 5.6.7

If $Z \sim N(0, 1)$, then $E[Z^n] = 0$ for odd n , and

$$E[Z^{2k}] = \frac{(2k)!}{2^k k!} = 1 \cdot 3 \cdot 5 \cdots (2k - 1).$$

In particular $E[Z^2] = 1$, $E[Z^4] = 3$ and $E[Z^6] = 15$.

Proof. Expand $\theta_Z(t) = e^{t^2/2}$ as a power series:

$$e^{t^2/2} = \sum_{k=0}^{\infty} \frac{(t^2/2)^k}{k!} = \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k!}.$$

Only even powers appear, so by (5.15) all odd moments vanish. Matching the coefficient of t^{2k} with $E[Z^{2k}]/(2k)!$ gives $E[Z^{2k}] = (2k)!/(2^k k!)$. Finally, $(2k)! = (2 \cdot 4 \cdots 2k)(1 \cdot 3 \cdots (2k - 1)) = 2^k k! \cdot (1 \cdot 3 \cdots (2k - 1))$. $\blacksquare$

The value $E[Z^4] = 3$ is the benchmark behind the word “kurtosis”: a distribution with $E[(X - \mu)^4]/\sigma^4 > 3$ has heavier tails than the normal, which is what [Figure 1.4](#) showed for real returns.

Note 5.6.8. The moment generating function does not always exist. A lognormal $Y = e^X$ with X normal has every moment finite, yet $E[e^{tY}] = \infty$ for every $t > 0$, because e^{te^x} grows far faster than the normal density $e^{-x^2/2}$ decays. So Proposition 5.6.2 cannot be run in reverse: finite moments do not imply an existing moment generating function. The usual remedy is the **characteristic function** $E[e^{itX}]$, with $i = \sqrt{-1}$, which always exists because $|e^{itX}| = 1$, and which also determines the distribution.

Remark 5.6.9. 這一節的重點可以濃縮成三句話，後面會一再用到。

(一) 動差生成函數把所有動差包在一起。把 e^{tX} 展開再取期望值，第 n 項的係數就是 $E[X^n]/n!$ ，所以在 0 微分 n 次就得到第 n 階動差 (Proposition 5.6.2)。實際在算的時候，對 $\ln \theta_X$ 微分通常比較方便：一次微分是平均數，二次微分是變異數。常態分佈的 $\ln \theta_X$ 剛好是 t 的二次式，所以它完全由平均數和變異數決定； $E[Z^4] = 3$ 這種高階動差，也是從同一個式子展開出來的。

(二) 獨立相加，在動差生成函數上就是相乘。這配合唯一性定理，就能直接證明「獨立常態相加還是常態」(Corollary 5.6.6)。下一節的多元常態分佈，以及後面把每一期報酬加起來得到整段報酬的論證，都是建立在這件事上。

(三) 在 $t = n$ 取值就得到對數常態的動差。由 (5.17)， $E[e^{nX}] = e^{n\mu + n^2\sigma^2/2}$ 。若 $Y = e^X$ ，這就是 $E[Y^n]$ ，所以 Section 5.10 的平均數與變異數 (5.25) 只是把 $n = 1, 2$ 代進去而已。但要注意， Y 自己的動差生成函數並不存在 (上面的 note)，所以處理對數常態時，永遠是透過 $\ln Y$ 這個常態變數來算，而不是直接算 Y 的動差生成函數。

§5.7. The Multivariate Normal Distribution

— Slides p. 174 —

A portfolio involves several assets at once, so a single normal variable is not enough: we need a joint distribution for several of them, with a prescribed covariance matrix. The independent case was settled in Section 5.6. If $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then

$$\sum_i X_i \sim N\left(\sum_i \mu_i, \sum_i \sigma_i^2\right).$$

This was proved in Corollary 5.6.6.

Dropping independence requires a definition rather than a theorem. Let $X_i \sim N(\mu_i, \sigma_i^2)$, which may not be independent, and suppose

$$\sum_{i=1}^n t_i X_i \sim N\left(\sum_{i=1}^n t_i \mu_i, \sum_{i=1}^n \sum_{j=1}^n t_i t_j \text{Cov}[X_i, X_j]\right) \quad (5.18)$$

for every linear combination $\sum_{i=1}^n t_i X_i$ with $\sum_{i=1}^n \sum_{j=1}^n t_i t_j \text{Cov}[X_i, X_j] \neq 0$.

— Slides p. 175 —

Then the X_i are said to have a **multivariate normal distribution**, or to be **jointly normal**.⁷ With $M \equiv C^{-1}$ and the (i, j) th entry of the matrix M being M_{ij} , the probability density function for the X_i is

$$\frac{1}{\sqrt{(2\pi)^n \det(C)}} \exp\left[-\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (X_i - \mu_i) M_{ij} (X_j - \mu_j)\right], \quad (5.19)$$

with a positive definite covariance matrix

$$C \triangleq [\text{Cov}[X_i, X_j]]_{1 \leq i, j \leq n}.$$

⁷Corrected by Mr. Huang, Guo-Hua (R98922107) on March 10, 2010.

The mean and the variance in (5.18) are not an extra assumption: they are forced by linearity and by (5.5). The real content of the definition is that every linear combination is *normal*. The variance condition only excludes combinations that are constant, which are “normal with variance 0” in a degenerate sense. The rest of this section explains where such random variables come from, where the density (5.19) comes from, and what is special about them.

§5.7.i. Construction from Independent Normals

Proposition 5.7.1

Let $Z_1, \dots, Z_k$ be independent standard normal variables, $\mathbf{Z} = (Z_1, \dots, Z_k)^\top$, and let A be an $n \times k$ matrix and $\boldsymbol{\mu} \in \mathbb{R}^n$. Then $\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance matrix $C = AA^\top$.

Proof. The covariance matrix of $\mathbf{Z}$ is the identity, so that of $\mathbf{X}$ is $AA^\top$ by item 3 of Proposition 5.3.2. For a weight vector $\mathbf{t}$, set $\mathbf{c} = A^\top \mathbf{t}$; then

$$\mathbf{t}^\top \mathbf{X} = \mathbf{t}^\top \boldsymbol{\mu} + \sum_{j=1}^k c_j Z_j.$$

Each $c_j Z_j$ with $c_j \neq 0$ is $N(0, c_j^2)$, and they are independent, so by Corollary 5.6.6 the sum is normal with variance $\sum_j c_j^2 = \mathbf{t}^\top AA^\top \mathbf{t}$, provided this is nonzero. Its mean is $\mathbf{t}^\top \boldsymbol{\mu}$. This is (5.18). ■

Conversely, the distribution of a multivariate normal vector is determined by $\boldsymbol{\mu}$ and C alone. Indeed, for every $\mathbf{t}$ the variable $\mathbf{t}^\top \mathbf{X}$ is normal, so by (5.17) at $t = 1$,

$$E[e^{\mathbf{t}^\top \mathbf{X}}] = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top C \mathbf{t}\right),$$

and the multivariate version of Theorem 5.6.4, which we quote, says that this function of $\mathbf{t}$ determines the joint distribution. Hence every multivariate normal vector with mean $\boldsymbol{\mu}$ and covariance matrix C has the same distribution as $\boldsymbol{\mu} + A\mathbf{Z}$, for any matrix A with $AA^\top = C$. Finding such an A is the subject of Section 5.9.

§5.7.ii. The Density

When C is positive definite, we can choose A square and invertible with $AA^\top = C$ (Section 5.9 shows how). The density (5.19) is then the result of a change of variables. In n dimensions, Proposition 5.5.1 becomes: if $\mathbf{X} = \mathbf{g}(\mathbf{Z})$ with $\mathbf{g}$ invertible and $\mathbf{h} = \mathbf{g}^{-1}$, then

$$f_{\mathbf{X}}(\mathbf{x}) = f_{\mathbf{Z}}(\mathbf{h}(\mathbf{x})) |\det D\mathbf{h}(\mathbf{x})|,$$

where $D\mathbf{h}$ is the Jacobian matrix of partial derivatives. The absolute determinant measures how $\mathbf{h}$ scales volumes, just as $|h'|$ measured how h scales lengths.

Proposition 5.7.2

If C is positive definite, the multivariate normal vector $\mathbf{X}$ has the density (5.19).

Proof. The Z_i are independent, so the density of $\mathbf{Z}$ is the product $\prod_i \phi(z_i) = (2\pi)^{-n/2} \exp(-\frac{1}{2} \mathbf{z}^\top \mathbf{z})$. Since $\mathbf{X}$ has the same distribution as $\boldsymbol{\mu} + A\mathbf{Z}$, we may take $\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}$, so $\mathbf{h}(\mathbf{x}) = A^{-1}(\mathbf{x} - \boldsymbol{\mu})$ and $D\mathbf{h} = A^{-1}$. Then

$$\mathbf{h}(\mathbf{x})^\top \mathbf{h}(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^\top (A^{-1})^\top A^{-1} (\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^\top C^{-1} (\mathbf{x} - \boldsymbol{\mu}),$$

because $(A^{-1})^\top A^{-1} = (AA^\top)^{-1} = C^{-1}$. Also $|\det A^{-1}| = 1/|\det A| = 1/\sqrt{\det C}$, since $\det C = \det A \det A^\top = (\det A)^2$. Substituting gives (5.19), whose double sum is $(\mathbf{x} - \boldsymbol{\mu})^\top M(\mathbf{x} - \boldsymbol{\mu})$. ■

When C is only positive semidefinite, some combination $\mathbf{t}^\top \mathbf{X}$ is constant (Proposition 5.3.2), so $\mathbf{X}$ lives on a lower-dimensional subspace and has no density; the definition (5.18) still applies.

§5.7.iii. Uncorrelated Jointly Normal Variables Are Independent

Section 5.1.iii showed that uncorrelated random variables need not be independent. For jointly normal variables they must be.

Proposition 5.7.3

If $X_1, \dots, X_n$ are jointly normal and pairwise uncorrelated, then they are independent.

Proof. Discard any X_i with zero variance; a constant is independent of everything. The remaining covariance matrix is diagonal with positive entries σ_i^2 , hence positive definite, with $C^{-1} = \text{diag}(1/\sigma_i^2)$ and $\det C = \prod_i \sigma_i^2$. The density (5.19) becomes

$$\prod_{i=1}^n \frac{1}{\sigma_i \sqrt{2\pi}} e^{-(x_i - \mu_i)^2 / (2\sigma_i^2)},$$

a product of one-variable densities, which is exactly independence. ■

The hypothesis is *joint* normality; normal marginals are not enough.

Example 5.7.4. Let $X \sim N(0, 1)$, and let S be independent of X with $P(S = 1) = P(S = -1) = \frac{1}{2}$. Put $Y = SX$. By the symmetry of ϕ , $Y \sim N(0, 1)$. Also $\text{Cov}[X, Y] = E[SX^2] = E[S]E[X^2] = 0$. But $|Y| = |X|$, so X and Y are far from independent. Proposition 5.7.3 is not contradicted, because (X, Y) is not jointly normal: $X + Y$ equals 0 with probability $\frac{1}{2}$, which no normal variable with positive variance does.

§5.7.iv. Conditional Expectation for Normal Variables

Proposition 5.7.3 has a striking consequence that ties together Section 5.2.iii and Section 5.4. In general the best predictor $E[Y | X]$ can be any function of X , and the best straight line can do much worse. For jointly normal variables the two coincide.

Proposition 5.7.5

Let (X, Y) be bivariate normal with $\sigma_X > 0$, and let $b^* = \text{Cov}[X, Y] / \text{Var}[X]$. Then

$$E[Y | X] = \mu_Y + b^*(X - \mu_X), \quad \text{Var}[Y | X] = \sigma_Y^2(1 - \rho_{X,Y}^2).$$

In fact, given X , the variable Y is normal with this mean and this variance.

Proof. Let $W = Y - b^*X$. Any combination $aX + cW = (a - cb^*)X + cY$ is a combination of X and Y , so (X, W) is again bivariate normal. By bilinearity, $\text{Cov}[X, W] = \text{Cov}[X, Y] - b^* \text{Var}[X] = 0$, so X and W are independent by Proposition 5.7.3. Now $Y = b^*X + W$. Given X , the first term is known and the second has the same distribution as without conditioning, namely normal with mean $E[W] = \mu_Y - b^*\mu_X$ and variance $\text{Var}[W]$. In the language of Proposition 5.4.6, items 2 and 3 give $E[Y | X] = b^*X + E[W]$. Finally $\text{Var}[W] = \text{Var}[Y - b^*X] = \sigma_Y^2(1 - \rho_{X,Y}^2)$ was computed in the proof of Proposition 5.2.5. ■

So for jointly normal variables, the regression line *is* the conditional expectation, and the residual uncertainty $\sigma_Y^2(1 - \rho^2)$ does not even depend on the observed value of X . All of the dependence between X and Y is captured by the single number ρ .

Remark 5.7.6. 這一節把前面幾節串在一起，重點有三個。

(一) 為什麼用「所有線性組合都是常態」來定義。這個定義比直接寫密度函數更廣：當 C 不可逆時（例如 $X_3 = X_1 + X_2$ ），密度不存在，但定義一樣適用。而且金融裡最常碰到的就是線性組合，因為投資組合就是資產的加權和；這個定義直接告訴你任何投資組合的分佈。要注意每個邊際分佈都是常態，並不代表聯合起來是多元常態，[Example 5.7.4](#) 就是反例。

(二) 多元常態都可以由獨立標準常態做出來。[Proposition 5.7.1](#) 說 $\mu + AZ$ 是多元常態，共變異數矩陣是 AA^T ；反過來，任何多元常態都長這樣。所以要模擬多元常態，只要產生獨立的標準常態（[Section 5.8](#)），再找一個 A 使 $AA^T = C$ （[Section 5.9](#)）就好。

(三) 在常態的世界裡，相關係數就是一切。[Section 5.2](#) 說過，相關係數只量得到直線關係，而且不相關不代表獨立。但如果變數是聯合常態，不相關就等於獨立（[Proposition 5.7.3](#)），而且最好的預測剛好就是回歸直線（[Proposition 5.7.5](#)），所以 ρ 已經描述了全部的相依關係。這也是聯合常態假設危險的地方：真實市場在危機時一起崩跌的那種非線性相依，在這個模型裡根本不存在。

§5.8. Generation of Univariate Normal Distributions

— Slides p. 176 —

Monte Carlo pricing needs normal variates by the million, and a computer supplies only uniform ones. This section explains how to turn the latter into the former.

§5.8.i. Uniform Variables and the Inverse Transform

A random variable ξ is **uniformly distributed** over $(0, 1)$ if it has density 1 on $(0, 1)$ and 0 elsewhere, that is, $P(\xi \leq x) = x$ for $0 < x \leq 1$. Its mean and variance are

$$E[\xi] = \int_0^1 x \, dx = \frac{1}{2}, \quad \text{Var}[\xi] = \int_0^1 x^2 \, dx - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

The most direct way to turn a uniform variable into any other distribution is the following.

Proposition 5.8.1 (Inverse transform)

Let F be a continuous, strictly increasing distribution function, and let ξ be uniform over $(0, 1)$. Then $F^{-1}(\xi)$ has distribution function F .

Proof. Since F is increasing, $F^{-1}(\xi) \leq x$ exactly when $\xi \leq F(x)$, which has probability $F(x)$. ■

So $N^{-1}(\xi)$ is a standard normal variable. The catch is that N^{-1} , like N itself, has no closed form and must be approximated numerically. The inverse transform works well for distributions whose F^{-1} is explicit. The one we need below is: if ξ is uniform over $(0, 1)$, then

$$P(-2 \ln \xi > s) = P(\xi < e^{-s/2}) = e^{-s/2}, \quad s \geq 0, \quad (5.20)$$

so $-2 \ln \xi$ has density $\frac{1}{2}e^{-s/2}$ on $(0, \infty)$, the **exponential distribution** with mean 2.

§5.8.ii. The Box–Muller Transform

The trick for avoiding N^{-1} is to generate two normal variables at a time. A pair of independent standard normals has a rotationally symmetric joint density, so it is natural to describe it in polar coordinates.

Theorem 5.8.2 (Box–Muller)

Let S and Θ be independent, with S exponential with mean 2 and Θ uniform over $[0, 2\pi)$. Then $\sqrt{S} \cos \Theta$ and $\sqrt{S} \sin \Theta$ are independent standard normal variables. In particular, if ω is uniform over $(0, 1)$ and independent of Θ , one may take $S = -2 \ln \omega$.

Proof. Start from the other end. Let Z_1, Z_2 be independent standard normal, with joint density $\frac{1}{2\pi} e^{-(z_1^2 + z_2^2)/2}$, and write them in polar coordinates, $Z_1 = R \cos \Theta'$ and $Z_2 = R \sin \Theta'$. For any region B of the (s, θ) plane, with $s = r^2$ and $dz_1 dz_2 = r dr d\theta = \frac{1}{2} ds d\theta$,

$$P((R^2, \Theta') \in B) = \iint_B \frac{1}{2\pi} e^{-s/2} \cdot \frac{1}{2} ds d\theta = \iint_B \left(\frac{1}{2} e^{-s/2} \right) \left(\frac{1}{2\pi} \right) ds d\theta.$$

The joint density of (R^2, Θ') factors into an exponential density with mean 2 in s times a uniform density on $[0, 2\pi)$ in θ . So R^2 and Θ' are independent with exactly the distributions of S and Θ . Since $(s, \theta) \mapsto (\sqrt{s} \cos \theta, \sqrt{s} \sin \theta)$ maps both pairs the same way, $(\sqrt{S} \cos \Theta, \sqrt{S} \sin \Theta)$ has the same joint distribution as (Z_1, Z_2) . The last claim is (5.20). ■

§5.8.iii. The Polar Method

The Box–Muller transform still needs $\cos$ and $\sin$. The **polar method** obtains the angle from a random point instead.

- Let X be uniformly distributed over $(0, 1]$ so that

$$\text{Prob}[X \leq x] = x, \quad 0 < x \leq 1.$$

- Repeatedly draw two samples x_1 and x_2 from X until

$$\omega \triangleq (2x_1 - 1)^2 + (2x_2 - 1)^2 < 1.$$

- Then $c(2x_1 - 1)$ and $c(2x_2 - 1)$ are independent standard normal variables, where⁸

$$c \triangleq \sqrt{-2(\ln \omega)/\omega}.$$

⁸As they are normally distributed, to prove independence it suffices to prove that they are uncorrelated, which is easy. Thanks to a lively class discussion on March 5, 2014. Strictly speaking, this shortcut needs them to be *jointly* normal (Proposition 5.7.3); normal marginals alone are not enough (Example 5.7.4). The proof below establishes independence directly.

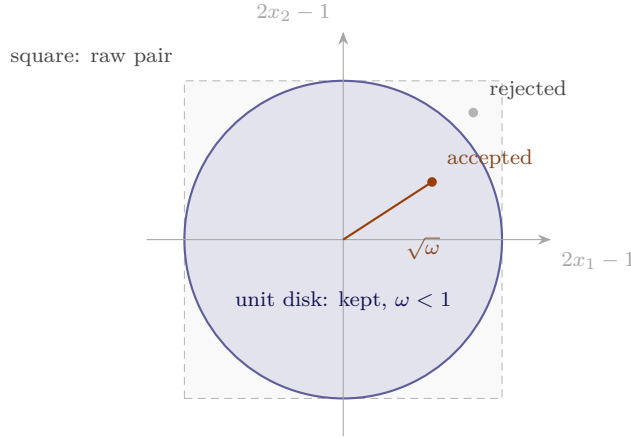


Figure 5.1.: The polar method. A raw pair lands uniformly in the square; the draw is retried unless it falls inside the unit disk. Accepted points are uniform on the disk, which makes the squared radius ω uniform on $(0, 1)$ and independent of the angle.

The acceptance step is **rejection sampling**. Its correctness rests on two facts.

- Lemma 5.8.3**
1. An accepted pair $(V_1, V_2) = (2x_1 - 1, 2x_2 - 1)$ is uniformly distributed on the unit disk, and the expected number of attempts per accepted pair is $4/\pi \approx 1.27$.
 2. If (V_1, V_2) is uniform on the unit disk, then its squared radius $\omega = V_1^2 + V_2^2$ is uniform over $(0, 1)$, its angle Θ is uniform over $[0, 2\pi)$, and ω and Θ are independent.

Proof. **1.** A raw pair is uniform on the square $(-1, 1]^2$ of area 4, so it lands in a region A with probability $\text{area}(A)/4$. For A inside the disk,

$$P(\text{pair} \in A \mid \text{accepted}) = \frac{\text{area}(A)/4}{\pi/4} = \frac{\text{area}(A)}{\pi},$$

which is the uniform distribution on the disk. Each attempt succeeds independently with probability $p = \pi/4$, and the number of attempts until the first success has mean $1/p$.

2. The event $\{\omega \leq s, \Theta \leq \alpha\}$ is a sector of radius $\sqrt{s}$ and angle α , with area $\frac{1}{2}\alpha s$. Therefore

$$P(\omega \leq s, \Theta \leq \alpha) = \frac{\alpha s/2}{\pi} = s \cdot \frac{\alpha}{2\pi}, \quad 0 \leq s \leq 1, \quad 0 \leq \alpha < 2\pi.$$

This is the product of the uniform distribution function on $(0, 1)$ and the uniform distribution function on $[0, 2\pi)$, which gives both the marginals and independence. ■

Theorem 5.8.4

The two numbers $c(2x_1 - 1)$ and $c(2x_2 - 1)$ produced by the polar method are independent standard normal variables.

Proof. By Lemma 5.8.3, the accepted pair is $(V_1, V_2) = (\sqrt{\omega} \cos \Theta, \sqrt{\omega} \sin \Theta)$ with ω uniform over $(0, 1)$ and independent of the uniform angle Θ . Hence

$$cV_1 = \sqrt{\frac{-2 \ln \omega}{\omega}} \sqrt{\omega} \cos \Theta = \sqrt{-2 \ln \omega} \cos \Theta, \quad cV_2 = \sqrt{-2 \ln \omega} \sin \Theta,$$

which are independent standard normals by Theorem 5.8.2. ■

Remark 5.8.5. 這個方法看起來很神祕，其實就是把 Box–Muller 裡「產生一個均勻角度再取 $\cos$ 、 $\sin$ 」這一步，換成「在圓裡隨機丟一個點」。點的方向直接給出 $\cos \Theta = V_1/\sqrt{\omega}$ 和 $\sin \Theta = V_2/\sqrt{\omega}$ ，而點到圓心距離的平方 ω 剛好又是一個獨立的均勻變數，可以拿來當 Box–Muller 裡的 ω ，一個點用了兩次。代價是大約 21% 的點會被丟掉，但整個過程不需要計算三角函數；在早期的硬體上，三角函數比開根號和取對數昂貴得多。而且每次都產生兩個獨立的常態變數。

§5.8.iv. A Dirty Trick and a Right Attitude

— Slides p. 177 —

Let ξ_i be independent and uniformly distributed over $(0, 1)$. A simple method to generate the standard normal variable is to calculate⁹

$$\left(\sum_{i=1}^{12} \xi_i \right) - 6. \quad (5.21)$$

But why use 12? Recall the mean and variance of ξ_i are $1/2$ and $1/12$, respectively.

— Slides p. 178 —

The general formula is

$$\frac{(\sum_{i=1}^n \xi_i) - (n/2)}{\sqrt{n/12}}, \quad (5.22)$$

which is the central limit theorem applied to n uniforms, standardised. By linearity and (5.6), the sum $\sum_{i=1}^n \xi_i$ has mean $n/2$ and variance $n/12$, so (5.22) always has mean 0 and variance 1; the theorem below says that its distribution approaches $N(0, 1)$ as n grows. Choosing $n = 12$ makes the denominator $\sqrt{12/12} = 1$ and the numerator's correction 6, yielding a formula without the need of division and square-root operations.¹⁰

Theorem 5.8.6 (Central limit theorem)

Let $X_1, X_2, \dots$ be independent and identically distributed with mean μ and variance $\sigma^2 > 0$. Then for every z ,

$$P\left(\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \leq z\right) \rightarrow N(z) \quad (n \rightarrow \infty).$$

Proof sketch. We assume in addition that the moment generating function of X_i exists; this covers the uniform case. Let $Y_i = (X_i - \mu)/\sigma$, which has mean 0 and variance 1, and let $T_n = \sum_{i=1}^n Y_i/\sqrt{n}$ be the standardised sum. By (5.15), $\theta_Y(t) = 1 + \frac{1}{2}t^2 + \varepsilon(t)t^2$ with $\varepsilon(t) \rightarrow 0$ as $t \rightarrow 0$. By Proposition 5.6.3,

$$\theta_{T_n}(t) = \left(\theta_Y(t/\sqrt{n})\right)^n = \left(1 + \frac{t^2/2 + \varepsilon(t/\sqrt{n})t^2}{n}\right)^n \rightarrow e^{t^2/2},$$

using $(1 + a_n/n)^n \rightarrow e^a$ whenever $a_n \rightarrow a$. The limit is the moment generating function of $N(0, 1)$. The *continuity theorem*, a strengthening of Theorem 5.6.4 that we quote, says that convergence of moment generating functions near 0 implies convergence of the distribution functions. ■

⁹Jäckel (2002): “this is not a highly accurate approximation and should only be used to establish ballpark estimates.”

¹⁰Contributed by Mr. Chen, Shih-Hang (R02723031) on March 5, 2014.

The theorem says nothing about how fast the convergence is. In relative terms the approximation is worst far out in the tails, where the probabilities are tiny to begin with, so a small absolute error can still be a large relative one. Figure 5.2 shows what this means for $n = 12$.

Always blame your random number generator last.^a Instead, check your programs first.

^a“The fault, dear Brutus, lies not in the stars but in ourselves that we are underlings.” William Shakespeare (1564–1616), *Julius Caesar*.

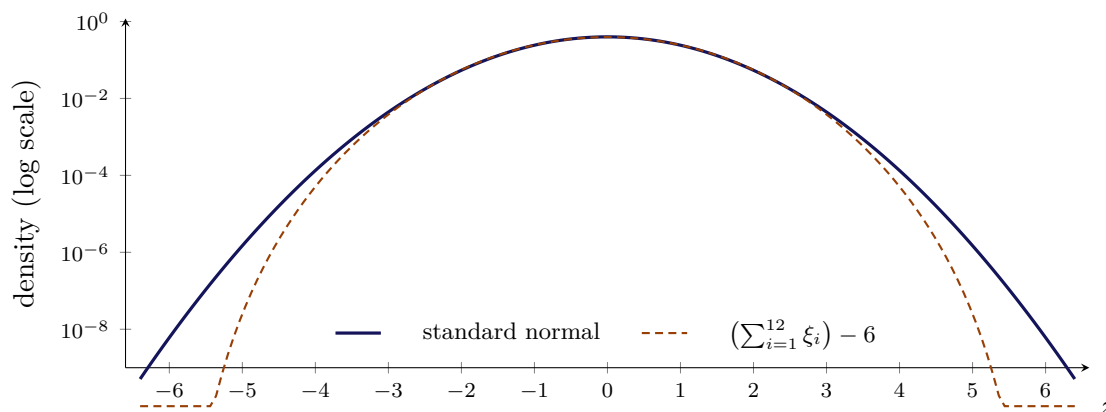


Figure 5.2.: Why (5.21) is only a ballpark approximation. On a logarithmic scale the two densities agree to the eye out to about $|z| = 2$ and then part company: at $z = 4$ the trick delivers 38% of the correct density, at $z = 5$ only 1.7%, and beyond $|z| = 6$ exactly zero, since a sum of twelve uniforms cannot leave $[-6, 6]$.

§5.9. Generation of Bivariate Normal Distributions

— Slides p. 179 —

Pairs of normally distributed variables with correlation ρ can be generated as follows. Let X_1 and X_2 be independent standard normal variables, and set

$$\begin{aligned} U &\triangleq aX_1, \\ V &\triangleq a\rho X_1 + a\sqrt{1-\rho^2} X_2. \end{aligned} \tag{5.23}$$

— Slides p. 180 —

Proposition 5.9.1

U and V are the desired random variables, with

$$\text{Var}[U] = \text{Var}[V] = a^2, \quad \text{Cov}[U, V] = \rho a^2.$$

Proof. $\text{Var}[U] = a^2 \text{Var}[X_1] = a^2$. Since X_1 and X_2 are uncorrelated, (5.6) gives

$$\text{Var}[V] = a^2 \rho^2 \text{Var}[X_1] + a^2(1-\rho^2) \text{Var}[X_2] = a^2 \rho^2 + a^2(1-\rho^2) = a^2.$$

Finally, using $\text{Cov}[X_1, X_2] = 0$ and $\text{Var}[X_1] = 1$,

$$\text{Cov}[U, V] = a^2 \rho \text{Var}[X_1] + a^2 \sqrt{1-\rho^2} \text{Cov}[X_1, X_2] = \rho a^2. \quad \blacksquare$$

Note that the mapping from (X_1, X_2) to (U, V) is a one-to-one correspondence for $a \neq 0$ and $|\rho| < 1$; when $\rho = \pm 1$, $V = \pm U$ and the pair has no density. By (5.3), the correlation of U and V is $\rho a^2 / (a \cdot a) = \rho$, as intended.

— Slides p. 181 —

The mapping in matrix form is

$$\begin{bmatrix} U \\ V \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}. \quad (5.24)$$

The proposition only checks variances and covariances, but more is true. This has the form $A\mathbf{X}$ with $\mathbf{X} = (X_1, X_2)^\top$, so by Proposition 5.7.1 the pair (U, V) is bivariate normal. Since a multivariate normal distribution is determined by its mean and covariance matrix, (U, V) has *exactly* the bivariate normal distribution with variances a^2 and correlation ρ , not merely the right second moments.

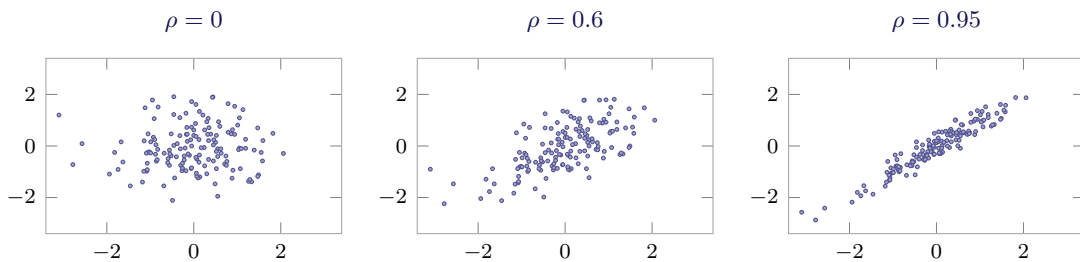


Figure 5.3.: Bivariate normal samples generated by (5.24) with $a = 1$, from the same underlying (X_1, X_2) draws. Raising ρ shears the cloud towards the diagonal without changing either marginal.

§5.9.i. The General Case: Cholesky Decomposition

The matrix in (5.24) was not pulled out of thin air. With $a = 1$ it is the lower-triangular matrix

$$L = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix} \quad \text{with} \quad LL^\top = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

the correlation matrix we wanted. By Proposition 5.7.1, to generate an n -dimensional normal vector with mean $\boldsymbol{\mu}$ and covariance matrix C we need, in general, a matrix L with $LL^\top = C$; then $\boldsymbol{\mu} + L\mathbf{Z}$, with $\mathbf{Z}$ a vector of independent standard normals, does the job. Such an L can always be found, and can be chosen lower triangular.

Theorem 5.9.2 (Cholesky decomposition)

If C is a symmetric positive definite $n \times n$ matrix, there is a lower-triangular matrix L with positive diagonal entries such that $C = LL^\top$.

Proof. Induction on n . For $n = 1$, $C = [c]$ with $c > 0$, and $L = [\sqrt{c}]$. For $n > 1$, write

$$C = \begin{bmatrix} c_{11} & \mathbf{c}^\top \\ \mathbf{c} & C' \end{bmatrix}, \quad L = \begin{bmatrix} \ell_{11} & \mathbf{0}^\top \\ \ell & L' \end{bmatrix},$$

where c_{11} is a number, $\mathbf{c}$ is a column of length $n - 1$ and C' is $(n - 1) \times (n - 1)$. Multiplying out, $LL^\top = C$ amounts to

$$\ell_{11}^2 = c_{11}, \quad \ell_{11}\ell = \mathbf{c}, \quad L'L'^\top = C' - \ell\ell^\top = C' - \frac{\mathbf{c}\mathbf{c}^\top}{c_{11}}.$$

Taking $\mathbf{a} = (1, 0, \dots, 0)^\top$ in the definition of positive definiteness gives $c_{11} > 0$, so the first two equations are solved by $\ell_{11} = \sqrt{c_{11}}$ and $\ell = \mathbf{c}/\sqrt{c_{11}}$. For the third, it suffices by induction that $C'' \triangleq C' - \mathbf{c}\mathbf{c}^\top/c_{11}$ be symmetric positive definite. Symmetry is clear. For $\mathbf{y} \neq \mathbf{0}$ of length $n-1$, put $\mathbf{a} = (-\mathbf{c}^\top\mathbf{y}/c_{11}, \mathbf{y})^\top \neq \mathbf{0}$; expanding,

$$0 < \mathbf{a}^\top C \mathbf{a} = \frac{(\mathbf{c}^\top \mathbf{y})^2}{c_{11}} - 2 \frac{(\mathbf{c}^\top \mathbf{y})^2}{c_{11}} + \mathbf{y}^\top C' \mathbf{y} = \mathbf{y}^\top C'' \mathbf{y}. \quad \blacksquare$$

The proof is also the algorithm: fill in L one column at a time, each step costing a square root and some divisions. It shows why positive definiteness is needed: the square roots taken along the way are square roots of variances of the form $\mathbf{a}^\top C \mathbf{a}$, and they break down as soon as some combination has zero or negative “variance”. If C is only positive semidefinite, a lower-triangular L with $LL^\top = C$ still exists but some diagonal entries are 0; one of the variables is then an exact combination of the others (Proposition 5.3.2).

Remark 5.9.3. 這一節的做法，是 Proposition 5.7.1 的具體實現：先產生獨立的標準常態，再乘上一個 L ，使 $LL^\top$ 等於想要的共變異數矩陣。Figure 5.3 可以直接看出這個線性變換在做什麼： ρ 變大時，點雲被往對角線方向「剪」過去，但每個座標自己的分佈都沒有改變。

實務上最常出問題的是正定性。由市場資料估出來的相關係數矩陣，理論上一定是半正定的 (Proposition 5.3.2)，但如果每一對資產是用不同的樣本期間分別估計的，或者手動調整過某些相關係數，拼起來的矩陣就可能不再是半正定。這時 Cholesky 分解會在某一步需要對負數開根號，模擬根本無法開始，必須先把矩陣修正成最接近的正定矩陣。這也呼應 Section 5.2 的提醒：相關係數本身就不太穩定，一整個矩陣的相關係數更需要小心處理。

§5.10. The Lognormal Distribution

— Slides p. 182 —

Definition 5.10.1. A random variable Y is said to have a **lognormal distribution** if $\ln Y$ has a normal distribution.

This is the distribution we will use for prices: if the log return $\ln(S_T/S_0)$ is normal, then S_T/S_0 is lognormal. Every question about a lognormal Y is answered by translating it into a question about the normal variable $\ln Y$, using the tools of Section 5.5 and Section 5.6.

§5.10.i. Density, Median and Moments

Let $X \sim N(\mu, \sigma^2)$ and $Y \triangleq e^X$.

Proposition 5.10.2 1. Y takes values in $(0, \infty)$ and has density

$$f(y) = \frac{1}{y \sigma \sqrt{2\pi}} e^{-(\ln y - \mu)^2 / (2\sigma^2)}, \quad y > 0.$$

2. The median of Y is e^μ : $P(Y \leq e^\mu) = \frac{1}{2}$.
3. $E[Y^n] = e^{n\mu + n^2\sigma^2/2}$ for every real n .

Proof. 1. Apply Proposition 5.5.1 with $g = \exp$, whose inverse $h = \ln$ has $h'(y) = 1/y$.

2. $\exp$ is increasing, so $P(Y \leq e^\mu) = P(X \leq \mu) = N(0) = \frac{1}{2}$ by Proposition 5.5.3 and the symmetry of ϕ .

3. $E[Y^n] = E[e^{nX}] = \theta_X(n)$, which is (5.17) at $t = n$. ■

The mean and variance of Y are

$$\begin{aligned}\mu_Y &= e^{\mu+\sigma^2/2}, \\ \sigma_Y^2 &= e^{2\mu+2\sigma^2}(e^{\sigma^2} - 1),\end{aligned}\tag{5.25}$$

respectively. They follow from $E[Y^n] = e^{n\mu+n^2\sigma^2/2}$ — which is just (5.17) evaluated at $t = n$. For the variance, $E[Y^2] - \mu_Y^2 = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2}$.

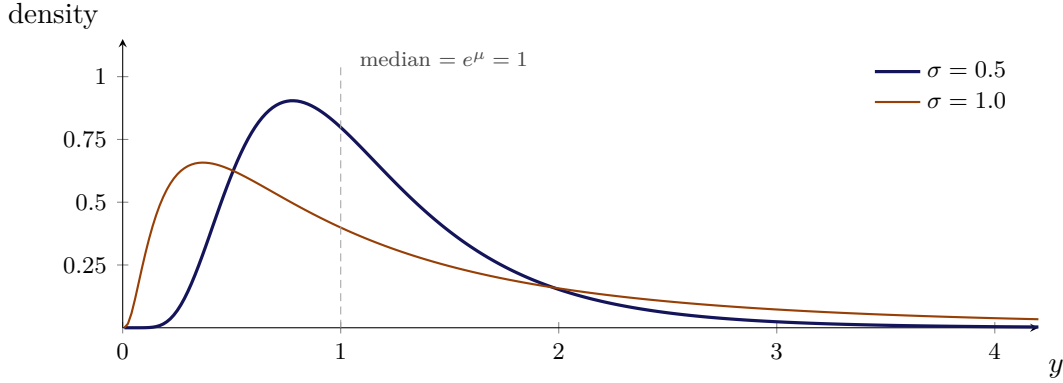


Figure 5.4.: Lognormal densities with $\mu = 0$. The support is $(0, \infty)$, the shape is right-skewed, and the right tail thickens rapidly with σ . The median sits at e^μ regardless of σ , while the mean $e^{\mu+\sigma^2/2}$ drifts to the right of it.

§5.10.ii. Jensen's Inequality

The mean $e^{\mu+\sigma^2/2}$ is strictly larger than the median e^μ , and the gap grows with σ . This is an instance of a general inequality, already used in Section 4.15 and needed again for options.

A function g is **convex** if through every point $(m, g(m))$ of its graph there is a straight line, a **supporting line**, that lies entirely below the graph. For differentiable g the supporting line is the tangent line, and for twice differentiable g convexity means $g'' \geq 0$. A convex function may have kinks, such as the call payoff $\max(x - a, 0)$ at $x = a$; there any line through the kink with slope between the left and right slopes supports the graph. The function is **strictly convex** if each supporting line touches the graph only at its own point, which holds when $g'' > 0$.

Theorem 5.10.3 (Jensen's inequality)

If g is convex and X and $g(X)$ have finite means, then

$$E[g(X)] \geq g(E[X]).$$

If g is strictly convex, equality holds only when X is constant with probability one.

Proof. Let $m = E[X]$ and let $\ell(x) = g(m) + k(x - m)$ be a supporting line of g at m (for differentiable g , $k = g'(m)$). Convexity says $g(x) \geq \ell(x)$ for all x , so by monotonicity and linearity

$$E[g(X)] \geq E[\ell(X)] = g(m) + k(E[X] - m) = g(m).$$

If equality holds, the non-negative random variable $g(X) - \ell(X)$ has mean 0, so it is 0 with probability one by Lemma 5.1.4. For strictly convex g that happens only at $X = m$. ■

The picture is simple: averaging the points of a convex curve lands *above* the curve. With $g(x) = e^x$ and $Y = e^X$,

$$E[Y] = E[e^X] > e^{E[X]} = e^\mu,$$

and (5.25) says that the ratio is exactly $e^{\sigma^2/2}$. Rare large values of Y pull the mean up, while small values of Y are bounded below by 0 and cannot pull it down as much.

§5.10.iii. Products of Lognormal Variables

The next result is why the lognormal distribution fits prices over many periods.

Proposition 5.10.4

If $Y_1, \dots, Y_n$ are independent and lognormal with $\ln Y_i \sim N(\mu_i, \sigma_i^2)$, then $\prod_i Y_i$ is lognormal with $\ln \prod_i Y_i \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$. In particular, if the log returns $r_t = \ln(S_t/S_{t-1})$, $t = 1, \dots, T$, are independent and normal, then $S_T = S_0 \exp(\sum_t r_t)$ is lognormal.

Proof. $\ln \prod_i Y_i = \sum_i \ln Y_i$ is a sum of independent normal variables, which is normal by Corollary 5.6.6. For the second claim, $S_T/S_0 = \prod_t S_t/S_{t-1}$, and multiplying by the constant S_0 only shifts the mean of the logarithm by $\ln S_0$. ■

Even without normality, Theorem 5.8.6 applied to the log returns suggests that $\ln(S_T/S_0)$ is approximately normal when T is large; this is the usual informal justification for the lognormal model.

Remark 5.10.5. 為什麼金融上用對數常態分佈 (lognormal distribution) 描述價格，用常態描述對數報酬？前面證明過的三件事就是理由。

(一) 價格不能為負。常態分佈的值域是整條實線，但股價不會是負的。 $Y = e^X$ 自動把值域限制在 $(0, \infty)$ (Proposition 5.10.2)。

(二) 報酬用加的，價格用乘的。Section 1.6.iii 的對數報酬 $r_t = \ln(S_t/S_{t-1})$ 有一個好處：跨期報酬就是逐期報酬的和。獨立常態相加還是常態，所以價格一定是對數常態 (Proposition 5.10.4)；就算每期報酬不是常態，中央極限定理也讓對數常態成為合理的近似。

(三) 平均數比中位數大。由 Jensen 不等式， $E[Y] = e^{\mu + \sigma^2/2}$ 嚴格大於中位數 e^μ ，多出來的 $\sigma^2/2$ 完全來自波動。這就是之後會一再出現的「 $-\sigma^2/2$ 修正項」的來源：想讓價格的期望成長率等於 r ，指數裡就要寫 $r - \sigma^2/2$ 。

§5.10.iv. Converting between the Two Parametrisations

— Slides p. 183 —

Proposition 5.10.6

Suppose Y is lognormally distributed with mean μ and variance σ^2 . Then $\ln Y$ has a normal distribution with

$$\begin{aligned} E[\ln Y] &= \ln \left[\mu / \sqrt{1 + (\sigma/\mu)^2} \right], \\ \text{Var}[\ln Y] &= \ln \left[1 + (\sigma/\mu)^2 \right]. \end{aligned} \quad (5.26)$$

Proof. Write $m = E[\ln Y]$ and $s^2 = \text{Var}[\ln Y]$. By (5.25), $\mu = e^{m+s^2/2}$ and $\sigma^2 = \mu^2(e^{s^2} - 1)$. The second equation gives $e^{s^2} = 1 + (\sigma/\mu)^2$, which is the formula for s^2 . Then $m = \ln \mu - s^2/2 = \ln \mu - \ln \sqrt{1 + (\sigma/\mu)^2}$, which is the formula for m . ■

Note 5.10.7. Beware the notation in Proposition 5.10.6: μ and σ^2 now denote the mean and variance of Y itself, not of $\ln Y$ as in (5.25). (5.26) is the inverse of (5.25), and the two must not be mixed.

We say X and Y are **joint-lognormally distributed** if $(\ln X, \ln Y)$ is bivariate normal.

Proposition 5.10.8

If X and Y are joint-lognormally distributed, then

$$\begin{aligned} E[XY] &= E[X]E[Y]e^{\text{Cov}[\ln X, \ln Y]}, \\ \text{Cov}[X, Y] &= E[X]E[Y] \left(e^{\text{Cov}[\ln X, \ln Y]} - 1 \right). \end{aligned} \quad (5.27)$$

Proof. Write $\ln X \sim N(m_X, s_X^2)$, $\ln Y \sim N(m_Y, s_Y^2)$ and $c = \text{Cov}[\ln X, \ln Y]$. Since $(\ln X, \ln Y)$ is bivariate normal, $\ln X + \ln Y$ is normal with mean $m_X + m_Y$ and, by (5.5), variance $s_X^2 + s_Y^2 + 2c$. So $XY = e^{\ln X + \ln Y}$ is lognormal, and by Proposition 5.10.2

$$E[XY] = e^{m_X + m_Y + (s_X^2 + s_Y^2 + 2c)/2} = e^{m_X + s_X^2/2} e^{m_Y + s_Y^2/2} e^c = E[X] E[Y] e^c.$$

The covariance is $E[XY] - E[X]E[Y]$. ■

In particular, X and Y are uncorrelated exactly when $\ln X$ and $\ln Y$ are, and the correlation of the prices always has the same sign as that of the log returns.

§5.10.v. Partial Expectations

— Slides p. 184 —

Option payoffs are not smooth: a call pays only when the price ends above the strike. Its expected payoff therefore involves the expectation of Y restricted to an event $\{Y > a\}$, and the probability of that event.

Theorem 5.10.9

Let Y be lognormally distributed such that $\ln Y \sim N(\mu, \sigma^2)$, with density f . Then

$$\int_a^\infty y f(y) dy = e^{\mu + \sigma^2/2} N\left(\frac{\mu - \ln a}{\sigma} + \sigma\right). \quad (5.28)$$

Proof. Substitute $y = e^x$, so that $\ln Y = X \sim N(\mu, \sigma^2)$ and the integral becomes

$$\int_{\ln a}^\infty e^x \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} dx.$$

Completing the square in the exponent,

$$x - \frac{(x - \mu)^2}{2\sigma^2} = \mu + \frac{\sigma^2}{2} - \frac{(x - \mu - \sigma^2)^2}{2\sigma^2},$$

so the integral equals

$$e^{\mu + \sigma^2/2} \int_{\ln a}^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{-(x - \mu - \sigma^2)^2/(2\sigma^2)} dx = e^{\mu + \sigma^2/2} \text{Prob}\left[Z > \frac{\ln a - \mu - \sigma^2}{\sigma}\right],$$

with Z standard normal. By the symmetry $1 - N(z) = N(-z)$ this is $e^{\mu + \sigma^2/2} N((\mu + \sigma^2 - \ln a)/\sigma)$, which is (5.28). ■

The other ingredient is the probability of the event itself.

Proposition 5.10.10

With Y as above and $a > 0$,

$$\text{Prob}[Y > a] = N\left(\frac{\mu - \ln a}{\sigma}\right).$$

Proof. $Y > a$ exactly when $\ln Y > \ln a$. By Proposition 5.5.3 this has probability $1 - N((\ln a - \mu)/\sigma)$, which equals $N((\mu - \ln a)/\sigma)$ by symmetry (Proposition 5.5.4). ■

Corollary 5.10.11

With Y as above and $a > 0$, let

$$d_2 = \frac{\mu - \ln a}{\sigma}, \quad d_1 = d_2 + \sigma.$$

Then

$$E[\max(Y - a, 0)] = e^{\mu + \sigma^2/2} N(d_1) - a N(d_2). \quad (5.29)$$

Proof. $\max(Y - a, 0) = (Y - a) \mathbf{1}_{\{Y > a\}}$, so its expectation is $\int_a^\infty y f(y) dy - a \text{Prob}[Y > a]$. Apply (5.28) to the first term and Proposition 5.10.10 to the second. ■

(5.28) is the **partial expectation** of a lognormal variable — the expected value of Y restricted to the event $Y > a$.

It is not a technical curiosity: it is the entire content of the first term of the Black–Scholes formula. A call option pays $Y - a$ exactly when $Y > a$, so its expected payoff splits as in Corollary 5.10.11, the first piece supplying $S_0 N(d_1)$ and the second $a N(d_2)$. Every remaining chapter of this course is, in one way or another, an elaboration of that split.

換句話說，(5.29) 就是 Black–Scholes 公式的骨架：買權在 $Y > a$ 時付 $Y - a$ ，期望報酬拆成兩塊，前者給出 $S_0 N(d_1)$ ，後者給出 $a N(d_2)$ 。之後要做的，只是決定該用哪一個機率分佈來取期望值，以及怎麼折現。

6. Option Basics

— Slides pp. 185–186 —

Lecture 3
September 24,
2026

*The shift toward options as
the center of gravity of finance [...]*
— Merton H. Miller¹ (1923–2000)

*Too many potential physicists and engineers
spend their careers shifting money around
in the financial sector,
instead of applying their talents to
innovating in the real economy.*
— Barack Obama (2016)

Every instrument so far has had a cash flow that is *fixed* the moment the contract is signed. A bond pays its coupons whether the world turns out well or badly; the only uncertainty is the rate at which we discount them, and all of Chapters 3 and 4 was about that rate. An option is the first instrument in this course whose cash flow at expiration is **contingent**: it depends on what the underlying asset happens to be worth on that day.

That single change is what makes the pricing problem hard, and it is the problem the rest of the course solves. This chapter, however, prices nothing. Its job is to fix

- (i) the vocabulary — call, put, strike, exercise, American, European;
- (ii) the *payoff functions*, which are the terminal conditions every later pricing model must reproduce;
- (iii) the market conventions that decide what the contract actually promises — dividends, splits, short selling; and
- (iv) the standard portfolios built out of options: hedges, spreads, and combinations.

The last item is worth more than it looks. Two of the spreads below, pushed to a limit, deliver model-*independent* prices for two exotic securities, and — pushed one step further — read the market's own probability distribution for S_T straight off the quoted call prices. All of that comes for free, before writing down any stochastic process at all.

§6.1. Calls and Puts

— Slides pp. 187–189 —

Definition 6.1.1. A **call** gives its holder the *right*, but not the obligation, to *buy* one unit of the underlying asset by paying a **strike price** (also called the exercise price).

¹Co-winner of the 1990 Nobel Prize in Economic Sciences.

The holder pays for that right up front. The price paid is the **option premium**, and it is the only cash flow that is certain: the cash flow at expiration is contingent, because the holder will simply walk away if buying at the strike is a bad deal. Figure 6.1 draws the two dates.

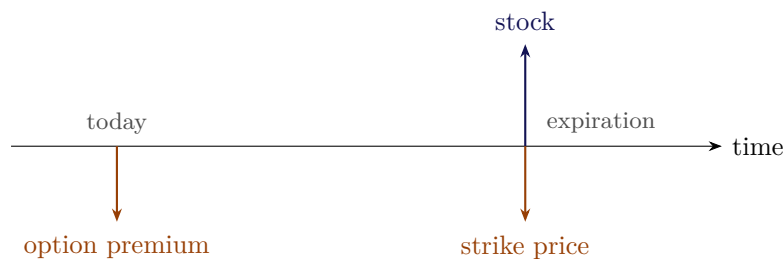


Figure 6.1.: Cash flows of a long call. Arrows below the axis are payments made by the holder, arrows above it are receipts. The premium leaves today with certainty; the exchange of the strike price for the stock happens at expiration *only if* the holder chooses to exercise.

Definition 6.1.2. A **put** gives its holder the right to *sell* one unit of the underlying asset for the strike price.

The put reverses the two contingent arrows: the holder delivers the stock and receives the strike price (Figure 6.2). The premium still leaves today.



Figure 6.2.: Cash flows of a long put — Figure 6.1 with the roles of stock and strike price interchanged.

Notation . Options sold on their own, as the two figures above assume, are called *standalone*. An **embedded option** is one that cannot be separated from its host: it has to be traded along with the underlying asset. The callable bond of Chapter 3 is the standing example — the issuer’s right to redeem early is an option, but nobody trades it apart from the bond.

Remark 6.1.3. How to price options? That is the question the next several chapters answer, and it is old: it can be traced to Aristotle’s (384 B.C.–322 B.C.) *Politics*, if not earlier.²

§6.2. Exercise

— Slides p. 190 —

To **exercise** an option is to invoke the right it grants, and the mechanics are exactly the two contingent arrows drawn above:

²Aristotle reports that Thales, foreseeing a large olive harvest, paid deposits in winter for the use of every olive press in Miletus and Chios — the deposits are option premiums, and the presses are the underlying asset.

- When a *call* is exercised, the holder pays the strike price in exchange for the stock.
- When a *put* is exercised, the holder receives the strike price from the writer in exchange for the stock.

The counterparty — the **writer**, who sold the option and pocketed the premium — has no choice in the matter: the writer must take the other side whenever the holder exercises. This asymmetry is the whole content of the word “option”.

Some options may be exercised *before* the expiration date. This is called **early exercise**, and deciding whether to do it is an optimization problem in its own right, not a detail of the contract. [Section 7.7](#) settles the question completely for calls; for puts it stays open for the rest of the course.

§6.3. American and European

— Slides p. 191 —

Definition 6.3.1. An **American option** can be exercised at any time up to the expiration date. A **European option** can only be exercised at expiration.

Proposition 6.3.2

An American option is worth at least as much as an otherwise identical European option.

Proof. The American contract contains every right the European contract contains, plus the right to exercise early. The holder of the American option may simply choose never to exercise early, thereby reproducing the European payoff exactly. A strictly larger set of available strategies cannot be worth less. ■

Note 6.3.3. The names are geographical only by accident; both types trade on both continents. Note also that [Proposition 6.3.2](#) is an inequality, not an equality — and the gap is genuinely hard to compute. European options on a non-dividend-paying stock admit a closed-form price (Black–Scholes); American puts do not, which is why so much of the numerical machinery later in the course exists — [Section 7.7](#) proves the gap is *zero* for calls on such a stock, and [Section 8.3.i](#) shows where it reopens for puts.

§6.4. Convenient Conventions

— Slides p. 192 —

Notation . The following symbols are used without further comment for the rest of the course.

Symbol	Meaning
C	call value
P	put value
X	strike price
S	stock price
D	dividend

We always assume $S \geq 0$.^a When the time matters we write S_t for the stock price at time t , and S_T for its value at expiration.

^aContributed by Mr. Tang, Bert (B08902102) on March 10, 2021.

§6.5. Payoff, Mathematically Speaking

— Slides pp. 193–194 —

At expiration the holder's decision is trivial, because there is no future left to speculate about: exercise if and only if doing so produces cash.

Definition 6.5.1. The **payoff** of a call at expiration is

$$C = \max(0, S - X), \quad (6.1)$$

and the payoff of a put at expiration is

$$P = \max(0, X - S). \quad (6.2)$$

The $\max(0, \cdot)$ is the contract's optionality made arithmetic. A call will be exercised only if the stock price is *higher* than the strike price — otherwise the stock can be bought more cheaply in the market, and the holder lets the option lapse. Symmetrically, a put will be exercised only if the stock price is *less* than the strike price.

Two facts about (6.1) and (6.2) drive everything that follows.
They are *non-negative*: an option can never impose a loss on its holder beyond the premium already paid. Hence $C \geq 0$ and $P \geq 0$ at *all* times, not just at expiration.
They are *convex* and *kinked* at $S = X$. Convexity is why options are worth more than their intrinsic value before expiration (Jensen's inequality again), and the kink is the single point where every numerical method in this course has to work hardest.

Figure 6.3 draws all four elementary positions with $X = 50$. The writer's payoff is the holder's payoff reflected through the horizontal axis — options are a zero-sum contract between the two parties, so the two pictures in each column must sum to zero pointwise.

§6.5.i. Intrinsic Value and Moneyness

— Slides pp. 195–196 —

Definition 6.5.2. At any time t *before* the expiration date, the **intrinsic value** of a call is

$$\max(0, S_t - X),$$

and the intrinsic value of a put is

$$\max(0, X - S_t).$$

The intrinsic value is what the option would pay if expiration were today. It is a *lower* bound on an American option's value (exercising is always available) and, as Lemma 7.6.1 will show, a lower bound on a European *call*'s value as well — though not, it turns out, on a European put's. Whatever the option is worth above its intrinsic value is its **time value** — the market's charge for the remaining chance that the stock moves further in the holder's favour.

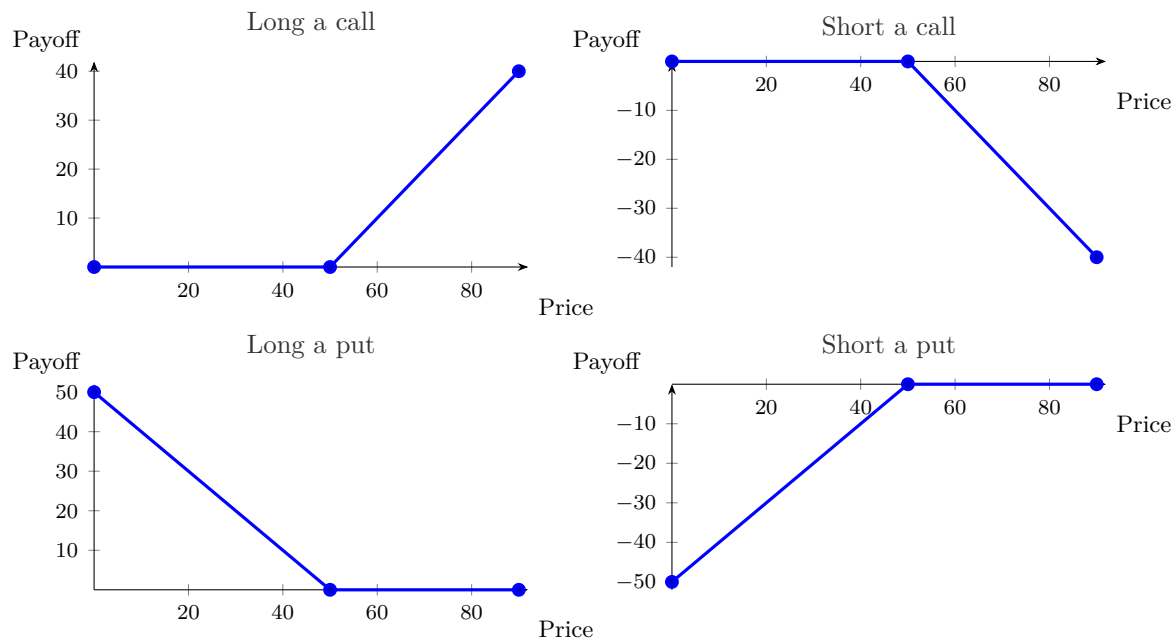


Figure 6.3.: Payoffs at expiration of the four elementary option positions, with strike $X = 50$. These are payoffs, not profits: the premium has not been subtracted. The kink always sits at $S = X$. Note the asymmetry in the writer's risk — a short call has unbounded loss as $S \rightarrow \infty$, whereas a short put loses at most X , since the stock cannot fall below zero.

Definition 6.5.3. A call is **in the money** if $S > X$, **at the money** if $S = X$, and **out of the money** if $S < X$. A put is in the money if $S < X$, at the money if $S = X$, and out of the money if $S > X$.

Note 6.5.4. The direction reverses between calls and puts, so the words describe *whether the option currently has positive intrinsic value*, not whether the stock is high or low.

Options that are in the money at expiration should be exercised — the payoff is positive and free. Astonishingly, they are not always.³

Finding an option's value at *any* time before expiration — that is, splitting it into intrinsic value plus time value with an actual number attached — is a major intellectual breakthrough. Chapter 8 makes it, and the rest of the course elaborates on it.

— Slides p. 197 —

Figure 6.4 shows what the answer looks like ahead of time. The dashed line in each panel is the intrinsic value, the payoff already drawn in Figure 6.3; the solid curve is the option's value one month before expiration. Three features are visible and will be proved later:

- The solid curve is *smooth*: the kink at $S = X$ has been rounded off, because with time left the market does not yet know which side of the strike the stock will land on.
- The call's curve lies *above* its intrinsic value everywhere — that gap is time value — and is monotone and convex, sandwiched between the intrinsic value and the stock price itself.
- The put's curve does *not*. Look at the left edge of the right-hand panel: for small S the solid curve dips *below* the dashed one, so a European put can be worth less than its intrinsic value. This is not a drawing error, and Lemma 7.6.2 explains it.

³About 11% of option holders let in-the-money options expire worthless.

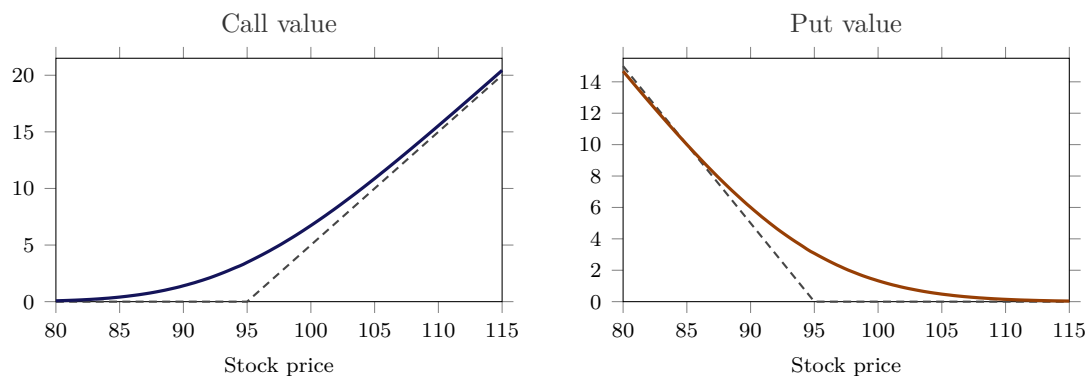


Figure 6.4.: Value of a European call and put with strike $X = 95$ one month before expiration (solid), against their intrinsic values (dashed). Computed from Black–Scholes with $r = 5\%$ and $\sigma = 30\%$; the formula itself is derived much later, and only the *shape* matters here. The vertical gap between the two lines is the time value, largest at the money and vanishing deep in or out of the money.

§6.6. Corporate Actions: Dividends and Splits

The payoff formulas above refer to “the stock”, but a share is not a fixed object: companies pay dividends and split their shares, and each event moves S for reasons that have nothing to do with the market’s opinion of the company. Whether the option contract is *adjusted* for such an event therefore changes what the option is worth. The convention differs by event type.

§6.6.i. Cash Dividends

— Slides p. 198 —

Exchange-traded stock options are *not* cash dividend-protected (or simply **protected**): the option contract is not adjusted for cash dividends.

Why this matters: on the **ex-dividend** date the stock price falls by an amount roughly equal to the cash dividend, because a buyer from that day on no longer receives it. The drop is mechanical, but the unprotected option absorbs it in full. Consequently

cash dividends are detrimental for calls, and the opposite is true for puts.

Intuition . The option holder owns exposure to the share price but not the share, and therefore collects no dividend. A dividend transfers value out of the share price and into the pockets of shareholders — over whom the call holder has no claim. So a call on a dividend-paying stock is worth less than a call on an otherwise identical stock that retains its cash, and a put is worth correspondingly more. This is also the reason early exercise of an American call can be optimal: exercising just before the ex-dividend date is the only way to capture the dividend. **Theorem 7.7.3** makes that precise — those dates are the *only* ones at which it can pay to exercise early.

§6.6.ii. Stock Splits and Stock Dividends

— Slides pp. 199–200 —

Options *are* adjusted for stock splits. After an **n -for- m stock split**, m shares become n shares. Accordingly, the option is rewritten so that

- the strike price becomes m/n times its previous value, and
- the number of shares covered by one option becomes n/m times its previous value.

Exchange-traded stock options are adjusted for **stock dividends** in the same way. The point of the adjustment is that the total strike outlay $X \times (\text{shares covered})$ is unchanged, so the split — which alters nothing real about the company — alters nothing about the option.

Example 6.6.1. Consider an option to buy 100 shares of a company for \$50 per share. A 2-for-1 split changes the terms to a strike price of \$25 per share for 200 shares. The outlay on exercise is $100 \times 50 = 200 \times 25 = \$5,000$ either way.

Note 6.6.2. Unless stated otherwise, **we assume options are unprotected** from here on. Cash dividends will therefore show up explicitly as a term in every pricing formula, while splits will never be mentioned again.

§6.7. Short Selling

— Slides pp. 201–203 —

Several of the portfolios below, and essentially every no-arbitrage argument later in the course, require taking a *negative* position in the stock. The mechanism that makes this possible is short selling.

Definition 6.7.1. **Short selling** (or **shorting**)⁴ means selling an asset that is not owned, with the intention of buying it back later.

Concretely, if you short 1,000 XYZ shares:

1. The broker borrows the 1,000 shares from another client and sells them in the market.
2. This action generates proceeds for the investor — cash now, and an obligation to return the shares later.
3. The investor closes out the short position by buying 1,000 XYZ shares and returning them.

Clearly, the investor profits if the stock price falls: the repurchase costs less than the sale brought in. The payoff of a short position is the payoff of a long position with the sign flipped (Figure 6.5), which is exactly the property that makes shorting the algebraic tool it is.

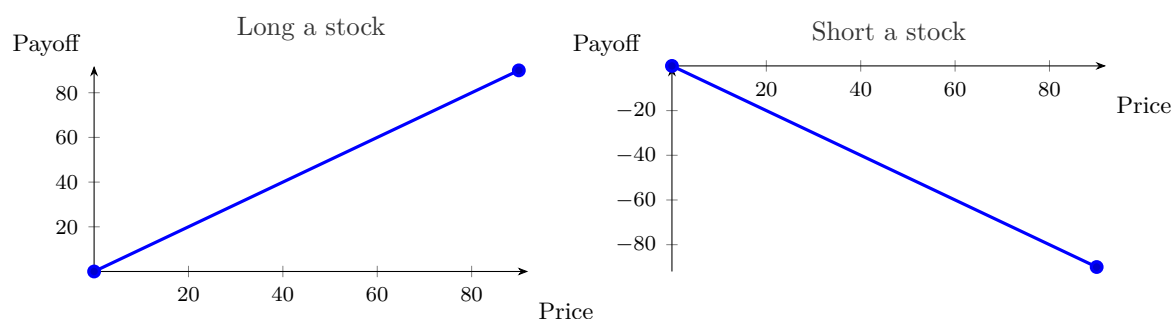


Figure 6.5.: Payoff of a stock position. Unlike the option payoffs of Figure 6.3 these are straight lines through the origin — the stock has no kink, and hence no optionality.

⁴Invented by Le Maire (1558–1624) in 1608.

Note 6.7.2. Two warnings, both of which the models we build will quietly ignore.

Not all assets can be shorted — there must be a lender, and lendable inventory is neither free nor guaranteed.

In reality, short selling is not simply the opposite of going long.⁵ The short seller pays a borrowing fee, can be forced to close the position when the lender recalls the shares, faces an unbounded loss, and may be barred from shorting altogether by regulators in a falling market.

§6.8. Covered Positions: Hedges

— Slides pp. 204–205 —

A **covered position** is an option combined with other securities. Three families are worth naming: *hedges*, which combine an option with its own underlying stock; *spreads* (Section 6.9), which combine options of the *same* type; and *combinations* (Section 6.10), which mix calls with puts.

Definition 6.8.1. A **hedge** combines an option with its underlying stock in such a way that one protects the other against loss.

Two hedges account for almost all practical use:

- **Covered call:** a long position in the stock together with a *short* call.⁶ It is called “covered” because the stock on hand can be delivered to the buyer of the call if the call is exercised — the writer is never forced to buy at an arbitrarily high market price.
- **Protective put:** a long position in the stock together with a *long* put. The put is an insurance policy on the shares: whatever happens, they can be sold for at least X .

Writing S_T for the stock price at expiration, the two payoffs are

$$\begin{aligned} \text{covered call: } S_T - \max(0, S_T - X) &= \min(S_T, X), \\ \text{protective put: } S_T + \max(0, X - S_T) &= \max(S_T, X). \end{aligned} \tag{6.3}$$

So the covered call *caps* the outcome at X and the protective put *floors* it at X .

Note 6.8.2. Both strategies break even only if the stock price rises above a certain level, so **both are bullish**. This is easy to miss, because a covered call is often described as “defensive”. The premium received merely offsets a fixed amount of downside; the position still loses roughly dollar-for-dollar with the stock once that cushion is gone, as the left-hand tail in Figure 6.6 shows.

⁵Kosowski & Neftci (2015). See <https://tw.news.appledaily.com/headline/daily/20180307/37950481/> for an example in Taiwan on February 6, 2018.

⁶A short position has a payoff opposite in sign to that of a long position, as in Figure 6.3. Some ETFs package this payoff directly, such as the Global X Nasdaq 100 Covered Call ETF (QYLD).

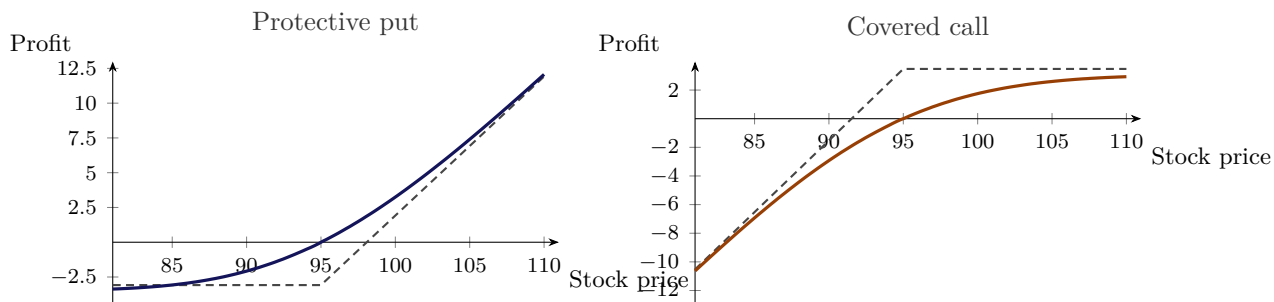


Figure 6.6.: Profit of the two hedges, with $X = 95$ and the portfolio set up when $S = 95$ one month before maturity. Dashed lines are profits at expiration, given by (6.3) less the setup cost. Solid lines are profits at the moment the portfolio is set up, as a function of where the stock price goes; they are smooth because the options still carry time value. Both pictures slope upward on the left, which is what “bullish” means.

§6.9. Covered Positions: Spreads

— Slides p. 206 —

Definition 6.9.1. A **spread** consists of options of the same type and on the same underlying asset, but with different strike prices or expiration dates.

Throughout this section we use X_L , X_M , and X_H to denote strike prices with

$$X_L < X_M < X_H.$$

All options in a spread are calls, or all are puts; mixing types produces a *combination* instead, which is Section 6.10.

§6.9.i. Bull Spreads

— Slides pp. 207–208 —

Definition 6.9.2. A **bull call spread** consists of a long X_L call and a short X_H call with the same expiration date.

Since $C_L > C_H$ (a lower strike is worth more), the position costs money to enter. Its accounting is completely determined by (6.1):

- The initial investment is $C_L - C_H$.
- The payoff is nonnegative; explicitly it is $\max(0, S_T - X_L) - \max(0, S_T - X_H)$.
- The maximum payoff is $X_H - X_L$, achieved when both calls are exercised at expiration, i.e. when $S_T \geq X_H$.
- The maximum profit is therefore $(X_H - X_L) - (C_L - C_H)$.
- The maximum loss is $C_L - C_H$, incurred when neither is exercised at expiration, i.e. when $S_T \leq X_L$.

Both the gain and the loss are bounded: the short call pays for part of the long call and, in exchange, sells away every gain above X_H . The investor is betting that the stock rises, but only so far.

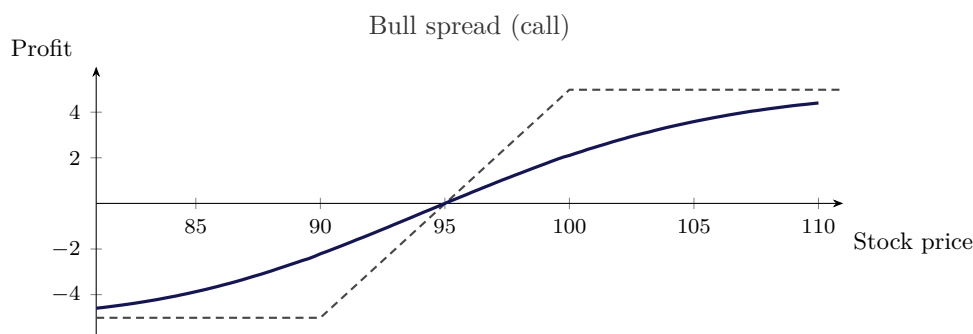


Figure 6.7.: Profit of a bull call spread with $X_L = 90$, $X_H = 100$, entered when $S = 95$ one month before expiration for a net debit of $C_L - C_H = 5.02$. Dashed: profit at expiration, a ramp clipped at both ends. Solid: profit at the moment of entry as the stock price varies. The maximum profit 4.98 and the maximum loss 5.02 are the two dashed plateaus.

§6.9.ii. The Binary Call: A Model-Free Price

— Slides p. 209 —

Now take the bull call spread to a limit. Buy $(X_H - X_L)^{-1}$ units of it, so that the ramp in Figure 6.7 always rises by exactly 1 over the interval $[X_L, X_H]$, and let $X_H - X_L \rightarrow 0$. The ramp becomes vertical, and a (Heaviside) step function emerges as the payoff:

$$\text{payoff} \rightarrow \begin{cases} 1, & S_T > X_L, \\ 0, & S_T < X_L. \end{cases}$$

Definition 6.9.3. The security with this payoff is the **binary** (or **digital**) **call**: it pays \$1 if the stock finishes above the strike, and nothing otherwise.

Its price is the limit of the prices of the spreads that converge to it. Writing $C(X)$ for the price of a standard call struck at X , that limit is a difference quotient:

$$\frac{C(X_L) - C(X_H)}{X_H - X_L} \xrightarrow{X_H \rightarrow X_L} -\frac{\partial C}{\partial X}.$$

So the binary call costs

$$-\frac{\partial C}{\partial X} \tag{6.4}$$

today, where C is the price of the standard call.

(6.4) is model independent.

We assumed no stochastic process, no volatility, no interest rate — only that standard calls of every strike trade at some price $C(X)$, and that a portfolio's price is the sum of its parts. Given the observed call prices, the binary call's price is *forced*.

This is the first appearance of a theme that organises the whole course: some facts follow from no-arbitrage alone, and only the remaining facts need a model.

§6.9.iii. Bear Spreads

— Slides p. 210 —

The same directional bet can be built out of puts, and reversed.

- Writing an X_H put and buying an X_L put with identical expiration date creates the **bull put spread**. This one is entered for a net *credit*, since $P_H > P_L$, but its profit profile has the same shape as Figure 6.7.
- A **bear spread** amounts to selling a bull spread. It profits from declining stock prices.

Note 6.9.4. A bull put spread collects cash up front and, most of the time, expires worthless — which is precisely what makes it dangerous. Its maximum loss, $X_H - X_L$ less the credit, is many times the credit received, so a single move can wipe out a long string of profitable months.⁷

§6.9.iv. Butterfly Spreads

— Slides pp. 211–212 —

Three calls or three puts with different strike prices and the same expiration date create a **butterfly spread**. In the call version the spread is

long one X_L call, long one X_H call, short two X_M calls.

Equivalently — and this is the useful way to see it — a butterfly is

$$\underbrace{[\text{long } X_L \text{ call} + \text{short } X_M \text{ call}]}_{\text{bull call spread } (X_L, X_M)} + \underbrace{[\text{short } X_M \text{ call} + \text{long } X_H \text{ call}]}_{\text{short a bull call spread } (X_M, X_H)},$$

that is, long a bull call spread with strikes X_L and X_M and short a bull call spread with strikes X_M and X_H . The first spread's ramp goes up over $[X_L, X_M]$ and the second's comes down over $[X_M, X_H]$, so the payoffs stack into a tent.

A butterfly spread has a positive payoff at expiration only if the asset price falls between X_L and X_H . It is therefore a bet on the stock *not moving*, as opposed to the directional bets above.

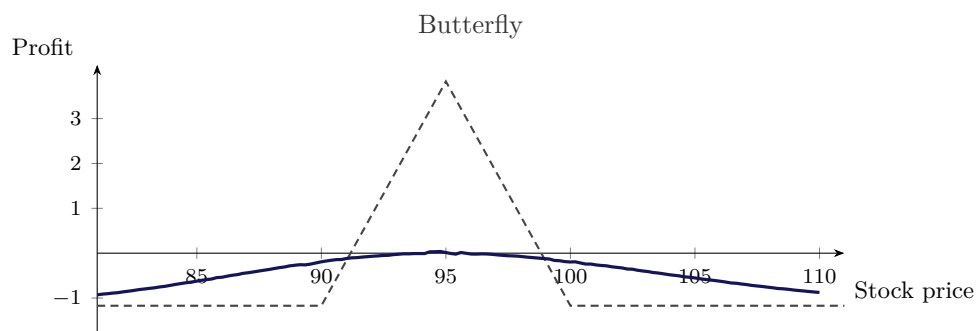


Figure 6.8.: Profit of a butterfly spread with $X_L = 90$, $X_M = 95$, $X_H = 100$, entered when $S = 95$ one month before expiration at a cost of 1.17. Dashed: the tent-shaped profit at expiration, positive only between X_L and X_H . Solid: profit at entry, which is nearly flat — with a month of time value still in the options, the market has not yet decided the bet.

⁷See <https://www.businesstoday.com.tw/article/category/80392/post/201803070> for a sad example in Taiwan on February 6, 2018.

§6.9.v. State Contingent Claims and State Prices

— Slides p. 213 —

The binary call came from squeezing a bull spread; squeezing a butterfly gives something sharper still.

Assume $X_M = (X_H + X_L)/2$, so the tent in Figure 6.8 is symmetric with half-width $X_M - X_L$ and height $X_M - X_L$. Take a position in $(X_M - X_L)^{-2}$ units of the butterfly spread: the tent now has unit *area* whatever its width. When $X_H - X_L \rightarrow 0$ the tent becomes infinitely tall and infinitely thin, and in the limit it approximates a security that pays \$1 only in the state $S = X_M$.⁸

Definition 6.9.5. A security that pays \$1 in one state of the world and nothing in any other is a **state contingent claim**, or an **Arrow security** in honour of Kenneth Arrow (1921–2017).⁹ Its price is called a **state price**.¹⁰

Its price follows from the same difference-quotient argument that produced (6.4), applied twice — for the butterfly is precisely a second difference of call prices in the strike. Writing $h = X_M - X_L$ and $X = X_M$, the cost of h^{-2} butterflies is

$$\frac{C(X-h) - 2C(X) + C(X+h)}{h^2} \xrightarrow{h \rightarrow 0} \frac{\partial^2 C}{\partial X^2}, \quad (6.5)$$

so the state price at X is $\partial^2 C / \partial X^2$.¹¹ Like (6.4), (6.5) is model independent.

State prices make pricing *linear*. Any payoff whatsoever can be chopped into state contingent claims — one for each terminal stock price — and priced by adding up the pieces, so pricing becomes integration against (6.5). Every valuation formula in the remainder of this course is an instance of that one idea.

§6.9.vi. The State Price Density

— Slides p. 214 —

(6.5) is a *price*, and a price is a present value. Undo the discounting — take the future value of $\partial^2 C / \partial X^2$ over the life of the option — and what is left is no longer a price at all.

Theorem 6.9.6 (Breedon & Litzenberger, 1978)

The future value of $\partial^2 C / \partial X^2$ is the probability density of the stock price $S_T = X$ at the option's maturity.

The reasoning is short once state prices are in hand. A claim that pays \$1 when S_T lands in a narrow band around X is worth the discounted value of what it is expected to pay, and what it is expected to pay is the probability of landing there. Hence

$$\underbrace{\frac{\partial^2 C}{\partial X^2}}_{\text{state price}} = e^{-r\tau} \underbrace{f(X)}_{\text{density}}, \quad \text{i.e.} \quad f(X) = e^{r\tau} \frac{\partial^2 C}{\partial X^2},$$

where τ is the time to maturity. Since (6.5) needed no model, neither does this: **the formula is model independent**, and a full set of quoted call prices determines the whole density.

⁸See Exercise 7.4.5 of the textbook.

⁹Co-winner of the 1972 Nobel Prize in Economic Sciences.

¹⁰As a function of the state it is the *state-price density* (Aït-Sahalia & Lo, 1998).

¹¹One can also use the put; see Exercise 9.3.6 of the textbook.

Note 6.9.7. The f recovered this way is not the density under which the stock “really” evolves. It is the density the market *prices with* — the **risk-neutral** density, which folds the market’s risk preferences into the probabilities. The distinction is made precise in [Section 8.6](#), where the risk-neutral probability is constructed explicitly; for now the point is only that a density is there and that it is observable.

The practical reading of [Theorem 6.9.6](#) closes the loop with [Figure 6.8](#). A butterfly spread centred at X is, up to scale, a bet on $S_T \approx X$, and [Theorem 6.9.6](#) says what the market charges for that bet. So:

buy a butterfly spread if you believe the probability of $S_T \approx X$ is higher than the probability the market is quoting.

An options market is not merely a collection of bets — it is a *published forecast*. Every strike quoted is one more point of the distribution, and [Theorem 6.9.6](#) turns the quotes into the density with nothing but arithmetic. Trading an option is therefore trading a disagreement with that forecast, and the butterfly is the instrument for disagreeing about one particular outcome.

§6.10. Covered Positions: Combinations

— Slides p. 215 —

Spreads bet on *direction* — where the stock goes. Combinations bet on *magnitude* — how far it goes, in either direction. Getting that distinction requires mixing the two option types, which is exactly what the definition permits.

Definition 6.10.1. A **combination** consists of options of different types on the same underlying asset. These options must be either all bought or all written.

Note 6.10.2. The “all bought or all written” clause is what separates a combination from the hedges of [Section 6.8](#) and the spreads of [Section 6.9](#). Buying a call and writing a put is not a combination; by (6.1) and (6.2) its payoff is $S_T - X$, a straight line with no kink at all — the position is a synthetic forward, not an option strategy. Keeping all the legs on the same side is what preserves the V-shape.

§6.10.i. Straddles

— Slides pp. 215–216 —

Definition 6.10.3. A **straddle** is a long call and a long put with the same strike price and expiration date.

By (6.1) and (6.2) the payoff is

$$\max(0, S_T - X) + \max(0, X - S_T) = |S_T - X|, \quad (6.6)$$

a V with its vertex at the strike. The holder is paid for movement and is indifferent to its direction; the loss is capped at the two premiums $C + P$, incurred only if the stock finishes exactly at X .

Since it profits from high volatility, a person who buys a straddle is said to be **long volatility**. Selling a straddle benefits from low volatility — the writer collects both premiums and keeps them if the stock stays put.

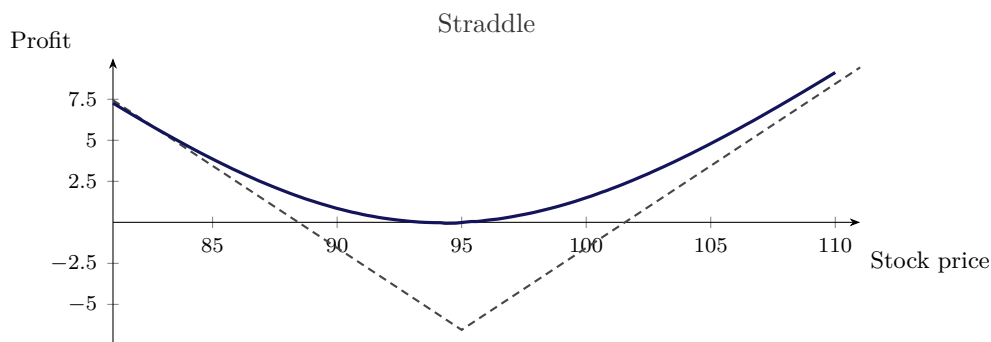


Figure 6.9.: Profit of a straddle with $X = 95$, bought when $S = 95$ one month before expiration for $C + P = 6.56$. Dashed: the V-shaped profit at expiration, (6.6) less the two premiums, with break-evens at $X \pm (C + P)$. Solid: profit at entry. The solid curve rises above the dashed one at both ends and sits above its vertex — a straddle held rather than exercised loses only time value, which is why the position is a bet on volatility and not on price.

§6.10.ii. Strangles

— Slides pp. 217–218 —

Definition 6.10.4. A **strangle** is identical to a straddle except that the call's strike price is higher than the put's.

Writing X_L for the put's strike and X_H for the call's, the payoff is

$$\max(0, X_L - S_T) + \max(0, S_T - X_H), \quad (6.7)$$

which is zero on the whole interval $[X_L, X_H]$ and rises with slope 1 outside it. The V of the straddle has been pulled apart into a flat-bottomed bucket.

Both options are now out of the money, so the strangle is cheaper than the straddle — but the stock has to move further before it pays anything. It is the same bet on volatility, bought at a discount and with a wider dead zone.

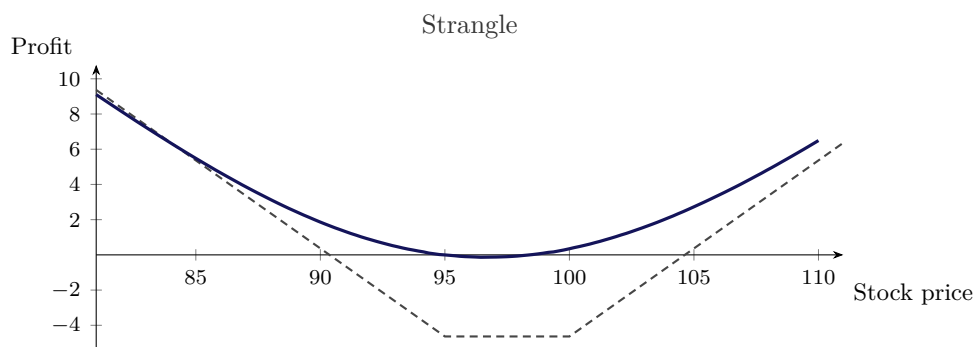


Figure 6.10.: Profit of a strangle with put strike $X_L = 95$ and call strike $X_H = 100$, bought when $S = 95$ one month before expiration for $P_L + C_H = 4.63$ — appreciably less than the straddle's 6.56. Dashed: profit at expiration, flat across $[X_L, X_H]$. Solid: profit at entry. Compared with Figure 6.9 the maximum loss is smaller but the break-evens are further apart.

Read [Figure 6.8](#), [Figure 6.9](#) and [Figure 6.10](#) side by side and the whole section collapses into one picture.

The straddle and the strangle are *long* volatility: they pay when the stock leaves a neighbourhood of the strike. The butterfly is *short* volatility: it pays when the stock stays inside one. Their profit diagrams are each other turned upside down, and [Theorem 6.9.6](#) says what both cost — the market's own probability of the stock staying put.

Direction is what spreads trade; magnitude is what combinations trade. Everything else in this section is bookkeeping around those two.

7 ■ Arbitrage in Option Pricing

— Slides pp. 219–220 —

*All general laws are
attended with inconveniences,
when applied to particular cases.*
— David Hume (1711–1776)

*The problem with QE is
it works in practice,
but it doesn't work in theory.*
— Ben Bernanke¹ (2014)

Chapter 6 described options without pricing any of them, and twice stopped just short of a price: the binary call cost $-\partial C/\partial X$ and the state contingent claim cost $\partial^2 C/\partial X^2$, both *relative* to quoted call prices, both obtained without a model. This chapter is that argument taken seriously.

Its single tool is the observation that a portfolio costing nothing and never losing money cannot exist. From that one sentence we will get the present-value formula we have been assuming since Chapter 2, a relation between every call and every put with the same terms, sharp bounds on both, and a complete answer to when an American call should be exercised early. None of it requires knowing how the stock price moves — which is exactly why these results survive every model the later chapters will propose and discard.

§7.1. Arbitrage

— Slides p. 221 —

The **no-arbitrage principle** says there is no free lunch. It supplies the argument for option pricing.

Definition 7.1.1. A **riskless arbitrage opportunity** is one that, without any initial investment, generates nonnegative returns under all circumstances and positive returns under some.

In an efficient market such opportunities do not exist — at least not for long.²

Note 7.1.2. Read the definition carefully, because the two quantifiers do all the work. *Non-negative under all circumstances* is what makes the profit riskless; *positive under some* is what makes it a profit. Drop the first and you have an ordinary speculation; drop the second and you have a portfolio that does nothing.

Note also what is *not* claimed: not that arbitrage never happens, only that it is competed away fast enough that prices may be computed as if it did not. That is an assumption about markets, not a theorem, and it is the one assumption every result below rests on.

¹Co-winner of the 2022 Nobel Prize in Economic Sciences.

²Heakal (2013): “In the real world of investments, however, there are obvious arguments against the EMH [efficient market hypothesis]. There are investors who have beaten the market—Warren Buffett.”

§7.2. The Portfolio Dominance Principle

— Slides pp. 222–223 —

The no-arbitrage principle is easier to apply in the following comparative form.

Proposition 7.2.1 (Portfolio dominance principle)

Consider two portfolios A and B . Suppose A 's payoff is at least as good as B 's under all circumstances, and better under some. Then A should be more valuable than B .

Proof. If not, buy A and short B . The initial cash flow is nonnegative, every future cash flow is nonnegative, and at least one is positive — a riskless arbitrage opportunity. ■

Two corollaries do almost all the work in this chapter.

Corollary 7.2.2

A portfolio yielding a zero return in every possible scenario must have a zero PV.^a

^aLyft, Inc. (2019): “We have incurred net losses each year since our inception and we may not be able to achieve or maintain profitability in the future.”

Proof. Short the portfolio if its PV is positive; buy it if its PV is negative. In both cases a free lunch is created, since the portfolio itself never pays or costs anything afterwards. ■

Corollary 7.2.3

Two portfolios that yield the same return at time T must have the same price before time T .^a

^aAristotle: “those who are equal should have everything alike.”

Proof. Their difference is a portfolio with zero return in every scenario; apply [Corollary 7.2.2](#). ■

Corollary 7.2.3 is the engine of everything that follows, and it deserves its own sentence: identical cash flows must carry identical prices.

Each proof below therefore has the same three steps — build a portfolio, show its payoff matches something whose price is known, conclude the prices agree. The ingenuity is never in the argument; it is in choosing the portfolio.

§7.3. The PV Formula Justified

— Slides pp. 224–226 —

We can now pay a debt outstanding since Chapter 2. Equations (2.6) and (4.5) were asserted, with a promise that they would be derived rather than assumed. Here is the derivation, and it uses nothing but [Section 7.1](#).

Theorem 7.3.1

For a certain cash flow $C_1, C_2, \dots, C_n$,

$$P = \sum_{i=1}^n C_i d(i).$$

Proof. Write P for the right-hand side and P^* for the security's market price, and suppose first that $P^* < P$.

Short the n zero-coupon bonds that match the security's n cash flows³ — that is, short C_i face value of the i -period zero for each i . Since the i -period zero sells for $d(i)$ per dollar of face value, the proceeds are exactly $\sum_i C_i d(i) = P$ dollars.

Then use P^* of those proceeds to buy the security (Figure 7.1). At every future date i the security pays C_i , which offsets exactly the obligation C_i owed on the zeros; the net future cash flow is zero at every date. A riskless profit of $P - P^*$ dollars has therefore been realised *now*.

If $P^* > P$, just reverse the trades: buy the zeros and short the security. Hence $P^* = P$. ■

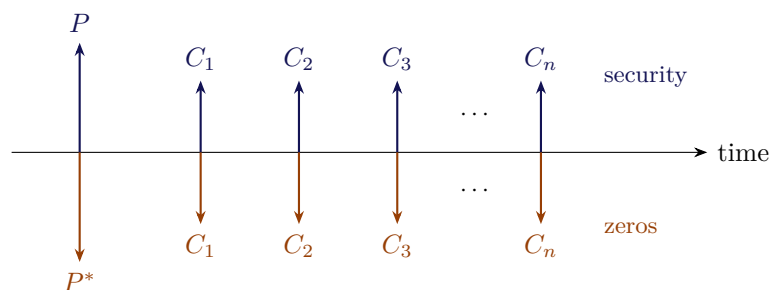


Figure 7.1.: The arbitrage of Theorem 7.3.1 when $P^* < P$. Above the axis are receipts, below it payments. Shorting the matching zeros raises P today against the obligations $C_1, \dots, C_n$; buying the security for P^* supplies exactly those C_i . Every future column cancels, leaving $P - P^* > 0$ in hand today.

Note 7.3.2. The argument is worth one more look, because it is the template for the whole chapter. The security and the matching bundle of zeros have *identical* cash flows at every future date, so by Corollary 7.2.3 they must cost the same today — and the bundle's cost is $\sum_i C_i d(i)$ by construction. The short sale is only the mechanism that converts a price discrepancy into cash; it is Corollary 7.2.3 that does the reasoning.

Here is a second example, on the instruments of Chapter 6.

Theorem 7.3.3

A put or a call must have a nonnegative value.

Proof. Suppose otherwise, so the option has a negative price. Buy it: that is a *positive* cash flow now. By (6.1) and (6.2) the position ends up with a nonnegative amount at expiration, whatever the stock does. So an arbitrage profit is realised now. ■

§7.4. Relative Option Prices

— Slides pp. 228–229 —

Everything from here to the end of the chapter is a *relative* pricing result: it ties the price

³A key assumption: that a zero maturing at each date exists and can be shorted. Compare Section 4.13.ii, where a missing zero is manufactured instead.

of one instrument to the price of another, without ever saying what either is worth in absolute terms. That is the source of their power — **these relations hold regardless of the model for stock prices.**

Notation . We assume throughout, among other things, that

- there are no transaction costs^a or margin requirements;
- borrowing and lending are available at the riskless interest rate;
- interest rates are nonnegative; and
- there are no arbitrage opportunities.

Let the current time be time zero, and write τ for the time to expiration. Then $PV(x)$ stands for the PV of x dollars at expiration, so that by (4.5)

$$PV(x) = x d(\tau). \quad (7.1)$$

^aSchwab cut the fees of online trades of stocks and ETFs to zero on October 7, 2019.

Note 7.4.1. The nonnegative-interest-rate assumption is doing more than it looks: it is what gives $d(\tau) \leq 1$, hence $PV(X) \leq X$. That single inequality is the reason a call is worth more than its intrinsic value and a put can be worth less — see [Lemma 7.6.1](#) and [Lemma 7.6.2](#).

§7.5. Put-Call Parity

— Slides pp. 230–232 —

Theorem 7.5.1 (Put-call parity)

For European options on a non-dividend-paying stock, with common strike price X and time to expiration τ ,^a

$$C = P + S - PV(X). \quad (7.2)$$

^aCastelli (1877).

Proof. Consider the portfolio consisting of

- one short European call,
- one long European put,
- one share of stock, and
- a loan of $PV(X)$.

All options carry the same strike price X and time to expiration τ . The initial cash flow is therefore

$$C - P - S + PV(X).$$

Now look at expiration, when the loan has grown to exactly X .

If $S_\tau \leq X$, the put is worth $X - S_\tau$ and the call expires worthless, so the net future cash flow is

$$0 + (X - S_\tau) + S_\tau - X = 0.$$

If instead $S_\tau > X$, the call is worth $S_\tau - X$ — a liability, since we are short — and the put expires worthless, so the net future cash flow is again

$$-(S_\tau - X) + 0 + S_\tau - X = 0.$$

The net future cash flow is zero in either case. By [Corollary 7.2.2](#) the initial investment must be nil as well, so $C - P - S + PV(X) = 0$, which is (7.2). ■

§7.5.i. Consequences

— Slides p. 233 —

There is only one kind of European option.
The other can be replicated from it in combination with stock and riskless lending or borrowing, so any statement about calls is a statement about puts in disguise. Combinations such as this create synthetic securities — the same move that built the synthetic bonds of Section 4.13.ii.

Rearranging (7.2) reads off the other synthetics:

- $S = C - P + \text{PV}(X)$: a stock is equivalent to a portfolio containing a long call, a short put, and lending $\text{PV}(X)$.
- $C - P = S - \text{PV}(X)$: a long call and a short put amount to a long position in stock together with borrowing the PV of the strike price — that is, buying stock on margin.

Note 7.5.2. The second identity is the one promised in Section 6.10: buying a call and writing a put is not a combination but a synthetic forward, and (7.2) is why its payoff $S_T - X$ carries no kink and costs $S - \text{PV}(X)$ today.

§7.6. Bounds from Parity

— Slides pp. 234–235 —

Parity immediately sharpens what Section 6.5 could only assert by picture.

Lemma 7.6.1

An American call, or a European call on a non-dividend-paying stock, is never worth less than its intrinsic value.

Proof. An American call cannot be worth less than its intrinsic value, since it may be exercised at once for exactly that amount.⁴

For European options, rearrange (7.2) as

$$C = (S - X) + (X - \text{PV}(X)) + P \geq S - X,$$

because $X - \text{PV}(X) \geq 0$ when interest rates are nonnegative and $P \geq 0$ by Theorem 7.3.3. Recalling $C \geq 0$, also from Theorem 7.3.3, it follows that

$$C \geq \max(S - X, 0),$$

the intrinsic value. ■

Such calls are said to have a nonnegative **time value**, defined as $C - \max(S - X, 0)$.

Puts are different, and the difference is visible in Figure 6.4.

Lemma 7.6.2

⁴See Lemma 8.3.1 of the textbook.

For European puts, $P \geq \max(\text{PV}(X) - S, 0)$.

Proof. Rearrange (7.2) as $P = C - S + \text{PV}(X)$ and use $C \geq 0$; combine with $P \geq 0$.⁵ ■

Note 7.6.3. The bound is $\text{PV}(X) - S$, *not* $X - S$, and since $\text{PV}(X) \leq X$ the two differ by the interest on the strike. So a European put on a non-dividend-paying stock may be worth less than its intrinsic value $X - S$, and this is exactly the dip at the left edge of the right-hand panel of Figure 6.4.

The reason is worth stating plainly. Deep in the money, a put is essentially a promise to receive X at expiration — but only *at* expiration. A European holder cannot take the money early, so the position is worth the *discounted* strike, $\text{PV}(X) - S$, not $X - S$. An American put has no such handicap, which is the first hint that early exercise of a put can be worth something.

§7.7. Early Exercise of American Calls

— Slides pp. 236–237 —

Proposition 6.3.2 said an American option is worth at least as much as its European twin but left the gap uncomputed. For calls on a non-dividend-paying stock, parity closes it entirely: **European calls and American calls are identical when the underlying stock pays no dividends.**

Theorem 7.7.1 (Merton, 1973)

An American call on a non-dividend-paying stock should not be exercised before expiration.

Proof. Sharpening Lemma 7.6.1 gives $C \geq \max(S - \text{PV}(X), 0)$.⁶ If the call is exercised, the value realised is $S - X$. But interest rates are nonnegative, so $\text{PV}(X) \leq X$ and therefore

$$\max(S - \text{PV}(X), 0) \geq S - X.$$

Holding is thus worth at least as much as exercising, at every moment before expiration. ■

Remark 7.7.2. The theorem does *not* mean American calls should be kept until maturity. What it does imply is that when early exercise is being considered, a better alternative is to *sell* the call — the market will pay at least the intrinsic value the holder was about to settle for, and generally more.

Early exercise may become optimal for American calls on a *dividend-paying* stock, however. Recall from Section 6.6 that options are assumed unprotected, and that the stock price declines as the stock goes ex-dividend: the holder who does not exercise watches that decline without being compensated for it.

— Slides p. 238 —

Surprisingly, even then an American call should be exercised only at a few dates.

Theorem 7.7.3 (Merton, 1973)

An American call will only be exercised at expiration or just before an ex-dividend date.^a

^aSee Theorem 8.4.2 of the textbook.

⁵See Lemma 8.3.2 of the textbook.

⁶By Exercise 8.3.2 of the textbook.

In contrast, it might be optimal to exercise an American *put* even if the underlying stock does not pay dividends.

The asymmetry between calls and puts is now complete, and it all traces to the sign of $X - PV(X)$.

A call holder pays the strike; waiting earns interest on it, so waiting is (weakly) optimal, and [Theorem 7.7.3](#) reduces the American call to a European one priced at finitely many candidate dates. A put holder *receives* the strike; waiting forgoes that interest, so exercising early can pay.

This is why the American put has no closed-form price and the American call effectively does — and why so much of the numerical machinery in the remaining chapters exists for puts.

§7.8. A General Result

— Slides p. 239 —

One last theorem says how far the replication arguments of this chapter reach.⁷

Theorem 7.8.1 (Ross, 1976; Cox & Rubinstein, 1985)

Any piecewise linear payoff function can be replicated using a portfolio of calls and puts.

Corollary 7.8.2

Any sufficiently well-behaved payoff function can be approximated by a portfolio of calls and puts.

Theorem 7.8.3 (Bakshi & Madan, 2000)

Any payoff function with bounded expectation can be replicated by a continuum of out-of-the-money European calls and puts.

Calls and puts are a *basis*.

[Theorem 7.8.1](#) is visible in every picture of Chapter 6: each kink in a payoff diagram is one option struck at that kink, and the spreads and combinations were nothing but small linear combinations of the basis. [Corollary 7.8.2](#) says a smooth payoff is the limit of such diagrams, and [Theorem 7.8.3](#) makes the limit exact.

Put this next to [Theorem 6.9.6](#) and the two halves of the story meet. Options span the payoffs; their prices reveal the density; and pricing an arbitrary claim is integrating one against the other. Everything after this chapter is the labour of computing that integral for a particular model of the stock price.

⁷See Exercise 8.3.6 of the textbook.

8. Option Pricing Models

— Slides pp. 240–241

*Black insisted that anything one could do
with a mouse could be done better
with macro redefinitions
of particular keys on the keyboard.*
— Emanuel Derman (2004), *My Life as a Quant*
*So we would bring in smart folks.
They didn't know anything about finance.*¹
— James Simons² (2015, May 13, 33:27)

Chapter 7 extracted everything no-arbitrage alone can give: parity, bounds, and the early-exercise rule for calls. It cannot give a number. This chapter supplies the missing ingredient — a model of how the stock price moves — and gets one.

§8.1. The Setting

— Slides p. 242

- The no-arbitrage principle is insufficient to pin down the exact option value.
- We need a model of the probabilistic behaviour of stock prices.
- An obstacle is that it seems a *risk-adjusted* interest rate is needed to discount the option's expected payoff — as in (5.28), where the expected payoff was computed but no rate was named.
- The breakthrough came in 1973, when Black (1938–1995) and Scholes, with help from Merton, published their celebrated option pricing model,³ known as the **Black–Scholes option pricing model**.

Note 8.1.1. The third bullet is the crux, and it is worth dwelling on before the machinery starts.

Discounting an uncertain payoff normally requires a rate that compensates for its risk, and that rate depends on investors' preferences — which nobody can observe. Worse, an option's risk changes continuously as the stock moves, so even a correct rate today would be wrong tomorrow. Every attempt at option pricing before 1973 foundered here.

The resolution, which Section 8.6 reaches, is that the question never has to be answered: the option can be *replicated* by stock and bonds, so its price is the cost of the replicating portfolio, and preferences drop out entirely.

¹<https://www.youtube.com/watch?v=QNznD9hMEh0>

²James Harris Simons (1938–2024) was the founder of Renaissance Technologies. Its Medallion Fund had a 66.1% average gross annual return rate in 1988–2018.

³The results were obtained as early as June 1969. Merton and Scholes were winners of the 1997 Nobel Prize in Economic Sciences.



Fischer Black (1938–1995)



Myron Scholes (1941–)



Robert C. Merton (1944–)

Figure 8.1.: The authors of the 1973 breakthrough. Black had died by 1997 and so could not share the Nobel Prize awarded to Scholes and Merton.

— Slides p. 246 —

Notation . The symbols C , P , X and S keep the meanings fixed in [Section 6.5](#). Two are new:

- $\hat{r} > 0$: the continuously compounded riskless rate *per period*;
- $R \triangleq e^{\hat{r}}$: the **gross return** over one period.

A dollar lent today is thus worth R dollars one period later. We start from the discrete-time binomial model.

§8.2. The Binomial Option Pricing Model

— Slides pp. 247–248 —

Definition 8.2.1. In the **binomial option pricing model** (BOPM), time is discrete and measured in periods. If the current stock price is S , one period later it is either

$$Su \quad \text{with probability } q, \quad \text{or} \quad Sd \quad \text{with probability } 1 - q,$$

where $0 < q < 1$ and $d < u$.

In fact $d \leq R \leq u$ must hold to rule out arbitrage.⁴

Intuition . If $R < d$ the stock beats the bond in *every* state: borrow, buy stock, and collect a riskless profit. If $R > u$ the reverse trade does it. So $d < R < u$ is not a modelling convenience — it is precisely the condition that [Section 7.1](#) is not violated by the model itself.

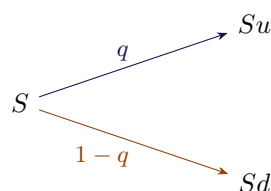


Figure 8.2.: One period of the binomial model. Only the two *multipliers* u and d matter; the current price S scales out.

⁴See Exercise 9.2.1 of the textbook. The sufficient condition is the strict one, $d < R < u$ (Björk, 2009), which we shall assume.

Six pieces of information suffice to determine the option value on arbitrage considerations alone:

S , u , d , X , $\hat{r}$, and the number of periods to expiration.

Conspicuously absent is q — the probability that the stock goes up. Section 8.4 explains why.

§8.3. A Call over a Single Period

— Slides pp. 249–253 —

Take the expiration date to be only one period from now, and write C_u and C_d for the call price at time 1 after an up and a down move. Since time 1 *is* expiration, (6.1) gives them at once:

$$\begin{aligned} C_u &= \max(0, Su - X), \\ C_d &= \max(0, Sd - X). \end{aligned} \tag{8.1}$$

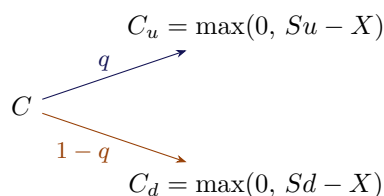


Figure 8.3.: The call written on Figure 8.2. The two terminal values are known; the problem is the value C at the root.

The trick is to build something with the same two terminal values out of instruments whose prices we already know.

Definition 8.3.1. Set up a portfolio of h shares of stock and B dollars in riskless bonds. It costs $hS + B$ today, and h is called the **hedge ratio** or **delta**.

The value of this portfolio at time one is

$$\begin{cases} hSu + RB, & \text{up move,} \\ hSd + RB, & \text{down move,} \end{cases}$$

so choosing h and B to replicate the payoff of the call means solving

$$\begin{aligned} hSu + RB &= C_u, \\ hSd + RB &= C_d. \end{aligned} \tag{8.2}$$

Two linear equations in two unknowns, with a nonsingular matrix because $u \neq d$:

$$h = \frac{C_u - C_d}{Su - Sd} \geq 0, \quad B = \frac{uC_d - dC_u}{(u - d)R}. \tag{8.3}$$

Theorem 8.3.2

By the no-arbitrage principle, the European call must cost the same as the equivalent portfolio:^a

$$C = hS + B. \tag{8.4}$$

^aAlso called the **replicating portfolio**, since it replicates the option.

Proof. By construction the portfolio and the call have identical payoffs in both states at time 1, so [Corollary 7.2.3](#) forces identical prices at time 0. ■

Note 8.3.3. Read off the signs. $h \geq 0$: the replicating portfolio is *long* stock, as one would expect of a call. And since $uC_d - dC_u < 0$ we have $B < 0$, so the equivalent portfolio is a **levered long position in stocks** — buy h shares, borrow $|B|$ to help pay for them.

This also explains an option's reputation for leverage: it packages exactly that trade, and its percentage swings are correspondingly larger than the stock's.

§8.3.i. American Calls and Puts over One Period

— Slides pp. 254–255 —

For an American call we have to consider immediate exercise, so

$$C = \max(hS + B, S - X).$$

When $hS + B \geq S - X$ the call should not be exercised immediately; when $hS + B < S - X$ it should. For non-dividend-paying stocks the first case always holds, by [Theorem 7.7.1](#), so once again $C = hS + B$.

Puts are priced the same way. Their delta is

$$h = \frac{P_u - P_d}{Su - Sd} \leq 0, \quad P_u = \max(0, X - Su), \quad P_d = \max(0, X - Sd),$$

and with $B = (uP_d - dP_u)/((u - d)R)$ the European put is worth $hS + B$, while the American put is worth

$$\max(hS + B, X - S).$$

Note 8.3.4. The max cannot be dropped this time: early exercise *can* be optimal with American puts, exactly as [Section 7.7](#) warned. Note also that $h \leq 0$ — replicating a put means *shorting* stock and lending, the mirror image of the call.

§8.4. Where Did the Risk Go?

— Slides p. 256 —

Look again at [\(8.3\)](#) and [\(8.4\)](#). Surprisingly, the option value is independent of q .⁵ Hence it is also independent of the stock's expected value,

$$qSu + (1 - q)Sd.$$

The option value depends instead on the *sizes* of the price changes, u and d , which the investors must agree upon. Given those, the set of possible stock prices is the same whatever q is.

⁵More precisely, not *directly* dependent on q . Thanks to a lively class discussion on March 16, 2011.

Two investors who violently disagree about whether the stock will rise must nevertheless quote the *same* option price, provided they agree on how far it can move.

The reason is (8.2): the replicating portfolio matches the call in *every* state, so how likely the states are never enters. This is what dissolves the risk-adjusted discount rate problem of Section 8.1 — there is no expectation to discount, only a portfolio to price.

§8.5. Pseudo Probability

— Slides pp. 257–258 —

The formula $C = hS + B$ is correct but opaque. Substituting (8.3) and rearranging gives

$$hS + B = \frac{\frac{R-d}{u-d} C_u + \frac{u-R}{u-d} C_d}{R},$$

whose two coefficients are nonnegative and sum to 1. Naming the first of them,

$$p \triangleq \frac{R-d}{u-d}, \quad (8.5)$$

turns the expression into

$$hS + B = \frac{p C_u + (1-p) C_d}{R}. \quad (8.6)$$

Since $d < R < u$ we have $0 < p < 1$, so p may be interpreted as a probability.

Note 8.5.1. The interpretation is optional. Read without it, the numerator of (8.6) simply *interpolates*: the line through the points (Su, C_u) and (Sd, C_d) , evaluated at SR , has height

$$\frac{R-d}{u-d} C_u + \frac{u-R}{u-d} C_d.$$

So (8.6) says the option is worth the linear interpolation of its payoffs at the stock's *forward* price, discounted back. Nothing probabilistic has happened yet — which is precisely why the next step is legitimate.

§8.6. The Risk-Neutral Probability

— Slides p. 259 —

The quantity p earns its name from one calculation. Under p , the expected rate of return of the stock is exactly the riskless rate:

$$p Su + (1-p) Sd = RS. \quad (8.7)$$

Substituting (8.5) verifies it in one line.

Now, the expected rates of return of *all* securities must equal the riskless rate when investors are risk-neutral — a risk-neutral investor demands no compensation for bearing risk. Equation (8.7) says the stock behaves that way under p , and for this reason p is called the **risk-neutral probability**.

The value of an option is the expectation of its discounted future payoff *in a risk-neutral economy*. So the rate used for discounting is the riskless rate — and the obstacle raised in Section 8.1 is gone, not by being solved but by being made irrelevant.

The risk-neutral economy is a computational fiction. Real investors are not risk-neutral, and $p \neq q$ in general. What (8.6) asserts is only that the *answer* is the same as if they were, because the replicating portfolio does not care either way.

This is also the f of Theorem 6.9.6: the density the market prices with, now constructed rather than merely observed.

§8.7. The Multi-Period Model

— Slides pp. 260–265 —

Consider a call with two periods remaining before expiration. Under the binomial model the stock can take on three possible prices at time two: S_{uu} , S_{ud} and S_{dd} . There are four *paths* — uu , ud , du , dd — but the tree **recombines**, since $ud = du$, so there are only three terminal prices.

At any node the next two stock prices depend only on the current price, not on the prices of earlier times; the model is **Markovian**.

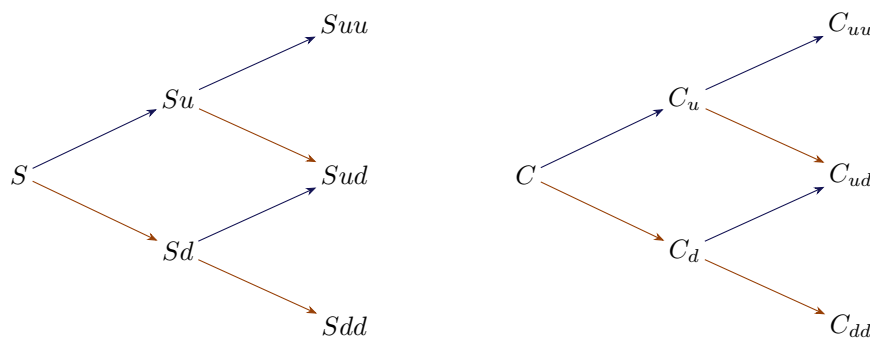


Figure 8.4.: Two periods. Four paths but three terminal states, because the tree recombines. The call tree on the right has exactly the same shape; only the numbers in the nodes differ.

At expiration the call's values are read off (6.1) as before:

$$C_{uu} = \max(0, S_{uu} - X), \quad C_{ud} = \max(0, S_{ud} - X), \quad C_{dd} = \max(0, S_{dd} - X).$$

The call values at time 1 are then obtained by applying (8.6) to each node:

$$C_u = \frac{p C_{uu} + (1-p) C_{ud}}{R}, \quad C_d = \frac{p C_{ud} + (1-p) C_{dd}}{R}. \quad (8.8)$$

Deltas come from (8.3); for example the delta at C_u is $(C_{uu} - C_{ud})/(S_{uu} - S_{ud})$.

We now reach the current period and compute

$$C = \frac{p C_u + (1-p) C_d}{R},$$

with h and B again from (8.3).

§8.7.i. Early Exercise, Again

— Slides p. 266 —

The tree lets us re-derive [Theorem 7.7.1](#) by direct computation. Since the call will not be exercised at time 1 even if it is American, $C_u \geq Su - X$ and $C_d \geq Sd - X$. Therefore

$$hS + B = \frac{pC_u + (1-p)C_d}{R} \geq \frac{[pu + (1-p)d]S - X}{R} = S - \frac{X}{R} > S - X,$$

using (8.7) for the middle equality and $R > 1$ for the last. So the call will not be exercised at present either, and

$$C = hS + B = \frac{pC_u + (1-p)C_d}{R}.$$

§8.8. Backward Induction

— Slides pp. 267–268 —

The expression above calculates C from the two successor nodes C_u and C_d and none beyond; and the same computation happened at C_u and C_d themselves, in (8.8). This recursive procedure is called **backward induction**.⁶

Unrolling it for two periods,

$$\begin{aligned} C &= \left[p^2 C_{uu} + 2p(1-p) C_{ud} + (1-p)^2 C_{dd} \right] / R^2 \\ &= \left[p^2 \max(0, Su^2 - X) + 2p(1-p) \max(0, Sud - X) \right. \\ &\quad \left. + (1-p)^2 \max(0, Sd^2 - X) \right] / R^2. \end{aligned} \quad (8.9)$$

The coefficient 2 on the middle term is the number of paths reaching that state, and it is the only trace the paths leave.

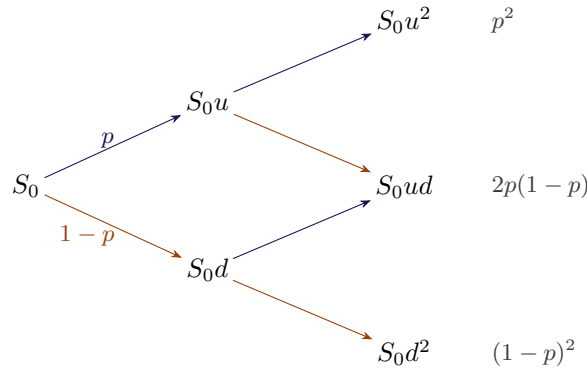


Figure 8.5.: The two-period tree with risk-neutral probabilities. The terminal weights in (8.9) are the binomial probabilities $\binom{2}{j} p^j (1-p)^{2-j}$.

— Slides p. 269 —

In the n -period case the same unrolling gives

$$C = \frac{\sum_{j=0}^n \binom{n}{j} p^j (1-p)^{n-j} \times \max(0, Su^j d^{n-j} - X)}{R^n}, \quad (8.10)$$

⁶Ernst Zermelo (1871–1953).

and similarly for the put,

$$P = \frac{\sum_{j=0}^n \binom{n}{j} p^j (1-p)^{n-j} \times \max(0, X - Su^j d^{n-j})}{R^n}. \quad (8.11)$$

The value of a call on a non-dividend-paying stock is the expected discounted payoff at expiration in a risk-neutral economy — literally the sum in (8.10), with $\binom{n}{j} p^j (1-p)^{n-j}$ as the probability of finishing in state j .

Note what (8.10) is *not*: it is not a shortcut that skips the tree. It is the tree, summed. For a European option the two agree; for an American one only the tree works, because the max against intrinsic value has to be taken at every node.

§8.8.i. Reading the Weights as State Prices

— Slides p. 270 —

Define

$$p_j \triangleq \frac{\binom{n}{j} p^j (1-p)^{n-j}}{R^n}, \quad j = 0, 1, \dots, n. \quad (8.12)$$

This is the *state price* of Section 6.9.vi, now for the state $Su^j d^{n-j}$: the price today of a claim paying \$1 there and nothing elsewhere. In general, then,

$$\text{option price} = \sum_j (p_j \times \text{payoff at state } j),$$

and (8.10) and (8.11) are two instances of it.

Note 8.8.1. The butterfly construction of Section 6.9.vi works here literally, and the discrete states simplify it.⁷ Buy $(X_M - X_L)^{-1}$ units of the butterfly spread with

$$X_L = Su^{j-1} d^{n-j+1}, \quad X_M = Su^j d^{n-j}, \quad X_H = Su^{j+1} d^{n-j-1},$$

that is, struck at the state itself and its two neighbours. No stock price strictly between X_L and X_H is attainable except X_M , so the tent's *whole* payoff is the single value 1 at state j ; the exponent -1 replaces the -2 of the continuous case, where the tent had to be scaled by area instead. Its price is therefore p_j , and $R^n p_j = \binom{n}{j} p^j (1-p)^{n-j}$ is the undiscounted state price — the risk-neutral probability.

§8.9. The Risk-Neutral Pricing Methodology

— Slides p. 271 —

Nothing in the argument used the payoff $\max(0, S - X)$ beyond the fact that it was a function of the terminal stock price. So the conclusion generalises at once.

Every derivative can be priced as if the economy were risk-neutral. For a European-style derivative with terminal payoff function D , its value is

$$e^{-\hat{r}n} E^\pi[D], \quad (8.13)$$

where E^π means the expectation is taken under the risk-neutral probability.

⁷Bahra (1997).

The “equivalence” between arbitrage freedom in a model and the existence of a risk-neutral probability is called the **(first) fundamental theorem of asset pricing**.⁸



Philip H. Dybvig (1955–)



Stephen Ross (1944–2017)

Figure 8.6.: Dybvig, co-winner of the 2022 Nobel Prize in Economic Sciences, and Ross — whose 1976 replication theorem already appeared as **Theorem 7.8.1**.

Note 8.9.1. The theorem closes the circle this course opened with. Chapter 7 assumed no arbitrage and extracted relations between prices; **Section 8.6** built a probability π out of a particular model. The fundamental theorem says these are two descriptions of the same thing: a model admits no arbitrage *exactly* when such a π exists. Pricing a derivative and checking a model for consistency are therefore one problem.

§8.10. Self-Financing

— Slides p. 274 —

One loose end remains. The delta h of (8.3) is computed node by node, and it changes as the stock moves, so the maintenance of an equivalent portfolio is *dynamic*: the replicating position must be rebalanced every period.

That sounds expensive, and it raises the obvious worry — if rebalancing requires fresh money, the option’s price is not really $hS + B$. It does not.

Proposition 8.10.1

The replicating strategy is **self-financing**: there is neither injection nor withdrawal of funds throughout.^a

^aExcept at the beginning, of course, when the option premium is paid before the replication starts.

The reason is already in (8.2). At the end of the current period the portfolio is worth $hSu + RB$ or $hSd + RB$, which *is* C_u or C_d — and C_u and C_d are, by (8.4) applied one period later, exactly the cost of the next portfolio. The portfolio’s value at the end of each period is therefore precisely the amount needed to set up the next one. Changes in value are due entirely to capital gains.

Rebalancing does not depend on predicting future stock prices.

The delta at each node is a function of where the stock *is*, not of where it is going; the strategy is told what to do by the tree, and it always has exactly enough money to do it. That is the whole content of “the option is worth its replicating portfolio”, and it is what makes (8.13) a hedging recipe rather than merely a formula.

⁸Dybvig & Ross (1987).